



CLINICAL FOCUS

Vaccination: Challenges in the U.S. and Developing Countries

Many previously common childhood diseases are seldom seen in the United States, a testament to the effectiveness of vaccination. A major barrier to similar success in the rest of the world is the difficulty of delivering vaccines to all children. However, even at home the U.S. is becoming a victim of its own success. Some parents who have never encountered diseases now nearly vanquished in the U.S. do not consider it important to have their infants vaccinated or they may be lax in adhering to recommended schedules of immunization. Others hold the uninformed belief that the risks associated with vaccination outweigh the risk of infection. This flawed reasoning is fueled by periodic allegations of linkage between vaccination and various disorders, such as the report circulating

recently of a causal relationship between vaccination and autism, a condition of unknown etiology. Most such reports are based solely on the coincidental timing of vaccination and onset of disease, or on limited sampling and poor statistical analyses. So far, no alleged associations have withstood scrutiny that included large population samples and acceptable statistical methods.

While children in this country are protected against a variety of once-deadly diseases, this protection depends on continuation of our immunization programs. Dependency on herd immunity is dangerous for both the individual and society. Adverse reactions to vaccines must be examined thoroughly, of course, and if a vaccine causes unacceptable side reactions, the vaccination program must be reconsidered. At the same time, anecdotal reports of disease brought on

by vaccines, and unsupported beliefs, such as the contention that vaccines weaken the immune system, must be countered by correct information from trusted sources. To retreat from our progress in immunization by noncompliance will return us to the age when measles, mumps, whooping cough, and polio were part of the risk of growing up.

Children in the developing world suffer from a problem different from those in the United States. Examination of infant deaths worldwide shows that existing vaccines could save the lives of millions of children. As seen in the table, there are safe, effective vaccines for five of the top ten killers of children. Although the list of diseases in the table includes HIV, TB, and malaria, for which no vaccines are available, administration of the vaccines that are recommended for infants in the United States could cut child mortality in the world by approximately half.

What barriers exist to the achievement of worldwide vaccination and complete eradication of many childhood diseases? The inability to achieve higher levels of

Genetic engineering techniques can be used to develop vaccines to maximize the immune response to selected epitopes and to simplify delivery of the vaccines. This chapter describes the vaccines now in use and describes vaccine strategies, including experimental designs that may lead to the vaccines of the future.

Active and Passive Immunization

Immunity to infectious microorganisms can be achieved by active or passive **immunization**. In each case, immunity can be acquired either by natural processes (usually by transfer from mother to fetus or by previous infection by the organism) or by artificial means such as injection of antibodies or vaccines (Table 18-1, on page 416). The agents used for inducing passive immunity include antibodies from humans or animals, whereas active immunization is achieved by inoculation with microbial pathogens that induce immunity but

do not cause disease or with antigenic components from the pathogens. This section describes current usage of passive and active immunization techniques.

Passive Immunization Involves Transfer of Preformed Antibodies

Jenner and Pasteur are recognized as the pioneers of vaccination, or induction of active immunity, but similar recognition is due to Emil von Behring and Hidesaburo Kitasato for their contributions to passive immunity. These investigators were the first to show that immunity elicited in one animal can be transferred to another by injecting it with serum from the first (see Clinical Focus, Chapter 4).

Passive immunization, in which preformed antibodies are transferred to a recipient, occurs naturally by transfer of maternal antibodies across the placenta to the developing fetus. Maternal antibodies to diphtheria, tetanus, streptococci, rubeola, rubella, mumps, and poliovirus all afford passively acquired protection to the developing fetus. Maternal

vaccination even in the United States is an indication of the difficulty of the task. Even if we assume that suitable vaccines have been developed and that compliance is universal, the ability to produce and deliver the vaccines everywhere is a profound challenge. The World Health Organization (WHO) has stated that the ideal vaccine would have the following properties:

- Affordable worldwide
- Heat stable
- Effective after a single dose
- Applicable to a number of diseases
- Administered by a mucosal route
- Suitable for administration early in life

Few, if any, vaccines in common use today conform to all of these properties. However, the WHO goals can guide us in the pursuit of vaccines useful for worldwide application. They further aid us in setting priorities, especially for development of the vaccines needed most in developing countries. For example, an HIV/AIDS vaccine that meets the WHO criteria could have an immediate effect on the world AIDS epidemic, whereas

one that does not will require further development before it reaches the populations most at risk.

Immunization saves millions of lives, and viable vaccines are increasingly avail-

able. The challenge to the biomedical research community is to develop better, safer, cheaper, easier-to-administer forms of these vaccines so that worldwide immunization becomes a reality.

Estimated annual deaths worldwide of children under 5 years of age, by pathogen

Pathogen	Deaths (millions)
<i>Pneumococcus</i> *	1.2
Measles	1.1
<i>Hemophilus (a-f, nst)</i>	0.9
Rotavirus**	0.8
Malaria	0.7
HIV	0.5
RSV	0.5
Pertussis	0.4
Tetanus	0.4
Tuberculosis	0.1

*Pathogens shown in bold are those for which an effective vaccine exists.

**A licensed vaccine is being tested for possible side-effects.

SOURCE: Adapted from Shann and Steinhoff, 1999, *Lancet* **354** (Suppl II):7–11.

antibodies present in colostrum and milk also provide passive immunity to the infant.

Passive immunization can also be achieved by injecting a recipient with preformed antibodies. In the past, before vaccines and antibiotics became available, passive immunization provided a major defense against various infectious diseases. Despite the risks (see Chapter 16) incurred by injecting animal sera, usually horse serum, this was the only effective therapy for otherwise fatal diseases. Currently, there are several conditions that warrant the use of passive immunization. These include:

- Deficiency in synthesis of antibody as a result of congenital or acquired B-cell defects, alone or together with other immunodeficiencies.
- Exposure or likely exposure to a disease that will cause complications (e.g., a child with leukemia exposed to varicella or measles), or when time does not permit adequate protection by active immunization.

- Infection by pathogens whose effects may be ameliorated by antibody. For example, if individuals who have not received up-to-date active immunization against tetanus suffer a puncture wound, they are given an injection of horse antiserum to tetanus toxin. The preformed horse antibody neutralizes any tetanus toxin produced by *Clostridium tetani* in the wound.

Passive immunization is routinely administered to individuals exposed to botulism, tetanus, diphtheria, hepatitis, measles, and rabies (Table 18-2). Passively administered antiserum is also used to provide protection from poisonous snake and insect bites. Passive immunization can provide immediate protection to travelers or health-care workers who will soon be exposed to an infectious organism and lack active immunity to it. Because passive immunization does not activate the immune system, it generates no memory response and the protection provided is transient.

For certain diseases such as the acute respiratory failure in children caused by respiratory syncytial virus (RSV), passive

TABLE 18-1

Acquisition of passive and active immunity

Type	Acquired through
Passive immunity	Natural maternal antibody
	Immune globulin*
	Humanized monoclonal antibody
	Antitoxin [†]
Active immunity	Natural infection
	Vaccines [‡]
	Attenuated organisms
	Inactivated organisms
	Purified microbial macromolecules
	Cloned microbial antigens
	Expressed as recombinant protein
	As cloned DNA alone or in virus vectors
	Multivalent complexes
	Toxoid [§]

*An antibody-containing solution derived from human blood, obtained by cold ethanol fractionation of large pools of plasma; available in intramuscular and intravenous preparations.

[†]An antibody derived from the serum of animals that have been stimulated with specific antigens.

[‡]A suspension of attenuated live or killed microorganisms, or antigenic portions of them, presented to a potential host to induce immunity and prevent disease.

[§]A bacterial toxin that has been modified to be nontoxic but retains the capacity to stimulate the formation of antitoxin.

immunization is the best preventative currently available. A monoclonal antibody or a combination of two monoclonal antibodies may be administered to children at risk for RSV disease. These monoclonal antibodies are prepared in mice but have been “humanized” by splicing the constant regions of human IgG to the mouse variable regions (see Chapter 5). This modification prevents many of the complications that may follow a second injection of the complete mouse antibody, which is a highly immunogenic foreign protein.

Although passive immunization may be an effective treatment, it should be used with caution because certain risks are associated with the injection of preformed antibody. If the antibody was produced in another species, such as a horse, the recipient can mount a strong response to the isotypic determinants of the foreign antibody. This anti-isotype response can cause serious complications. Some individuals, for example, produce IgE antibody

TABLE 18-2

Common agents used for passive immunization

Disease	Agent
Black widow spider bite	Horse antivenin
Botulism	Horse antitoxin
Diphtheria	Horse antitoxin
Hepatitis A and B	Pooled human immune gamma globulin
Measles	Pooled human immune gamma globulin
Rabies	Pooled human immune gamma globulin
Respiratory disease	Monoclonal anti-RSV*
Snake bite	Horse antivenin
Tetanus	Pooled human immune gamma globulin or horse antitoxin

*Respiratory syncytial virus

specific for determinants on the injected antibody. Immune complexes of this IgE bound to the passively administered antibody can mediate systemic mast cell degranulation, leading to systemic anaphylaxis. Other individuals produce IgG or IgM antibodies specific for the foreign antibody, which form complement-activating immune complexes. The deposition of these complexes in the tissues can lead to type III hypersensitive reactions. Even when human gamma globulin is administered passively, the recipient can generate an anti-allotype response to the human immunoglobulin, although its intensity is usually much less than that of an anti-isotype response.

Active Immunization Elicits Long-Term Protection

Whereas the aim of passive immunization is transient protection or alleviation of an existing condition, the goal of active immunization is to elicit protective immunity and immunologic memory. When active immunization is successful, a subsequent exposure to the pathogenic agent elicits a heightened immune response that successfully eliminates the pathogen or prevents disease mediated by its products. Active immunization can be achieved by natural infection with a microorganism, or it can be acquired artificially by administration of a **vaccine** (see Table 18-1). In active immunization, as the name implies, the immune system plays an active role—proliferation of antigen-reactive T and B cells results in the formation of memory cells. Active immunization with various types of vaccines has played an important

role in the reduction of deaths from infectious diseases, especially among children.

Vaccination of children is begun at about 2 months of age. The recommended program of childhood immunizations in this country, updated in 2002 by the American Academy of Pediatrics, is outlined in Table 18-3. The program includes the following vaccines:

- Hepatitis B vaccine
- Diphtheria-pertussis (acellular)-tetanus (DPaT) combined vaccine
- Inactivated (Salk) polio vaccine (IPV); the oral (Sabin) vaccine is no longer recommended for use in the United States
- Measles-mumps-rubella (MMR) combined vaccine
- *Haemophilus influenzae* (Hib) vaccine
- Varicella zoster (Var) vaccine for chickenpox

- Pneumococcal conjugate vaccine (PCV); a new addition to the list.

In addition, hepatitis A vaccine at 18 months and influenza vaccines after 6 months are recommended for infants in high-risk populations.

The introduction and spreading use of various vaccines for childhood immunization has led to a dramatic decrease in the incidence of common childhood diseases in the United States (Figure 18-1). The comparisons of disease incidence in 1999 to that reported in the peak years show dramatic drops and, in one case, complete elimination of the disease in the United States. As long as widespread, effective immunization programs are maintained, the incidence of these childhood diseases should remain low. However, the occurrence of side reactions to a vaccine may cause a drop in its use, which can lead to re-emergence of that disease. For example, the side effects from the pertussis attenuated bacterial vaccine included seizures, encephalitis, brain damage, and even death. Decreased usage of the vaccine led to an increase in the inci-

TABLE 18-3 Recommended childhood immunization schedule in the United States, 2002

Vaccine*	AGE							
	Birth	1 mo	2 mos	4 mos	6 mos	12 mos	15 mos	18 mos 4–6 yrs
Hepatitis B [†]		+						
Diphtheria, tetanus, pertussis [‡]			+	+	+		+	+
<i>H. influenzae</i> , type b			+	+	+	+		
Inactivated polio [§]			+	+	+			+
Pneumococcal conjugate			+	+	+	+		
Measles, mumps, rubella						+		+
Varicella [#]						+		

*This schedule indicates the recommended ages for routine administration of currently licensed childhood vaccines.

Bars indicate ranges of recommended ages. Any dose not given at the recommended age should be given as a "catch-up" immunization at any subsequent visit when indicated and feasible.

[†]Different schedules exist depending upon the HBsAg status of the mother. A first vaccination after the first month is recommended only if the mother is HBsAg negative.

[‡]DTaP (diphtheria and tetanus toxoids and acellular pertussis vaccine) is the preferred vaccine for all doses in the immunization series. Td (tetanus and diphtheria toxoids) is recommended at 11–12 years of age if at least 5 years have elapsed since the last dose.

[§]Only inactivated poliovirus (IPV) vaccine is now recommended for use in the United States. However, OPV remains the vaccine of choice for mass immunization campaigns to control outbreaks due to wild poliovirus.

[#]Varicella (Var) vaccine is recommended at any visit on or after the first birthday for susceptible children, i.e., those who lack a reliable history of chickenpox (as judged by a health-care provider) and who have not been immunized. Susceptible persons 13 years of age or older should receive 2 doses, given at least 4 weeks apart.

SOURCE: Adapted from the ECBT Web site (see references); approved by the American Academy of Pediatrics.

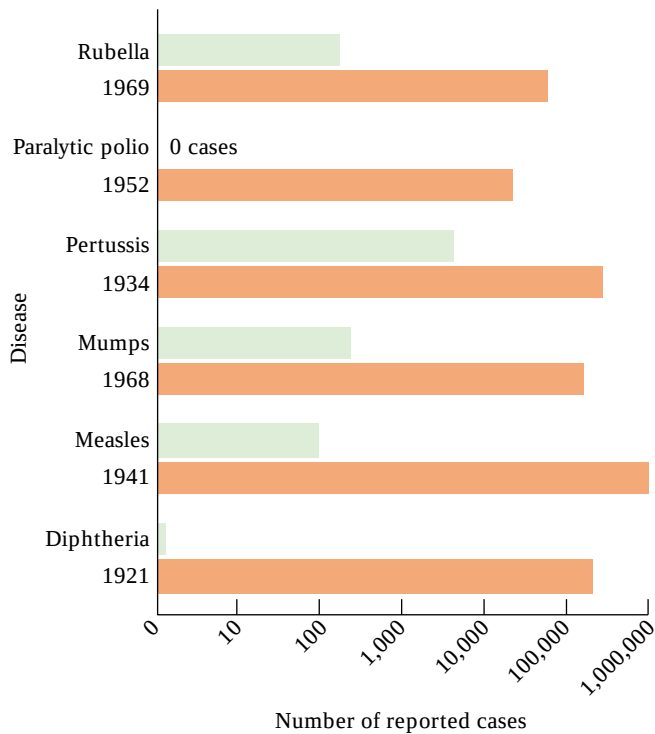


FIGURE 18-1 Reported annual number of cases of rubella (German measles), polio, pertussis (whooping cough), mumps, measles, and diphtheria in the United States in the peak year for which data are available (orange) compared with the number of cases of each disease in 1999 (green). Currently, vaccines are available for each of these diseases, and vaccination is recommended for all children in the United States. [Data from Centers for Disease Control.]

dence of whooping cough, with 7405 cases in 1998. The recent development of an acellular pertussis vaccine that is as effective as the older vaccine, but with none of the side effects, is expected to reverse this trend.

As indicated in Table 18-3, children typically require multiple boosters (repeated inoculations) at appropriately timed intervals to achieve effective immunity. In the first months of life, the reason for this may be persistence of circulating maternal antibodies in the young infant. For example, passively acquired maternal antibodies bind to epitopes on the DPT vaccine and block adequate activation of the immune system; therefore, this vaccine must be given several times after the maternal antibody has been cleared from an infant's circulation in order to achieve adequate immunity. Passively acquired maternal antibody also interferes with the effectiveness of the measles vaccine; for this reason, the MMR vaccine is not given before 12–15 months of age. In Third World countries, however, the measles vaccine is administered at 9 months, even though maternal antibodies are still present, because 30%–50% of young children in these countries contract the disease before 15 months of age.

Multiple immunizations with the polio vaccine are required to ensure that an adequate immune response is generated to each of the three strains of poliovirus that make up the vaccine.

Recommendations for vaccination of adults depend on the risk group. Vaccines for meningitis, pneumonia, and influenza are often given to groups living in close quarters (e.g., military recruits) or to individuals with reduced immunity (e.g., the elderly). Depending on their destination, international travelers are also routinely immunized against such endemic diseases as cholera, yellow fever, plague, typhoid, hepatitis, meningitis, typhus, and polio. Immunization against the deadly disease anthrax had been reserved for workers coming into close contact with infected animals or products from them. Recently, however, suspected use of anthrax spores by terrorists or in biological warfare has widened use of the vaccine to military personnel and civilians in areas at risk of attack with this deadly agent.

Vaccination is not 100% effective. With any vaccine, a small percentage of recipients will respond poorly and therefore will not be adequately protected. This is not a serious problem if the majority of the population is immune to an infectious agent. In this case, the chance of a susceptible individual contacting an infected individual is so low that the susceptible one is not likely to become infected. This phenomenon is known as *herd immunity*. The appearance of measles epidemics among college students and unvaccinated preschool-age children in the United States during the mid-to late 1980s resulted partly from an overall decrease in vaccinations, which had lowered the herd immunity of the population (Figure 18-2). Among preschool-age children, 88% of those who developed measles were unvaccinated. Most of the college students who contracted measles had been vaccinated as children, but only once; the failure of the single vaccination to protect them may have resulted from the presence of passively acquired maternal antibodies that reduced their overall response to the vaccine. The increase in the incidence of measles prompted the recommendation that children receive two immunizations with the combined measles-mumps-rubella vaccine, one at 12–15 months of age and the second at 4–6 years.

The Centers for Disease Control (CDC) has called attention to the decline in vaccination rates and herd immunity among American children. For example, a 1995 publication reported that in California nearly one-third of all infants are unvaccinated and about half of all children under the age of 2 are behind schedule on their vaccinations. Such a decrease in herd immunity portends serious consequences, as illustrated by recent events in the newly independent states of the former Soviet Union. By the mid-1990s, a diphtheria epidemic was raging in many regions of these new countries, linked to a decrease in herd immunity resulting from decreased vaccination rates after the breakup of the Soviet Union. This epidemic, which led to over 157,000 cases of diphtheria and 5000 deaths, is now controlled by mass immunization programs.

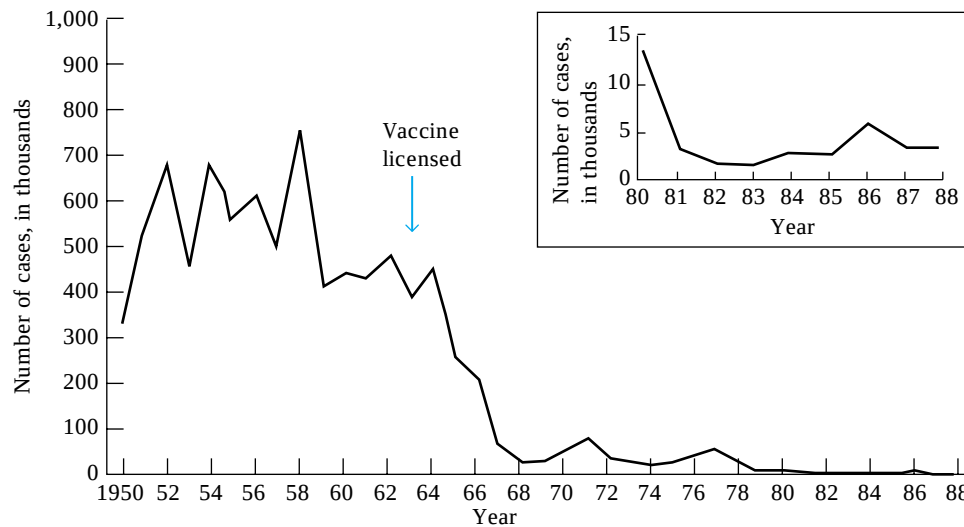


FIGURE 18-2 Introduction of the measles vaccine in 1962 led to a dramatic decrease in the annual incidence of this disease in the United States. Occasional outbreaks of measles in the 1980s (*inset*)

occurred mainly among unvaccinated young children and among college students; most of the latter had been vaccinated, but only once, when they were young. [Data from Centers for Disease Control.]

Designing Vaccines for Active Immunization

Several factors must be kept in mind in developing a successful vaccine. First and foremost, the development of an immune response does not necessarily mean that a state of protective immunity has been achieved. What is often critical is which branch of the immune system is activated, and therefore vaccine designers must recognize the important differences between activation of the humoral and the cell-mediated branches. A second factor is the development of immunologic memory. For example, a vaccine that induces a protective primary response may fail to induce the formation of memory cells, leaving the host unprotected after the primary response to the vaccine subsides.

The role of memory cells in immunity depends, in part, on the incubation period of the pathogen. In the case of influenza virus, which has a very short incubation period (1 or 2 days), disease symptoms are already under way by the time memory cells are activated. Effective protection against influenza therefore depends on maintaining high levels of neutralizing antibody by repeated immunizations; those at highest risk are immunized each year. For pathogens with a longer incubation period, maintaining detectable neutralizing antibody at the time of infection is not necessary. The poliovirus, for example, requires more than 3 days to begin to infect the central nervous system. An incubation period of this length gives the memory B cells time to respond by producing high levels of serum antibody. Thus, the vaccine for polio is designed to induce high levels of immunologic memory. After immunization with the Salk vaccine, serum antibody levels peak within 2 weeks and then decline, but the

memory response continues to climb, reaching maximal levels at 6 months and persisting for years (Figure 18-3). If an immunized individual is later exposed to the poliovirus, these memory cells will respond by differentiating into plasma cells that produce high levels of serum antibody, which defend the individual from the infection.

In the remainder of this chapter, various approaches to the design of vaccines—both currently used vaccines and experimental ones—are described, with an examination of their ability to induce humoral and cell-mediated immunity and the production of memory cells.

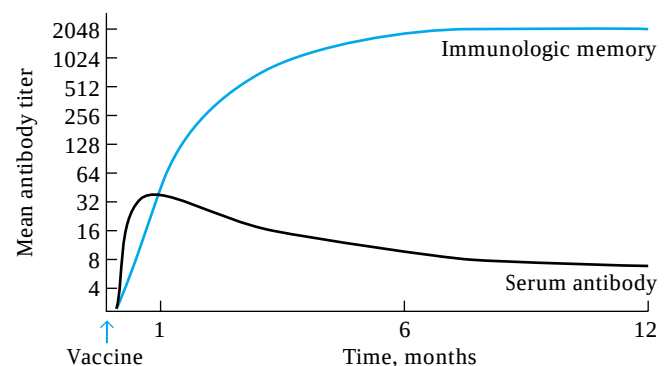


FIGURE 18-3 Immunization with a single dose of the Salk polio vaccine induces a rapid increase in serum antibody levels, which peak by 2 weeks and then decline. Induction of immunologic memory follows a slower time course, reaching maximal levels 6 months after vaccination. The persistence of the memory response for years after primary vaccination is responsible for immunity to poliomyelitis. [From M. Zanetti et al., 1987, *Immunol. Today* 8:18.]

Whole-Organism Vaccines

As Table 18-4 indicates, many of the common vaccines currently in use consist of inactivated (killed) or live but attenuated (avirulent) bacterial cells or viral particles. The primary characteristics of these two types of vaccines are compared in Table 18-5 to one another and to DNA vaccines that are currently being tested for use in humans.

Attenuated Viruses and Bacteria Cause Immunity Without Disease

In some cases, microorganisms can be attenuated so that they lose their ability to cause significant disease (pathogenicity) but retain their capacity for transient growth within an inoculated host. Attenuation often can be achieved by growing a pathogenic bacterium or virus for prolonged periods under abnormal culture conditions. This procedure selects mutants that are better suited to growth in the abnormal culture conditions and are therefore less capable of growth in the natural host. For example, an attenuated strain of *Mycobacterium bovis* called **Bacillus Calmette-Guerin (BCG)** was developed by growing *M. bovis* on a medium containing increasing concentrations of bile. After 13 years, this strain had adapted to growth in strong bile and had become sufficiently attenuated that it was suitable as a vaccine for tuberculosis. The Sabin polio vaccine and the measles vaccine both consist of attenuated viral strains. The poliovirus used in the Sabin vaccine was attenuated by growth in monkey kidney epithelial cells. The measles vaccine contains a strain of rubella virus that was grown in duck embryo cells and later in human cell lines.

Attenuated vaccines have advantages and disadvantages. Because of their capacity for transient growth, such vaccines provide prolonged immune-system exposure to the individual epitopes on the attenuated organisms, resulting in increased immunogenicity and production of memory cells. As a consequence, these vaccines often require only a single immunization, eliminating the need for repeated boosters. This property is a major advantage in Third World countries, where epidemiologic studies have shown that roughly 20% of individuals fail to return for each subsequent booster. The ability of many attenuated vaccines to replicate within host cells makes them particularly suitable for inducing a cell-mediated response.

The Sabin polio vaccine, consisting of three attenuated strains of poliovirus, is administered orally to children on a sugar cube or in sugar liquid. The attenuated viruses colonize the intestine and induce protective immunity to all three strains of virulent poliovirus. Sabin vaccine in the intestines induces production of secretory IgA, which serves as an important defense against naturally acquired poliovirus. The vaccine also induces IgM and IgG classes of antibody. Unlike most other attenuated vaccines, which require a single immunizing dose, the Sabin polio vaccine requires boosters,

TABLE 18-4 Classification of common vaccines for humans

Disease or pathogen	Type of vaccine
WHOLE ORGANISMS	
<i>Bacterial cells</i>	
Anthrax	Inactivated
Cholera	Inactivated
Pertussis*	Inactivated
Plague	Inactivated
Tuberculosis	Live attenuated BCG [†]
Typhoid	Live attenuated
<i>Viral particles</i>	
Hepatitis A	Inactivated
Influenza	Inactivated
Measles	Live attenuated
Mumps	Live attenuated
Polio (Sabin)	Live attenuated
Polio (Salk)	Inactivated
Rabies	Inactivated
Rotavirus	Live attenuated
Rubella	Inactivated
Varicella zoster (chickenpox)	Live attenuated
Yellow fever	Live attenuated
PURIFIED MACROMOLECULES	
<i>Toxoids</i>	
Diphtheria	Inactivated exotoxin
Tetanus	Inactivated exotoxin
<i>Capsular polysaccharides</i>	
<i>Haemophilus influenzae</i> type b	Polysaccharide + protein carrier
<i>Neisseria meningitidis</i>	Polysaccharide
<i>Streptococcus pneumoniae</i>	23 distinct capsular polysaccharides
<i>Surface antigen</i>	
Hepatitis B	Recombinant surface antigen (HBsAg)

*There is now also an acellular pertussis vaccine consisting of toxoids and inactivated bacteria components.

[†]Bacillus Calmette-Guerin (BCG) is an avirulent strain of *Mycobacterium bovis*.

TABLE 18-5 Comparison of attenuated (live), inactivated (killed), and DNA vaccines

Characteristic	Attenuated vaccine	Inactivated vaccine	DNA vaccine
Production	Selection for avirulent organisms: virulent pathogen is grown under adverse culture conditions or prolonged passage of a virulent human pathogen through different hosts	Virulent pathogen is inactivated by chemicals or irradiation with γ -rays	Easily manufactured and purified
Booster requirement	Generally requires only a single booster	Requires multiple boosters	Single injection may suffice
Relative stability	Less stable	More stable	Highly stable
Type of immunity induced	Humoral and cell-mediated	Mainly humoral	Humoral and cell-mediated
Reversion tendency	May revert to virulent form	Cannot revert to virulent form	Cannot revert

because the three strains of attenuated poliovirus in the vaccine interfere with each other's replication in the intestine. With the first immunization, one strain will predominate in its growth, inducing immunity to that strain. With the second immunization, the immunity generated by the previous immunization will limit the growth of the previously predominant strain in the vaccine, enabling one of the two remaining strains to predominate and induce immunity. Finally, with the third immunization, immunity to all three strains is achieved.

A major disadvantage of attenuated vaccines is the possibility that they will revert to a virulent form. The rate of reversion of the Sabin polio vaccine (OPV) leading to subsequent paralytic disease is about one case in 2.4 million doses of vaccine. This reversion implies that pathogenic forms of the virus are being passed by a few immunized individuals and can find their way into the water supply, especially in areas where sanitation standards are not rigorous or where waste water must be recycled. This possibility has led to the exclusive use of the inactivated polio vaccine in this country (see Table 18-3). The projected eradication of paralytic polio (Figure 18-4) will be impossible as long as OPV is used anywhere in the world. The alternative inactivated Salk vaccine should be substituted as the number of cases decrease, although there are problems in delivering this vaccine in developing countries.

Attenuated vaccines also may be associated with complications similar to those seen in the natural disease. A small percentage of recipients of the measles vaccine, for example, develop post-vaccine encephalitis or other complications. As shown in Table 18-6 (page 423), however, the risk of vaccine-related complications is much lower than risks from infection. An independent study showed that 75 million doses of measles vaccine were given between 1970 and 1993, with an incidence of 48 cases of vaccine-related encephalopathy. The low incidence of this side effect compared with the rate of encephalopathy associated with infection argues for the effi-

cacy of the vaccine. A more convincing argument for vaccination is the high death rate associated with measles infection even in developed countries.

Genetic engineering techniques provide a way to attenuate a virus irreversibly by selectively removing genes that are necessary for virulence. This has been done with a herpesvirus vaccine for pigs, in which the thymidine kinase gene was removed. Because thymidine kinase is required for the virus to grow in certain types of cells (e.g., neurons), removal of this gene rendered the virus incapable of causing disease. It is possible that similar genetic engineering techniques could eliminate the risk of reversion of the attenuated polio vaccine. More recently, a vaccine against rotavirus, a major cause of infant diarrhea, was developed using genetic engineering techniques to modify an animal rotavirus to contain antigens present on the human viruses.

Pathogenic Organisms Are Inactivated by Heat or Chemical Treatment

Another common approach in vaccine production is inactivation of the pathogen by heat or by chemical means so that it is no longer capable of replication in the host. It is critically important to maintain the structure of epitopes on surface antigens during inactivation. Heat inactivation is generally unsatisfactory because it causes extensive denaturation of proteins; thus, any epitopes that depend on higher orders of protein structure are likely to be altered significantly. Chemical inactivation with formaldehyde or various alkylating agents has been successful. The Salk polio vaccine is produced by formaldehyde inactivation.

Attenuated vaccines generally require only one dose to induce long-lasting immunity. Killed vaccines, on the other hand, often require repeated boosters to maintain the immune status of the host. In addition, killed vaccines induce a predominantly humoral antibody response; they are less effective than attenuated vaccines in inducing

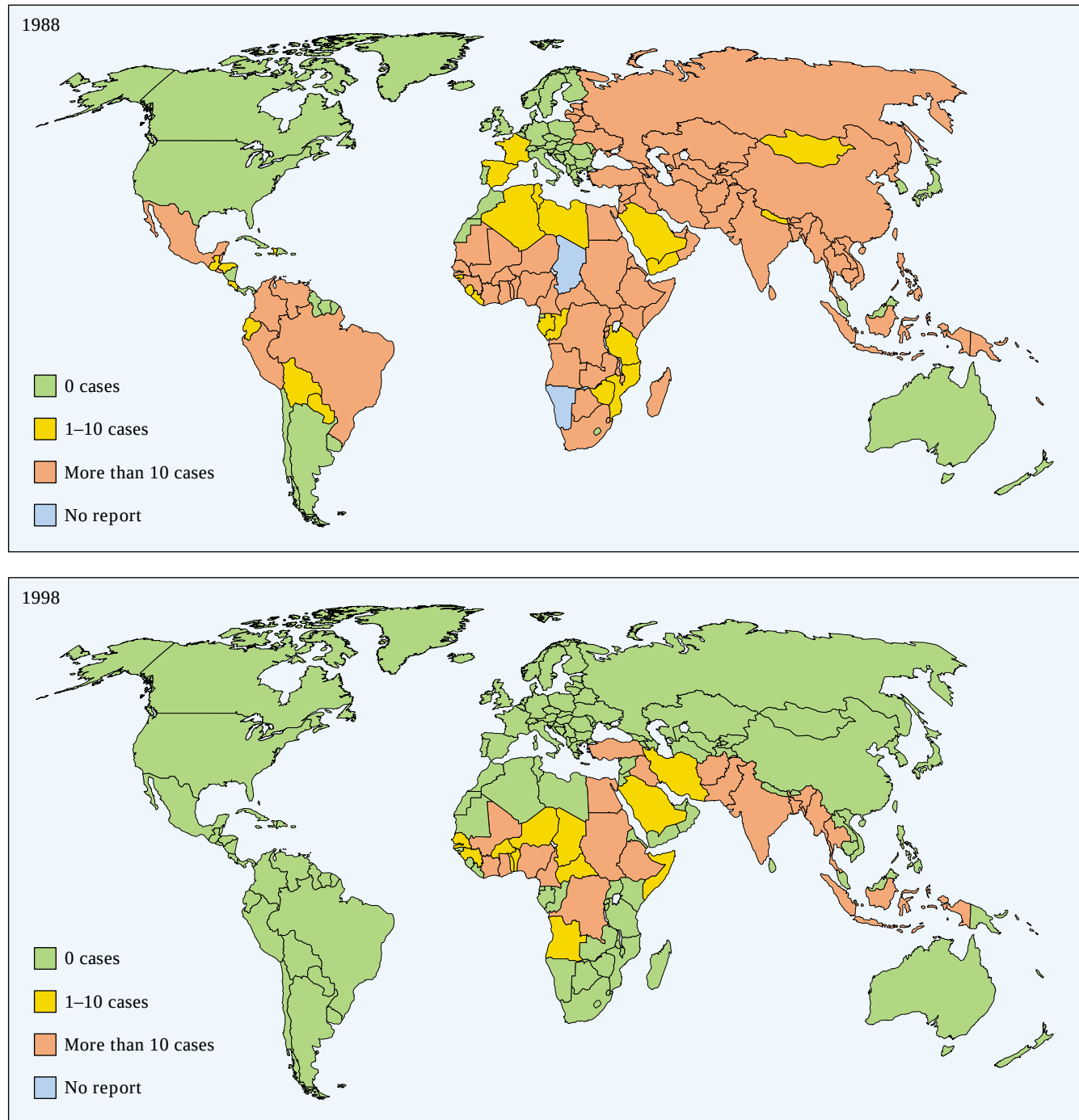
Reported polio cases

FIGURE 18-4 Progress toward the worldwide eradication of polio. Comparison of infection numbers for 1988 with those for 1998 show considerable progress in most parts of the world. Some experts question whether the use of live attenuated oral polio vaccine will

cause reversion to pathogenic forms at a rate sufficiently high to prevent total eradication of this once prevalent crippling disease. [Data from WHO.]

cell-mediated immunity and in eliciting a secretory IgA response.

Even though they contain killed pathogens, inactivated whole-organism vaccines are still associated with certain

risks. A serious complication with the first Salk vaccines arose when formaldehyde failed to kill all the virus in two vaccine lots, which caused paralytic polio in a high percentage of recipients.

TABLE 18-6

Risk of complications from natural measles infection compared with known risks of vaccination with a live attenuated virus in immunocompetent individuals

Complication	Risk after natural disease*	Risk after vaccination†
Otitis media	7–9%	0
Pneumonia	1–6%	0
Diarrhea	66%	0
Post-infectious encephalomyelitis	0.5–1 per 1000	1 per 1,000,000
SSPE	1 per 100,000	0
Thrombocytopenia	—‡	1 per 30,000§
Death	0.1–1 per 1000 (up to 5–15% in developing countries)	0

*Risk after natural measles are calculated in terms of events per number of cases.

†Risks after vaccination are calculated in terms of events per number of doses.

‡Although there have been several reports of thrombocytopenia occurring after measles including bleeding, the risk has not been properly quantified.

§This risk has been reported after MMR vaccination and cannot only be attributed to the measles component.

MMR = measles, mumps, and rubella.

SSPE = subacute sclerosing panencephalitis.

Purified Macromolecules as Vaccines

Some of the risks associated with attenuated or killed whole-organism vaccines can be avoided with vaccines that consist of specific, purified macromolecules derived from pathogens. Three general forms of such vaccines are in current use: inactivated exotoxins, capsular polysaccharides, and recombinant microbial antigens (see Table 18-4).

Bacterial Polysaccharide Capsules Are Used as Vaccines

The virulence of some pathogenic bacteria depends primarily on the antiphagocytic properties of their hydrophilic polysaccharide capsule. Coating of the capsule with antibodies and/or complement greatly increases the ability of macrophages and neutrophils to phagocytose such pathogens. These findings provide the rationale for vaccines consisting of purified capsular polysaccharides.

The current vaccine for *Streptococcus pneumoniae*, which causes pneumococcal pneumonia, consists of 23 antigenically different capsular polysaccharides. The vaccine induces formation of opsonizing antibodies and is now on the list of vaccines recommended for all infants. The vaccine for *Neisseria meningitidis*, a common cause of bacterial meningitis, also consists of purified capsular polysaccharides.

One limitation of polysaccharide vaccines is their inability to activate T_H cells. They activate B cells in a thymus-independent type 2 (TI-2) manner, resulting in IgM production but little class switching, no affinity maturation, and little, if any, development of memory cells. Several investigators have

reported the induction of IgA-secreting plasma cells in humans receiving subcutaneous immunization with the pneumococcal polysaccharide vaccine. In this case, since T_H cells are not involved in the response, the vaccine may activate IgA-specific memory B cells previously generated by naturally-occurring bacterial antigens at mucosal surfaces. Because these bacteria have both polysaccharide and protein epitopes, they would activate T_H cells, which in turn could mediate class switching and memory-cell formation.

One way to involve T_H cells directly in the response to a polysaccharide antigen is to conjugate the antigen to some sort of protein carrier. For example, the vaccine for *Haemophilus influenzae* type b (Hib), the major cause of bacterial meningitis in children less than 5 years of age, consists of type b capsular polysaccharide covalently linked to a protein carrier, tetanus toxoid. The polysaccharide-protein conjugate is considerably more immunogenic than the polysaccharide alone, and because it activates T_H cells, it enables class switching from IgM to IgG. Although this type of vaccine can induce memory B cells, it cannot induce memory T cells specific for the pathogen. In the case of the Hib vaccine, it appears that the memory B cells can be activated to some degree in the absence of a population of memory T_H cells, thus accounting for the efficacy of this vaccine.

Toxoids Are Manufactured from Bacterial Toxins

Some bacterial pathogens, including those that cause diphtheria and tetanus, produce exotoxins. These exotoxins produce many of the disease symptoms that result from

infection. Diphtheria and tetanus vaccines, for example, can be made by purifying the bacterial exotoxin and then inactivating the toxin with formaldehyde to form a **toxoid**. Vaccination with the toxoid induces anti-toxoid antibodies, which are also capable of binding to the toxin and neutralizing its effects. Conditions for the production of toxoid vaccines must be closely controlled to achieve detoxification without excessive modification of the epitope structure. The problem of obtaining sufficient quantities of the purified toxins to prepare the vaccines has been overcome by cloning the exotoxin genes and then expressing them in easily grown host cells. In this way, large quantities of the exotoxin can be produced, purified, and subsequently inactivated.

Proteins from Pathogens Are Produced by Recombinant Techniques

Theoretically, the gene encoding any immunogenic protein can be cloned and expressed in bacterial, yeast, or mammalian cells using recombinant DNA technology. A number of genes encoding surface antigens from viral, bacterial, and protozoan pathogens have been successfully cloned into bacterial, yeast, insect, or mammalian expression systems, and the expressed antigens used for vaccine development. The first such recombinant antigen vaccine approved for human use is the hepatitis B vaccine. This vaccine was developed by cloning the gene for the major surface antigen of hepatitis B virus (HBsAg) and expressing it in yeast cells. The recombinant yeast cells are grown in large fermenters, and HBsAg accumulates in the cells. The yeast cells are harvested and disrupted by high pressure, releasing the recombinant HBsAg, which is then purified by conventional biochemical techniques. This recombinant hepatitis B vaccine has been shown to induce the production of protective antibodies. This vaccine holds much promise for the 250 million carriers of chronic hepatitis B worldwide!

Use of Synthetic Peptides as Vaccines Has Progressed Slowly

Although once considered very promising, the use of synthetic peptides as vaccines has not progressed as originally projected. Peptides are not as immunogenic as proteins, and it is difficult to elicit both humoral and cellular immunity to them. The use of conjugates and adjuvants can assist in raising protective immunity to peptides, but barriers to the widespread use of peptide vaccines remain and pose an interesting problem for immunologists. Most importantly, advances in techniques to produce recombinant proteins or fragments of proteins in transfected cell culture have removed the impetus to develop vaccines based on synthetic peptides. Nonetheless, there remains theoretical interest in immunity to them, and studies of peptide immunity may generate insights leading to new vaccines.

Construction of synthetic peptides for use as vaccines to induce either humoral or cell-mediated immunity requires

an understanding of the nature of T-cell and B-cell epitopes. Ideally, vaccines for inducing humoral immunity should include peptides that form immunodominant B-cell epitopes. Such epitopes can be identified by determining the dominant antibody in the sera of individuals who are recovering from a disease and then testing various synthetic peptides for their ability to react with that antibody with high affinity. A successful vaccine must also generate a population of memory T_H cells; therefore the peptide should include immunodominant T-cell epitopes. Since MHC molecules differ in their ability to present peptides to T cells, MHC polymorphism within a species influences the level of T-cell response by different individuals to different peptides. Moreover, different subpopulations of T cells probably recognize different epitopes. Experiments by E. Sercarz have identified nonoverlapping amino acid sequences within hen egg-white lysozyme that induce a strong helper response to an antigen and other peptides that induce immunologic suppression. For example, immunization with the amino-terminal residues 1–17 of hen egg-white lysozyme suppressed the response to native lysozyme. By identifying suppressor peptides and eliminating them from synthetic vaccines, it might be possible to generate enhanced immunity.

Recombinant-Vector Vaccines

Genes that encode major antigens of especially virulent pathogens can be introduced into attenuated viruses or bacteria. The attenuated organism serves as a vector, replicating within the host and expressing the gene product of the pathogen. A number of organisms have been used for vector vaccines, including vaccinia virus, the canarypox virus, attenuated poliovirus, adenoviruses, attenuated strains of *Salmonella*, the BCG strain of *Mycobacterium bovis*, and certain strains of streptococcus that normally exist in the oral cavity.

Vaccinia virus, the attenuated vaccine used to eradicate smallpox, has been widely employed as a vector vaccine. This large, complex virus, with a genome of about 200 genes, can be engineered to carry several dozen foreign genes without impairing its capacity to infect host cells and replicate. The procedure for producing a vaccinia vector that carries a foreign gene from a pathogen is outlined in Figure 18-5. The genetically engineered vaccinia expresses high levels of the inserted gene product, which can then serve as a potent immunogen in an inoculated host. Like the smallpox vaccine, genetically engineered vaccinia vector vaccines can be administered simply by scratching the skin, causing a localized infection in host cells. If the foreign gene product expressed by the vaccinia is a viral envelope protein, it is inserted into the membrane of the infected host cell, inducing development of cell-mediated immunity as well as antibody-mediated immunity.

Other attenuated-vector vaccines may prove to be safer than the vaccinia vaccine. The canarypox virus has recently been tried as a vector vaccine. Like its relative vaccinia, the canarypox virus is large and easily engineered to carry multiple

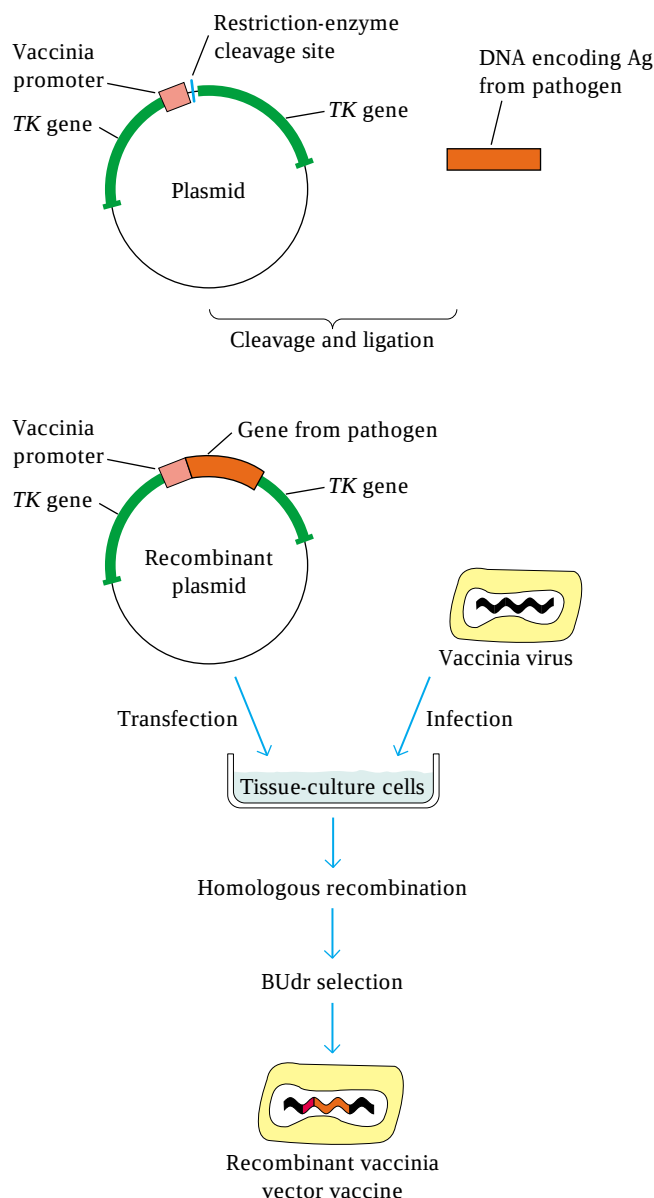


FIGURE 18-5 Production of vaccinia vector vaccine. The gene that encodes the desired antigen (orange) is inserted into a plasmid vector adjacent to a vaccinia promoter (pink) and flanked on either side by the vaccinia thymidine kinase (TK) gene (green). When tissue-culture cells are incubated simultaneously with vaccinia virus and the recombinant plasmid, the antigen gene and promoter are inserted into the vaccinia virus genome by homologous recombination at the site of the nonessential TK gene, resulting in a TK^{-} recombinant virus. Cells containing the recombinant vaccinia virus are selected by addition of bromodeoxyuridine (BUdr), which kills TK^{+} cells. [Adapted from B. Moss, 1985, *Immunol. Today* 6:243.]

genes. Unlike vaccinia, the canarypox virus does not appear to be virulent even in individuals with severe immune suppression. Another possible vector is an attenuated strain of *Salmonella typhimurium*, which has been engineered with genes

from the bacterium that causes cholera. The advantage of this vector vaccine is that *Salmonella* infects cells of the mucosal lining of the gut and therefore will induce secretory IgA production. Effective immunity against a number of diseases, including cholera and gonorrhea, depends on increased production of secretory IgA at mucous membrane surfaces. Similar strategies using bacteria that are a normal part of oral flora are in development. The strategy would involve introduction of genes encoding antigens from pathogenic organisms into bacterial strains that inhabit the oral cavity or respiratory tract. Eliciting immunity at the mucosal surface could provide excellent protection at the portal used by the pathogen.

DNA Vaccines

In a recently developed vaccination strategy, plasmid DNA encoding antigenic proteins is injected directly into the muscle of the recipient. Muscle cells take up the DNA and the encoded protein antigen is expressed, leading to both a humoral antibody response and a cell-mediated response. What is most surprising about this finding is that the injected DNA is taken up and expressed by the muscle cells with much greater efficiency than in tissue culture. The DNA appears either to integrate into the chromosomal DNA or to be maintained for long periods in an episomal form. The viral antigen is expressed not only by the muscle cells but also by dendritic cells in the area that take up the plasmid DNA and express the viral antigen. The fact that muscle cells express low levels of class I MHC molecules and do not express costimulatory molecules suggests that local dendritic cells may be crucial to the development of antigenic responses to DNA vaccines (Figure 18-6).

DNA vaccines offer advantages over many of the existing vaccines. For example, the encoded protein is expressed in the host in its natural form—there is no denaturation or modification. The immune response is therefore directed to the antigen exactly as it is expressed by the pathogen. DNA vaccines also induce both humoral and cell-mediated immunity; to stimulate both arms of the immune response with non-DNA vaccines normally requires immunization with a live attenuated preparation, which introduces additional elements of risk. Finally, DNA vaccines cause prolonged expression of the antigen, which generates significant immunological memory.

The practical aspects of DNA vaccines are also very promising (Table 18-5). Refrigeration is not required for the handling and storage of the plasmid DNA, a feature that greatly lowers the cost and complexity of delivery. The same plasmid vector can be custom tailored to make a variety of proteins, so that the same manufacturing techniques can be used for different DNA vaccines, each encoding an antigen from a different pathogen. An improved method for administering these vaccines entails coating microscopic gold beads with the plasmid DNA and then delivering the coated particles through the skin into the underlying muscle with an air gun (called a *gene gun*). This will allow rapid delivery of a

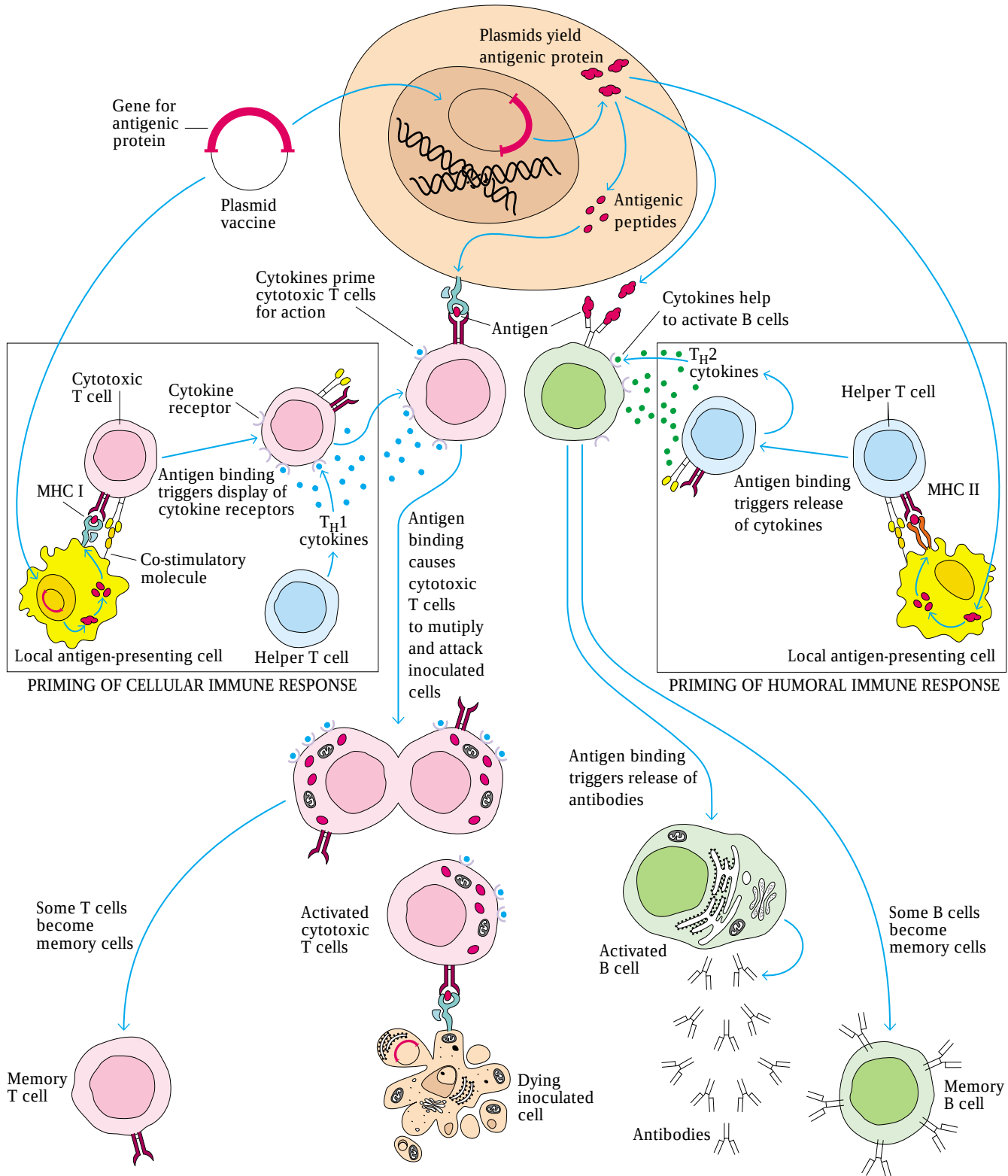


FIGURE 18-6 Use of DNA vaccines raises both humoral and cellular immunity. The injected gene is expressed in the injected muscle cell and in nearby APCs. The peptides from the protein encoded by the DNA are expressed on the surface of both cell types after processing as an endogenous antigen by the MHC class I pathway. Cells that present the antigen in the context of class I MHC molecules

stimulate development of cytotoxic T cells. The protein encoded by the injected DNA is also expressed as a soluble, secreted protein, which is taken up, processed, and presented in the context of class II MHC molecules. This pathway stimulates B-cell immunity and generates antibodies and B-cell memory against the protein. [Adapted from D. B. Weiner and R. C. Kennedy, 1999, *Sci. Am.* **281**:50.]

vaccine to large populations without the requirement for huge supplies of needles and syringes.

Tests of DNA vaccines in animal models have shown that these vaccines are able to induce protective immunity against a number of pathogens, including the influenza virus. It has been further shown that the inclusion of certain DNA sequences in the vector leads to enhanced immune response. At present, there are human trials underway with several different DNA vaccines, including those for malaria, AIDS, influenza, and herpesvirus. Future experimental trials of DNA vaccines will mix genes for antigenic proteins with those for cytokines or chemokines that direct the immune response to the optimum pathway. For example, the IL-12 gene may be included in a DNA vaccine; expression of IL-12 at the site of immunization will stimulate TH₁-type immunity induced by the vaccine.

DNA vaccines will likely be used for human immunization within the next few years. However, they are not a universal solution to the problems of vaccination; for example, only protein antigens can be encoded—certain vaccines,

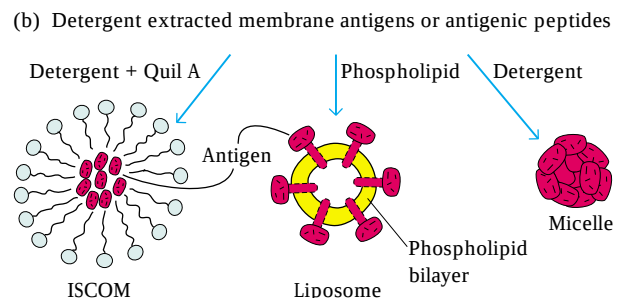
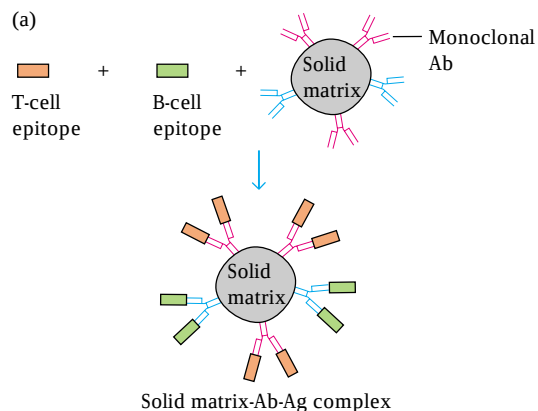
such as those for pneumococcal and meningococcal infections, use protective polysaccharide antigens.

Multivalent Subunit Vaccines

One of the limitations of synthetic peptide vaccines and recombinant protein vaccines is that they tend to be poorly immunogenic; in addition, they tend to induce a humoral antibody response but are less likely to induce a cell-mediated response. What is needed is a method for constructing synthetic peptide vaccines that contain both immunodominant B-cell and T-cell epitopes. Furthermore, if a CTL response is desired, the vaccine must be delivered intra-cellularly so that the peptides can be processed and presented together with class I MHC molecules. A number of innovative techniques are being applied to develop multivalent vaccines that can present multiple copies of a given peptide or a mixture of peptides to the immune system (Figure 18-7).



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(c) ISCOM delivery of antigen into cell

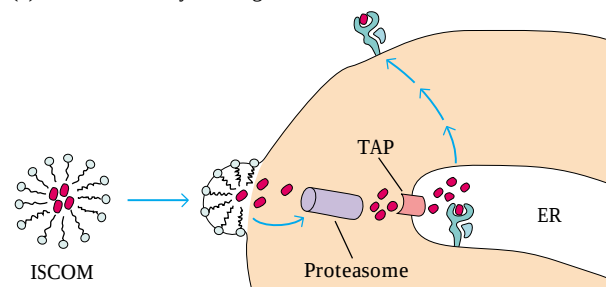


FIGURE 18-7 Multivalent subunit vaccines. (a) Solid matrix-antibody-antigen complexes can be designed to contain synthetic peptides representing both T-cell epitopes and B-cell epitopes. (b) Protein micelles, liposomes, and immunostimulating complexes (ISCOMs) can all be prepared with extracted antigens or antigenic peptides. In micelles and liposomes, the hydrophilic residues of the antigen molecules are oriented outward. In ISCOMs, the long fatty-acid tails of the external detergent layer are adjacent to the hydrophobic residues of the centrally located antigen molecules. (c) ISCOMs and liposomes can deliver antigens inside cells, so they mimic endogenous antigens. Subsequent processing by the cytosolic pathway and presentation with class I MHC molecules induces a cell-mediated response.

One approach is to prepare solid matrix–antibody–antigen (SMAA) complexes by attaching monoclonal antibodies to particulate solid matrices and then saturating the antibody with the desired antigen. The resulting complexes are then used as vaccines. By attaching different monoclonal antibodies to the solid matrix, it is possible to bind a mixture of peptides or proteins, composing immunodominant epitopes for both T cells and B cells, to the solid matrix (see Figure 18-7a). These multivalent complexes have been shown to induce vigorous humoral and cell-mediated responses. Their particulate nature contributes to their increased immunogenicity by facilitating phagocytosis by phagocytic cells.

Another means of producing a multivalent vaccine is to use detergent to incorporate protein antigens into protein micelles, lipid vesicles (called liposomes), or immunostimulating complexes (see Figure 18-7b). Mixing proteins in detergent and then removing the detergent forms micelles. The individual proteins orient themselves with their hydrophilic residues toward the aqueous environment and the hydrophobic residues at the center so as to exclude their interaction with the aqueous environment. Liposomes containing protein antigens are prepared by mixing the proteins with a suspension of phospholipids under conditions that form vesicles bounded by a bilayer. The proteins are incorporated into the bilayer with the hydrophilic residues exposed. Immunostimulating complexes (ISCOMs) are lipid carriers prepared by mixing protein with detergent and a glycoside called Quil A.

Membrane proteins from various pathogens, including influenza virus, measles virus, hepatitis B virus, and HIV have been incorporated into micelles, liposomes, and ISCOMs and are currently being assessed as potential vaccines. In addition to their increased immunogenicity, liposomes and ISCOMs appear to fuse with the plasma membrane to deliver the antigen intracellularly, where it can be processed by the cytosolic pathway and thus induce a cell-mediated response (see Figure 18-7c).

SUMMARY

- A state of immunity can be induced by passive or active immunization
 - a) Short-term passive immunization is induced by transfer of preformed antibodies.
 - b) Infection or inoculation achieves long-term active immunization.
- Three types of vaccines are currently used in humans: attenuated (avirulent) microorganisms, inactivated (killed) microorganisms, or purified macromolecules.
- Protein components of pathogens expressed in cell culture may be effective vaccines.
- Recombinant vectors, including viruses or bacteria, engineered to carry genes from infectious microorganisms, maximize cell-mediated immunity to the encoded antigens.
- Plasmid DNA encoding a protein antigen from a pathogen can serve as an effective vaccine inducing both humoral and cell-mediated immunity.
- Realizing the optimum benefit of vaccines will require cheaper manufacture and improved delivery methods for existing vaccines.

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USEFUL WEB SITES

<http://www.VaccineAlliance.org/>

Homepage of global alliance for vaccines and immunization (GAVI), a source of information about vaccines in developing countries and worldwide efforts at disease eradication.

<http://www.ecbt.org/>

Every Child by Two offers information on childhood vaccination.

Study Questions

CLINICAL FOCUS QUESTION A connection between the new pneumococcus vaccine and a relatively rare form of arthritis has been reported. What data would you need to validate this report? How would you proceed to evaluate this possible connection?

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. Transplacental transfer of maternal IgG antibodies against measles confers short-term immunity on the fetus.
 - b. Attenuated vaccines are more likely to induce cell-mediated immunity than killed vaccines are.
 - c. Multivalent subunit vaccines generally induce a greater response than synthetic peptide vaccines.
 - d. One disadvantage of DNA vaccines is that they don't generate significant immunologic memory.
 - e. Macromolecules generally contain a large number of potential epitopes.
 - f. A DNA vaccine only induces a response to a single epitope.
2. What are the advantages and disadvantages of using attenuated organisms as vaccines?
3. A young girl who had never been immunized to tetanus stepped on a rusty nail and got a deep puncture wound. The doctor cleaned out the wound and gave the child an injection of tetanus antitoxin.
 - a. Why was antitoxin given instead of a booster shot of tetanus toxoid?
 - b. If the girl receives no further treatment and steps on a rusty nail again 3 years later, will she be immune to tetanus?
4. What are the advantages of the Sabin polio vaccine compared with the Salk vaccine? Why is the Sabin vaccine no longer recommended for use in the United States?
5. In an attempt to develop a synthetic peptide vaccine, you have analyzed the amino acid sequence of a protein antigen for (a) hydrophobic peptides and (b) strongly hydrophilic peptides. How might peptides of each type be used as a vaccine to induce different immune responses?
6. Explain the phenomenon of herd immunity. How does this phenomenon relate to the appearance of certain epidemics?
7. You have identified a bacterial protein antigen that confers protective immunity to a pathogenic bacterium and have cloned the gene that encodes it. The choices are either to express the protein in yeast and use this recombinant protein as a vaccine, or to use the gene for the protein to prepare a DNA vaccine. Which approach would you take and why?
8. Explain the relationship between the incubation period of a pathogen and the approach needed to achieve effective active immunization.
9. List the three types of purified macromolecules that are currently used as vaccines.

chapter 19

AIDS and Other Immunodeficiencies

LIKE ANY COMPLEX MULTI-COMPONENT SYSTEM, THE immune system is subject to failure of some or all of its parts. This failure can have dire consequences. When the system loses its sense of self and begins to attack host cells and tissues, the result is **autoimmunity**, which is described in Chapter 20. When the system errs by failing to protect the host from disease-causing agents or from malignant cells, the result is **immunodeficiency**, which is the subject of this chapter.

A condition resulting from a genetic or developmental defect in the immune system is called a primary immunodeficiency. In such a condition, the defect is present at birth although it may not manifest itself until later in life. Secondary immunodeficiency, or acquired immunodeficiency, is the loss of immune function and results from exposure to various agents. By far the most common secondary immunodeficiency is **acquired immunodeficiency syndrome**, or **AIDS**, which results from infection with the human immunodeficiency virus 1 (HIV-1). In the year 2000, AIDS killed approximately 3 million persons, and HIV infection continues to spread to an estimated 15,000 persons per day. AIDS patients, like other individuals with severe immunodeficiency, are at risk of infection with so-called opportunistic agents. These are microorganisms that healthy individuals can harbor with no ill consequences but that cause disease in those with impaired immune function.

The first part of this chapter describes the common primary immunodeficiencies, examines progress in identifying the genetic defects that underlie these disorders, and considers approaches to their treatment, including innovative uses of gene therapy. Animal models of primary immunodeficiency are also described. The rest of this chapter describes acquired immunodeficiency, with a strong focus on HIV infection, AIDS, and the current status of therapeutic and prevention strategies for combating this fatal acquired immunodeficiency.

Primary Immunodeficiencies

A primary immunodeficiency may affect either adaptive or innate immune functions. Deficiencies involving components of adaptive immunity, such as T or B cells, are thus



Nude Mouse (*nu/nu*)

- Primary Immunodeficiencies
- AIDS and Other Acquired or Secondary Immunodeficiencies

differentiated from immunodeficiencies in which the non-specific mediators of innate immunity, such as phagocytes or complement, are impaired. Immunodeficiencies are conveniently categorized by the type or the developmental stage of the cells involved. Figure 19-1 reviews the overall cellular development in the immune system, showing the locations of defects that give rise to primary immunodeficiencies. As Chapter 2 explained, the two main cell lineages important to immune function are lymphoid and myeloid. Most defects that lead to immunodeficiencies affect either one or the other. The lymphoid cell disorders may affect T cells, B cells, or, in combined immunodeficiencies, both B and T cells. The myeloid cell disorders affect phagocytic function. Most of the primary immunodeficiencies are inherited, and the precise molecular variations and the genetic defects that lead to many of these dysfunctions have been determined (Table 19-1 and Figure 19-2). In addition, there are immunodeficiencies that stem from developmental defects that impair proper function of an organ of the immune system.

The consequences of primary immunodeficiency depend on the number and type of immune system components involved. Defects in components early in the hematopoietic developmental scheme affect the entire immune system. In this category is reticular dysgenesis, a stem-cell defect that affects the maturation of all leukocytes; the resulting general failure of immunity leads to susceptibility to infection by a variety of microorganisms. Without aggressive treatment, the affected individual usually dies young from severe infection. In the



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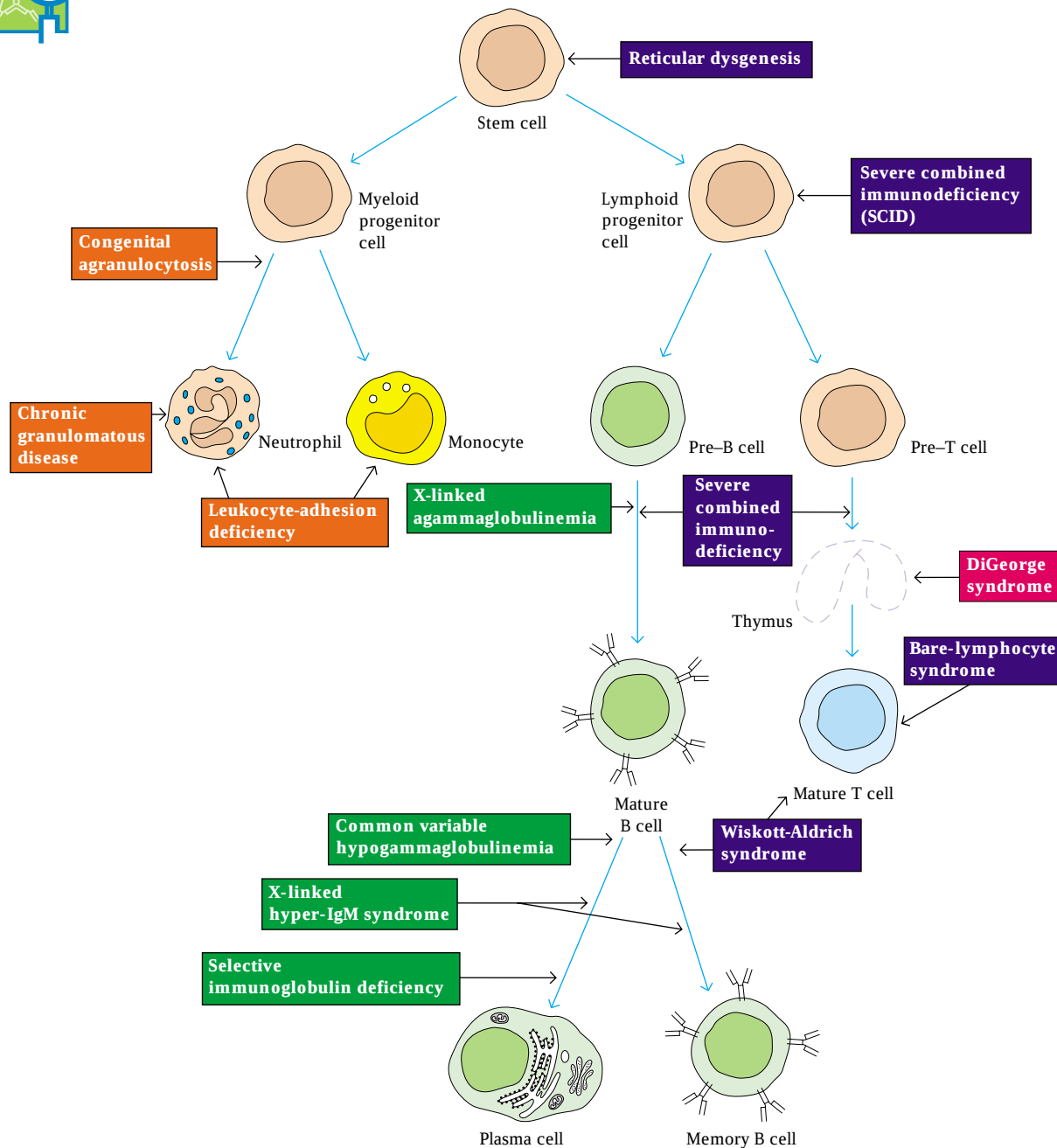


FIGURE 19-1 Congenital defects that interrupt hematopoiesis or impair functioning of immune-system cells result in various immunodeficiency diseases. (Orange boxes = phagocytic defi-

ciencies, green = humoral deficiencies, red = cell-mediated deficiencies, and purple = combined immunodeficiencies, defects that affect more than one cell lineage.)

more restricted case of defective phagocytic function, the major consequence is susceptibility to bacterial infection. Defects in more highly differentiated compartments of the immune system have consequences that are more specific

and usually less severe. For example, an individual with selective IgA deficiency may enjoy a full life span, troubled only by a greater than normal susceptibility to infections of the respiratory and genitourinary tracts.

TABLE 19-1 Some primary human immunodeficiency diseases and underlying genetic defects

Immunodeficiency disease	Specific defect	Impaired function	Inheritance mode*	Chromosomal defect
Severe combined immunodeficiency (SCID)	RAG-1/RAG-2 deficiency	No TCR or Ig gene rearrangement	AR	11p13
	ADA deficiency } PNP deficiency }	Toxic metabolite in T and B cells	{ AR AR	20q13 14q13
	JAK-3 deficiency } IL-2R γ -deficiency }	Defective signals from IL-2, 4, 7, 9, 15,	{ AR XL	19p13 Xq13
	ZAP-70 deficiency	Defective signal from TCR	AR	2q12
Bare lymphocyte syndrome	Defect in MHC class II gene promoter	No class II MHC molecules	AR	16p13
Wiskott-Aldrich syndrome (WAS)	Cytoskeletal protein (CD43)	Defective T cells and platelets	XL	Xp11
Interferon gamma receptor	IFN- γ -receptor defect	Impaired immunity to mycobacteria	AR	6q23
DiGeorge syndrome	Thymic aplasia	T- and B-cell development	AD	22q11
Ataxia telangiectasia	Defective cell-cycle kinase	Low IgA, IgE	AR	11q22
Gammaglobulinemias	X-linked agammaglobulinemia	Bruton's tyrosine kinase (Btk); no mature B cells	XL	Xq21
	X-linked hyper-IgM syndrome	Defective CD40 ligand	XL	Xq26
	Common variable immunodeficiency	Low IgG, IgA; variable IgM		Complex
	Selective IgA deficiency	Low or no IgA		Complex
Chronic granulomatous disease	Cyt p91 ^{phox} } Cyt p67 ^{phox} } Cyt p22 ^{phox} }	No oxidative burst for bacterial killing	{ XL AR AR	Xp21 1q25 16q24
Chediak-Higashi syndrome	Defective intracellular transport protein (LYST)	Inability to lyse bacteria	AR	1q42
Leukocyte-adhesion defect	Defective integrin β 2 (CD18)	Leukocyte extravasation	AR	21q22

*AR = autosomal recessive; AD = autosomal dominant; XL = X linked; "Complex" indicates conditions for which precise genetic data are not available and that may involve several interacting loci.

Lymphoid Immunodeficiencies May Involve B Cells, T Cells, or Both

The combined forms of lymphoid immunodeficiency affect both lineages and are generally lethal within the first few years of life; these arise from defects early in developmental pathways. They are less common than conditions, usually less severe, that result from defects in more highly differentiated lymphoid cells.

B-cell immunodeficiency disorders make up a diverse spectrum of diseases ranging from the complete absence of mature recirculating B cells, plasma cells, and immunoglobulin to the selective absence of only certain classes of

immunoglobulins. Patients with these disorders usually are subject to recurrent bacterial infections but display normal immunity to most viral and fungal infections, because the T-cell branch of the immune system is largely unaffected. Most common in patients with humoral immunodeficiencies are infections by such encapsulated bacteria as staphylococci, streptococci, and pneumococci, because antibody is critical for the opsonization and clearance of these organisms.

Because of the central role of T cells in the immune system, a T-cell deficiency can affect both the humoral and the cell-mediated responses. The impact on the cell-mediated system can be severe, with a reduction in both delayed-type hypersensitive responses and cell-mediated cytotoxicity.

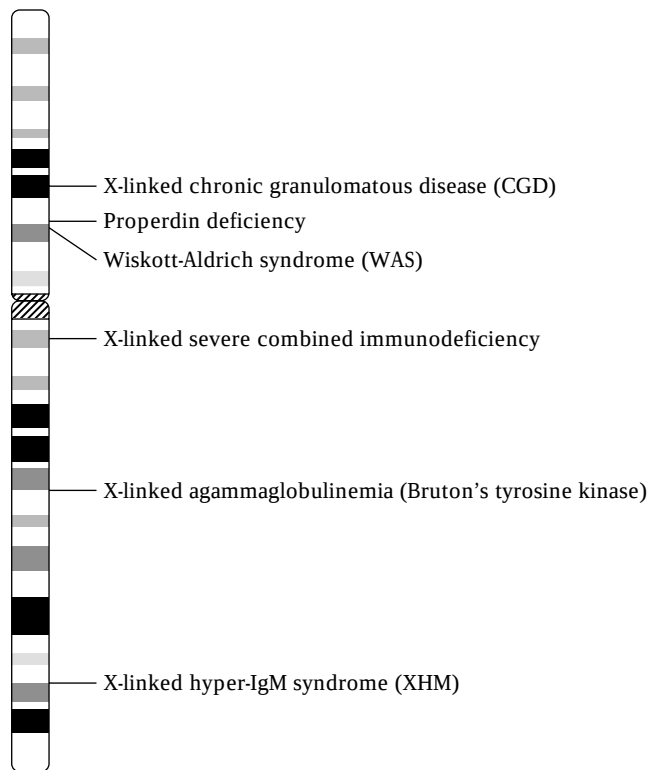


FIGURE 19-2 Several X-linked immunodeficiency diseases result from defects in loci on the X chromosome. [Data from the Natl. Center for Biotechnology Information Web site.]

Immunoglobulin deficiencies are associated primarily with recurrent infections by extracellular bacteria, but those affected have normal responses to intracellular bacteria, as well as viral and fungal infections. By contrast, defects in the cell-mediated system are associated with increased susceptibility to viral, protozoan, and fungal infections. Intracellular pathogens such as *Candida albicans*, *Pneumocystis carinii*, and *Mycobacteria* are often implicated, reflecting the importance of T cells in eliminating intracellular pathogens. Infections with viruses that are rarely pathogenic for the normal individual (such as cytomegalovirus or even an attenuated measles vaccine) may be life threatening for those with impaired cell-mediated immunity. Defects that cause decreased T-cell counts generally also affect the humoral system, because of the requirement for T_H cells in B-cell activation. Generally there is some decrease in antibody levels, particularly in the production of specific antibody after immunization.

As one might expect, combined deficiencies of the humoral and cell-mediated branches are the most serious of the immunodeficiency disorders. The onset of infections begins early in infancy, and the prognosis for these infants is early death unless therapeutic intervention reconstitutes their defective immune system. As described below, there are increasing numbers of options for the treatment of immunodeficiencies.

The immunodeficiencies that affect lymphoid function have in common the inability to mount or sustain a complete

immune response against specific agents. A variety of failures can lead to such immunodeficiency. Defective intercellular communication may be rooted in deleterious mutations of genes that encode cell-surface receptors or signal-transduction molecules; defects in the mechanisms of gene rearrangement and other functions may prevent normal B- or T-cell responses. Figure 19-3 is an overview of the molecules involved in the more well-described interactions among T cells and B cells that give rise to specific responses, with a focus on proteins in which defects leading to immunodeficiency have been identified.

SEVERE COMBINED IMMUNODEFICIENCY (SCID)

The family of disorders termed SCID stems from defects in lymphoid development that affect either T cells or both T and B cells. All forms of SCID have common features despite differences in the underlying genetic defects. Clinically, SCID is characterized by a very low number of circulating lymphocytes. There is a failure to mount immune responses mediated by T cells. The thymus does not develop, and the few circulating T cells in the SCID patient do not respond to stimulation by mitogens, indicating that they cannot proliferate in response to antigens. Myeloid and erythroid (red-blood-cell precursors) cells appear normal in number and function, indicating that only lymphoid cells are depleted in SCID.

SCID results in severe recurrent infections and is usually fatal in the early years of life. Although both the T and B lineages may be affected, the initial manifestation of SCID in infants is almost always infection by agents, such as fungi or viruses, that are normally dealt with by T-cell immunity. The B-cell defect is not evident in the first few months of the affected infant's life because antibodies are passively obtained from transplacental circulation or from mother's milk. SCID infants suffer from chronic diarrhea, pneumonia, and skin, mouth, and throat lesions as well as a host of other opportunistic infections. The immune system is so compromised that even live attenuated vaccines (such as the Sabin polio vaccine) can cause infection and disease. The life span of a SCID patient can be prolonged by preventing contact with all potentially harmful microorganisms, for example by confinement in a sterile atmosphere. However, extraordinary effort is required to prevent direct contact with other persons and with unfiltered air; any object, including food, that comes in contact with the sequestered SCID patient must first be sterilized. Such isolation is feasible only as a temporary measure, pending treatment.

The search for defects that underlie SCID has revealed several different causes for this general failure of immunity. A survey of 141 patients by Rebecca Buckley indicated that the most common cause (64 cases) was deficiency of the common gamma chain of the IL-2 receptor (IL-2R γ ; see Figure 12-7). Defects in this chain impede signaling through receptors for IL-4, -7, -9, and -15 as well as the IL-2 receptor, because the chain is present in receptors for all of these cytokines. Deficiency in the kinase JAK-3, which has a similar

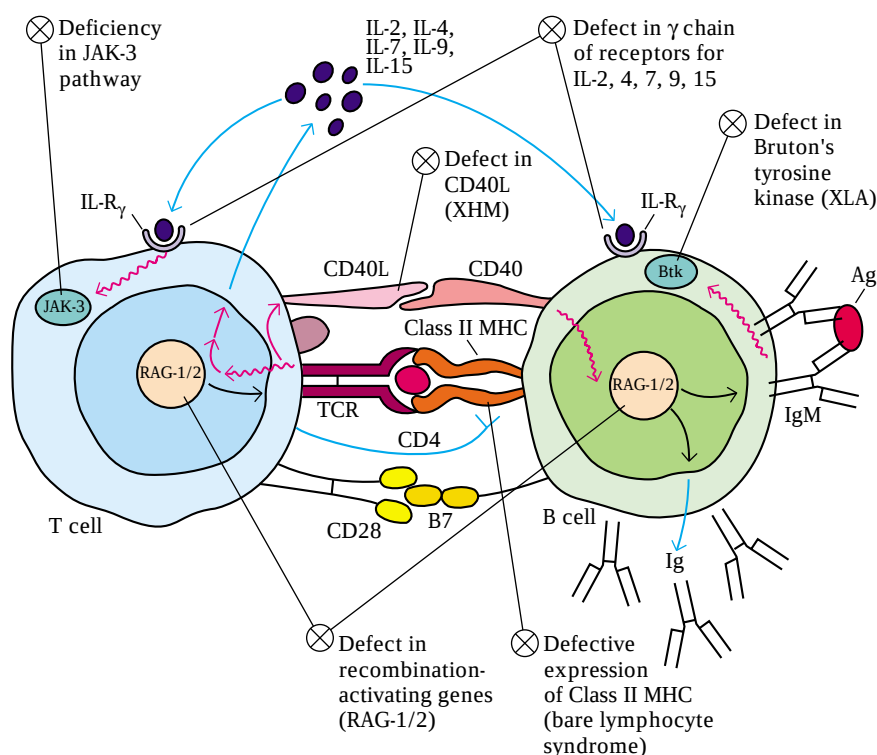


FIGURE 19-3 Defects in cell interaction and signaling can lead to severe immunodeficiency. The interaction of T cell and B cell is shown here with a number of the components important to the intra- and extracellular signaling pathways. A number of primary immunodeficiencies are rooted in defects in these interactions. SCID may result from defects in (1) the recombination-activating genes (*RAG-1* and *-2*) required for synthesis of the functional immunoglobulins and T-cell receptors that characterize mature B and T cells; (2) the γ chain

of receptors for IL-2, 4, 7, 9, and 15 (*IL-R γ*); (3) JAK-3, which transduces signals from the gamma chain of the cytokine receptor; or (4) expression of the class II MHC molecule (bare lymphocyte syndrome). XLA results from defective transduction of activating signals from the cell-surface IgM by Bruton's tyrosine kinase (*Btk*). XHM results from defects in CD40L that preclude normal maturation of B cells. [Adapted from B. A. Smart and H. D. Ochs, 1997, *Curr. Opin. Pediatr.* 9:570.]

phenotype because the IL receptors signal through this molecule, accounted for 9 of the cases (see Figure 12-10). A rare defect found in only 2 of the patients involved the IL-7 receptor; these patients have impaired T and B cells but normal NK cells. Another common defect is the adenosine deaminase or ADA deficiency found in 22 patients. Adenosine deaminase catalyzes conversion of adenosine to inosine, and its deficiency results in accumulation of adenosine, which interferes with purine metabolism and DNA synthesis. The remaining cases included single instances of reticular dysgenesis and cartilage hair dysplasia or were classified as autosomal recessive defects not related to known *IL-2R γ* or JAK-3 mutations. Thirteen of the 141 cases were of unknown origin, with no apparent genetic defect or family history of immunodeficiency.

There are other known defects that give rise to SCID. There is a defect characterized by depletion of CD8⁺ T cells that involves the tyrosine kinase ZAP-70, an important element in T-cell signal transduction (see Figures 10-11 and 10-12). Infants with defects in ZAP-70 may have normal levels of immunoglobulin and CD4⁺ lymphocytes, but their CD4⁺ T cells are nonfunctional. A deficiency in the enzyme purine

nucleoside phosphorylase (PNP) causes immunodeficiency by a mechanism similar to the ADA defect. As described in Chapters 5 and 9, both immunoglobulin and T-cell receptor genes undergo rearrangement to express the active forms of these molecules. A defect in the genes that encode mediators of the rearrangement processes (recombination-activating proteins *RAG-1* and *RAG-2*) precludes development of B and T cells with functional receptors and leads to SCID.

A defect leading to general failure of immunity similar to SCID is failure to transcribe the genes that encode class II MHC molecules. Without these molecules, the patient's lymphocytes cannot participate in cellular interactions with T helper cells. This type of immunodeficiency is also called the *bare-lymphocyte syndrome*. Molecular studies of a class II MHC deficiency revealed a defective interaction between a 5' promoter sequence of the gene for the class II MHC molecule and a DNA-binding protein necessary for gene transcription. Other patients with SCID-like symptoms lack class I MHC molecules. This rare variant of immunodeficiency was ascribed to mutation in the *TAP* genes that are vital to antigen processing by class I MHC molecules (see Clinical Focus Chapter 8). This defect causes a deficit in CD8-mediated

immunity, characterized by susceptibility to viral infection. A recent case of SCID uncovered a defect in the gene for the cell-surface phosphatase CD45. Interestingly, this defect caused lack of $\alpha\beta$ T-cells but spared the $\gamma\delta$ lineage.

WISKOTT-ALDRICH SYNDROME (WAS)

The severity of this X-linked disorder increases with age and usually results in fatal infection or lymphoid malignancy. Initially, T and B lymphocytes are present in normal numbers. WAS first manifests itself by defective responses to bacterial polysaccharides and by lower-than-average IgM levels. Other responses and effector mechanisms are normal in the early stages of the syndrome. As the WAS sufferer ages, there are recurrent bacterial infections and a gradual loss of humoral and cellular responses. The syndrome includes thrombocytopenia (lowered platelet count; the existing platelets are smaller than usual and have a short half-life), which may lead to fatal bleeding. Eczema (skin rashes) in varying degrees of severity may also occur, usually beginning around one year of age. The defect in WAS has been mapped to the short arm of the X chromosome (see Table 19-1 and Figure 19-2) and involves a cytoskeletal glycoprotein present in lymphoid cells called sialophorin (CD43). The WAS protein is required for assembly of actin filaments required for the formation of microvesicles.

INTERFERON-GAMMA–RECEPTOR DEFECT

A recently described immunodeficiency that falls into the mixed-cell category involves a defect in the receptor for interferon gamma (IFN- γ , see Chapter 12). This deficiency was found in patients suffering from infection with atypical mycobacteria (intracellular organisms related to the bacteria that cause tuberculosis and leprosy). Most of those carrying this autosomal recessive trait are from families with a history of inbreeding. The susceptibility to infection with mycobacteria is selective in that those who survive these infections are not unusually susceptible to other agents, including other intracellular bacteria. This immunodeficiency points to a specific role for IFN- γ and its receptor in protection from infection with mycobacteria.

Whereas SCID and the related combined immunodeficiencies affect T cells or all lymphoid cells, other primary immunodeficiencies affect B-cell function and result in the reduction or absence of some or all classes of immunoglobulins. While the underlying defects have been identified for some of these, little information exists concerning the exact cause of some of the more common deficiencies, such as common variable immunodeficiency and selective IgA deficiency.

X-LINKED AGAMMAGLOBULINEMIA

A B-cell defect called X-linked agammaglobulinemia (XLA) or Bruton's hypogammaglobulinemia is characterized by extremely low IgG levels and by the absence of other immunoglobulin classes. Individuals with XLA have no peripheral B cells and suffer from recurrent bacterial infections, beginning at about nine months of age. A palliative

treatment for this condition is periodic administration of immunoglobulin, but patients seldom survive past their teens. There is a defect in B-cell signal transduction in this disorder, due to a defect in a transduction molecule called Bruton's tyrosine kinase (Btk), after the investigator who described the syndrome. B cells in the XLA patient remain in the pre-B stage with H chains rearranged but L chains in their germ-line configuration. (The Clinical Focus in Chapter 11 describes the discovery of this immunodeficiency and its underlying defect in detail.)

X-LINKED HYPER-IgM SYNDROME

A peculiar immunoglobulin deficiency first thought to result from a B-cell defect has recently been shown to result instead from a defect in a T-cell surface molecule. X-linked hyper-IgM (XHM) syndrome is characterized by a deficiency of IgG, IgA, and IgE, and elevated levels of IgM, sometimes as high as 10 mg/ml (normal IgM concentration is 1.5 mg/ml). Although individuals with XHM have normal numbers of B cells expressing membrane-bound IgM or IgD, they appear to lack B cells expressing membrane-bound IgG, IgA, or IgE. XHM syndrome is generally inherited as an X-linked recessive disorder (see Figure 19-2), but some forms appear to be acquired and affect both men and women. Affected individuals have high counts of IgM-secreting plasma cells in their peripheral blood and lymphoid tissue. In addition, XHM patients often have high levels of autoantibodies to neutrophils, platelets, and red blood cells. Children with XHM suffer recurrent infections, especially respiratory infections; these are more severe than expected for a deficiency characterized by low levels of immunoglobulins.

The defect in XHM is in the gene encoding the CD40 ligand (CD40L), which maps to the X chromosome. T_H cells from patients with XHM fail to express functional CD40L on their membrane. Since an interaction between CD40 on the B cell and CD40L on the T_H cell is required for B-cell activation, the absence of this co-stimulatory signal inhibits the B-cell response to T-dependent antigens (see Figures 19-3 and 11-10). The B-cell response to T-independent antigens, however, is unaffected by this defect, accounting for the production of IgM antibodies. As described in Chapter 11, class switching and formation of memory B cells both require contact with T_H cells by a CD40–CD40L interaction. The absence of this interaction in XHM results in the loss of class switching to IgG, IgA, or IgE isotypes and in a failure to produce memory B cells. In addition, XHM individuals fail to produce germinal centers during a humoral response, which highlights the role of the CD40–CD40L interaction in the generation of germinal centers.

COMMON VARIABLE IMMUNODEFICIENCY (CVI)

CVI is characterized by a profound decrease in numbers of antibody-producing plasma cells, low levels of most immunoglobulin isotypes (hypogammaglobulinemia), and recurrent infections. The condition is usually manifested later

in life than other deficiencies and is sometimes called late-onset hypogammaglobulinemia or, incorrectly, acquired hypogammaglobulinemia. However, CVI has a genetic component and is considered a primary immunodeficiency, although the exact pattern of inheritance is not known. Because the manifestations are very similar to those of acquired hypogammaglobulinemia, there is some confusion between the two forms (see below). Infections in CVI sufferers are most frequently bacterial and can be controlled by administration of immunoglobulin. In CVI patients, B cells fail to mature into plasma cells; however in vitro studies show that CVI B cells are capable of maturing in response to appropriate differentiation signals. The underlying defect in CVI is not known, but must involve either an in vivo blockage of the maturation of B cells to the plasma-cell stage or their inability to produce the secreted form of immunoglobulins.

HYPER-IgE SYNDROME (JOB SYNDROME)

A primary immunodeficiency characterized by skin abscesses, recurrent pneumonia, eczema, and elevated levels of IgE accompanies facial abnormalities and bone fragility. This multi-system disorder is autosomal dominant and has variable expressivity. The gene for hyper IgE syndrome, or HIES, maps to chromosome 4. HIES immunologic signs include recurrent infection and eosinophilia in addition to elevated IgE levels.

SELECTIVE DEFICIENCIES OF IMMUNOGLOBULIN CLASSES

A number of immunodeficiency states are characterized by significantly lowered amounts of specific immunoglobulin isotypes. Of these, IgA deficiency is by far the most common. There are family-association data showing that IgA deficiency prevails in the same families as CVI, suggesting a relationship between these conditions. The spectrum of clinical symptoms of IgA deficiency is broad; many of those affected are asymptomatic, while others suffer from an assortment of serious problems. Recurrent respiratory and genitourinary tract infections resulting from lack of secreted IgA on mucosal surfaces are common. In addition, problems such as intestinal malabsorption, allergic disease, and autoimmune disorders may also be associated with low IgA levels. The reasons for this variability in the clinical profile of IgA deficiency are not clear but may relate to the ability of some, but not all, patients to substitute IgM for IgA as a mucosal antibody. The defect in IgA deficiency is related to the inability of IgA B cells to undergo normal differentiation to the plasma-cell stage. IgG2 and IgG4 may also be deficient in IgA-deficient patients. No causative defect in IgA genes has been identified, and the surface IgA molecules on these patients' B cells appear to be expressed normally. A gene outside of the immunoglobulin gene complex is suspected to be responsible for this fairly common syndrome.

Other immunoglobulin deficiencies have been reported, but these are rarer. An IgM deficiency has been identified as an autosomal recessive trait. Victims of this condition are

subject to severe infection by agents such as meningococcus, which causes fatal disease. IgM deficiency may be accompanied by various malignancies or by autoimmune disease. IgG deficiencies are also rare. These are often not noticed until adulthood and can be effectively treated by administration of immunoglobulin.

ATAXIA TELANGIECTASIA

Although not classified primarily as an immunodeficiency, ataxia telangiectasia is a disease syndrome that includes deficiency of IgA and sometimes of IgE. The syndrome is characterized by difficulty in maintaining balance (ataxia) and by the appearance of broken capillaries (telangiectasia) in the eyes. The primary defect appears to be in a kinase involved in regulation of the cell cycle. The relationship between the immune deficiency and the other defects in ataxia telangiectasia remains obscure.

IMMUNE DISORDERS INVOLVING THE THYMUS

Several immunodeficiency syndromes are grounded in failure of the thymus to undergo normal development. Thymic malfunction has a profound effect on T-cell function; all populations of T cells, including helper, cytolytic, and regulatory varieties, are affected. Immunity to viruses and fungi is especially compromised in those suffering from these conditions.

DiGeorge syndrome, or congenital thymic aplasia, in its most severe form is the complete absence of a thymus. This developmental defect, which is associated with the deletion in the embryo of a region on chromosome 22, causes immunodeficiency along with characteristic facial abnormalities, hypoparathyroidism, and congenital heart disease (Figure 19-4). The stage at which the causative developmental defect occurs has been determined, and the syndrome is sometimes called the *third and fourth pharyngeal pouch syndrome* to reflect its precise embryonic origin. The immune defect includes a profound depression of T-cell numbers and absence of T-cell responses. Although B cells are present in normal numbers, affected individuals do not produce antibody in response to immunization with specific antigens. Thymic transplantation is of some value for correcting the T-cell defects, but many DiGeorge patients have such severe heart disease that their chances for long-term survival are poor, even if the immune defects are corrected.

Whereas the DiGeorge syndrome results from an intrauterine or developmental anomaly, thymic hypoplasia, or the Nezelof syndrome, is an inherited disorder. The mode of inheritance for this rare disease is not known and its presentation varies, making it somewhat difficult to diagnose. As the name implies, thymic hypoplasia is a defect in which a vestigial thymus is unable to serve its function in T-cell development. In some patients, B cells are normal, whereas in others a B-cell deficiency is secondary to the T-cell defect. Affected individuals suffer from chronic diarrhea, viral and fungal infections, and a general failure to thrive.



FIGURE 19-4 A child with DiGeorge syndrome showing characteristic dysplasia of ears and mouth and abnormally long distance between the eyes. [R. Kretschmer *et al.*, 1968, *New Engl. J. Med.* **279**:1295; photograph courtesy of F. S. Rosen.]

Immunodeficiencies of the Myeloid Lineage Affect Innate Immunity

Immunodeficiencies of the lymphoid lineage affect adaptive immunity. By contrast, defects in the myeloid cell lineage affect the innate immune functions (see Figure 19-1). Most of these defects result in impaired phagocytic processes that are manifested by recurrent microbial infection of greater or lesser severity. There are several stages at which the phagocytic processes may be faulty; these include cell motility, adherence to and phagocytosis of organisms, and killing by macrophages.

REDUCTION IN NEUTROPHIL COUNT

As described in Chapter 2, neutrophils are circulating granulocytes with phagocytic function. Quantitative deficiencies in neutrophils can range from an almost complete absence of cells, called agranulocytosis, to a reduction in the concentration of peripheral blood neutrophils below $1500/\text{mm}^3$, called granulocytopenia or neutropenia. These quantitative deficiencies may result from congenital defects or may be acquired through extrinsic factors. Acquired neutropenias are much more common than congenital ones.

Congenital neutropenia is often due to a genetic defect that affects the myeloid progenitor stem cell; it results in reduced production of neutrophils during hematopoiesis. In congenital agranulocytosis, myeloid stem cells are present

in the bone marrow but rarely differentiate beyond the promyelocyte stage. As a result, children born with this condition show severe neutropenia, with counts of less than 200 neutrophils/ mm^3 . These children suffer from frequent bacterial infections beginning as early as the first month of life; normal infants are protected at this age by maternal antibody as well as by innate immune mechanisms, including neutrophils. Experimental evidence suggests that this genetic defect results in decreased production of granulocyte colony-stimulating factor (G-CSF) and thus in a failure of the myeloid stem cell to differentiate along the granulocytic lineage (see Figure 2-1).

Neutrophils have a short life span, and their precursors must divide rapidly in the bone marrow to maintain levels of these cells in the circulation. For this reason, agents such as radiation and certain drugs (e.g., chemotherapeutic drugs) that specifically damage rapidly dividing cells are likely to cause neutropenia. Occasionally, neutropenia develops in such autoimmune diseases as Sjögren's syndrome or systemic lupus erythematosus; in these conditions, autoantibodies destroy the neutrophils. Transient neutropenia often develops after certain bacterial or viral infections, but neutrophil counts return to normal as the infection is cleared.

CHRONIC GRANULOMATOUS DISEASE (CGD)

CGD is a genetic disease that has at least two distinct forms: an X-linked form that occurs in about 70% of patients and an autosomal recessive form found in the rest. This disease is rooted in a defect in the oxidative pathway by which phagocytes generate hydrogen peroxide and the resulting reactive products, such as hypochlorous acid, that kill phagocytosed bacteria. CGD sufferers undergo excessive inflammatory reactions that result in gingivitis, swollen lymph nodes, and nonmalignant granulomas (lumpy subcutaneous cell masses); they are also susceptible to bacterial and fungal infection. CGD patients are not subject to infection by those bacteria, such as pneumococcus, that generate their own hydrogen peroxide. In this case, the myeloperoxidase in the host cell can use the bacterial hydrogen peroxide to generate enough hypochlorous acid to thwart infection. Several related defects may lead to CGD; these include a missing or defective cytochrome (cyt b_{558}) that functions in an oxidative pathway and defects in proteins (phagocyte oxidases, or phox) that stabilize the cytochrome. In addition to the general defect in the killer function of phagocytes, there is also a decrease in the ability of mononuclear cells to serve as APCs. Both processing and presentation of antigen are impaired. Increased amounts of antigen are required to trigger T-cell help when mononuclear cells from CGD patients are used as APCs.

The addition of IFN- γ has been shown to restore function to CGD granulocytes and monocytes *in vitro*. This observation prompted clinical trials of IFN- γ for CGD patients. Encouraging increases in oxidative function and restoration of cytoplasmic cytochrome have been reported in these

patients. In addition, knowledge of the precise gene defects underlying CGD makes it a candidate for gene therapy, and replacement of the defective cytochrome has had promising results (see below).

CHEDIAK-HIGASHI SYNDROME

This autosomal recessive disease is characterized by recurrent bacterial infections, partial oculo-cutaneous albinism (lack of skin and eye pigment), and aggressive but nonmalignant infiltration of organs by lymphoid cells. Phagocytes from patients with this immune defect contain giant granules but do not have the ability to kill bacteria. The molecular basis of the defect is a mutation in a protein (LYST) involved in the regulation of intracellular trafficking. The mutation impairs the targeting of proteins to secretory lysosomes, which makes them unable to lyse bacteria.

LEUKOCYTE ADHESION DEFICIENCY (LAD)

As described in Chapter 15, cell-surface molecules belonging to the integrin family of proteins function as adhesion molecules and are required to facilitate cellular interaction. Three of these, LFA-1, Mac-1, and gp150/95 (CD11a, b, and c, respectively) have a common β chain (CD18) and are variably present on different monocytic cells; CD11a is also expressed on B cells (Table 19-2). An immunodeficiency related to dysfunction of the adhesion molecules is rooted in a defect

localized to the common β chain and affects expression of all three of the molecules that use this chain. This defect, called leukocyte adhesion deficiency (LAD), causes susceptibility to infection with both gram-positive and gram-negative bacteria as well as various fungi. Impairment of adhesion of leukocytes to vascular endothelium limits recruitment of cells to sites of inflammation. Viral immunity is somewhat impaired, as would be predicted from the defective T-B cell cooperation arising from the adhesion defect. LAD varies in its severity; some affected individuals die within a few years, others survive into their forties. The reason for the variable disease phenotype in this disorder is not known. LAD is the subject of a Clinical Focus in Chapter 15.

Complement Defects Result in Immunodeficiency or Immune-Complex Disease

Immunodeficiency diseases resulting from defects in the complement system are described in Chapter 13. Many complement deficiencies are associated with increased susceptibility to bacterial infections and/or immune-complex diseases. One of these complement disorders, a deficiency in properdin, which stabilizes the C3 convertase in the alternative complement pathway, is caused by a defect in a gene located on the X chromosome (see Figure 19-2).

TABLE 19-2 Properties of integrin molecules that are absent in leukocyte-adhesion deficiency

Property	INTEGRIN MOLECULES*		
	LFA-1	CR3	CR4
CD designation	CD11a/CD18	CD11b/CD18	CD11c/CD18
Subunit composition	α L β 2	α M β 2	α X β 2
Subunit molecular mass (kDa)			
α chain	175,000	165,000	150,000
β chain	95,000	95,000	95,000
Cellular expression	Lymphocytes	Monocytes	Monocytes
	Monocytes	Macrophages	Macrophages
	Macrophages	Granulocytes	Granulocytes
	Granulocytes	Natural killer cells	
	Natural killer cells		
Ligand	ICAM-1	C3bi	C3bi
	ICAM-2		
Functions inhibited with monoclonal antibody	Extravasation CTL killing T-B conjugate formation ADCC	Opsonization Granulocyte adherence, aggregation, and chemotaxis ADCC	Granulocyte adherence and aggregation

*CR3 = type 3 complement receptor, also known as Mac-1; CR4 = type 4 complement receptor, also known as gp150/95;

LFA-1, CR3, and CR4 are heterodimers containing a common β chain but different α chains designated L, M, and X, respectively.

Immunodeficiency Disorders Are Treated by Replacement of the Defective Element

Although there are no cures for immunodeficiency disorders, there are several treatment possibilities. In addition to the drastic option of total isolation from exposure to any microbial agent, treatment options for the immunodeficiencies include:

- replacement of a missing protein
- replacement of a missing cell type or lineage
- replacement of a missing or defective gene

For disorders that impair antibody production, the classic course of treatment is administration of the missing protein immunoglobulin. Pooled human gamma globulin given either intravenously or subcutaneously protects against recurrent infection in many types of immunodeficiency. Maintenance of reasonably high levels of serum immunoglobulin (5 mg/ml serum) will prevent most common infections in the agammaglobulinemic patient. This is generally accomplished by the administration of immunoglobulin that has been selected for antibodies directed against a particular organism. Recent advances in the preparation of human monoclonal antibodies and in the ability to genetically engineer chimeric antibodies with mouse V regions and human-derived C regions make it possible to prepare antibodies specific for important pathogens (see Chapter 5).

Advances in molecular biology make it possible to clone the genes that encode other immunologically important proteins, such as cytokines, and to express these genes *in vitro*, using bacterial or eukaryotic expression systems. The availability of such proteins allows new modes of therapy in which immunologically important proteins may be replaced or their concentrations increased in the patient. For example, the administration of recombinant IFN- γ has proven effective for patients with CGD, and the use of recombinant IL-2 may help to restore immune function in AIDS patients. Recombinant adenosine deaminase has been successfully administered to ADA deficient SCID patients.

Cell replacement as therapy for immunodeficiencies has been made possible by recent progress in bone-marrow transplantation (see Chapter 21). Replacement of stem cells with those from an immunocompetent donor allows development of a functional immune system (see Clinical Focus Chapter 2). High rates of success have been reported for those who are fortunate enough to have an HLA-identical donor. Careful matching of patients with donors and the ability to manipulate stem-cell populations to select CD34⁺ precursor cells continues to minimize the risk in this procedure, even when no ideal donor exists. These procedures have been highly successful with SCID infants when haploidentical (complete match of one HLA gene set or haplotype) donor marrow is used. T cells are depleted and CD34⁺ stem cells are enriched before introducing the donor bone marrow into the SCID infant. Because this therapy has been used only in recent years,

it is not known whether transplantation cures the immunodeficiency permanently. A variation of bone-marrow transplantation is the injection of paternal CD34⁺ cells *in utero* when the birth of an infant with SCID is expected. Two infants born after this procedure had normal T-cell function and did not develop the infections that characterize SCID.

If a single gene defect has been identified, as in adenosine deaminase deficiency or chronic granulomatous disease, replacement of the defective gene may be a treatment option. Clinical tests of such therapy are underway for SCID caused by ADA deficiency and for chronic granulomatous disease with defective p67^{phox}, with promising initial results. Disease remission for up to 18 months was seen in the SCID patients and up to 6 months in the CGD patients. A similar procedure was used in both trials. It begins with obtaining cells (CD34⁺ stem cells are usually selected for these procedures) from the patient and transfecting them with a normal copy of the defective gene. The transfected cells are then returned to the patient. As this treatment improves, it will become applicable to a number of immunodeficiencies for which a genetic defect is well defined. As mentioned above, these include defects in genes that encode the γ chain of the IL-2 receptor, JAK-3, and ZAP-70, all of which give rise to SCID.

Experimental Models of Immunodeficiency Include Genetically Altered Animals

Immunologists use two well-studied animal models of primary immunodeficiency for a variety of experimental purposes. One of these is the athymic, or nude, mouse; the other is the severe combined immunodeficiency, or SCID, mouse.

NUDE (ATHYMIC) MICE

A genetic trait designated *nu*, which is controlled by a recessive gene on chromosome 11, was discovered in certain mice. Mice homozygous for this trait (*nu/nu*) are hairless and have a vestigial thymus (Figure 19-5). Heterozygotic, *nu/+*, litter mates have hair and a normal thymus. It is not known whether the hairlessness and the thymus defect are caused by the same gene. It is possible that two very closely linked genes control these defects, which, although unrelated, appear together in this mutant mouse. A gene that controls development may be involved, since the pathway that leads to the differential development of the thymus is related to the one that controls the skin epithelial cells. The *nu/nu* mouse cannot easily survive; under normal conditions, the mortality is 100% within 25 weeks and 50% die within the first two weeks after birth. Therefore, when these animals are to be used for experimental purposes, they must be maintained under conditions that protect them from infection. Precautions include use of sterilized food, water, cages, and bedding. The cages are protected from dust by placing them in a laminar flow rack or by the use of air filters fitted over the individual cages.

Nude mice lack cell-mediated immune responses, and they are unable to make antibodies to most antigens. The



FIGURE 19-5 A nude mouse (*nu/nu*). This defect leads to absence of a thymus or a vestigial thymus and cell-mediated im-

munodeficiency. [Courtesy of the Jackson Laboratory, Bar Harbor, Maine.]

immunodeficiency in the nude mouse can be reversed by a thymic transplant. Because they can permanently tolerate both allografts and xenografts, they have a number of practical experimental uses. For example, hybridomas or solid tumors from any origin may be grown as ascites or as implanted tumors in a nude mouse. It is known that the nude mouse does not completely lack T cells; rather, it has a limited population that increases with age. The source of these T cells is not known; an intriguing possibility is that there is an extrathymic source of mature T cells. However, it is more likely that the T cells arise from the vestigial thymus. The majority of cells in the circulation of a nude mouse carry T-cell receptors of the $\gamma\delta$ type instead of the $\alpha\beta$ type that prevails in the circulation of a normal mouse.

THE SCID MOUSE

In 1983, Melvin and Gayle Bosma and their colleagues described an autosomal recessive mutation in mice that gave rise to a severe deficiency in mature lymphocytes. They designated the trait SCID because of its similarity to human severe combined immunodeficiency. The SCID mouse was shown to have early B- and T-lineage cells, but there was a virtual absence of lymphoid cells in the thymus, spleen, lymph nodes, and gut tissue, the usual locations of functional T and B cells. The precursor T and B cells in the SCID mouse appeared to be unable to differentiate into mature functional B and T lymphocytes. Inbred mouse lines carrying the SCID defect have been derived and studied in great detail. The SCID mouse can neither make antibody nor carry out delayed-type hypersensitivity (DTH) or graft-rejection reactions. If the animals are not kept in an extremely clean environment, they succumb to infection early in life. Cells other than lymphocytes develop normally in the SCID mouse; red blood cells, monocytes, and granulocytes are present and functional. SCID mice may be rendered immunologically competent by transplantation of stem cells from normal mice.

The mutation in a DNA protein kinase that causes mouse SCID is a so-called “leaky” mutation, because a certain number of SCID mice do produce immunoglobulin. About half of these leaky SCID mice can also reject skin allografts. This

finding suggests that the defective enzyme can function partly in T- and B-cell development, allowing normal differentiation of a small percentage of precursor cells. More recently, immunodeficient SCID-like mice have been developed by deletion of the recombination-activating enzymes (RAG-1 and RAG-2) responsible for the rearrangement of immunoglobulin or T-cell–receptor genes in both B- and T-cell precursors (RAG knockout mice). This gives rise to a defect in both B and T cells of the mouse; neither can rearrange the genes for their receptor and thus neither proceeds along a normal developmental path. Because cells with abnormal rearrangements are eliminated *in vivo*, both B and T cells are absent from the lymphoid organs of the RAG knockout mouse. In addition to providing a window into possible causes of combined T- and B-cell immunodeficiency, the SCID mouse has proven extremely useful in studies of cellular immunology. Because its rejection mechanisms do not operate, the SCID mouse can be used for studies on cells or organs from various sources. For example, immune precursor cells from human sources may be used to reestablish the SCID mouse’s immune system. These human cells can develop in a normal fashion and, as a result, the SCID mouse circulation will contain immunoglobulin of human origin. In one important application, these SCID mice are infected with HIV-1. Although normal mice are not susceptible to HIV-1 infection, the SCID mouse reconstituted with human lymphoid tissue (SCID-Hu mouse) provides an animal model in which to test therapeutic or prophylactic strategies against HIV infection of the transplanted human lymphoid tissue.

AIDS and Other Acquired or Secondary Immunodeficiencies

As described above, a variety of defects in the immune system give rise to immunodeficiency. In addition to the primary immunodeficiencies, there are also acquired, or secondary, immunodeficiencies. One that has been known for some time is called acquired hypogammaglobulinemia.

(As mentioned above, this condition is sometimes confused with common variable immunodeficiency, a condition that shows genetic predisposition.) The origin of acquired hypogammaglobulinemia is unknown, and its major symptom, recurrent infection, manifests itself in young adults. The patients generally have very low but detectable levels of total immunoglobulin. T-cell numbers and function may be normal, but there are some cases with T-cell defects and these may grow more severe as the disease progresses. The disease is generally treated by immunoglobulin therapy, allowing patients to survive into their seventh and eighth decades. Unlike similar deficiencies described above, there is no evidence for genetic transmission of this disease. Mothers with acquired hypogammaglobulinemia deliver normal infants. However, at birth the infants will be deficient in circulating immunoglobulin, because the deficiency in maternal circulation is reflected in the infant.

Another form of secondary immunodeficiency, known as agent-induced immunodeficiency, results from exposure to any of a number of chemical and biological agents that induce an immunodeficient state. Certain of these are drugs used to combat autoimmune diseases such as rheumatoid arthritis or lupus erythematosus. Corticosteroids, which are commonly used for autoimmune disorders, interfere with the immune response in order to relieve disease symptoms. Similarly, a state of immunodeficiency is deliberately induced in transplantation patients who are given immunosuppressive drugs, such as cyclosporin A, in order to blunt the attack of the immune system on transplanted organs. As will be described in Chapter 21, there are recent efforts to use more specific means of inducing tolerance to allografts to circumvent the unwanted side effects of general immunosuppression. The mechanism of action of the immunosuppressive agents varies, although T cells are a common target. In addition, cytotoxic drugs or radiation treatments given to treat various forms of cancer frequently damage the dividing cells in the body, including those of the immune system, and induce a state of immunodeficiency as an unwanted consequence. Patients undergoing such therapy must be monitored closely and treated with antibiotics or immunoglobulin if infection appears.

HIV/AIDS Has Claimed Millions of Lives Worldwide

In recent years, all other forms of immunodeficiency have been overshadowed by an epidemic of severe immunodeficiency caused by the infectious agent called human immunodeficiency virus 1, or HIV-1. The disease that HIV-1 causes, acquired immunodeficiency syndrome (AIDS) was first reported in the United States in 1981 in Los Angeles, New York, and San Francisco. A group of patients displayed unusual infections, including the opportunistic fungal pathogen *Pneumocystis carinii*, which causes a pneumonia called PCP (*P. carinii* pneumonia) in persons with immunodeficiency. In addition to PCP, some patients had Kaposi's sarcoma, an extremely rare skin tumor, as well as other, rarely encoun-

tered opportunistic infections. More complete evaluation of the patients showed that they had in common a marked deficiency in cellular immune responses and a significant decrease in the subpopulation of T cells that carry the CD4 marker (T helper cells.) When epidemiologists examined the background of the first patients with this new syndrome, it was found that the majority of those afflicted were homosexual males. As the number of AIDS cases increased and the disease was recognized throughout the world, persons found to be at high risk for AIDS were homosexual males, promiscuous heterosexual individuals of either sex and their partners, intravenous drug users, persons who received blood or blood products prior to 1985, and infants born to HIV-infected mothers.

Since its discovery in 1981, AIDS has increased to epidemic proportions throughout the world. As of December 2000, the cumulative total number of persons in the United States reported to have AIDS was 688,200, and of these approximately 420,000 have died. Although reporting of AIDS cases is mandatory, many states do not require reporting of cases of HIV infection that have not yet progressed to AIDS. Therefore, there is no official count of the number of HIV-infected individuals; as many as 1 million Americans are estimated to be infected. Although the death rate from AIDS has decreased in recent years because of improved treatments, AIDS remains among the leading killers of persons in the 25–44-year-old age range in this country (Figure 19-6). The fact that the number of yearly AIDS deaths has leveled off is encouraging, but does not indicate an end to the epidemic in this country; there were an estimated 45,000 persons newly infected in 2000.

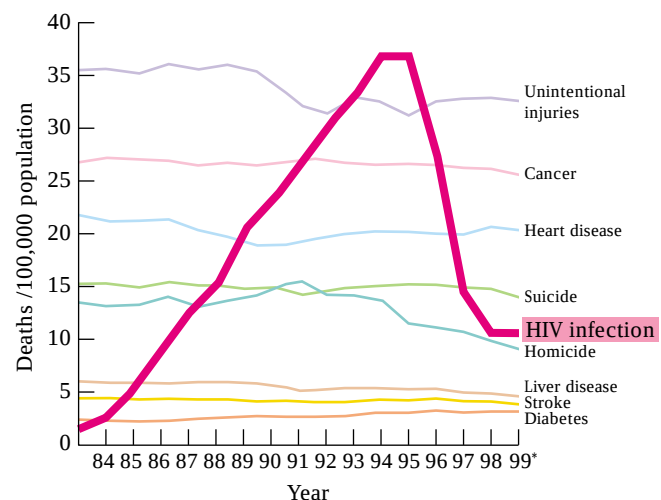


FIGURE 19-6 Death rates of the leading causes of death in persons aged 25–44 years in the United States for the years 1982–99 (* = preliminary data). The heavy line shows that the death rate per 100,000 persons caused by AIDS surpassed any other single cause of death in this age range during the period 1993 to 1995. The recent decrease in AIDS deaths in the United States is attributed to improvements in anti-HIV drug therapy, which prolongs the lives of patients. [National Vital Statistics Report.]

The magnitude of the AIDS epidemic in the United States is dwarfed by figures for other parts of the world. The global distribution of those afflicted with AIDS is shown in Figure 19-7. In sub-Saharan Africa an estimated 25.3 million persons were living with AIDS at the end of 2000, and in South and Southeast Asia there were another 5.8 million. There are an estimated 36.1 million persons worldwide with AIDS, including over 5 million children. In addition, there are over 8 million children who have been orphaned by the death of their parents from AIDS. Recent estimates from the World Health Organization indicate that there were 5.3 million new HIV infections in 2000, or an average of almost 15,000 persons infected each day during that year. This number includes a daily infection toll of 1700 children under 15 years of age.

The initial group of AIDS patients in the United States and Western Europe was predominantly white and male. Although this remains the group predominantly affected in these areas, more recently the distribution in the United States has shifted to include a larger proportion of women (20% in 2000 versus 6% in 1985) and an increasing proportion of minorities (39% black and Hispanic in 1996 versus

11% in 1985). Worldwide, the number of AIDS patients distributes more equally between males and females, and in sub-Saharan Africa, which has the highest incidence of AIDS, about 50% of those afflicted are females.

HIV-1 Spreads by Sexual Contact, Infected Blood, and from Mother to Infant

Although the precise mechanism by which HIV-1 infects an individual is not known, epidemiological data indicate that common means of transmission include homosexual and heterosexual intercourse, receipt of infected blood or blood products, and passage from mothers to infants. Before tests for HIV in the blood supply were routinely used, patients who received blood transfusions and hemophiliacs who received blood products were at risk for HIV-1 infection. Exposure to infected blood accounts for the high incidence of AIDS among intravenous drug users who normally share hypodermic needles. Infants born to mothers who are infected with HIV-1 are at high risk of infection. Unless infected mothers are treated with anti-viral agents before delivery, approximately

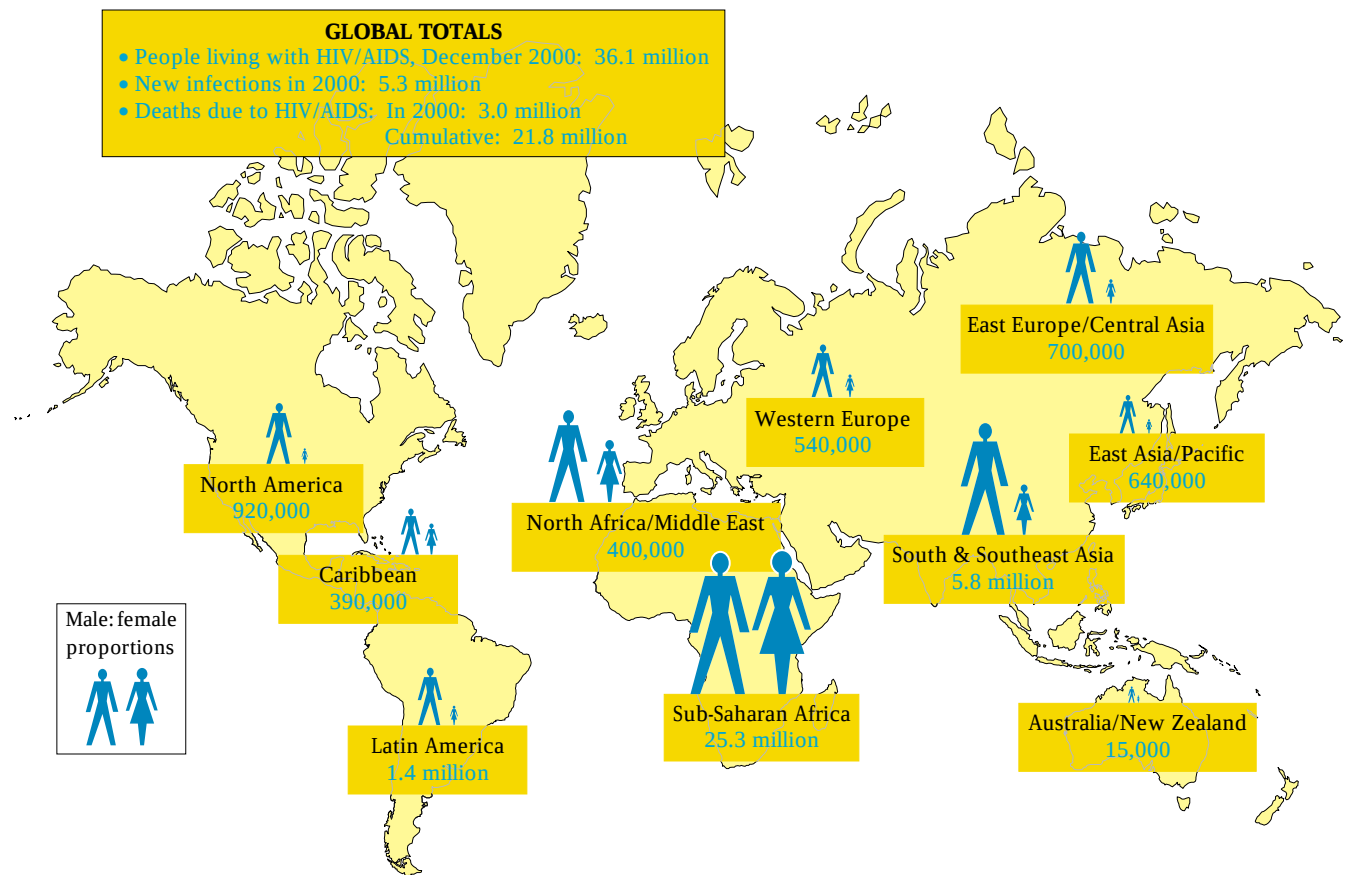


FIGURE 19-7 The global AIDS epidemic. The estimated worldwide distribution of AIDS cases as of December 2000. There were approximately 36.1 million persons living with AIDS as of December 2000; most of these were in sub-Saharan Africa and Southeast Asia. In

North America and Western Europe, about 80% of those affected were men, whereas in Africa nearly equal numbers of women and men have AIDS. [HIV/AIDS UNAIDS: Report on the Global Epidemic, 2000.]

30% of infants born to them will become infected with the virus (see Clinical Focus). Possible vehicles of passage from mother to infant include blood transferred in the birth process and milk in the nursing period. Transmission from an infected to an uninfected individual is most likely by transmission of HIV-infected cells—in particular, macrophages, dendritic cells, and lymphocytes.

In the worldwide epidemic, it is estimated that 75% of the cases of HIV transmission are attributable to heterosexual contact. While the probability of transmission by vaginal intercourse is lower than by other means, such as IV drug use or receptive anal intercourse, the likelihood of infection is greatly enhanced by the presence of other sexually transmitted diseases (STDs). In populations where prostitution is rampant, STDs flourish and provide a powerful cofactor for the heterosexual transmission of HIV-1. Reasons for this increased infection rate include the lesions and open sores present in many STDs, which favor the transfer of HIV-infected blood during intercourse.

While the AIDS epidemic has engendered an understandable fear of infection among most informed individuals, there are also exaggerated claims of the ease with which HIV infection may be passed on. At present, there is no evidence that casual contact with or touching an infected person can spread HIV-1 infection. Airborne transmission has never been observed to cause infection. In virtually every well-documented case of HIV-1 infection, there is evidence for contact with blood, milk, semen, or vaginal fluid from an infected individual. Research workers and medical professionals who take reasonable precautions have a very low incidence of AIDS, despite repeated contact with infected materials. The risk of transmitting HIV infection can be minimized by simple precautionary measures, including the avoidance of any practice that could allow exposure of broken or abraded skin or any mucosal membrane to blood from a potentially infected person. The use of condoms when having sex with individuals of unknown infection status is highly recommended. One factor contributing to the spread of HIV is the long period after infection during which no clinical signs may appear but during which the infected individual may infect others. Thus, universal use of precautionary measures is important whenever and wherever infection status is uncertain.

It is a sobering thought that the epidemic of AIDS came at a time when many believed that infectious diseases no longer posed a serious threat to people in the United States and other industrialized nations. Vaccines and antibiotics controlled most serious infectious agents. The eradication of smallpox in the world had recently been celebrated, and polio was yielding to widespread vaccination efforts; these were considered milestones on the road to elimination of most infectious diseases. The outbreak of AIDS shattered this complacency and triggered a massive effort to combat this disease. In addition, the immunodeficiency that characterizes AIDS has allowed re-emergence of other infectious diseases, such as tuberculo-

sis, which have the potential to spread into populations not infected with HIV.

A Retrovirus, HIV-1, Is the Causative Agent of AIDS

Within a few years after recognition of AIDS as an infectious disease, the causative agent was discovered and characterized by efforts in the laboratories of Luc Montagnier in Paris and Robert Gallo in Bethesda (Figure 19-8). This immunodeficiency syndrome was novel at the time in that the type of virus causing it was a **retrovirus**. Retroviruses carry their genetic information in the form of RNA. When the virus enters a cell, the RNA is reverse transcribed to DNA by a virally encoded enzyme, reverse transcriptase (RT). As the name implies, RT reverses the normal transcription process and makes a DNA copy of the viral RNA genome. This copy, which is called a **provirus**, is integrated into the cell genome and is replicated along with the cell DNA. When the provirus is expressed to form new virions, the cell lyses. Alternatively, the provirus may remain latent in the cell until some regulatory signal starts the expression process.

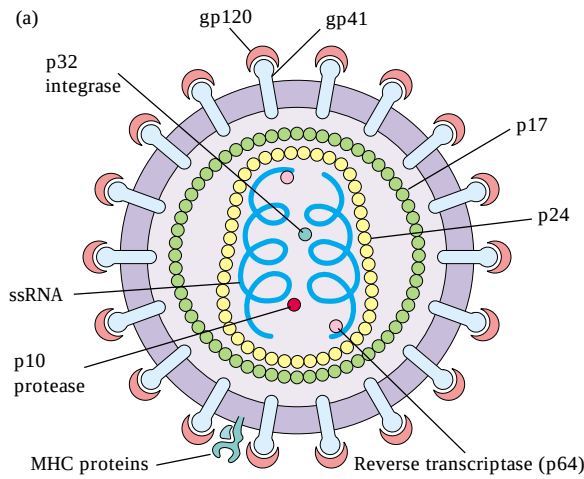
Only one other human retrovirus, human T-cell lymphotropic virus I, or HTLV-I, had been described before HIV-1. This retrovirus is endemic in the southern part of Japan and in the Caribbean. Although most individuals infected with HTLV-I display no clinical signs of disease, a small percentage develop serious illness, either adult T-cell leukemia, which is aggressive and usually fatal, or a disabling progressive neurologic disorder called HTLV-I-associated myelopathy (called tropical spastic paraparesis in early reports). Although comparisons of their genomic sequences revealed that HIV-1 is not a close relative of HTLV-I, similarities in overall characteristics led to use of the name HTLV-III for the AIDS virus in early reports. There is also a related human virus called HIV-2, which is less pathogenic in humans than HIV-1. HIV-2 is similar to viruses isolated from monkeys; it infects certain nonhuman primates that are not infected by HIV-1.

Viruses related to HIV-1 have been found in nonhuman primates. These viruses, variants of simian immunodeficiency virus, or SIV, cause immunodeficiency disease in certain infected monkeys. Normally, SIV strains cause no disease in their normal host but produce immunodeficiency similar to AIDS when injected into another species. For example, the virus from African green monkeys (SIV_{agm}) is present in a high percentage of normal healthy African green monkeys in the wild. However, when SIV_{agm} is injected into macaques, it causes a severe, often lethal, immunodeficiency.

A number of other animal retroviruses more or less similar to HIV-1 have been reported. These include the feline and bovine immunodeficiency viruses and the mouse leukemia virus. Study of these animal viruses has yielded information concerning the general nature of retrovirus action, but specific information about HIV-1 cannot be gained by infecting ani-



VISUALIZING CONCEPTS



(b)

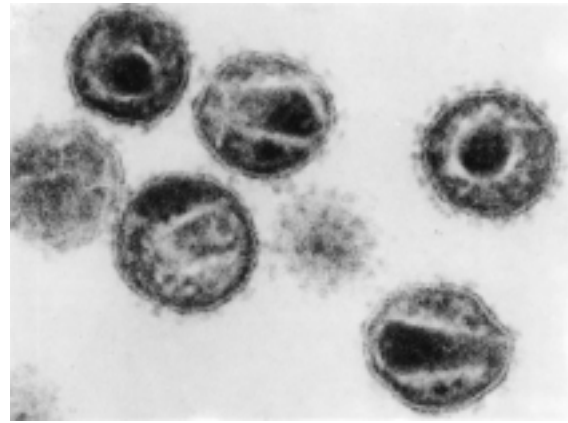


FIGURE 19-8 Structure of HIV. (a) Cross-sectional schematic diagram of HIV virion. Each virion expresses 72 glycoprotein projections composed of gp120 and gp41. The gp41 molecule is a transmembrane molecule that crosses the lipid bilayer of the viral envelope. Gp120 is associated with gp41 and serves as the viral receptor for CD4 on host cells. The viral envelope derives from the host cell and contains some host-cell membrane proteins, including class I and class II MHC molecules. Within the envelope is the viral core, or nucleocapsid, which includes a layer of a protein called p17 and an inner layer of a protein called p24. The HIV

genome consists of two copies of single-stranded RNA, which are associated with two molecules of reverse transcriptase (p64) and nucleoid proteins p10, a protease, and p32, an integrase. (b) Electron micrograph of HIV virions magnified 200,000 times. The glycoprotein projections are faintly visible as “knobs” extending from the periphery of each virion. [Part (a) adapted from B. M. Peterlin and P. A. Luciw, 1988, *AIDS* 2:S29; part (b) from a micrograph by Hans Geldenblom of the Robert Koch Institute (Berlin), in R. C. Gallo and L. Montagnier, 1988, *Sci. Am.* 259(6):41.]

imals because HIV-1 does not replicate in them. Only the chimpanzee supports infection with HIV-1 at a level sufficient to be useful in vaccine trials, but infected chimpanzees only rarely develop AIDS, which limits the value of this model in the study of viral pathogenesis. In addition, the number of chimpanzees available for such studies is low and both the expense and the ethical issues involved in experiments with chimpanzees preclude widespread use of this infection model. The SCID mouse (see above) reconstituted with human lymphoid tissue for infection with HIV-1 has been useful for certain studies of HIV-1 infection, especially in the development of drugs to combat viral replication.

Reasons for the limited host range of HIV-1 include not only the cell-surface receptors required for entry of the virus into the host cell but dependence of the virus on host-cell factors for early events in its replication process, such as transcription and splicing of viral messages. For example, mouse cells transfected with genes that mediate expression of the human receptors for HIV-1 will not support HIV-1 replication because they lack other host factors. By contrast, cells

from hamsters or rabbits transfected to express the human receptors support levels of virus replication similar to those seen in human cells. Despite some progress in understanding the factors needed for HIV-1 infection, no clear candidate for an animal model of HIV-1 infection exists. This lack of a suitable infection model hampers efforts to develop both drugs and vaccines to combat AIDS.

Recent publicity focused on activists claiming that there is no connection between HIV and AIDS and that antiretroviral drugs are useless to combat the disease. The so-called AIDS denialists believe that precautions against infection are not necessary, and that testing for HIV infection has no value because treatment is worthless or harmful. Some even deny the existence of an epidemic or that AIDS is an actual disease. While science requires that all ideas should be tested, denial of medical care to infected individuals based on this fringe group's notions is not an option. All relevant studies support a near perfect correlation between HIV infection and disease; drugs that lower the amount of virus in a patient (viral load) prevent opportunistic infections.



CLINICAL FOCUS

Prevention of Infant HIV Infection by Anti-Retroviral Treatment

Approximately

500,000 infants become infected with HIV each year. The majority of these infections result from transmission of virus from HIV-infected mothers during childbirth or by transfer of virus from milk during breast-feeding. The incidence of maternal acquired infection can be reduced as much as 67% by treatment of the infected mother with a course of Zidovudine (AZT) for several months prior to delivery, and treatment of her infant for 6 weeks after birth. This treatment regimen is widely used in the U.S. However, the majority of worldwide HIV infection of infants occurs in sub-Saharan Africa and other less developed areas, where the cost and timing of the Zidovudine regimen render it an impractical solution to the problem of maternal-infant HIV transmission.

A 1999 clinical trial of the anti-retroviral Nevirapine (viramune) brings hope for a practical way to combat infant HIV infection under less than ideal conditions of clinical care. The trial took place at Mulago Hospital in Kampala, Uganda, and enrolled 645 mothers who tested positive for HIV infection. About half of the mothers were given a single dose of Nevirapine at the onset of labor and their infants were given a single dose 24–30 hours after birth. The dose and timing were dictated by the customary rapid discharge at the hospital. The control arm of the study involved a more extensive course of Zidovudine, but in-country conditions did not allow exact replication of the full course administered to infected mothers in the U.S.

The overall rate of infection for infants born to untreated mothers is esti-

mated to be about 37%. When the full course of Zidovudine is used, the rate drops to 20%. The highly encouraging results of the Uganda study revealed infection in only 13.1% of the babies in the Nevirapine group when tested at 16 weeks of age. Of those given a short course of Zidovudine, 25.1% were infected at this age compared to 40.2% in a small group given placebo. From this study it appears that the single dose of Nevirapine is the most effective means found thus far to prevent maternal-infant transmission of HIV infection—even better than the more extensive and costly regimen currently used in developed countries. These results must be verified

and the possibility of unexpected side effects must be explored. However, this result gives hope for reduction of infant infection in parts of the world where access to medical care is limited.

As mentioned above, the study was designed to conform to the reality of maternal health care in Kampala; it fits this system perfectly. The use of Nevirapine has other significant advantages, including stability of the drug at room temperature and reasonable cost. The dose of Nevirapine administered to the mother and infant costs about 200 times less than the Zidovudine regimen in current use in the U.S. In fact, the treatment is sufficiently inexpensive to suggest that it may be cost-effective to treat all mothers at the time of delivery in those areas where rates of infection are high, because the Nevirapine treatment costs less than the tests used to determine HIV infection. Obviously, such a strategy must be embarked upon cautiously, given the danger of long-term side effects and other unexpected problems.



Mural showing mother and child on an outside wall of Mulago Hospital Complex in Kampala, Uganda, site of the clinical trial demonstrating that maternal-infant HIV-1 transmission was greatly reduced by Nevirapine. [Courtesy of Thomas Quinn, Johns Hopkins University.]

In Vitro Studies Revealed the HIV-1 Replication Cycle

The AIDS virus can infect human T cells in culture, replicating itself and in many cases causing the lysis of the cell host (Figure 19-9). Much has been learned about the life cycle of HIV-1 from in vitro studies. The various proteins encoded by the viral genome have been characterized and the functions of most of them are known (Figure 19-10).

The first step in HIV infection is viral attachment and entry into the target cell. HIV-1 infects T cells that carry the CD4 antigen on their surface; in addition, certain HIV strains will infect monocytes and other cells that have CD4 on their surface. The preference for CD4⁺ cells is due to a high-affinity interaction between a coat (envelope or env) protein of HIV-1 and cell-surface CD4. Although the virus binds to CD4 on the cell surface, this interaction alone is not sufficient for entry and productive infection. Expression of other cell-surface molecules, coreceptors present on T cells and monocytes, is required for HIV-1 infection. The infection of a T cell, depicted in Figure 19-11a, is assisted by the T-cell coreceptor CXCR4 (in initial reports, this molecule was called fusin). An analogous receptor called CCR5 functions for the monocyte or macrophage.

After the virus has entered the cell, the RNA genome of the virus is reverse transcribed and a cDNA copy (provirus) integrates into the host genome. The integrated provirus is

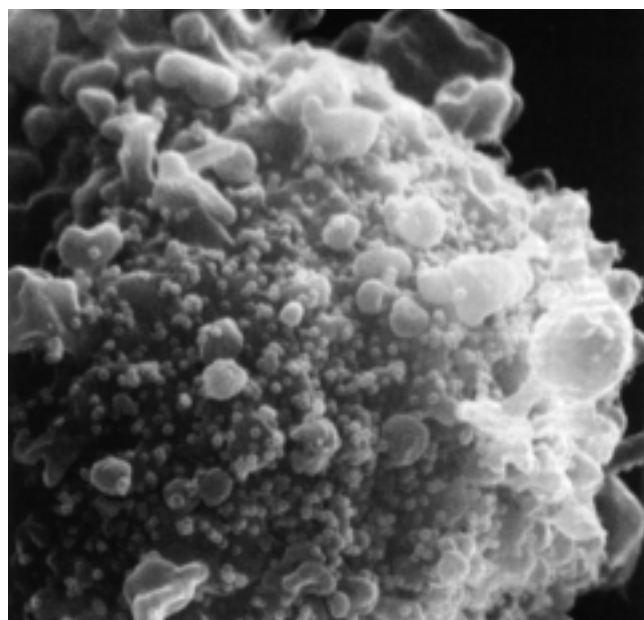


FIGURE 19-9 Once the HIV provirus has been activated, buds representing newly formed viral particles can be observed on the surface of an infected T cell. The extensive cell damage resulting from budding and release of virions leads to the death of infected cells. [Courtesy of R. C. Gallo, 1988, *J. Acquired Immune Deficiency Syndromes* 1:521.]

transcribed and the various viral RNA messages spliced and translated into proteins, which along with a complete new copy of the RNA genome are used to form new viral particles (Figure 19-11b). The gag proteins of the virus are cleaved by the viral protease into the forms that make up the nuclear capsid (see Figure 19-10) in a mature infectious viral particle. As will be described below, different stages in this viral replication process can be targeted by antiviral drugs.

The discovery that CXCR4 and CCR5 serve as coreceptors for HIV-1 on T cells and macrophages, respectively, explained why some strains of HIV-1 preferentially infect T cells (T-tropic strains) while others prefer macrophages (M-tropic strains). A T-tropic strain uses CXCR4, while the M-tropic strains use CCR5. This use of different coreceptors also helped to explain the different roles of cytokines and chemokines in virus replication. It was known from in vitro studies that certain chemokines had a negative effect on virus replication while certain pro-inflammatory cytokines had a positive effect. Both of the HIV coreceptors, CCR5 and CXCR4, function as receptors for chemokines (see Table 15-2). Because the receptors cannot bind simultaneously to HIV-1 and to their chemokine ligand, there is competition for the receptor between the virus and the normal ligand (Figure 19-11c), and the chemokine can block viral entry into the host cell. Whereas the chemokines compete with HIV for usage of the coreceptor and thus inhibit viral entry, the pro-inflammatory cytokines induce greater expression of the chemokine receptors on the cell surface, making the cells more susceptible to viral entry.

HIV-1 infection of T cells with certain strains of virus leads to the formation of giant cells or syncytia. These are formed by the fusion of a group of cells caused by the interaction of the viral envelope protein gp120 on the surface of infected cells with CD4 and the coreceptors on the surface of other cells, infected or not. After the initial binding, the action of other cell-adhesion molecules welds the cells together in a large multinuclear mass with a characteristic fused ballooning membrane which eventually bursts. Formation of syncytia may be blocked by antibodies to some of the epitopes of the CD4 molecule, by soluble forms of the CD4 molecule (prepared by in vitro expression of a CD4 gene genetically engineered to lack the transmembrane portion), and by antibodies to cell-adhesion molecules. Individual isolates of HIV-1 differ in their ability to induce syncytia formation.

Isolates of HIV-1 from different sources were formerly classified as syncytia-inducing (SI) or non-syncytium-inducing (NSI). In most cases, these differences correlated with the ability of the virus to infect T cells or macrophages: T-tropic strains were SI, whereas M-tropic strains were NSI. More recent classifications of HIV-1 are based on which coreceptor the virus uses; there is good but not absolute correlation between the use of CXCR4, which is present on T cells, and syncytia-inducing ability. The NSI strains use

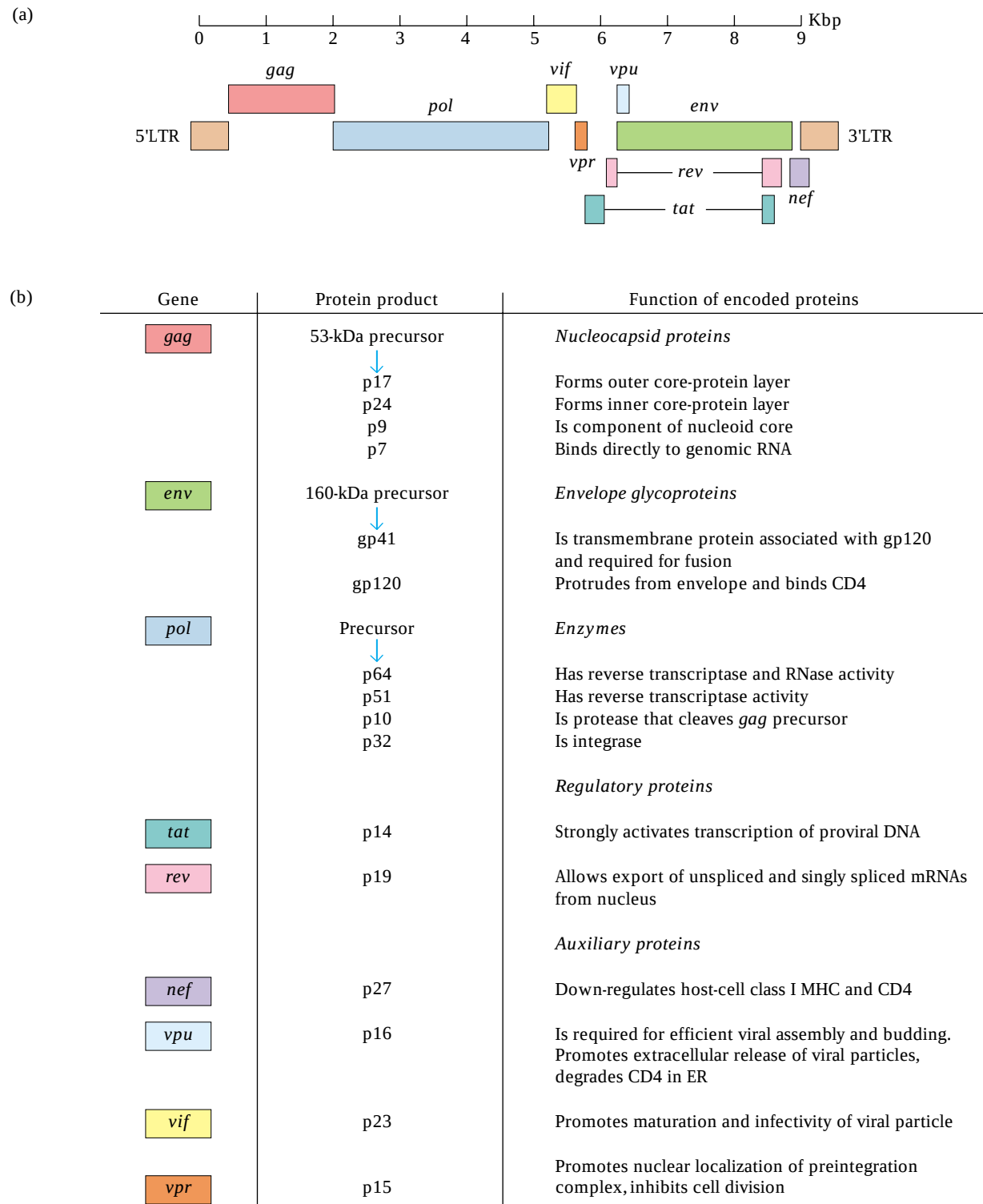


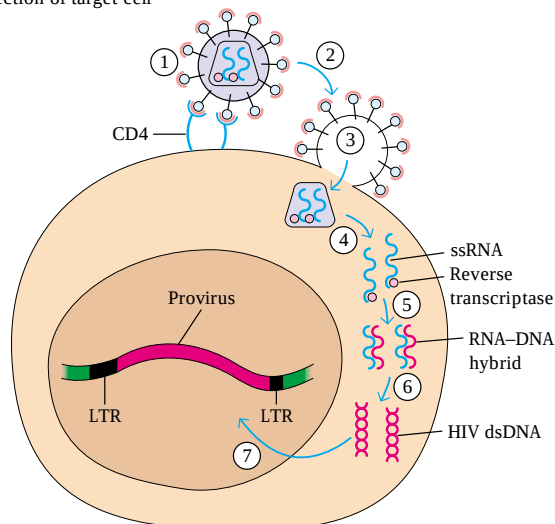
FIGURE 19-10 Genetic organization of HIV-1 (a) and functions of encoded proteins (b). The three major genes—*gag*, *pol*, and *env*—encode polyprotein precursors that are cleaved to yield the nucleocapsid core proteins, enzymes required for replication, and envelope core proteins. Of the remaining six genes, three (*tat*, *rev*, and *nef*) encode regulatory proteins that play a major role in controlling expres-

sion; two (*vif* and *vpu*) encode proteins required for virion maturation; and one (*vpr*) encodes a weak transcriptional activator. The 5' long terminal repeat (LTR) contains sequences to which various regulatory proteins bind. The organization of the HIV-2 and SIV genomes are very similar, except that the *vpu* gene is replaced by *vpx* in both of these.



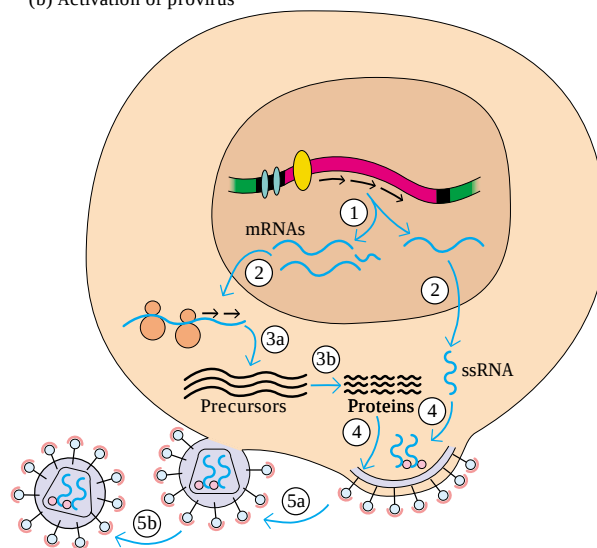
VISUALIZING CONCEPTS

(a) Infection of target cell



- ① HIV gp120 binds to CD4 on target cell.
- ② Fusogenic domain in gp41 and CXCR4, a G-protein-linked receptor in the target-cell membrane, mediate fusion.
- ③ Nucleocapsid containing viral genome and enzymes enters cells.
- ④ Viral genome and enzymes are released following removal of core proteins.
- ⑤ Viral reverse transcriptase catalyzes reverse transcription of ssRNA, forming RNA-DNA hybrids.
- ⑥ Original RNA template is partially degraded by ribonuclease H, followed by synthesis of second DNA strand to yield HIV dsDNA.
- ⑦ The viral dsDNA is then translocated to the nucleus and integrated into the host chromosomal DNA by the viral integrase enzyme.

(b) Activation of provirus



- ① Transcription factors stimulate transcription of proviral DNA into genomic ssRNA and, after processing, several mRNAs.
- ② Viral RNA is exported to cytoplasm.
- ③a Host-cell ribosomes catalyze synthesis of viral precursor proteins.
- ③b Viral protease cleaves precursors into viral proteins.
- ④ HIV ssRNA and proteins assemble beneath the host-cell membrane, into which gp41 and gp120 are inserted.
- ⑤a The membrane buds out, forming the viral envelope.
- ⑤b Released viral particles complete maturation; incorporated precursor proteins are cleaved by viral protease present in viral particles.

(c)

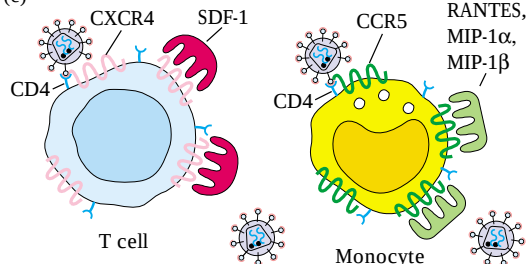


FIGURE 19-11 Overview of HIV infection of target cells and activation of provirus. (a) Following entry of HIV into cells and formation of dsDNA, integration of the viral DNA into the host-cell genome creates the provirus. (b) The provirus remains latent until events in the infected cell trigger its activation, leading to formation and release of

viral particles. (c) Although CD4 binds to the envelope glycoprotein of HIV-1, a second receptor is necessary for entry and infection. The T-cell-tropic strains of HIV-1 use the coreceptor CXCR4, while the macrophage-tropic strains use CCR5. Both are receptors for chemokines, and their normal ligands can block HIV infection of the cell.



CCR5, which is present on monocytes. Studies of the viral envelope protein gp120 identified a region called the V3 loop, which plays a role in the choice of receptors used by the virus. A study by Mark Goldsmith and Bruce Chesebro and their colleagues indicates that a single amino acid difference in this region of gp120 may be sufficient to determine which receptor is used.

HIV-1 Infection Leads to Opportunistic Infections

Isolation of HIV-1 and its growth in culture has allowed purification of viral proteins and the development of tests for infection with the virus. The most commonly used test is for the presence of antibodies directed against proteins of HIV-1. These generally appear in the serum of infected individuals by three months after the infection has occurred. When the antibodies appear, the individual is said to have seroconverted or to be seropositive for HIV-1. Although the precise course of HIV-1 infection and disease onset varies considerably in different patients, a general scheme for the progression of AIDS can be constructed (Figure 19-12). The course of HIV-1 infection begins with no detectable anti-HIV-1 an-

tibodies or virus and progresses to the full AIDS syndrome. Diagnosis of AIDS includes evidence for infection with HIV-1 (presence of antibodies or virus in blood), greatly diminished numbers of CD4⁺ T cells (< 200 cells/mm³), impaired or absent delayed-hypersensitivity reactions, and the occurrence of opportunistic infections (Table 19-3). Patients with AIDS generally succumb to tuberculosis, pneumonia, severe wasting diarrhea, or various malignancies. The time between acquisition of the virus and death from the immunodeficiency averages nine to twelve years. In the period between infection and severe disease, there may be few symptoms. Primary infection in a minority of patients may be symptomatic with fever, lymphadenopathy (swollen lymph nodes), and a rash, but these symptoms generally do not persist more than a few weeks. Most commonly, primary infection goes unnoticed and is followed by a long chronic phase, during which the infected individual shows little or no overt sign of HIV-1 infection.

The first overt indication of AIDS may be opportunistic infection with the fungus *Candida albicans*, which causes the appearance of sores in the mouth (thrush) and, in women, a vulvovaginal yeast infection that does not respond to treatment. A persistent hacking cough caused by *P. carinii* infection of the lungs may also be an early indicator. A rise in the level of circulating HIV-1 (viral load) in the plasma and concomitant drop in the number of CD4⁺ T cells generally previews this first appearance of symptoms. Some relation between the CD4⁺ T-cell number and the type of infection experienced by the patient has been established (see Table 19-3). Of intense interest to immunologists are the events that take place between the initial confrontation with HIV-1 and the takeover and collapse of the host immune system. Understanding how the immune system holds HIV-1 in check during this chronic phase can lead to the design of effective therapeutic and preventive strategies.

Research into the process that underlies the progression of HIV infection to AIDS has revealed a dynamic interplay between the virus and the immune system. The initial infection event causes dissemination of virus to lymphoid organs and a resultant strong immune response. This response, which involves both antibody and cytotoxic CD8⁺ T lymphocytes, keeps viral replication in check; after the initial burst of viremia (high levels of virus in the circulation), the viral level in the circulation achieves a steady state. Although the infected individual normally has no clinical signs of disease at this stage, viral replication continues and virus can be detected in circulation by sensitive PCR assays for viral RNA. These PCR-based assays, which measure **viral load** (the number of copies of viral genome in the plasma), have assumed a major role in determination of the patient's status and prognosis. Even when the level of virus in the circulation is stable, large amounts of virus are produced in infected CD4⁺ T cells; as many as 10⁹ virions are released every day and continually infect and destroy additional host T cells (Figure 19-13a). Despite this high rate of replication, the virus is kept in check by the immune system throughout the

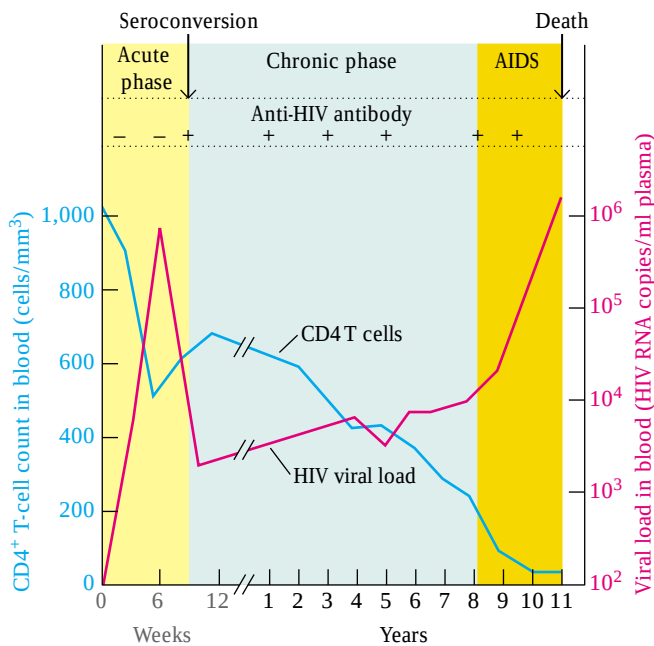


FIGURE 19-12 Serologic profile of HIV infection showing three stages in the infection process. Soon after infection, viral RNA is detectable in the serum. However, HIV infection is most commonly detected by the presence of anti-HIV antibodies after seroconversion, which normally occurs within a few months after infection. Clinical symptoms indicative of AIDS generally do not appear for at least 8 years after infection, but this interval is variable. The onset of clinical AIDS is usually signaled by a decrease in T-cell numbers and an increase in viral load. [Adapted from A. Fauci et al., 1996, *Annals Int. Med.* 124:654.]

TABLE 19-3 Clinical diagnosis of HIV-infected individuals

CD4 ⁺ T-cell count	CLINICAL CATEGORIES*		
	A	B	C
≥ 500/ μ l	A1	B1	C1
200–499/ μ l	A2	B2	C2
< 200/ μ l	A3	B3	C3

CLASSIFICATION OF AIDS INDICATOR DISEASE	
Category A	Category C
Asymptomatic: no symptoms at the time of HIV infection	Candidiasis of bronchi, tracheae, or lungs
Acute primary infection: glandular fever-like illness lasting a few weeks at the time of infection	Candidiasis, esophageal
Persistent generalized lymphadenopathy (PGL): lymph-node enlargement persisting for 3 or more months with no evidence of infection	Cervical cancer (invasive)
	Coccidioidomycosis, disseminated or extrapulmonary
	Cryptococcosis, extrapulmonary
	Cryptosporidiosis, chronic intestinal (> 1 month duration)
	Cytomegalovirus disease (other than liver, spleen, or nodes)
	Cytomegalovirus retinitis (with loss of vision)
	Encephalopathy, HIV-related
	Herpes simplex: chronic ulcer(s) (> 1 month duration), bronchitis, pneumonitis, or esophagitis
	Histoplasmosis, disseminated or extrapulmonary
	Isosporiasis, chronic intestinal (> 1 month duration)
	Kaposi's sarcoma
	Lymphoma, Burkitt's
	Lymphoma, immunoblastic
	Lymphoma, primary of brain
	<i>Mycobacterium avium</i> complex or <i>M. Kansaii</i> , disseminated or extrapulmonary
	<i>Mycobacterium tuberculosis</i> , any site
	<i>Mycobacterium</i> , other or unidentified species, disseminated or extrapulmonary
	<i>Pneumocystis carinii</i> pneumonia
	Progressive multifocal leukoencephalopathy
	<i>Salmonella</i> septicemia (recurrent)
	Toxoplasmosis of brain
	Wasting syndrome due to HIV

*All categories shown in bold type are considered AIDS. For Category A diagnosis, no condition in categories B or C can be present; for category B, no category C condition can be present.

SOURCE: CDC guidelines for AIDS diagnosis, 1993 revision.

chronic phase of infection, and the level of virus in circulation from about six months after infection is a good predictor of the course of disease. Low levels of virus in this period correlate with a longer time in which the infected individual remains free of opportunistic infection. But the virus eventually breaks through host immune defenses, resulting in an in-

crease in viral load, a decrease in CD4⁺ T cell numbers, increased opportunistic infection, and death of the patient.

While the viral load in plasma remains fairly stable throughout the period of chronic HIV infection, examination of the lymph nodes has revealed a different story. Fragments of nodes obtained by biopsy from infected subjects

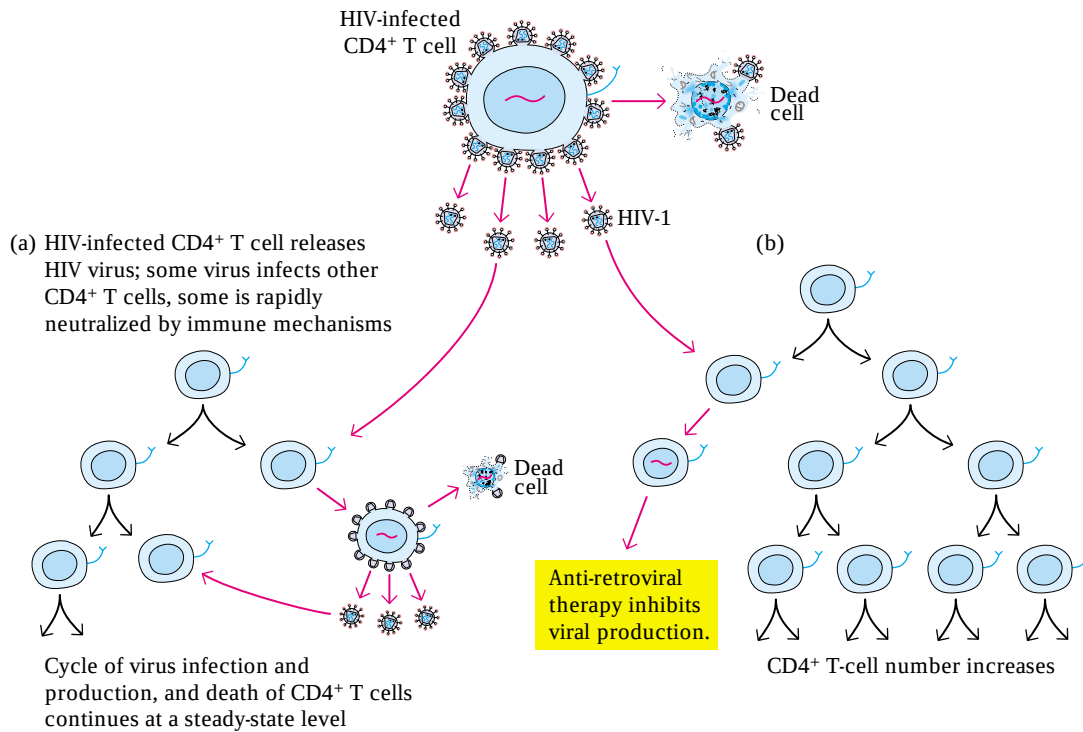


FIGURE 19-13 Production of virus by CD4⁺ T cells and maintenance of a steady state of viral load and T-cell number. (a) A dynamic relationship exists between the number of CD4⁺ cells and the amount of virus produced. As virus is produced, new CD4⁺ cells are infected, and these

infected cells have a half-life of 1.5 days. In progression to full AIDS, the viral load increases and the CD4⁺ T-cell count decreases before onset of opportunistic infections. (b) If the viral load is decreased by anti-retroviral treatment, the CD4⁺ T-cell number increases almost immediately.

showed high levels of infected cells at all stages of infection; in many cases, the structure of the lymph node had been completely destroyed by virus long before plasma viral load increased above the steady-state level.

The decrease in CD4⁺ T cells is the hallmark of AIDS. Several explanations have been advanced for the depletion of these cells in patients. In early studies, direct viral infection and destruction of CD4⁺ T cells was discounted as the primary cause, because the large numbers of circulating HIV-infected T cells predicted by the model were not found. More recent studies indicate that the reason for the difficulty in finding the infected cells is that they are so rapidly killed by HIV; the half-life of an actively infected CD4⁺ T cell is less than 1.5 days. There are smaller numbers of CD4⁺ T cells that become infected but do not actively replicate virus. These latently infected cells persist for long periods, and the integrated proviral DNA replicates in cell division along with cell DNA. Studies in which viral load is decreased by anti-retroviral therapy show a concurrent increase in CD4⁺ T cell numbers (Figure 19-13b). These data support a model of dynamic interaction between virus and T cells, with simultaneous high levels of viral production and rapid depletion of infected CD4⁺ T cells. While other mechanisms for depletion of CD4⁺ T cells may be envisioned, infection with HIV remains the prime suspect.

Not only depletion of CD4⁺ T cells but other immunologic consequences can be measured in HIV-infected individuals during the progression to AIDS. These include a decrease or absence of delayed hypersensitivity to antigens to which the individual normally reacts. Serum levels of immunoglobulins, especially IgG and IgA, show a sharp increase in the AIDS patient. This increase may be due to increased levels in HIV-infected individuals of a B-cell subpopulation with low CD21 expression and enhanced immunoglobulin secretion. This population proliferates poorly in response to B-cell mitogens. Cellular parameters of immunologic response, such as the proliferative response to mitogens, to antigens, or to alloantigens, all show a marked decrease. Generally, the HIV-infected individual loses the ability to mount T-cell responses in a predictable sequence: responses to specific antigens (for example, influenza virus) are first lost, then response to alloantigens declines, and lastly, the response to mitogens such as concanavalin A or phytohemagglutinin can no longer be detected. Table 19-4 lists some immune abnormalities in AIDS.

HIV-1 infected individuals often display dysfunction of the central and peripheral nervous systems. Specific viral DNA and RNA sequences have been detected by HIV-1 probes in the brains of children and adults with AIDS, suggesting that viral replication occurs there. Quantitative comparison of speci-

TABLE 19-4 Immunologic abnormalities associated with HIV infection

Stage of infection	Typical abnormalities observed
LYMPH NODE STRUCTURE	
Early	Infection and destruction of dendritic cells; some structural disruption
Late	Extensive damage and tissue necrosis; loss of follicular dendritic cells and germinal centers; inability to trap antigens or support activation of T and B cells
T HELPER (T _H) CELLS	
Early	No in vitro proliferative response to specific antigen
Late	Decrease in T _H -cell numbers and corresponding helper activities; no response to T-cell mitogens or alloantigens
ANTIBODY PRODUCTION	
Early	Enhanced nonspecific IgG and IgA production but reduced IgM synthesis
Late	No proliferation of B cells specific for HIV-1; no detectable anti-HIV antibodies in some patients; increased numbers of B cells with low CD21 and enhanced Ig secretion.
CYTOKINE PRODUCTION	
Early	Increased levels of some cytokines
Late	Shift in cytokine production from T _H 1 subset to T _H 2 subset
DELAYED-TYPE HYPERSENSITIVITY	
Early	Highly significant reduction in proliferative capacity of T _{DTH} cells and reduction in skin-test reactivity
Late	Elimination of DTH response; complete absence of skin-test reactivity
T CYTOTOXIC (T _C) CELLS	
Early	Normal reactivity
Late	Reduction but not elimination of CTL activity due to impaired ability to generate CTLs from T _C cells

mens from brain, lymph node, spleen, and lung of AIDS patients with progressive encephalopathy indicated that the brain was heavily infected. A frequent complication in later stages of HIV infection is AIDS dementia complex, a neurological syndrome characterized by abnormalities in cognition, motor performance, and behavior. Whether AIDS dementia and other clinical and histopathological effects observed in the central nervous systems of HIV-infected individuals are a direct effect of viral antigens on the brain, a consequence of immune responses to the virus, or a result of infection by opportunistic agents remains unknown.

Therapeutic Agents Inhibit Retrovirus Replication

Development of a vaccine to prevent the spread of AIDS is the highest priority for immunologists, but it is also critical to develop drugs and therapies that can reverse the effects of HIV-1 in infected individuals. The number of HIV-infected persons is estimated to be close to 1 million in the United

States alone; for all of these individuals to develop AIDS would be an enormous tragedy. There are several strategies for development of effective anti-viral drugs. The life cycle of HIV shows several susceptible points that might be blocked by pharmaceutical agents (Figure 19-14). The key to success of such therapies is that they must be specific for HIV-1 and interfere minimally with normal cell processes. Thus far, two types of antiviral agents have found their way into common usage. The first success in treatment was with drugs that interfere with the reverse transcription of viral RNA to cDNA; several drugs in common use operate at this step. A second stage of viral replication that has proved amenable to blockade is the step at which precursor proteins are cleaved into the units needed for construction of a new mature virion. This step requires the action of a specific viral protease, which can be inhibited by chemical agents; this precludes the formation of infectious viral particles.

Several antiretroviral drugs are now in widespread use (Table 19-5) that either interfere with reverse transcription or inhibit the viral protease. The prototype of the drugs that

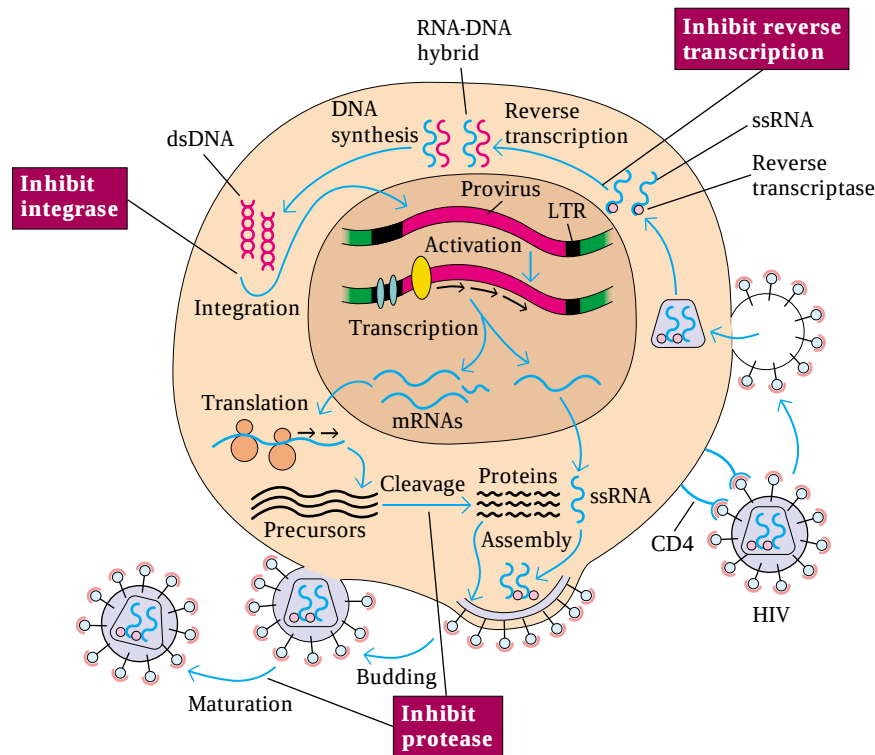


FIGURE 19-14 Stages in the viral replication cycle that provide targets for therapeutic antiretroviral drugs. At present, the licensed drugs with anti-HIV activity block the step of reverse transcription of

viral RNA to cDNA or inhibit the viral protease necessary to cleave viral precursor proteins into the proteins needed to assemble a new virion and complete its maturation to infectious virus.

interfere with reverse transcription is zidovudine, or AZT (azidothymidine). The introduction of AZT, a nucleoside analog, into the growing cDNA chain of the retrovirus causes termination of the chain. AZT is effective in some but not all patients, and its efficacy is further limited because long-term use has several adverse side effects and because resistant viral mutants develop in treated patients. The administered AZT is used not only by the HIV-1 reverse transcriptase but also by human DNA polymerase. The incorporation of AZT into the DNA of host cells kills them. Precursors of red blood cells are especially sensitive to AZT, which thus causes anemia in addition to other side effects. A different approach to blocking reverse transcription employs drugs such as Nevirapine, which inhibit the action of the reverse transcriptase enzyme (see Table 19-5).

A second class of drugs called protease inhibitors has proven effective when used in conjunction with AZT and/or other nucleoside analogs. Current treatment for AIDS is a combination therapy, using regimens designated HAART (highly active anti-retroviral therapy). In most cases, this combines the use of two nucleoside analogs and one protease inhibitor. The combination strategy appears to overcome the ability of the virus to rapidly produce mutants that are drug resistant. In many cases, HAART has lowered viral load to

levels that are not detectable by current methods and has improved the health of AIDS patients to the point that they can again function at a normal level. The decrease in the number of AIDS deaths in the United States in recent years (see Figure 19-6) is attributed to this advance in therapy. Despite the optimism engendered by success with HAART, present drawbacks include a strict time schedule of administration and the large number of pills to be taken every day. In addition, there may be serious side effects (see Table 19-5) that, in some patients, may be too severe to allow use of HAART.

The success of HAART in treating AIDS has opened discussion of whether it might be possible to eradicate all virus from an infected individual and thus actually cure AIDS. Most AIDS experts are not convinced that this is possible, mainly because of the persistence of latently infected $CD4^+$ T cells and macrophages, which can serve as a reservoir of infectious virus if the provirus should be activated. Even with a viral load beneath the level of detection by PCR assays, the immune system may not recover sufficiently to clear virus should it begin to replicate in response to some activation signal. In addition, virus may persist in sites such as the brain, not readily penetrated by the antiretroviral drugs, even though the virus in circulation is undetectable. The use of immune modulators, such as recombinant IL-2, in conjunction with HAART is be-

TABLE 19-5 Some anti-HIV drugs in clinical use

Generic name (other names)	Typical dosage	Some potential side effects
REVERSE TRANSCRIPTASE INHIBITORS: NUCLEOSIDE ANALOG		
Didanosine (Videx, ddl)	2 pills, 2 times a day on empty stomach	Nausea, diarrhea, pancreatic inflammation, peripheral neuropathy
Lamivudine (Epivir, 3TC)	1 pill, 2 times a day	Usually none
Stavudine (Zerit, d4T)	1 pill, 2 times a day	Peripheral neuropathy
Zalcitabine (HIVID, ddC)	1 pill, 3 times a day	Peripheral neuropathy, mouth inflammation, pancreatic inflammation
Zidovudine (Retrovir, AZT)	1 pill, 2 times a day	Nausea, headache, anemia, neutropenia (reduced levels of neutrophil white blood cells), weakness, insomnia
Pill containing lamivudine and zidovudine (Combivir)	1 pill, 2 times a day	Same as for zidovudine
REVERSE TRANSCRIPTASE INHIBITORS: NONNUCLEOSIDE ANALOGUES		
Delavirdine (Rescriptor)	4 pills, 3 times a day (mixed into water); not within an hour of antacids or didanosine	Rash, headache, hepatitis
Nevirapine (Viramune)	1 pill, 2 times a day	Rash, hepatitis
PROTEASE INHIBITORS		
Indinavir (Crixivan)	2 pills, 3 times a day on empty stomach or with a low-fat snack and not within 2 hours of didanosine	Kidney stones, nausea, headache, blurred vision, dizziness, rash, metallic taste in mouth, abnormal distribution of fat, elevated triglyceride and cholesterol levels, glucose intolerance
Nelfinavir (Viracept)	3 pills, 3 times a day with some food	Diarrhea, abnormal distribution of fat, elevated triglyceride and cholesterol levels, glucose intolerance
Ritonavir (Norvir)	6 pills, 2 times a day (or 4 pills, 2 times a day if taken with saquinavir) with food and not within 2 hours of didanosine	Nausea, vomiting, diarrhea, abdominal pain, headache, prickling sensation in skin, hepatitis, weakness, abnormal distribution of fat, elevated triglyceride and cholesterol levels, glucose intolerance
Saquinavir (Invirase, a hard-gel capsule; Fortovase, a soft-gel capsule)	6 pills, 3 times a day (or 2 pills, 2 times a day if taken with ritonavir) with a large meal	Nausea, diarrhea, headache, abnormal distribution of fat, elevated triglyceride and cholesterol levels, glucose intolerance

SOURCE: J. G. Bartlett and R. D. Moore, 1998, Improving HIV therapy, *Sci. Am.* **279**(1):87.

ing examined as a strategy to help reconstitute the immune system and restore normal immune function.

New drugs are in various stages of development. One promising class of drugs interferes with integration of the viral DNA into the host genome (see Figure 19-14). Others drugs being considered act at the stage of viral attachment to the host cell. It should be stressed that the development of any drug to the point at which it can be used for patients is a long and arduous procedure. The drugs that pass the rigor-

ous tests for safety and efficacy represent a small fraction of those that receive initial consideration.

A Vaccine May Be the Only Way to Stop the HIV/AIDS Epidemic

The AIDS epidemic continues to rage despite the advances in therapeutic approaches outlined above. The present expense of HAART (as much as \$15,000 per year), the strict regimen

TABLE 19-6 Why AIDS does not fit the paradigm for classic vaccine development

Classic vaccines mimic natural immunity against reinfection generally seen in individuals recovered from infection; there are no recovered AIDS patients.

Most vaccines protect against disease, not against infection; HIV infection may remain latent for long periods before causing AIDS.

Most vaccines protect for years against viruses that change very little over time; HIV-1 mutates at a rapid rate and efficiently selects mutant forms that evade immunity.

Most effective vaccines are whole-killed or live-attenuated organisms; killed HIV-1 does not retain antigenicity and the use of a live retrovirus vaccine raises safety issues.

Most vaccines protect against infections that are infrequently encountered; HIV may be encountered daily by individuals at high risk.

Most vaccines protect against infections through mucosal surfaces of the respiratory or gastrointestinal tract; the great majority of HIV infection is through the genital tract.

Most vaccines are tested for safety and efficacy in an animal model before trials with human volunteers; there is no suitable animal model for HIV/AIDS at present.

SOURCE: Adapted from A. S. Fauci, 1996, An HIV vaccine: breaking the paradigms, *Proc. Am. Assoc. Phys.* **108**:6.

required, and the possibility of side effects precludes universal application. Even if eradication of the virus in individuals treated with combination therapy becomes possible, it will not greatly influence the epidemic in the developing countries, which include the majority of AIDS victims. It is likely that effective, inexpensive, and well-tolerated drugs will be developed in the future, but at present it appears that the best option to stop the spread of AIDS is a safe, effective vaccine that prevents infection and progression to disease. Why do we not have an AIDS vaccine? The best answer to this question is to examine the special conditions that must be addressed in developing a safe, effective vaccine for this disease (Table 19-6).

Most effective vaccines mimic the natural state of infection. Individuals who recover from most diseases are immune from subsequent attacks. The infection by HIV-1 and progression to immunodeficiency syndrome flourishes even in the presence of circulating antibodies directed against proteins of the virus. Immunity may hold the virus in check for a time, but as mentioned above, it rarely exceeds 12 years. In a rare subset of infected individuals called long-term nonprogressors, the period of infection without disease is longer and even indefinite. Another group for whom immunity seems to function are those who are persistently exposed but who remain seronegative. In this category are a low percentage of commercial sex workers in areas of high endemic infection, such as Nairobi, who have not become infected despite multiple daily exposures to infected individuals. Because the state of immunity (which antibodies are present and what type of cellular immunity is active) in these individuals is not clear or consistent, it is difficult to duplicate for vaccine development. Certain of the long-term nonprogressors or exposed and noninfected individuals have mutations and deletions in genes encoding cell coreceptors that slow the progress of viral attack on their immune sys-

tem, rather than an immune response that is holding HIV replication in check.

Most vaccines prevent disease, not infection. Polio and influenza vaccines hold the virus produced by infected cells in check so that it does not cause harm to the host, and it is then cleared. HIV-1 does not fit this model, because it integrates into the host genome and may remain latent for long periods. As described above in the context of treatment strategies, eradication of a retrovirus is not a simple matter. Clearance of a retrovirus is a difficult goal for a vaccine; every copy of the virus and every infected cell, including those latently infected, must be eradicated from the host. However even without complete eradication, an HIV vaccine may benefit the infected individual; furthermore, a vaccine that caused a lowered viral load would help to control the spread of infection. A recent study in Uganda of sexual partners unmatched for infection showed that low viral load in the infected partner inhibited spread to the uninfected mate.

Most vaccines prevent infection by viruses that show little variation. The instability of its genome differentiates HIV-1 from most viruses for which successful vaccines have been developed. With the exception of influenza, for which the vaccines must be changed periodically, most viruses that can be controlled by immunization show only minor variability in structure. For comparison, consider that the rhinoviruses that cause the common cold have more than 100 subtypes; therefore no effective vaccine has been developed. HIV-1 shows variation in most viral antigens; and the rate of replication may be as high as 10^9 viruses per day. This variability along with the high rate of replication allows the production of viruses with multiple mutations; some of these allow escape from immunity. The fact that significant differences in viral-envelope protein sequences have been seen in viral iso-

lates taken from the same patient at different times indicates that variation occurs and that some of the variants replicate, presumably because they have learned to evade host immune defenses. Data showing that antibody from advanced AIDS patients will not neutralize virus isolated from that patient, but will kill other strains of HIV-1, argues that HIV-1 does evade the immune system by mutation of proteins targeted by antibody.

The majority of successful vaccines are live-attenuated or heat-killed organisms. While there are exceptions to this, notably the recombinant protein used for hepatitis B vaccine and the conjugate used for *Haemophilus influenzae* B vaccine (see Chapter 18), most of the widely used vaccines are attenuated organisms. The development of a live-attenuated retrovirus vaccine from animal viruses engineered to include HIV antigens is a possible route. However, the use of live vaccines is predicated on the supposition that the immunity raised will clear the vaccine virus from the host. This is not easily done for a retrovirus, which integrates into the host genome. A massive testing effort would be required to assure that a live retroviral vaccine was safe and did not cause chronic host infection. On the positive side, clinical studies using other viruses such as attenuated vaccinia or canarypox as carriers for genes encoding HIV proteins have passed phase I (safety) trials and have advanced to phase II (efficacy) trials.

For most viruses, the frequency of exposure to infection is rare or seasonal. In many high-risk individuals, such as commercial sex workers, monogamous sexual partners of HIV-infected subjects, and intravenous drug users, the virus is encountered frequently and, potentially, in large doses. An AIDS vaccine is thus asked to prevent infection against a constant attack by the virus and/or massive doses of virus; this is not normally the case with other viruses for which immunization has proved successful.

Most vaccines protect against respiratory or gastrointestinal infection. In addition to the frequency of exposure to HIV, which may be extraordinarily high for some high-risk individuals, there is also the question of route. The majority of successful vaccines protect against viruses that are encountered in the respiratory and gastrointestinal tracts; the most common route of HIV-1 infection is by the genital tract. It is not known whether the immunity established by conventional vaccination procedures will protect against infection by this route. Although the lack of a completely relevant animal model precludes an in-depth test of protection, preliminary vaccine studies using rectal or vaginal challenge of immunized primates with HIV-SIV chimeric viruses (SHIV) show protection to this challenge route.

Development of most vaccines through to clinical trials relies upon animal experiments. Testing a vaccine for safety and efficacy normally involves challenge of an animal with the virus under conditions similar to those encountered in the human. In this way, the correlates of protective immunity are established. For example, if high titers of CD8⁺ T cells and neutralizing antibody are necessary for protection in an ani-

mal, then CD8⁺ T-cell immunity and antibody should be measured in human trials of the vaccine. Thus far, animal studies of HIV infection and disease have yielded only a few hard facts about immune responses that are protective against infection or that prevent progression to disease. Many results involve a specific virus in a particular host and are not easily extrapolated to universal concepts, because they depend upon host factors as well the relationships between the immunizing and challenge strains of virus. However, experiments have shown that passive immunization with antibodies taken from HIV-infected chimpanzees protect macaques from challenge with SHIV strains bearing HIV-envelope glycoprotein. Further indication that antibodies can prevent infection is given by studies in which monoclonal antibodies protected macaques from vaginal challenge with SHIV. In all cases the antibodies needed to be present at the time of challenge. Post-challenge administration of antibodies was not effective in preventing infection.

Although there are no reports of great success in human HIV vaccine trials, research in this difficult area continues to be active. At the end of the year 2000, there were 60 phase I trials in progress involving recombinant proteins, peptides, DNA vaccines, and poxvirus/recombinant protein combination trials. At the same time, only 6 phase II trials were in progress and only 2 candidates advanced to phase III—the 3rd, or final, phase of clinical trials—the test of efficacy. Despite a massive effort, progress remains slow. There is now hope that a vaccine can emerge from the accumulating knowledge on human responses to the vaccine candidates.

In addition to developing a scientific rationale, behavioral and social issues influence the development and testing of candidate AIDS vaccines. Counseling concerning safe sexual practice must be part of the care given to volunteers in a vaccine trial. Will this influence the results? Would a lowering of the infection rate in all groups taking part in the trial preclude seeing a meaningful difference in the infection rate between the vaccine and the placebo groups? A further consideration is the fact that anyone successfully immunized against the AIDS virus will become seropositive and will test positive in the standard screening assays for infection. What ramifications will this have? Will the more complex viral-load assays be needed to ascertain whether an immunized individual is actually infected?

It is clear that development of an AIDS vaccine is not a simple exercise in classic vaccinology. More research is needed to understand how this viral attack against the immune system can be thwarted. While much has been written about the subject and large-scale initiatives are proposed, the path to an effective vaccine is not obvious. It is certain only that all data must be carefully analyzed and that all possible means of creating immunity must be tested. This is one of the greatest public health challenges of our time. An intense and cooperative effort must be launched to devise, test, and deliver a safe and effective vaccine for AIDS. The status of current efforts in AIDS vaccine development is summarized in Table 19-7.

TABLE 19-7 Vaccine strategies under study

Vaccine constituents	Status	Advantages	Disadvantages
VACCINES ELICITING ANTI-HIV ANTIBODIES			
Viral surface proteins, such as gp120	In phase I and II trials, which examine safety	Safe and simple to prepare	Vaccine-elicited antibodies have failed to recognize HIV from patients
Whole, killed HIV	Not under study in humans	Should present HIV surface proteins in a relatively natural conformation; simple to prepare	Slight risk that preparations might include some active virus; inactivated virus might shed its proteins and become ineffective
Pseudovirions (artificial viruses containing HIV surface proteins)	Close to phase I trials	Present HIV surface proteins in a relatively natural conformation	Difficult to produce and to ensure long-term stability
VACCINES ELICITING CELLULAR RESPONSES			
Live vector viruses (non-HIV viruses engineered to carry genes encoding HIV proteins)	In phase II trials	Makers can control amount and kinds of viral proteins produced	Complicated to prepare; current vaccines elicit modest immune response
Naked DNA containing one or more HIV genes	In phase I trials	Simple and inexpensive to prepare	Some worry that integration of HIV genes into human cells could harm patients
HIV peptides (protein fragments)	In phase I trials	Simple to prepare	Do not elicit strong immune response
VACCINES ELICITING ANTIBODY AND CELLULAR RESPONSES			
Combinations of elements, such as pure gp120 protein plus canarypox vector	In phase II trials	Should stimulate both arms of the immune response at once	Complicated to prepare
Live, attenuated HIV	Not under study in humans; being assessed in nonhuman primates	Most closely mimics HIV; may interfere with ability of infectious HIV to replicate	Vaccine virus could potentially cause AIDS

SOURCE: D. Baltimore and C. Heilman, HIV vaccines: prospects and challenges, 1998, *Sci. Am.* **279**(1):101.

SUMMARY

- Immunodeficiency results from the failure of one or more components of the immune system. Primary immunodeficiencies are present at birth, secondary or acquired immunodeficiencies arise from a variety of causes.
- Immunodeficiencies may be classified by the cell types involved and may affect either the lymphoid or the myeloid cell lineage or both.
- The gene defects that underlie primary immunodeficiency allow precise classification. Genetic defects in molecules

involved in signal transduction or in cellular communication are found in many immunodeficiencies.

- Lymphoid immunodeficiencies affect T cells, B cells, or both. Failure of thymic development results in severe immunodeficiency and can hinder normal development of B cells, because of the lack of cellular cooperation.
- Myeloid immunodeficiency causes impaired phagocytic function. Those affected suffer from increased susceptibility to bacterial infection.



- Severe combined immunodeficiency, or SCID, may result from a number of different defects in the lymphoid lineage and is usually fatal.
- Selective immunoglobulin deficiencies are a less severe form of immunodeficiency and result from defects in more highly differentiated cell types.
- Immunodeficiency may be treated by replacement of the defective or missing protein, cells, or gene. Administration of human immunoglobulin is a common treatment.
- Animal models for immunodeficiency include nude and SCID mice. Gene knockout mice provide a means to study the role of specific genes on immune function.
- Secondary immunodeficiency results from injury or infection; the most common form is HIV/AIDS caused by a retrovirus, human immunodeficiency virus-1.
- HIV-1 infection is spread mainly by sexual contact, passage of blood, and from HIV-infected mother to infant.
- Infection with HIV-1 results in severe impairment of immune function marked by depletion of CD4⁺ T cells and death from opportunistic infection, usually within 10 years of infection.
- Treatment of HIV infection with anti-retroviral drugs can cause lowering of viral load and relief from infection, but this is temporary and no cures have been documented.
- Efforts to develop a vaccine for HIV/AIDS have not yet been successful. The millions of new infections in the year 2000 emphasize the need for an effective vaccine.

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USEFUL WEB SITES

<http://www.scid.net/>

The SCID home page contains links to periodicals and databases with information about SCID.

<http://www.nhgri.nih.gov/DIR/LGT/SCID/>

This site from the National Institute for Human Genome Research includes a database of mutations in X-linked SCID.

<http://hivinsite.ucsf.edu>

Information about the global AIDS epidemic can be accessed from this site.

<http://www.cdc.gov>

Up-to-date information concerning AIDS epidemiology in the United States can be obtained at this site.

<http://hiv-web.lanl.gov>

Web site maintained by the Los Alamos National laboratories containing all available sequence data on HIV and SIV along with up-to-date reviews on topics of current interest to AIDS research.

<ftp://nlmpubs.nlm.nih.gov/aids/adatabases/drugs.txt>

A listing with detailed information for several hundred drugs under development for HIV infection and opportunistic infections associated with AIDS; maintained by the National Library of Medicine.

<http://www.niaid.nih.gov/daids/vaccine/abtvaccines.htm>

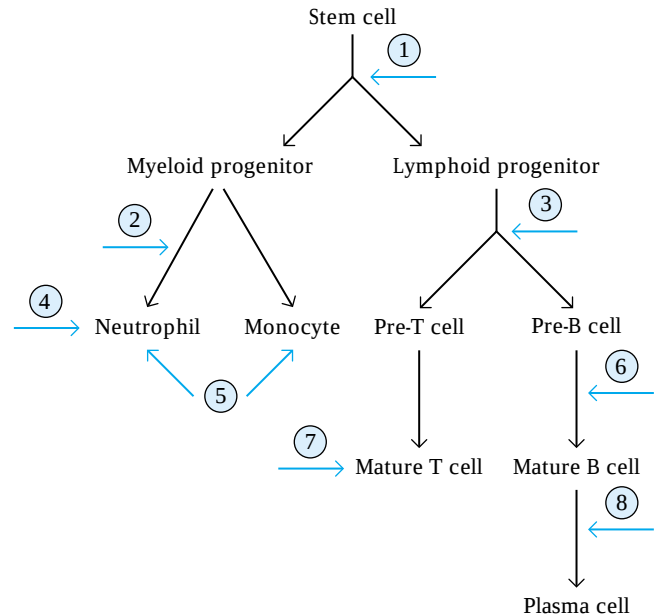
Information about AIDS vaccines from the National Institute of Allergy and Infectious Diseases, NIH. Includes links to documents about vaccines in general.

Study Questions

CLINICAL FOCUS QUESTION The spread of HIV/AIDS from infected mothers to infants can be reduced by single-dose regimens of the reverse transcriptase inhibitor Nevirapine. What would you want to know before giving this drug to all mothers and infants (without checking infection status) at delivery in areas of high endemic infection?

- Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - DiGeorge syndrome is a congenital birth defect resulting in absence of the thymus.
 - X-linked agammaglobulinemia (XLA) is a combined B-cell and T-cell immunodeficiency disease.
 - The hallmark of a phagocytic deficiency is increased susceptibility to viral infections.
 - In chronic granulomatous disease, the underlying defect is in a cytochrome or an associated protein.
 - Injections of immunoglobulins are given to treat individuals with X-linked agammaglobulinemia.
 - Multiple defects have been identified in human SCID.
 - Mice with the SCID defect lack functional B and T lymphocytes.
 - Mice with SCID-like phenotype can be produced by knockout of *RAG* genes.
 - Children born with SCID often manifest increased infections by encapsulated bacteria in the first months of life.
 - Failure to express class II MHC molecules in bare-lymphocyte syndrome affects cell-mediated immunity only.
- Granulocytes from patients with leukocyte-adhesion deficiency (LAD) express greatly reduced amounts of three integrin molecules designated CR3, CR4, and LFA-1.
 - What is the nature of the defect that results in decreased expression or in no expression of these receptors in LAD patients?
 - What is the normal function of the integrin molecule LFA-1? Give specific examples.
- Immunologists have studied the defect in SCID mice in an effort to understand the molecular basis for severe combined immunodeficiency in humans. In both SCID mice and humans with this disorder, mature B and T cells fail to develop.
 - In what way do rearranged Ig heavy-chain genes in SCID mice differ from those in normal mice?
 - In SCID mice, rearrangement of κ light-chain DNA is not attempted. Explain why.
 - If you introduced a rearranged, functional μ heavy-chain gene into progenitor B cells of SCID mice, would the κ light-chain DNA undergo a normal rearrangement? Explain your answer.

- The accompanying figure outlines some of the steps in the development of immune-system cells. The numbered arrows indicate the cell type whose function is defective or the developmental step that does not occur in particular immunodeficiency diseases. Identify the defective cell type or developmental step associated with each of the following diseases. Use each number only once.



- Chronic granulomatous disease
 - Severe combined immunodeficiency disease (SCID)
 - Congenital agranulocytosis
 - Reticular dysgenesis
 - Common variable hypogammaglobulinemia
 - X-linked agammaglobulinemia
 - Leukocyte-adhesion deficiency (LAD)
 - Bare-lymphocyte syndrome
- Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - HIV-1 and HIV-2 are more closely related to each other than to SIV.
 - HIV-1 causes immune suppression in both humans and chimpanzees.
 - SIV is endemic in the African green monkey.
 - The anti-HIV drugs zidovudine and indinavir both act on the same point in the viral replication cycle.
 - T-cell activation increases transcription of the HIV proviral genome.
 - Patients with advanced stages of AIDS always have detectable antibody to HIV.
 - The polymerase chain reaction is a sensitive test used to detect antibodies to HIV.
 - If HAART is successful, viral load will decrease.
 - Various mechanisms have been proposed to account for the decrease in the numbers of $CD4^+$ T cells in HIV-infected individuals. What seems to be the most likely reason for depletion of $CD4^+$ T cells?

7. Would you expect the viral load in the blood of HIV-infected individuals in the chronic phase of HIV-1 infection to vary?
8. If viral load begins to increase in the blood of an HIV-infected individual and the level of $CD4^+$ T cells decrease, what would this indicate about the infection?
9. Why do clinicians monitor the level of skin-test reactivity in HIV-infected individuals? What change might you expect to see in skin-test reactivity with progression into AIDS?
10. Certain chemokines have been shown to suppress infection of cells by HIV, and pro-inflammatory cytokines enhance cell infection. What is the explanation for this?
11. Treatments with combinations of anti-HIV drugs (HAART) have reduced virus levels significantly in some treated patients and delayed the onset of AIDS. If an AIDS patient becomes free of opportunistic infection and has no detectable virus in the circulation, can that person be considered cured?
12. Suppose you are a physician who has two HIV-infected patients. Patient B. W. has a fungal infection (candidiasis) in the mouth, and patient L. S. has a *Mycobacterium* infection. The $CD4^+$ T-cell counts of both patients are about 250 per mm^3 . Would you diagnose either patient or both of them as having AIDS?

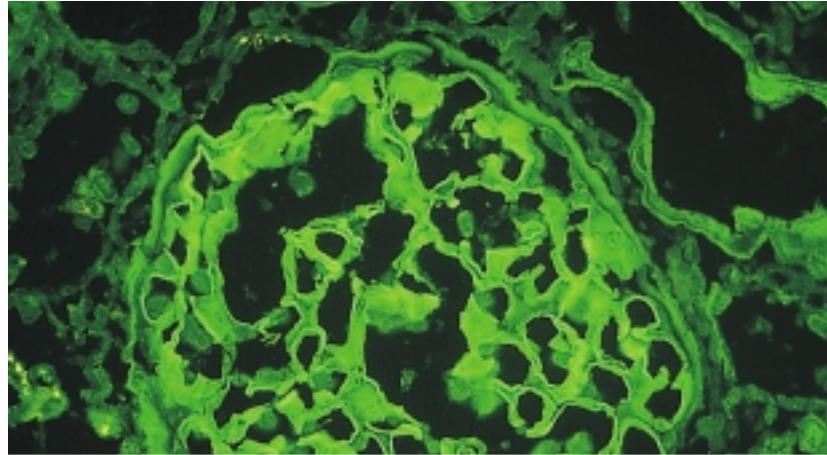
Autoimmunity

EARLY IN THE LAST CENTURY, PAUL EHRLICH realized that the immune system could go awry and, instead of reacting against foreign antigens, could focus its attack on self-antigens. He termed this condition “horror autotoxicus.” We now understand that, while mechanisms of self-tolerance normally protect an individual from potentially self-reactive lymphocytes, there are failures. They result in an inappropriate response of the immune system against self-components termed **autoimmunity**. In the 1960s, it was believed that all self-reactive lymphocytes were eliminated during their development in the bone marrow and thymus and that a failure to eliminate these lymphocytes led to autoimmune consequences. Since the late 1970s, a broad body of experimental evidence has countered that belief, revealing that not all self-reactive lymphocytes are deleted during T-cell and B-cell maturation. Instead, normal healthy individuals have been shown to possess mature, recirculating, self-reactive lymphocytes. Since the presence of these self-reactive lymphocytes in the periphery does not inevitably result in autoimmune reactions, their activity must be regulated in normal individuals through clonal anergy or clonal suppression. A breakdown in this regulation can lead to activation of self-reactive clones of T or B cells, generating humoral or cell-mediated responses against self-antigens. These reactions can cause serious damage to cells and organs, sometimes with fatal consequences.

Sometimes the damage to self-cells or organs is caused by antibodies; in other cases, T cells are the culprit. For example, a common form of autoimmunity is tissue injury by mechanisms similar to type II hypersensitivity reactions. As Chapter 16 showed, type II hypersensitivity reactions involve antibody-mediated destruction of cells. Autoimmune hemolytic anemia is an excellent example of such an autoimmune disease. In this disease, antigens on red blood cells are recognized by auto-antibodies, which results in the destruction of the blood cells, which in turn results in anemia. Auto-antibodies are also the major offender in Hashimoto’s thyroiditis, in which antibodies reactive with tissue-specific antigens such as thyroid peroxidase and thyroglobulin cause severe tissue destruction. Other autoimmune diseases that involve auto-antibodies are listed in Table 20-1.

Many autoimmune diseases are characterized by tissue destruction mediated directly by T cells. A well-known example is rheumatoid arthritis, in which self-reactive T cells attack the tissue in joints, causing an inflammatory response that results in swelling and tissue destruction. Other examples include insulin-dependent diabetes mellitus and multiple sclerosis (see Table 20-1).

chapter 20



Kidney Biopsy from Goodpasture's Syndrome

- Organ-Specific Autoimmune Diseases
- Systemic Autoimmune Diseases
- Animal Models for Autoimmune Diseases
- Evidence Implicating the CD4⁺ T Cell, MHC, and TCR in Autoimmunity
- Proposed Mechanisms for Induction of Autoimmunity
- Treatment of Autoimmune Diseases

This chapter describes some common human autoimmune diseases. These can be divided into two broad categories: organ-specific and systemic autoimmune disease (Table 20-1). Such diseases affect 5%–7% of the human population, often causing chronic debilitating illnesses. Several experimental animal models used to study autoimmunity and various mechanisms that may contribute to induction of autoimmune reactions also are described. Finally, current and experimental therapies for treating autoimmune diseases are described.

Organ-Specific Autoimmune Diseases

In an organ-specific autoimmune disease, the immune response is directed to a target antigen unique to a single organ or gland, so that the manifestations are largely limited to that organ. The cells of the target organs may be damaged di-

TABLE 20-1 Some autoimmune diseases in humans

Disease	Self-antigen	Immune response
ORGAN-SPECIFIC AUTOIMMUNE DISEASES		
Addison's disease	Adrenal cells	Auto-antibodies
Autoimmune hemolytic anemia	RBC membrane proteins	Auto-antibodies
Goodpasture's syndrome	Renal and lung basement membranes	Auto-antibodies
Graves' disease	Thyroid-stimulating hormone receptor	Auto-antibody (stimulating)
Hashimoto's thyroiditis	Thyroid proteins and cells	T _{DTH} cells, auto-antibodies
Idiopathic thrombocytopenia purpura	Platelet membrane proteins	Auto-antibodies
Insulin-dependent diabetes mellitus	Pancreatic beta cells	T _{DTH} cells, auto-antibodies
Myasthenia gravis	Acetylcholine receptors	Auto-antibody (blocking)
Myocardial infarction	Heart	Auto-antibodies
Pernicious anemia	Gastric parietal cells; intrinsic factor	Auto-antibody
Poststreptococcal glomerulonephritis	Kidney	Antigen-antibody complexes
Spontaneous infertility	Sperm	Auto-antibodies
SYSTEMIC AUTOIMMUNE DISEASES		
Ankylosing spondylitis	Vertebrae	Immune complexes
Multiple sclerosis	Brain or white matter	T _H 1 cells and T _C cells, auto-antibodies
Rheumatoid arthritis	Connective tissue, IgG	Auto-antibodies, immune complexes
Scleroderma	Nuclei, heart, lungs, gastrointestinal tract, kidney	Auto-antibodies
Sjogren's syndrome	Salivary gland, liver, kidney, thyroid	Auto-antibodies
Systemic lupus erythematosus (SLE)	DNA, nuclear protein, RBC and platelet membranes	Auto-antibodies, immune complexes

rectly by humoral or cell-mediated effector mechanisms. Alternatively, the antibodies may overstimulate or block the normal function of the target organ.

Some Autoimmune Diseases Are Mediated by Direct Cellular Damage

Autoimmune diseases involving direct cellular damage occur when lymphocytes or antibodies bind to cell-membrane antigens, causing cellular lysis and/or an inflammatory response in the affected organ. Gradually, the damaged cellular structure is replaced by connective tissue (scar tissue), and the function of the organ declines. This section briefly describes a few examples of this type of autoimmune disease.

HASHIMOTO'S THYROIDITIS

In Hashimoto's thyroiditis, which is most frequently seen in middle-aged women, an individual produces auto-antibodies

and sensitized T_H1 cells specific for thyroid antigens. The DTH response is characterized by an intense infiltration of the thyroid gland by lymphocytes, macrophages, and plasma cells, which form lymphocytic follicles and germinal centers (Figure 20-1). The ensuing inflammatory response causes a goiter, or visible enlargement of the thyroid gland, a physiological response to hypothyroidism. Antibodies are formed to a number of thyroid proteins, including thyroglobulin and thyroid peroxidase, both of which are involved in the uptake of iodine. Binding of the auto-antibodies to these proteins interferes with iodine uptake and leads to decreased production of thyroid hormones (hypothyroidism).

AUTOIMMUNE ANEMIAS

Autoimmune anemias include pernicious anemia, autoimmune hemolytic anemia, and drug-induced hemolytic anemia. Pernicious anemia is caused by auto-antibodies to intrinsic factor, a membrane-bound intestinal protein on gastric parietal cells. Intrinsic factor facilitates uptake of vitamin B₁₂

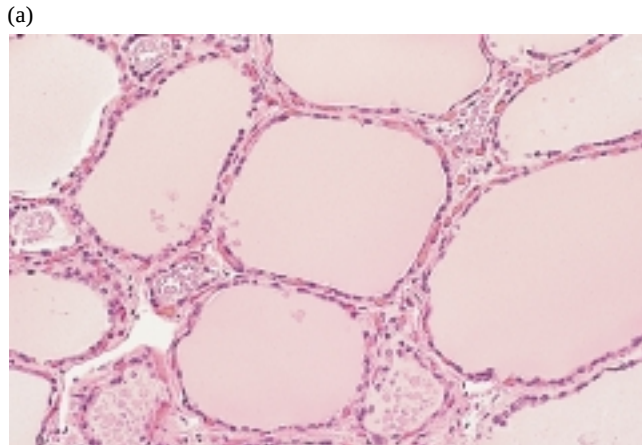
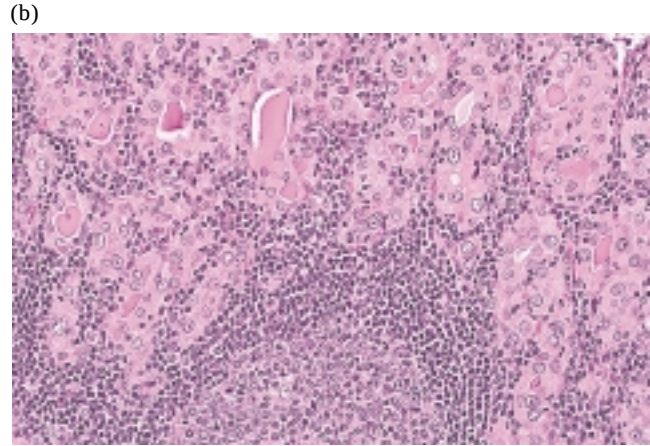


FIGURE 20-1 Photomicrographs of (a) normal thyroid gland showing a follicle lined by cuboidal follicular epithelial cells and (b) gland in



Hashimoto's thyroiditis showing intense lymphocyte infiltration. [From *Web Path*, courtesy of E. C. Klatt, University of Utah.]

from the small intestine. Binding of the auto-antibody to intrinsic factor blocks the intrinsic factor–mediated absorption of vitamin B₁₂. In the absence of sufficient vitamin B₁₂, which is necessary for proper hematopoiesis, the number of functional mature red blood cells decreases below normal. Pernicious anemia is treated with injections of vitamin B₁₂, thus circumventing the defect in its absorption.

An individual with autoimmune hemolytic anemia makes auto-antibody to RBC antigens, triggering complement-mediated lysis or antibody-mediated opsonization and phagocytosis of the red blood cells. One form of autoimmune anemia is drug-induced: when certain drugs such as penicillin or the anti-hypertensive agent methyldopa interact with red blood cells, the cells become antigenic. The immunodiagnostic test for autoimmune hemolytic anemias generally involves a Coombs test, in which the red cells are incubated with an anti-human IgG antiserum. If IgG auto-antibodies are present on the red cells, the cells are agglutinated by the antiserum.

GOODPASTURE'S SYNDROME

In **Goodpasture's syndrome**, auto-antibodies specific for certain basement-membrane antigens bind to the basement membranes of the kidney glomeruli and the alveoli of the lungs. Subsequent complement activation leads to direct cellular damage and an ensuing inflammatory response mediated by a buildup of complement split products. Damage to the glomerular and alveolar basement membranes leads to progressive kidney damage and pulmonary hemorrhage. Death may ensue within several months of the onset of symptoms. Biopsies from patients with Goodpasture's syndrome stained with fluorescent-labeled anti-IgG and anti-C3b reveal linear deposits of IgG and C3b along the basement membranes (Figure 20-2).

INSULIN-DEPENDENT DIABETES MELLITUS

A disease afflicting 0.2% of the population, **insulin-dependent diabetes mellitus (IDDM)** is caused by an autoimmune attack on the pancreas. The attack is directed against specialized insulin-producing cells (beta cells) that are located in spherical clusters, called the islets of Langerhans, scattered throughout the pancreas. The autoimmune attack destroys beta cells, resulting in decreased production of insulin and consequently increased levels of blood glucose. Several factors are important in the destruction of beta cells. First, activated CTLs migrate into an islet and begin to attack the insulin-producing cells. Local cytokine production during this

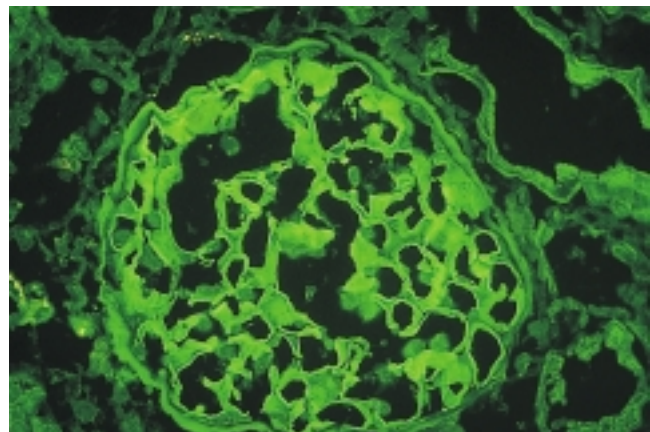
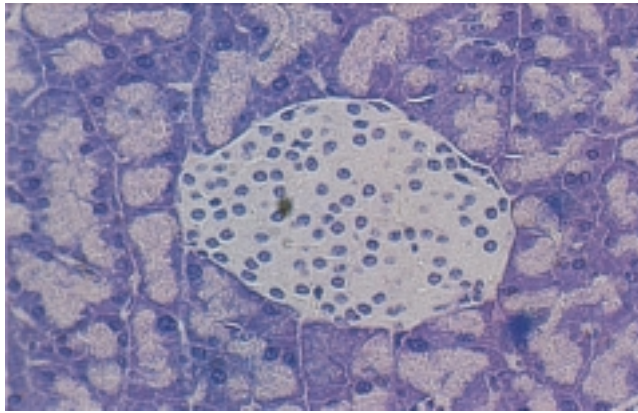


FIGURE 20-2 Fluorescent anti-IgG staining of a kidney biopsy from a patient with Goodpasture's syndrome reveals linear deposits of auto-antibody along the basement membrane. [From *Web Path*, courtesy of E. C. Klatt, University of Utah.]

(a)



(b)

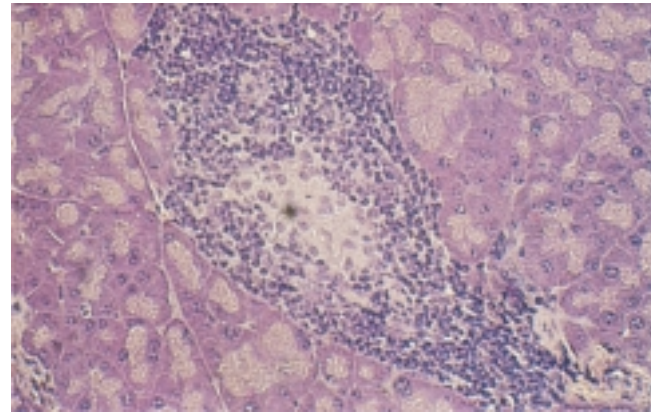


FIGURE 20-3 Photomicrographs of an islet of Langerhans (a) in pancreas from a normal mouse and (b) one in pancreas from a mouse with a disease resembling insulin-dependent diabetes mellitus. Note

the lymphocyte infiltration into the islet (insulinitis) in (b). [From M. A. Atkinson and N. K. Maclaren, 1990, *Sci. Am.* **263**(1):62.]

response includes IFN- γ , TNF- α , and IL-1. Auto-antibody production can also be a contributing factor in IDDM. The first CTL infiltration and activation of macrophages, frequently referred to as insulinitis (Figure 20-3), is followed by cytokine release and the presence of auto-antibodies, which leads to a cell-mediated DTH response. The subsequent beta-cell destruction is thought to be mediated by cytokines released during the DTH response and by lytic enzymes released from the activated macrophages. Auto-antibodies to beta cells may contribute to cell destruction by facilitating either antibody-plus-complement lysis or antibody-dependent cell-mediated cytotoxicity (ADCC).

The abnormalities in glucose metabolism that are caused by the destruction of islet beta cells result in serious metabolic problems that include ketoacidosis and increased urine production. The late stages of the disease are often characterized by atherosclerotic vascular lesions—which in turn cause gangrene of the extremities due to impeded vascular flow—renal failure, and blindness. If untreated, death can result. The most common therapy for diabetes is daily administration of insulin. This is quite helpful in managing the disease, but, because sporadic doses are not the same as metabolically regulated continuous and controlled release of the hormone, periodically injected doses of insulin do not totally alleviate the problems caused by the disease. Another complicating feature of diabetes is that the disorder can go undetected for several years, allowing irreparable loss of pancreatic tissue to occur before treatment begins.

Some Autoimmune Diseases Are Mediated by Stimulating or Blocking Auto-Antibodies

In some autoimmune diseases, antibodies act as agonists, binding to hormone receptors in lieu of the normal ligand

and stimulating inappropriate activity. This usually leads to an overproduction of mediators or an increase in cell growth. Conversely, auto-antibodies may act as antagonists, binding hormone receptors but blocking receptor function. This generally causes impaired secretion of mediators and gradual atrophy of the affected organ.

GRAVES' DISEASE

The production of thyroid hormones is carefully regulated by thyroid-stimulating hormone (TSH), which is produced by the pituitary gland. Binding of TSH to a receptor on thyroid cells activates adenylate cyclase and stimulates the synthesis of two thyroid hormones, thyroxine and triiodothyronine. A patient with **Graves' disease** produces auto-antibodies that bind the receptor for TSH and mimic the normal action of TSH, activating adenylate cyclase and resulting in production of the thyroid hormones. Unlike TSH, however, the auto-antibodies are not regulated, and consequently they overstimulate the thyroid. For this reason these auto-antibodies are called long-acting thyroid-stimulating (LATS) antibodies (Figure 20-4).

MYASTHENIA GRAVIS

Myasthenia gravis is the prototype autoimmune disease mediated by blocking antibodies. A patient with this disease produces auto-antibodies that bind the acetylcholine receptors on the motor end-plates of muscles, blocking the normal binding of acetylcholine and also inducing complement-mediated lysis of the cells. The result is a progressive weakening of the skeletal muscles (Figure 20-5). Ultimately, the antibodies destroy the cells bearing the receptors. The early signs of this disease include drooping eyelids and inability to retract the corners of the mouth, which gives the appearance of snarling. Without treatment, progressive weakening of the

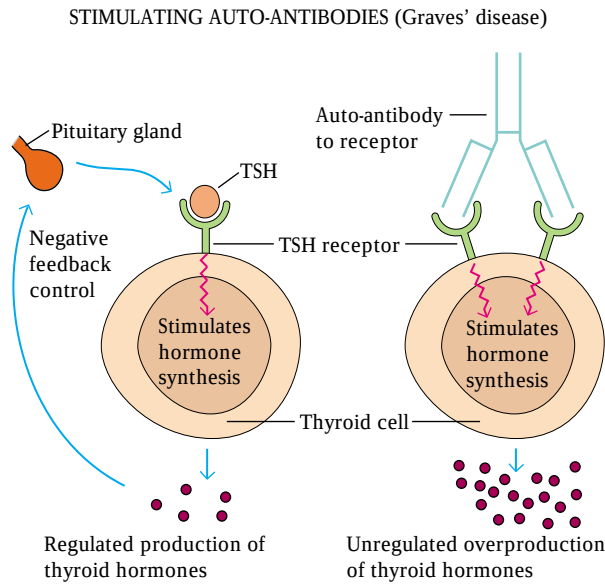


FIGURE 20-4 In Graves' disease, binding of auto-antibodies to the receptor for thyroid-stimulating hormone (TSH) induces unregulated activation of the thyroid, leading to overproduction of the thyroid hormones (purple dots).

muscles can lead to severe impairment of eating as well as problems with movement. However, with appropriate treatment, this disease can be managed quite well and afflicted individuals can lead a normal life.

Systemic Autoimmune Diseases

In systemic autoimmune diseases, the response is directed toward a broad range of target antigens and involves a number of organs and tissues. These diseases reflect a general defect in immune regulation that results in hyperactive T cells and B cells. Tissue damage is widespread, both from cell-mediated immune responses and from direct cellular damage caused by auto-antibodies or by accumulation of immune complexes.

Systemic Lupus Erythematosus Attacks Many Tissues

One of the best examples of a systemic autoimmune disease is **systemic lupus erythematosus (SLE)**, which typically appears in women between 20 and 40 years of age; the ratio of female to male patients is 10:1. SLE is characterized by fever, weakness, arthritis, skin rashes, pleurisy, and kidney dysfunction (Figure 20-6). Lupus is more frequent in African-American and Hispanic women than in Caucasians, although it is not known why this is so. Affected individuals may produce auto-antibodies to a vast array of tissue antigens, such as DNA, histones, RBCs, platelets, leukocytes, and clotting factors; interaction of these auto-antibodies with their specific antigens produces various symptoms. Auto-antibody specific for RBCs and platelets, for example, can lead to complement-mediated lysis, resulting in hemolytic anemia and thrombocytopenia, respectively. When immune complexes of auto-antibodies with various nuclear antigens are deposited along the walls of

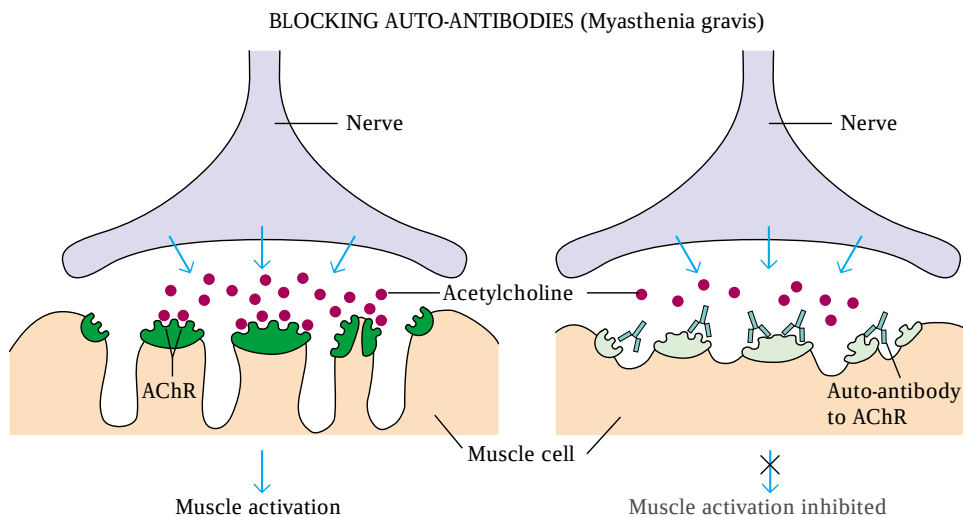


FIGURE 20-5 In myasthenia gravis, binding of auto-antibodies to the acetylcholine receptor (*right*) blocks the normal binding of acetylcholine (burgandy dots) and subsequent muscle activation (*left*). In

addition, the anti-AChR auto-antibody activates complement, which damages the muscle end-plate; the number of acetylcholine receptors declines as the disease progresses. AChR = acetylcholine receptor.



FIGURE 20-6 Characteristic “butterfly” rash over the cheeks of a young girl with systemic lupus erythematosus. [From L. Steinman, 1993, *Sci. Am.* **269**(3):80.]

small blood vessels, a type III hypersensitive reaction develops. The complexes activate the complement system and generate membrane-attack complexes and complement split products that damage the wall of the blood vessel, resulting in vasculitis and glomerulonephritis.

Excessive complement activation in patients with severe SLE produces elevated serum levels of the complement split products C3a and C5a, which may be three to four times higher than normal. C5a induces increased expression of the type 3 complement receptor (CR3) on neutrophils, facilitating neutrophil aggregation and attachment to the vascular endothelium. As neutrophils attach to small blood vessels, the number of circulating neutrophils declines (neutropenia) and various occlusions of the small blood vessels develop (vasculitis). These occlusions can lead to widespread tissue damage.

Laboratory diagnosis of SLE focuses on the characteristic antinuclear antibodies, which are directed against double-stranded or single-stranded DNA, nucleoprotein, histones, and nucleolar RNA. Indirect immunofluorescent staining with serum from SLE patients produces various characteristic nucleus-staining patterns.

Multiple Sclerosis Attacks the Central Nervous System

Multiple sclerosis (MS) is the most common cause of neurologic disability associated with disease in Western countries. The symptoms may be mild, such as numbness in the limbs, or severe, such as paralysis or loss of vision. Most people with MS are diagnosed between the ages of 20 and 40. Individuals with this disease produce autoreactive T cells that participate in the formation of inflammatory lesions along the myelin sheath of nerve fibers. The cerebrospinal fluid of patients

with active MS contains activated T lymphocytes, which infiltrate the brain tissue and cause characteristic inflammatory lesions, destroying the myelin. Since myelin functions to insulate the nerve fibers, a breakdown in the myelin sheath leads to numerous neurologic dysfunctions.

Epidemiological studies indicate that MS is most common in the Northern hemisphere and, interestingly, in the United States. Populations who live north of the 37th parallel have a prevalence of 110–140 cases per 100,000, while those who live south of the 37th parallel show a prevalence of 57–78 per 100,000. And individuals from south of the 37th parallel who move north assume a new risk if the move occurs before 15 years of age. These provocative data suggest that there is an environmental component of the risk of contracting MS. This is not the entire story, however, since genetic influences also are important. While the average person in the United States has about one chance in 1000 of developing MS, close relatives of people with MS, such as children or siblings, have 1 chance in 50 to 100 of developing MS. The identical twin of a person with MS has a 1 in 3 chance of developing the disease. These data point strongly to the genetic component of the disease. And, as is described in the Clinical Focus of this chapter, MS affects women two to three times more frequently than men.

The cause of MS, like most autoimmune diseases, is not well understood. However, there are some suggestions that infection by certain viruses may predispose a person to MS. Certainly some viruses can cause demyelinating diseases, and it is tempting to speculate that virus infection plays a significant role in MS, but at present there is no definitive data implicating a particular virus.

Rheumatoid Arthritis Attacks Joints

Rheumatoid arthritis is a common autoimmune disorder, most often affecting women from 40 to 60 years old. The major symptom is chronic inflammation of the joints, although the hematologic, cardiovascular, and respiratory systems are also frequently affected. Many individuals with rheumatoid arthritis produce a group of auto-antibodies called **rheumatoid factors** that are reactive with determinants in the Fc region of IgG. The classic rheumatoid factor is an IgM antibody with that reactivity. Such auto-antibodies bind to normal circulating IgG, forming IgM-IgG complexes that are deposited in the joints. These immune complexes can activate the complement cascade, resulting in a type III hypersensitive reaction, which leads to chronic inflammation of the joints.

Animal Models for Autoimmune Diseases

Animal models for autoimmune diseases have contributed valuable insights into the mechanism of autoimmunity, to

TABLE 20-2 Experimental animal models of autoimmune diseases

Animal model	Possible human disease counterpart	Inducing antigen	Disease transferred by T cells
SPONTANEOUS AUTOIMMUNE DISEASES			
Nonobese diabetic (NOD) mouse	Insulin-dependent diabetes mellitus (IDDM)	Unknown	Yes
(NZB $\times$ NZW) F ₁ mouse	Systemic lupus erythematosus (SLE)	Unknown	Yes
Obese-strain chicken	Hashimoto's thyroiditis	Thyroglobulin	Yes
EXPERIMENTALLY INDUCED AUTOIMMUNE DISEASES*			
Experimental autoimmune myasthenia gravis (EAMG)	Myasthenia gravis	Acetylcholine receptor	Yes
Experimental autoimmune encephalomyelitis (EAE)	Multiple sclerosis (MS)	Myelin basic protein (MBP); proteolipid protein (PLP)	Yes
Autoimmune arthritis (AA)	Rheumatoid arthritis	<i>M. tuberculosis</i> (proteoglycans)	Yes
Experimental autoimmune thyroiditis (EAT)	Hashimoto's thyroiditis	Thyroglobulin	Yes

* These diseases can be induced by injecting appropriate animals with the indicated antigen in complete Freund's adjuvant. Except for autoimmune arthritis, the antigens used correspond to the self-antigens associated with the human-disease counterpart. Rheumatoid arthritis involves reaction to proteoglycans, which are self-antigens associated with connective tissue.

our understanding of autoimmunity in humans, and to potential treatments. Autoimmunity develops spontaneously in certain inbred strains of animals and can also be induced by certain experimental manipulations (Table 20-2).

Autoimmunity Can Develop Spontaneously in Animals

A number of autoimmune diseases that develop spontaneously in animals exhibit important clinical and pathologic similarities to certain autoimmune diseases in humans. Certain inbred mouse strains have been particularly valuable models for illuminating the immunologic defects involved in the development of autoimmunity.

New Zealand Black (NZB) mice and F₁ hybrids of NZB and New Zealand White (NZW) mice spontaneously develop autoimmune diseases that closely resemble systemic lupus erythematosus. NZB mice spontaneously develop autoimmune hemolytic anemia between 2 and 4 months of age, at which time various auto-antibodies can be detected, including antibodies to erythrocytes, nuclear proteins, DNA, and T lymphocytes. F₁ hybrid animals develop glomerulonephritis from immune-complex deposits in the kidney and die prematurely by 18 months. As in human SLE, the incidence of autoimmunity in the (NZB $\times$ NZW)F₁ hybrids is greater in females.

An accelerated and severe form of systemic autoimmune disease resembling systemic lupus erythematosus develops in

a mouse strain called MRL/*lpr/lpr*. These mice are homozygous for a gene called *lpr*, which has been identified as a defective *fas* gene. The *fas*-gene product is a cell-surface protein belonging to the TNF family of cysteine-rich membrane receptors (see Figure 12-6d). When the normal Fas protein interacts with its ligand, it transduces a signal that leads to apoptotic death of the Fas-bearing cells. This mechanism may operate in destruction of target cells by some CTLs (see Figure 14-9). Fas is known also to be essential in the death of hyperactivated peripheral CD4⁺ cells. Normally, when mature peripheral T cells become activated, they are induced to express both Fas antigen and Fas ligand. When Fas-bearing cells come into contact with a neighboring activated cell bearing Fas ligand, the Fas-bearing cell is induced to die. It is also possible that Fas ligand can engage Fas from the same cell, inducing a cellular suicide. In the absence of Fas, mature peripheral T cells do not die, and these activated cells continue to proliferate and produce cytokines that result in grossly enlarged lymph nodes and spleen. Defects in *fas* expression similar to that found in the *lpr* mouse are observed in humans, and these can have severe consequences. However there is no link between *fas* expression and SLE in humans, which suggests that the *lpr* mouse may not be a true model for SLE.

Another important animal model is the nonobese diabetic (NOD) mouse, which spontaneously develops a form of diabetes that resembles human insulin-dependent dia-

betes mellitus (IDDM). Like the human disease, the NOD mouse disease begins with lymphocytic infiltration into the islets of the pancreas. Also, as in IDDM, there is a strong association between certain MHC alleles and the development of diabetes in these mice. Experiments have shown that T cells from diabetic mice can transfer diabetes to nondiabetic recipients. For example, when the immune system of normal mice is destroyed by lethal doses of x-rays and then is reconstituted with an injection of bone-marrow cells from NOD mice, the reconstituted mice develop diabetes. Conversely, when the immune system of still healthy NOD mice is destroyed by x-irradiation and then reconstituted with normal bone-marrow cells, the NOD mice do not develop diabetes. Various studies have demonstrated a pivotal role for CD4⁺ T cells in the NOD mouse, and recent evidence implicates the T_H1 subset in disease development.

Several other spontaneous autoimmune diseases have been discovered in animals that have served as models for similar human diseases. Among these are *Obese*-strain chickens, which develop both humoral and cell-mediated reactivity to thyroglobulin resembling that seen in Hashimoto's thyroiditis.

Autoimmunity Can Be Induced Experimentally in Animals

Autoimmune dysfunctions similar to certain human autoimmune diseases can be induced experimentally in some animals (see Table 20-2). One of the first such animal models was discovered serendipitously in 1973 when rabbits were immunized with acetylcholine receptors purified from electric eels. The animals soon developed muscular weakness similar to that seen in myasthenia gravis. This experimental autoimmune myasthenia gravis (EAMG) was shown to result when antibodies to the acetylcholine receptor blocked muscle stimulation by acetylcholine in the synapse. Within a year, this animal model had proved its value with the discovery that auto-antibodies to the acetylcholine receptor were the cause of myasthenia gravis in humans.

Experimental autoimmune encephalomyelitis (EAE) is another animal model that has greatly improved understanding of autoimmunity. This is one of the best-studied models of autoimmune disease. EAE is mediated solely by T cells and can be induced in a variety of species by immunization with myelin basic protein (MBP) or proteolipid protein (PLP) in complete Freund's adjuvant (Figure 20-7). Within 2–3 weeks the animals develop cellular infiltration of the myelin sheaths of the central nervous system, resulting in demyelination and paralysis. Most of the animals die, but others have milder symptoms, and some animals develop a chronic form of the disease that resembles chronic relapsing and remitting MS in humans. Those that recover are resistant to the development of disease from a subsequent injection of MBP and adjuvant.

The mouse EAE model provides a system for testing treatments for human MS. For example, because MBP- or PLP-specific T-cell clones are found in the periphery, it is

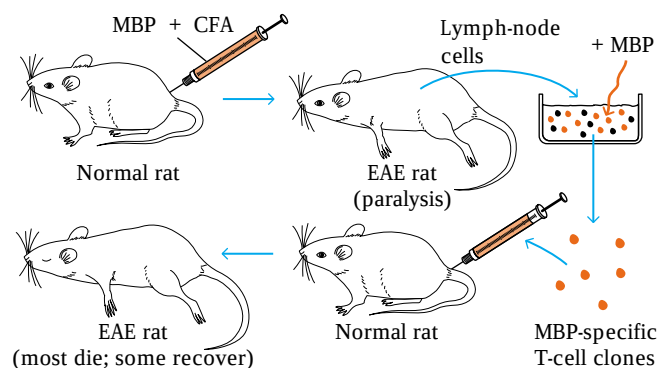


FIGURE 20-7 Experimental autoimmune encephalomyelitis (EAE) can be induced in rats by injecting them with myelin basic protein (MBP) in complete Freund's adjuvant (CFA). MBP-specific T-cell clones can be generated by culturing lymph-node cells from EAE rats with MBP. When these T cells are injected into normal animals, most develop EAE and die, although a few recover.

assumed that these clones must have escaped negative selection in the thymus. Recent mouse experiments have suggested that orally administered MBP may make these antigen-specific peripheral T-cell clones self-tolerant. These studies have paved the way for clinical trials in MS patients.

Experimental autoimmune thyroiditis (EAT) can be induced in a number of animals by immunizing with thyroglobulin in complete Freund's adjuvant. Both humoral antibodies and T_H1 cells directed against the thyroglobulin develop, resulting in thyroid inflammation. EAT appears to best mimic Hashimoto's thyroiditis. In contrast to both EAE and EAT, which are induced by immunizing with self-antigens, autoimmune arthritis (AA) is induced by immunizing rats with *Mycobacterium tuberculosis* in complete Freund's adjuvant. These animals develop an arthritis whose features are similar to those of rheumatoid arthritis in humans.

Evidence Implicating the CD4⁺ T Cell, MHC, and TCR in Autoimmunity

The inappropriate response to self-antigens that characterizes all autoimmune diseases can involve either the humoral or cell-mediated branches of the immune system. Identifying the defects underlying human autoimmune diseases has been difficult; more success has been achieved in characterizing the immune defects in the various animal models. Each of the animal models has implicated the CD4⁺ T cell as the primary mediator of autoimmune disease. For example, the evidence is quite strong that, in mice, EAE is caused by CD4⁺ T_H1 cells specific for the immunizing antigen. The disease can be transferred from one animal into another by T cells from animals immunized with either MBP or PLP or by

cloned T-cell lines from such animals. It also has been shown that disease can be prevented by treating animals with anti-CD4 antibodies. These data are compelling evidence for the involvement of CD4 in the establishment of EAE.

T-cell recognition of antigen, of course, involves a trimolecular complex of the T-cell receptor, an MHC molecule, and antigenic peptide (see Figure 9-16). Thus, an individual susceptible to autoimmunity must possess MHC molecules and T-cell receptors capable of binding self-antigens.

CD4⁺ T Cells and T_H1/T_H2 Balance Plays an Important Role in Autoimmunity in Some Animal Models

Autoimmune T-cell clones have been obtained from all of the animal models listed in Table 20-2 by culturing lymphocytes from the autoimmune animals in the presence of various T-cell growth factors and by inducing proliferation of specific autoimmune clones with the various autoantigens. For example, when lymph-node cells from EAE rats are cultured in vitro with myelin basic protein (MBP), clones of activated T cells emerge. When sufficient numbers of these MBP-specific T-cell clones are injected intravenously into normal syngeneic animals, the cells cross the blood-brain barrier and induce demyelination; EAE develops very quickly, within 5 days (see Figure 20-7).

A similar experimental protocol has been used to isolate T-cell clones specific for thyroglobulin and for *M. tuberculosis* from EAT and AA animals, respectively. In each case, the T-cell clone induces the experimental autoimmune disease in normal animals. Examination of these T cells has revealed that they bear the CD4 membrane marker. In a number of animal models for autoimmune diseases it has been possible to reverse the autoimmunity by depleting the T-cell population with antibody directed against CD4. For example, weekly injections of anti-CD4 monoclonal antibody abolished the autoimmune symptoms in (NZB × NZW) F₁ mice and in mice with EAE.

Most cases of organ-specific autoimmune disease develop as a consequence of self-reactive CD4⁺ T cells. Analysis of these cells has revealed that the T_H1/T_H2 balance can affect whether autoimmunity develops. T_H1 cells have been implicated in the development of autoimmunity, whereas, in a number of cases, T_H2 cells not only protect against the induction of disease but also against progression of established disease. In EAE, for example, immunohistologic studies revealed the presence of T_H1 cytokines (IL-2, TNF- α , and IFN- γ) in the central nervous system tissues at the height of the disease. In addition, the MBP-specific CD4⁺ T-cell clones generated from animals with EAE, as shown in Figure 20-7, can be separated into T_H1 and T_H2 clones. Experiments have shown that only the T_H1 clones transfer EAE to normal healthy mice, whereas the T_H2 clones not only do not transfer EAE to normal healthy mice but also protect the mice against induction of EAE by subsequent immunization with MBP plus adjuvant.

Experiments that assessed the role of various cytokines or cytokine inhibitors on the development of EAE have provided

further evidence for the different roles of T_H1 and T_H2 cells in autoimmunity. When mice were injected with IL-4 at the time of immunization with MBP plus adjuvant, the development of EAE was inhibited, whereas administration of IL-12 had the opposite effect, promoting the development of EAE. As noted in Chapter 12, IL-4 promotes development of T_H2 cells and IFN- γ , in addition to other cytokines such as IL-12, promotes development of T_H1 cells (see Figure 12-12). Thus, the observed effects of IL-4 and IL-12 on EAE development are consistent with a role for T_H1 cells in the genesis of autoimmunity.

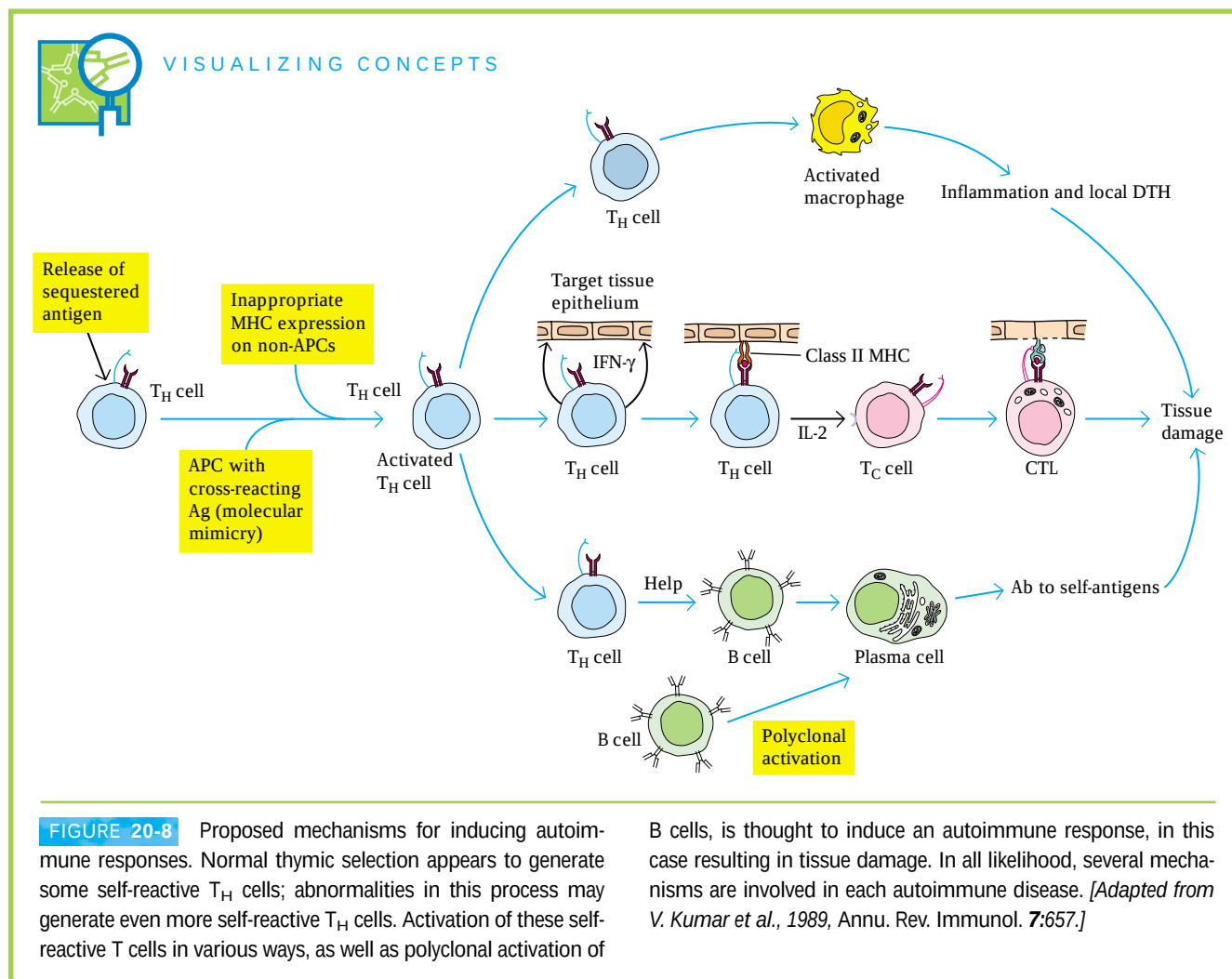
Autoimmunity Can Be Associated with the MHC or with Particular T-Cell Receptors

Several types of studies have supported an association between expression of a particular MHC allele and susceptibility to autoimmunity, an issue covered in detail in Chapter 7. The strongest association between an HLA allele and an autoimmune disease is seen in ankylosing spondylitis, an inflammatory disease of vertebral joints. Individuals who have *HLA-B27* have a 90 times greater likelihood of developing ankylosing spondylitis than individuals with a different *HLA-B* allele. However, the existence of such an association should not be interpreted to imply that the expression of a particular MHC allele has caused the disease, because the relationship between MHC alleles and development of autoimmune disease is complex. It is interesting to note that, unlike many other autoimmune diseases, 90% of the cases of ankylosing spondylitis are male.

The presence of T-cell receptors containing particular V α and V β domains also has been linked to a number of autoimmune diseases, including experimental EAE and its human counterpart, multiple sclerosis. In one approach, T cells specific for various encephalitogenic peptides of MBP were cloned and their T-cell receptors analyzed. For example, T-cell clones were obtained from PL/J mice by culturing their T cells with the acetylated amino-terminal nonapeptide of MBP presented in association with a class II IA^u MHC molecule. Analysis of the T-cell receptors on these clones revealed a restricted repertoire of V α and V β domains: 100% of the T-cell clones expressed V α 4.3, and 80% of the T-cell clones expressed V β 8.2. In human autoimmune diseases, evidence for restricted TCR expression has been obtained for both multiple sclerosis and myasthenia gravis. The preferential expression of TCR variable-region genes in these autoimmune T-cell clones suggests that a single epitope might induce the clonal expansion of a small number of pathogenic T cells.

Proposed Mechanisms for Induction of Autoimmunity

A variety of mechanisms have been proposed to account for the T-cell-mediated generation of autoimmune diseases (Figure 20-8). Evidence exists for each of these mechanisms,



and it is likely that autoimmunity does not develop from a single event but rather from a number of different events.

In addition, susceptibility to many autoimmune diseases differs between the two sexes. As noted earlier, Hashimoto's thyroiditis, systemic lupus erythematosus, multiple sclerosis, rheumatoid arthritis, and scleroderma preferentially affect women. Factors that have been proposed to account for this preferential susceptibility, such as hormonal differences between the sexes and the potential effects of fetal cells in the maternal circulation during pregnancy, are discussed in the Clinical Focus.

Release of Sequestered Antigens Can Induce Autoimmune Disease

As discussed in Chapter 10, the induction of self-tolerance in T cells results from exposure of immature thymocytes to self-antigens and the subsequent clonal deletion of those that are self-reactive. Any tissue antigens that are sequestered from the circulation, and are therefore not seen by the developing

T cells in the thymus, will not induce self-tolerance. Exposure of mature T cells to such normally sequestered antigens at a later time might result in their activation.

Myelin basic protein (MBP) is an example of an antigen normally sequestered from the immune system, in this case by the blood-brain barrier. In the EAE model, animals are injected directly with MBP, together with adjuvant, under conditions that maximize immune exposure. In this type of animal model, the immune system is exposed to sequestered self-antigens under nonphysiologic conditions; however, trauma to tissues following either an accident or a viral or bacterial infection might also release sequestered antigens into the circulation. A few tissue antigens are known to fall into this category. For example, sperm arise late in development and are sequestered from the circulation. However, after a vasectomy, some sperm antigens are released into the circulation and can induce auto-antibody formation in some men. Similarly, the release of lens protein after eye damage or of heart-muscle antigens after myocardial infarction has been shown to lead on occasion to the formation of auto-antibodies.



CLINICAL FOCUS

Why Are Women More Susceptible Than Men to Autoimmunity? Gender Differences in Autoimmune Disease

Of the nearly

9 million individuals in the United States with autoimmune disease, approximately 6.7 million are women. This predisposition to autoimmunity is more apparent in some diseases than others. For example, the female:male ratio of individuals who suffer from diseases such as multiple sclerosis (MS) or rheumatoid arthritis (RA) is approximately two or three females to one male, and there are nine women for every one male afflicted with systemic lupus erythematosus (SLE). However, these statistics do not tell the entire story, since, in some diseases, MS for example, the severity of the disease can be worse in men than in women. The fact that women are more susceptible to autoimmune disease has been recognized for several years but the reasons for this increased risk are not entirely understood. Here some of the possible explanations are considered.

Although it may seem unlikely, considerable evidence suggests there are significant gender differences in immune responses in both humans and mice. Immunization studies in both species suggest that females produce a higher titer of antibodies than males. In fact, females in general tend to mount more vigorous immune responses. In humans, this is particularly apparent in

young females. Women tend to have higher levels of CD4⁺ T cells and significantly higher levels of serum IgM.

In mice, whose gender differences are easier to study, there is a large body of literature documenting gender differences in immune responses. Female mice are much more likely than male mice to develop T_H1 responses and, in infections to which pro-inflammatory T_H1 responses are beneficial, are more likely to be resistant to the infection. An excellent example is infection by viruses such as vesicular stomatitis virus (VSV), herpes simplex virus (HSV), and Theiler's murine encephalomyelitis virus (TMEV). Clearance of these viruses is enhanced by T_H1 responses. In some cases, however, a pro-inflammatory response can be deleterious. For example, a T_H1 response to lymphocytic choriomeningitis virus (LCMV) correlates with more severe disease and significant pathology. Thus, female mice are more likely to succumb to infection with LCMV. The fact that gender is important in LCMV infection is underscored by experiments demonstrating that castrated male mice behave like females and are more likely to succumb to infection than their un-castrated male littermates.

Another disease in which gender plays a role is infection by coxsackie virus Type B-3 (CVB-3), an etiological agent of immune myocarditis. Male

mice are much more susceptible to this disease than females. CVB-3 induces a predominant T_H1 response in males, while females, contrary to the situations described above, respond by mounting a protective T_H2 response. The response by females can be altered by injecting them with testosterone, which makes them susceptible to the disease. Additionally, the male response can be altered by injecting them with estradiol, which makes them resistant to the virus. These data in mice are consistent with the possibility that basic differences may well exist between men and women in their responses to pathogens. We must stress, however, that the particular gender differences observed in mice may not extend to human populations.

How do these gender differences arise? The evidence cited above that estradiol or testosterone can alter the outcome of infection by CVB-3 suggests a critical role for sex hormones. In humans it appears that estrogen on its own does not play a significant role in the etiology of either RA or MS, but there are indications that it may be important in SLE. This is suggested by data indicating that estrogen can stimulate autoantibody production in SLE-prone mice and these effects can be modulated by an anti-estrogenic compound. Such data imply that, at least in mice, estrogen is capable of triggering SLE-like autoimmunity. Additionally, androgens such as testosterone clearly play an important role in some autoimmune diseases. Female NOD mice are much more susceptible to spontaneous diabetes, and castration significantly increases the susceptibility of male NOD mice. Female SJL mice are more likely to be susceptible to EAE, a mouse MS-like disease. This indicates that testosterone may well be effective in ameliorating some autoimmune responses and so

Data indicate that injection of normally sequestered antigens directly into the thymus can reverse the development of tissue-specific autoimmune disease in animal models. For instance, intrathymic injection of pancreatic islet beta cells prevented development of autoimmunity in NOD mice.

Moreover, EAE was prevented in susceptible rats by prior injection of MBP directly into the thymus. In these experiments, exposure of immature T cells to self-antigens that normally are not present in the thymus presumably led to tolerance to these antigens.

may be protective against several autoimmune diseases, including MS, diabetes, SLE, and Sjogren's syndrome.

Why do sex steroids affect immune responses? This is not well understood, but it is likely that these hormones, which circulate throughout the body, alter immune responses by altering patterns of gene expression. The sex steroids, a highly lipophilic group of compounds, function by passing through the cell membrane and binding a cytoplasmic receptor. Each hormone has a cognate receptor and binding of hormone to receptor leads to the activation or, in some instances, repression of gene expression. This is mediated by the binding of the receptor/hormone complex to a specific DNA sequence. Thus, estrogen enters a cell, binds to the estrogen receptor, and induces the binding of the estrogen receptor to a specific DNA sequence, which in turn results in the modulation of transcription. Therefore, in cells that contain hormone receptors, sex hormones can regulate gene expression, and it is highly likely that sex steroids play an important role in the immune system through their receptors. Whether various cells of the immune system contain hormone receptors is not known at present; to understand how sex hormones mediate immune responses, clearly we must determine which cells express which hormone receptors.

Hormonal effects on immune responses may not be limited to steroidal sex hormones. Prolactin, a hormone that is expressed in higher levels in women than in men, is not a member of the lipophilic sex steroid family that includes estrogen, progesterone, and testosterone. But prolactin secretion (by the anterior pituitary) is stimulated by estrogen, thus explaining the higher levels of prolactin in women and the very high levels observed during pregnancy. Prolactin

can have a profound influence on immune responses, as demonstrated in mice by removal of the anterior pituitary: this results in a severe immunosuppression, which can be entirely reversed by treatment with exogenous prolactin. The presence of prolactin receptors on peripheral T and B cells in humans is further evidence that this hormone may play a role in regulating immune responses. In fact, some evidence suggests that prolactin may tend to turn cells towards T_H1 -dominated immune responses.

Pregnancy may give us a clue to how sex plays a role in regulating immune response. It is clear that, while women normally mount a normal response to foreign antigens, during pregnancy it is critical that the mother tolerate the fetus (which is, in fact, a foreign graft). This makes it very likely that the female immune system undergoes important modifications during pregnancy. Recall that women normally tend to mount more T_H1 -like responses than T_H2 responses. During pregnancy, however, women mount more T_H2 -like responses. It is thought that pregnancy-associated levels of sex steroids may promote an anti-inflammatory environment. In this regard, it is notable that diseases enhanced by T_H2 -like responses, such as SLE, which has a strong antibody-mediated component, can be exacerbated during pregnancy, while diseases that involve inflammatory responses, such as RA and MS, sometimes are ameliorated in pregnant women.

Another effect of pregnancy is the presence of fetal cells in the maternal circulation (see the description of scleroderma on page 000). It is known that fetal cells can persist in the maternal circulation for decades, so these long-lived fetal cells may play a significant role in the development of autoimmune disease. Furthermore, the exchange of cells

during pregnancy is bi-directional (cells of the mother may also appear in the fetal circulatory system), and this has led some to postulate that the presence of mother's cells in the male circulation could be a contributing factor in autoimmune disease.

In summary, women and men differ significantly in their ability to mount an immune response. Women mount more robust immune responses, and these responses tend to be more T_H1 -like. It has been reported that estrogen is immunostimulatory; this may be due, in part, to the ability of the hormone to regulate specific gene expression through the estrogen receptor. Furthermore, the incidence of autoimmune diseases is sharply higher in women than in men. These observations have generated the compelling hypothesis that the tendency of females to mount more T_H1 -like responses may, in part, explain differences in susceptibility to autoimmunity. Since this type of response is pro-inflammatory, it may enhance the development of autoimmunity. Whether the bias towards a T_H1 response is due to differences in sex steroids between males and females is less certain, but surely, in the next several years, experiments that explore this idea are likely to be pursued vigorously.

NOTE: The data discussed in this Clinical Focus were extracted from a letter to *Science* (C. C. Whitacre, S. C. Reingold, and P. A. O'Looney, 1999, *Science* **283**:1277) from the Task Force on Gender, MS, and Autoimmunity, a group convened by the National Multiple Sclerosis Society to begin a dialog on issues of gender and autoimmune disease. *Science* also has established a Web site (<http://www.sciencemag.org/feature/data/983519.shl>) that contains more detailed data concerning autoimmunity and gender.

Molecular Mimicry May Contribute to Autoimmune Disease

For several reasons, the notion that microbial or viral agents might play a role in autoimmunity is very attractive. It is well

accepted that migrant human populations acquire the diseases of the area to which they move and that the incidence of autoimmunity has increased dramatically as populations have become more mobile. This, coupled with the fact that a number of viruses and bacteria have been shown to possess

TABLE 20-3 Molecular mimicry between proteins of infectious organisms and human host proteins

Protein*	Residue [†]	Sequence [‡]
Human cytomegalovirus IE2	79	P D P L G R P D E D
HLA-DR molecule	60	V T E L G R P D A E
Poliovirus VP2	70	S T T K E S R G T T
Acetylcholine receptor	176	T V I K E S R G T K
Papilloma virus E2	76	S L H L E S L K D S
Insulin receptor	66	V Y G L E S L K D L
Rabies virus glycoprotein	147	T K E S L V I I S
Insulin receptor	764	N K E S L V I S E
<i>Klebsiella pneumoniae</i> nitrogenase	186	S R Q T D R E D E
HLA-B27 molecule	70	K A Q T D R E D L
Adenovirus 12 E1B	384	L R R G M F R P S Q C N
α-Gliadin	206	L G Q G S F R P S Q Q N
Human immunodeficiency virus p24	160	G V E T T T P S
Human IgG constant region	466	G V E T T T P S
Measles virus P3	13	L E C I R A L K
Corticotropin	18	L E C I R A C K
Measles virus P3	31	E I S D N L G Q E
Myelin basic protein	61	E I S F K L G Q E

*In each pair, the human protein is listed second. The proteins in each pair have been shown to exhibit immunologic cross-reactivity.

[†]Each number indicates the position on the intact protein of the amino-terminal amino acid in the listed sequence.

[‡]Amino acid residues are indicated by single-letter code. Identical residues are shown in blue.

SOURCE: Adapted from M. B. A. Oldstone, 1987, *Cell* 50:819.

antigenic determinants that are identical or similar to normal host-cell components led Michael Oldstone to propose that a pathogen may express a region of protein that resembles a particular self-component in conformation or primary sequence. Such molecular mimicry appears in a wide variety of organisms (Table 20-3). In one study, 600 different monoclonal antibodies specific for 11 different viruses were tested to evaluate their reactivity with normal tissue antigens. More than 3% of the virus-specific antibodies tested also bound to normal tissue, suggesting that molecular mimicry is a fairly common phenomenon.

Molecular mimicry has been suggested as one mechanism that leads to autoimmunity. One of the best examples of this type of autoimmune reaction is post-rabies encephalitis, which used to develop in some individuals who had received the rabies vaccine. In the past, the rabies virus was grown in rabbit brain-cell cultures, and preparations of the vaccine included antigens derived from the rabbit brain cells. In a vaccinated person, these rabbit brain-cell antigens could induce formation of antibodies and activated T cells, which could cross-react with the recipient's own brain cells, leading

to encephalitis. Cross-reacting antibodies are also thought to be the cause of heart damage in rheumatic fever, which can sometimes develop after a *Streptococcus* infection. In this case, the antibodies are to streptococcal antigens, but they cross-react with the heart muscle.

There Is Evidence for Mimicry Between MBP and Viral Peptides

Since the encephalitogenic MBP peptides are known, the extent to which they are mimicked by proteins from other organisms can be assessed. For example, one MBP peptide (amino acid residues 61–69) is highly homologous with a peptide in the P3 protein of the measles virus (see Table 20-3). In one study, the sequence of another encephalitogenic MBP peptide (66–75) was compared with the known sequences of a large number of viral proteins. This computer analysis revealed sequence homologies between this MBP peptide and a number of peptides from animal viruses, including influenza, polyoma, adenovirus, Rous sarcoma, Abelson leukemia, poliomyelitis, Epstein-Barr, and hepatitis B viruses.

One peptide from the polymerase enzyme of the hepatitis B virus was particularly striking, exhibiting 60% homology with a sequence in the encephalitogenic MBP peptide. To test the hypothesis that molecular mimicry can generate autoimmunity, rabbits were immunized with this hepatitis B virus peptide. The peptide was shown to induce both the formation of antibody and the proliferation of T cells that cross-reacted with MBP; in addition, central nervous system tissue from the immunized rabbits showed cellular infiltration characteristic of EAE.

These findings suggest that infection with certain viruses expressing epitopes that mimic sequestered self-components, such as myelin basic protein, may induce autoimmunity to those components. Susceptibility to this type of autoimmunity may also be influenced by the MHC haplotype of the individual, since certain class I and class II MHC molecules may be more effective than others in presenting the homologous peptide for T-cell activation.

Another particularly compelling example of molecular mimicry comes from studies of herpes stromal keratitis (HSK). In these studies, investigators showed that prior infection of mice with herpes simplex virus Type 1 leads to a disease known as herpes stromal keratitis (HSK), an autoimmune-like disease in which T cells specific for a particular viral peptide attack corneal tissue, thus causing blindness. These data demonstrated very clearly that a particular epitope of HSV-1 is responsible for the disease and that mutant strains of HSV-1 that lack this epitope do not cause HSK. The data provide strong evidence for molecular mimicry in the development of a particular autoimmune disease.

Inappropriate Expression of Class II MHC Molecules Can Sensitize Autoreactive T Cells

The pancreatic beta cells of individuals with insulin-dependent diabetes mellitus (IDDM) express high levels of both class I and class II MHC molecules, whereas healthy beta cells express lower levels of class I and do not express class II at all. Similarly, thyroid acinar cells from those with Graves' disease have been shown to express class II MHC molecules on their membranes. This inappropriate expression of class II MHC molecules, which are normally expressed only on antigen-presenting cells, may serve to sensitize T_H cells to peptides derived from the beta cells or thyroid cells, allowing activation of B cells or T_C cells or sensitization of T_H1 cells against self-antigens.

Other evidence suggests that certain agents can induce some cells that should not express class II MHC molecules to express them. For example, the T-cell mitogen phytohemagglutinin (PHA) has been shown to induce thyroid cells to express class II molecules. In vitro studies reveal that IFN- γ also induces increases in class II MHC molecules on a wide variety of cells, including pancreatic beta cells, intestinal epithelial cells, melanoma cells, and thyroid acinar cells. It was hypothesized that trauma or viral infection in an organ

may induce a localized inflammatory response and thus increased concentrations of IFN- γ in the affected organ. If IFN- γ induces class II MHC expression on non-antigen-presenting cells, inappropriate T_H -cell activation might follow, with autoimmune consequences. It is noteworthy that SLE patients with active disease have higher serum titers of IFN- γ than patients with inactive disease. These data suggested that the increase in IFN- γ in these patients may lead to inappropriate expression of class II MHC molecules and thus to T-cell activation against a variety of autoantigens.

An interesting transgenic mouse system implicates IFN- γ and inappropriate class II MHC expression in autoimmunity. In this system, an IFN- γ transgene was genetically engineered with the insulin promoter, so that the transgenic mice secreted IFN- γ from their pancreatic beta cells (Figure 20-9a). Since IFN- γ up-regulates class II MHC expression, these transgenic mice also expressed class II MHC molecules on their pancreatic beta cells. The mice developed diabetes, which was associated with cellular infiltration of lymphocytes and inflammatory cells like the infiltration seen in autoimmune NOD mice and in patients with insulin-dependent diabetes mellitus (Figure 20-9b).

Although inappropriate class II MHC expression on pancreatic beta cells may be involved in the autoimmune reaction in these transgenic mice, other factors also may play a role. For example, IFN- γ is known to induce production of several other cytokines, including IL-1 and TNF. Therefore, the development of autoimmunity in this transgenic system may involve antigen presentation by class II MHC molecules on pancreatic beta cells together with a co-stimulatory signal, such as IL-1, that may activate self-reactive T cells. There is also some evidence to suggest that IL-1, IFN- γ , and TNF may directly impair the secretory function of human beta cells.

Polyclonal B-Cell Activation Can Lead to Autoimmune Disease

A number of viruses and bacteria can induce nonspecific polyclonal B-cell activation. Gram-negative bacteria, cytomegalovirus, and Epstein-Barr virus (EBV) are all known to be such polyclonal activators, inducing the proliferation of numerous clones of B cells that express IgM in the absence of T_H cells. If B cells reactive to self-antigens are activated by this mechanism, auto-antibodies can appear. For instance, during infectious mononucleosis, which is caused by EBV, a variety of auto-antibodies are produced, including auto-antibodies reactive to T and B cells, rheumatoid factors, and antinuclear antibodies. Similarly, lymphocytes from patients with SLE produce large quantities of IgM in culture, suggesting that they have been polyclonally activated. Many AIDS patients also show high levels of nonspecific antibody and auto-antibodies to RBCs and platelets. These patients are often coinfectd with other viruses such as EBV and cytomegalovirus, which may induce the polyclonal B-cell activation that results in auto-antibody production.

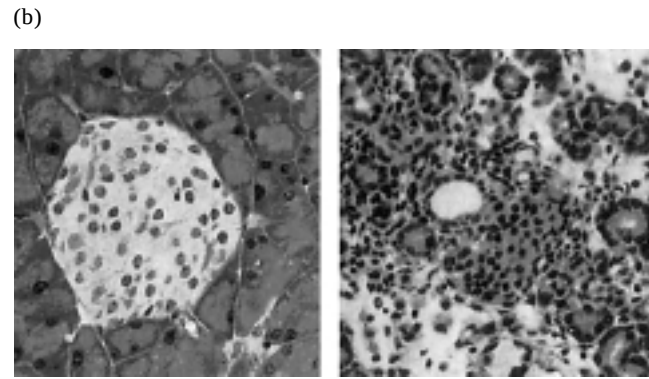
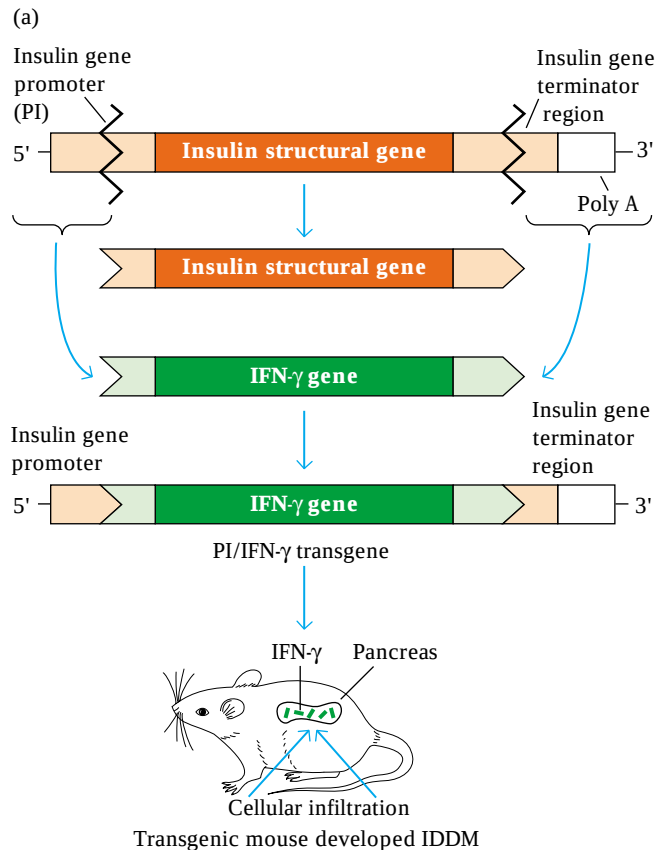


FIGURE 20-9 Insulin-dependent diabetes mellitus (IDDM) in transgenic mice. (a) Production of transgenic mice containing an IFN- γ transgene linked to the insulin promoter (PI). The transgenics, which expressed the PI/IFN- γ transgene only in the pancreas, developed symptoms characteristic of IDDM. (b) Pancreatic islets of Langerhans from a normal BALB/c mouse (*left*) and from PI/IFN- γ transgenics at 3 weeks (*right*) showing infiltration of inflammatory cells. [Part (b) from N. Sarvetnick, 1988, *Cell* **52**:773.]

Treatment of Autoimmune Diseases

Ideally, treatment for autoimmune diseases should be aimed at reducing only the autoimmune response while leaving the rest of the immune system intact. To date, this ideal has not been reached.

Current therapies for autoimmune diseases are not cures but merely palliatives, aimed at reducing symptoms to provide the patient with an acceptable quality of life. For the most part, these treatments provide nonspecific suppression of the immune system and thus do not distinguish between a pathologic autoimmune response and a protective immune response. Immunosuppressive drugs (e.g., corticosteroids, azathioprine, and cyclophosphamide) are often given with the intent of slowing proliferation of lymphocytes. By depressing the immune response in general, such drugs can reduce the severity of autoimmune symptoms. The general reduction in immune responsiveness, however, puts the patient at greater risk for infection or the development of cancer. A somewhat more selective approach employs **cyclosporin A** or FK506 to treat autoimmunity. These agents block signal transduction mediated by the T-cell receptor; thus, they inhibit only antigen-activated T cells while sparing nonactivated ones.

Another therapeutic approach that has produced positive results in some cases of myasthenia gravis is removal of the

thymus. Because patients with this disease often have thymic abnormalities (e.g., thymic hyperplasia or thymomas), adult thymectomy often increases the likelihood of remission of symptoms. Patients with Graves' disease, myasthenia gravis, rheumatoid arthritis, or systemic lupus erythematosus may experience short-term benefit from plasmapheresis. In this process, plasma is removed from a patient's blood by continuous-flow centrifugation. The blood cells are then resuspended in a suitable medium and returned to the patient. Plasmapheresis has been beneficial to patients with autoimmune diseases involving antigen-antibody complexes, which are removed with the plasma. Removal of the complexes, although only temporary, can result in a short-term reduction in symptoms.

On the positive side, studies with experimental autoimmune animal models have provided evidence that it is indeed possible to induce specific immunity to the development of autoimmunity. Several of these approaches are described below and outlined in Figure 20-10.

T-Cell Vaccination Is a Possible Therapy

The basis for T-cell vaccination as a therapy for some autoimmune diseases came from experiments with the EAE animal model. When rats were injected with low doses ($<10^{-4}$) of cloned T cells specific for MBP, they did not develop

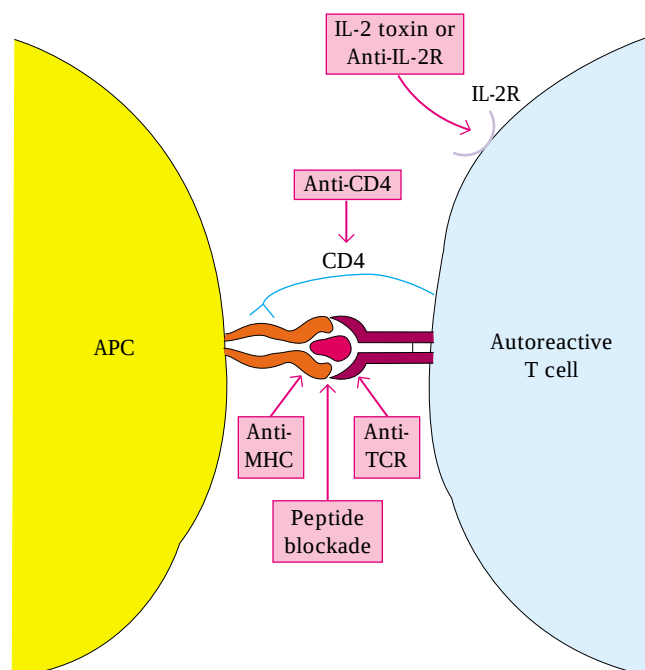


FIGURE 20-10 Some experimental agents for immunointervention in autoimmune disease.

symptoms of EAE. Instead they became resistant to the development of EAE when later challenged with a lethal dose of activated MBP-specific T cells or MBP in adjuvant. Later findings revealed that the efficacy of these autoimmune T-cell clones as a vaccine could be enhanced by crosslinking the cell-membrane components with formaldehyde or glutaraldehyde. When crosslinked T cells were injected into animals with active EAE, permanent remission of symptoms was observed. The crosslinked T cells apparently elicit regulatory T cells specific for TCR variable-region determinants of the autoimmune clones. Presumably these regulatory T cells act to suppress the autoimmune T cells that mediate EAE.

Peptide Blockade of MHC Molecules Can Modulate Autoimmune Responses

Identification and sequencing of various autoantigens has led to the development of new approaches to modulate autoimmune T-cell activity. In EAE, for example, the encephalitogenic peptides of MBP have been well characterized. Synthetic peptides differing by only one amino acid from their MBP counterpart have been shown to bind to the appropriate MHC molecule. Moreover, when sufficient amounts of such a peptide were administered along with the corresponding encephalitogenic MBP peptide, the clinical development of EAE was blocked. Presumably, the synthetic peptide acts as

a competitor, occupying the antigen-binding cleft on MHC molecules and thus preventing binding of the MBP peptide.

In other studies, blocking peptides complexed to soluble class II MHC molecules reversed the clinical progression of EAE in mice, presumably by inducing a state of clonal anergy in the autoimmune T cells.

Monoclonal Antibodies May Be Used to Treat Autoimmunity

Monoclonal antibodies have been used successfully to treat autoimmune disease in several animal models. For example, a high percentage of (NZB $\times$ NZW) F₁ mice given weekly injections of high doses of monoclonal antibody specific for the CD4 membrane molecule recovered from their autoimmune lupus-like symptoms (Figure 20-11). Similar positive results were observed in NOD mice, in which treatment with an anti-CD4 monoclonal antibody led to disappearance of the lymphocytic infiltration and diabetic symptoms.

Because anti-CD4 monoclonal antibodies block or deplete all T_H cells, regardless of their specificity, they can threaten the overall immune responsiveness of the recipient. One remedy for this disadvantage is to try to block antigen-activated T_H cells only, since these cells are involved in the autoimmune state. To do this, researchers have used monoclonal antibody directed against the α subunit of the high-affinity IL-2 receptor, which is expressed only by antigen-activated T_H cells. Because the IL-2R α subunit is expressed at higher levels on autoimmune T cells, monoclonal antibody to the α subunit (anti-TAC) might preferentially block autoreactive T cells. This approach was tested in adult rats injected with activated MBP-specific T cells in the presence or absence of anti-TAC. All the control rats died of EAE, whereas six of the nine treated with anti-TAC had no symptoms, and the symptoms in the other three were mild.

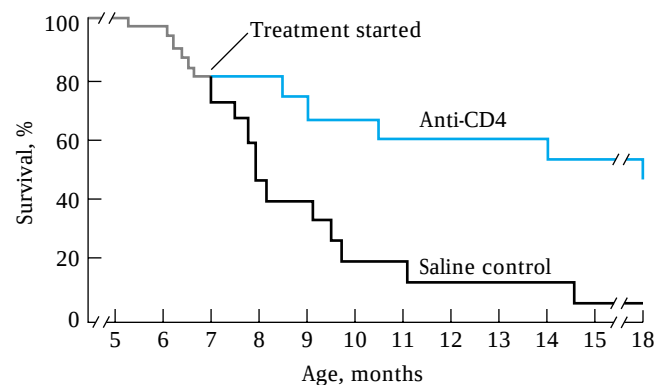


FIGURE 20-11 Weekly injections of anti-CD4 monoclonal antibody into (NZB $\times$ NZW) F₁ mice exhibiting autoimmune lupus-like symptoms significantly increased their survival rate. [Adapted from D. Wofsy, 1988, *Prog. Allergy* 45:106.]

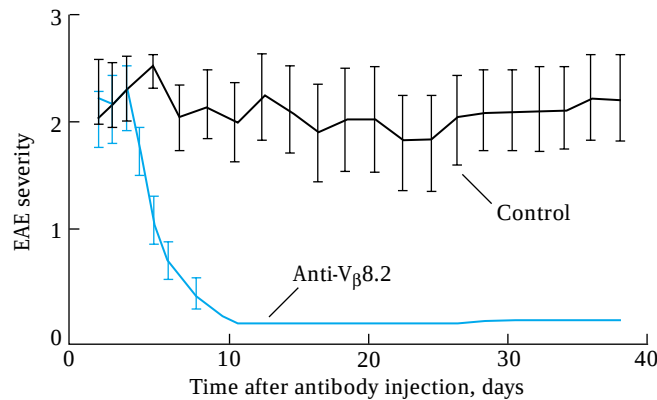


FIGURE 20-12 Injection of monoclonal antibody to the V_{β} 8.2 T-cell receptor into PL/J mice exhibiting EAE symptoms produced nearly complete remission of symptoms. EAE was induced by injecting mice with MBP-specific T-cell clones. EAE severity scale: 3 = total paralysis of lower limbs; 2 = partial paralysis of lower limbs; 1 = limb tail; 0 = normal (no symptoms). [Adapted from H. Acha-Orbea et al., 1989, *Annu. Rev. Immunol.* 7:371.]

The association of autoimmune disease with restricted TCR expression in a number of animal models has prompted researchers to see if blockage of the preferred receptors with monoclonal antibody might be therapeutic. Injection of PL/J mice with monoclonal antibody specific for the V_{β} 8.2 T-cell receptor prevented induction of EAE by MBP in adjuvant. Even more promising was the finding that the V_{β} 8.2 monoclonal antibody could also reverse the symptoms of autoimmunity in mice manifesting induced EAE (Figure 20-12) and that these mice manifested long-term remission. Clearly, the use of monoclonal antibodies as a treatment for human autoimmune diseases presents exciting possibilities.

Similarly, the association of various MHC alleles with autoimmunity (see Table 7-4), as well as the evidence for increased or inappropriate MHC expression in some autoimmune disease, offers the possibility that monoclonal antibodies against appropriate MHC molecules might retard development of autoimmunity. Moreover, since antigen-presenting cells express many different class II MHC molecules, it should theoretically be possible to selectively block an MHC molecule that is associated with autoimmunity while sparing the others. In one study, injecting mice with monoclonal antibodies to class II MHC molecules before injecting MBP blocked the development of EAE. If, instead, the antibody was given after the injection of MBP, development of EAE was delayed but not prevented. In nonhuman primates, monoclonal antibodies to HLA-DR and HLA-DQ have been shown to reverse EAE.

Oral Antigens Can Induce Tolerance

When antigens are administered orally, they tend to induce the state of immunologic unresponsiveness called **tolerance**.

For example, as mentioned earlier in this chapter, mice fed MBP do not develop EAE after subsequent injection of MBP. This finding led to a double-blind pilot trial in which 30 individuals with multiple sclerosis were fed either a placebo or 300 mg of bovine myelin every day for a year. The results of this study revealed that T cells specific for MBP were reduced in the myelin-fed group; there also was some suggestion that MS symptoms were reduced in the male recipients (although the reduction fell short of statistical significance) but not in the female recipients. While the results of oral tolerance induction in mice were promising, the data from humans do not appear to be as beneficial. However, the human clinical trials are in the early stages, and it may be that the peptides used so far were not the most effective, or perhaps the doses were not correct. Because of the promise of this approach as shown in animal studies, it is likely that more clinical trials will be conducted over the next few years.

SUMMARY

- Human autoimmune diseases can be divided into organ-specific and systemic diseases. The organ-specific diseases involve an autoimmune response directed primarily against a single organ or gland. The systemic diseases are directed against a broad spectrum of tissues and have manifestations in a variety of organs resulting from cell-mediated responses and cellular damage caused by auto-antibodies or immune complexes.
- There are both spontaneous and experimental animal models for autoimmune diseases. Spontaneous autoimmune diseases resembling systemic lupus erythematosus occur in NZB and (NZB $\times$ NZW) F_1 mice and in MRL/*lpr/lpr* mice, which have a defective *fas* gene. Several experimental animal models have been developed by immunizing animals with self-antigens in the presence of adjuvant.
- Studies with experimental autoimmune animal models have revealed a central role for $CD4^+$ T_H cells in the development of autoimmunity. In each of the experimentally induced autoimmune diseases, autoimmune T-cell clones can be isolated that induce the autoimmune disease in normal animals. The relative number of T_H1 and T_H2 cells appears to play a pivotal role in determining whether autoimmunity develops: T_H1 cells promote the development of autoimmunity, whereas T_H2 cells appear to block development of autoimmune disease and also block the progression of the disease once it is established. The MHC haplotype of the experimental animal determines the ability to present various autoantigens to T_H cells.
- A variety of mechanisms have been proposed for induction of autoimmunity, including release of sequestered antigens, molecular mimicry, inappropriate class II MHC expression on cells (in some cases stimulated by $IFN-\gamma$, and polyclonal B-cell activation. Evidence exists for each of these mecha-

nisms, reflecting the many different pathways leading to autoimmune reactions.

- Current therapies for autoimmune diseases include treatment with immunosuppressive drugs, thymectomy, and plasmapheresis for diseases involving immune complexes. Other therapies include vaccination with T cells specific for a given autoantigen, administration of synthetic blocking peptides that compete with autoantigen for binding to MHC molecules, treatment with monoclonal antibodies that react with some component specifically involved in an autoimmune reaction, and induction of tolerance to autoantigens by administering them orally.

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USEFUL WEB SITES

<http://www.lupus.org/index.html>

The site for the Lupus Foundation of America contains valuable information for patients and family members as well as current information about research in this area.

<http://www.nih.gov/niams/>

Home page for the National Institute for Arthritis and Musculoskeletal and Skin Diseases. This site contains links to other arthritis sites.

<http://www.niddk.nih.gov/>

Home page for the National Institute for Diabetes and Digestive and Kidney Diseases. This site contains an exhaustive list of links to other diabetes health-related sites.

<http://www.sciencemag.org/feature/data/983519.shl>

Link to a Web site that provides specific information concerning the role of gender in autoimmune disease.

Study Questions

CLINICAL FOCUS QUESTION What are some of the possible reasons why females are more susceptible to autoimmune diseases than males?

1. For each of the following autoimmune diseases (a–m), select the most appropriate characteristic (1–13) listed below.

Diseases

- a. _____ Experimental autoimmune encephalitis (EAE)
- b. _____ Goodpasture's syndrome
- c. _____ Graves' disease
- d. _____ Systemic lupus erythematosus (SLE)
- d. _____ Insulin-dependent diabetes mellitus (IDDM)



- f. _____ Rheumatoid arthritis
- g. _____ Hashimoto's thyroiditis
- h. _____ Experimental autoimmune myasthenia gravis (EAMG)
- i. _____ Myasthenia gravis
- j. _____ Pernicious anemia
- k. _____ Multiple sclerosis
- l. _____ Autoimmune hemolytic anemia

Characteristics

- (1) Auto-antibodies to intrinsic factor block vitamin B₁₂ absorption
 - (2) Auto-antibodies to acetylcholine receptor
 - (3) T_H1-cell reaction to thyroid antigens
 - (4) Auto-antibodies to RBC antigens
 - (5) T-cell response to myelin
 - (6) Induced by injection of myelin basic protein plus complete Freund's adjuvant
 - (7) Auto-antibody to IgG
 - (8) Auto-antibodies to basement membrane
 - (9) Auto-antibodies to DNA and DNA-associated protein
 - (10) Auto-antibodies to receptor for thyroid-stimulating hormone
 - (11) Induced by injection of acetylcholine receptors
 - (12) T_H1-cell response to pancreatic beta cells
2. Experimental autoimmune encephalitis (EAE) has proved to be a useful animal model of autoimmune disorders.
- a. Describe how this animal model is made.
 - b. What is unusual about the animals that recover from EAE?
 - c. How has this animal model indicated a role for T cells in the development of autoimmunity?

3. Molecular mimicry is one mechanism proposed to account for the development of autoimmunity. How has induction of EAE with myelin basic protein contributed to the understanding of molecular mimicry in autoimmune disease?

4. Describe at least three different mechanisms by which a localized viral infection might contribute to the development of an organ-specific autoimmune disease.

5. Transgenic mice expressing the IFN- γ transgene linked to the insulin promoter developed diabetes.

- a. Why was the insulin promoter used?
- b. What is the evidence that the diabetes in these mice is due to autoimmune damage?
- c. What is unusual about MHC expression in this system?
- d. How might this system mimic events that might be caused by a localized viral infection in the pancreas?

6. Monoclonal antibodies have been administered for therapy in various autoimmune animal models. Which monoclonal antibodies have been used and what is the rationale for these approaches?

7. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.

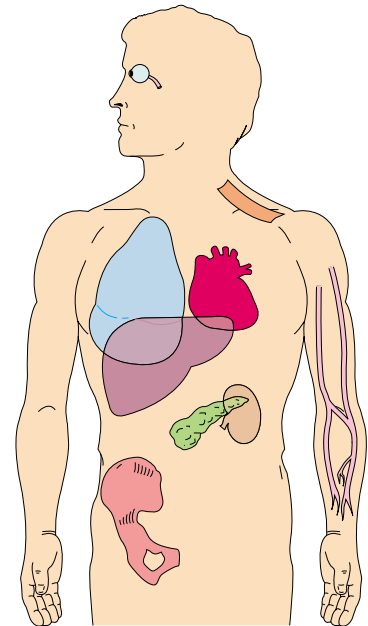
- a. T_H1 cells have been associated with development of autoimmunity.
- b. Immunization of mice with IL-12 prevents induction of EAE by injection of myelin basic protein plus adjuvant.
- c. The presence of the *HLA B27* allele is diagnostic for ankylosing spondylitis, an autoimmune disease affecting the vertebrae.
- d. Individuals with pernicious anemia produce antibodies to intrinsic factor.
- e. A defect in the gene encoding Fas can reduce programmed cell death by apoptosis.

Transplantation Immunology

TRANSPLANTATION, AS THE TERM IS USED IN immunology, refers to the act of transferring cells, tissues, or organs from one site to another. The desire to accomplish transplants stems from the realization that many diseases can be cured by implantation of a healthy organ, tissue, or cells (a graft) from one individual (the donor) to another in need of the transplant (the recipient or host). The development of surgical techniques that allow the facile reimplantation of organs has removed one barrier to successful transplantation, but others remain. One is the lack of organs for transplantation. Although a supply of organs is provided by accident victims and, in some cases, living donors, there are more patients in need of transplants than there are organs available. The seriousness of the donor-organ shortage is reflected in the fact that, as of November 2000, an estimated 73,000 patients in the United States were on the waiting list for an organ transplantation. The majority of those on the list (~70%) require a kidney; at present, the waiting period for this organ averages over 800 days. While the lack of organs for transplantation is a serious issue, the most formidable barrier to making transplantation a routine medical treatment is the immune system. The immune system has evolved elaborate and effective mechanisms to protect the organism from attack by foreign agents, and these same mechanisms cause rejection of grafts from anyone who is not genetically identical to the recipient.

Alexis Carrel reported the first systematic study of transplantation in 1908; he interchanged both kidneys in a series of nine cats. Some of those receiving kidneys from other cats maintained urinary output for up to 25 days. Although all the cats eventually died, the experiment established that a transplanted organ could carry out its normal function in the recipient. The first human kidney transplant, attempted in 1935 by a Russian surgeon, failed because there was a mismatch of blood types between donor and recipient. This incompatibility caused almost immediate rejection of the kidney, and the patient died without establishing renal function. The rapid immune response experienced here, termed hyperacute rejection, is mediated by antibodies and will be described in this chapter. The first successful human kidney transplant, which was between identical twins, was accomplished in Boston in 1954. Today, kidney, pancreas, heart, lung, liver, bone-marrow, and cornea transplantations are performed among nonidentical individuals with ever-increasing frequency and success.

chapter 21



Transplantations Routinely Used in Clinical Practice

- Immunologic Basis of Graft Rejection
- Clinical Manifestations of Graft Rejection
- General Immunosuppressive Therapy
- Specific Immunosuppressive Therapy
- Immune Tolerance to Allografts
- Clinical Transplantation

A variety of immunosuppressive agents can aid in the survival of the transplants, including drugs and specific antibodies developed to diminish the immunologic attack on grafts, but the majority of these agents have an overall immunosuppressive effect, and their long-term use is deleterious. New methods of inducing specific tolerance to the graft without suppressing other immune responses are being developed and promise longer survival of transplants without compromise of host immunity. This chapter describes the mechanisms underlying graft rejection, various procedures that are used to prolong graft survival, and a summary of the current status of transplantation as a clinical tool. A Clinical Focus section examines the use of organs from non-human species (xenotransplants) to circumvent the shortage of organs available for patients in need of them.

Immunologic Basis of Graft Rejection

The degree of immune response to a graft varies with the type of graft. The following terms are used to denote different types of transplants:

- **Autograft** is self-tissue transferred from one body site to another in the same individual. Transferring healthy skin to a burned area in burn patients and use of healthy blood vessels to replace blocked coronary arteries are examples of frequently used autografts.
- **Isograft** is tissue transferred between genetically identical individuals. In inbred strains of mice, an isograft can be performed from one mouse to another syngeneic mouse. In humans, an isograft can be performed between genetically identical (monozygotic) twins.
- **Allograft** is tissue transferred between genetically different members of the same species. In mice, an allograft is performed by transferring tissue or an organ from one strain to another. In humans, organ grafts from one individual to another are allografts unless the donor and recipient are identical twins.
- **Xenograft** is tissue transferred between different species (e.g., the graft of a baboon heart into a human). Because of significant shortages in donated organs, raising animals for the specific purpose of serving as organ donors for humans is under serious consideration.

Autografts and isografts are usually accepted, owing to the genetic identity between graft and host (Figure 21-1a). Because an allograft is genetically dissimilar to the host, it is often recognized as foreign by the immune system and is rejected. Obviously, xenografts exhibit the greatest genetic disparity and therefore engender a vigorous graft rejection.

Allograft Rejection Displays Specificity and Memory

The rate of allograft rejection varies according to the tissue involved. In general, skin grafts are rejected faster than other tissues such as kidney or heart. Despite these time differences, the immune response culminating in graft rejection always displays the attributes of specificity and memory. If an inbred mouse of strain A is grafted with skin from strain B, primary graft rejection, known as first-set rejection, occurs (Figure 21-1b). The skin first becomes revascularized between days 3 and 7; as the reaction develops, the vascularized transplant becomes infiltrated with lymphocytes, monocytes, neutrophils, and other inflammatory cells. There is decreased vascularization of the transplanted tissue by 7–10 days, visible necrosis by 10 days, and complete rejection by 12–14 days.

Immunologic memory is demonstrated when a second strain-B graft is transferred to a previously grafted strain-A

mouse. In this case, a graft-rejection reaction develops more quickly, with complete rejection occurring within 5–6 days; this secondary response is designated second-set rejection (Figure 21-1c). The specificity of second-set rejection can be demonstrated by grafting an unrelated strain-C graft at the same time as the second strain-B graft. Rejection of the strain-C graft proceeds according to first-set rejection kinetics, whereas the strain-B graft is rejected in an accelerated second-set fashion.

T Cells Play a Key Role in Allograft Rejection

In the early 1950s, Avron Mitchison showed in adoptive-transfer experiments that lymphocytes, but not serum antibody, could transfer allograft immunity. Later studies implicated T cells in allograft rejection. For example, nude mice, which lack a thymus and consequently lack functional T cells, were found to be incapable of allograft rejection; indeed, these mice even accept xenografts. In other studies, T cells derived from an allograft-primed mouse were shown to transfer second-set allograft rejection to an unprimed syngeneic recipient, as long as that recipient was grafted with the same allogeneic tissue (Figure 21-2).

Analysis of the T-cell subpopulations involved in allograft rejection has implicated both CD4⁺ and CD8⁺ populations. In one study, mice were injected with monoclonal antibodies to deplete one or both types of T cells and then the rate of graft rejection was measured. As shown in Figure 21-3, removal of the CD8⁺ population alone had no effect on graft survival, and the graft was rejected at the same rate as in control mice (15 days). Removal of the CD4⁺ T-cell population alone prolonged graft survival from 15 days to 30 days. However, removal of both the CD4⁺ and the CD8⁺ T cells resulted in long-term survival (up to 60 days) of the allografts. This study indicated that both CD4⁺ and CD8⁺ T-cells participated in rejection and that the collaboration of both subpopulations resulted in more pronounced graft rejection.

Similar Antigenic Profiles Foster Allograft Acceptance

Tissues that are antigenically similar are said to be **histocompatible**; such tissues do not induce an immunologic response that leads to tissue rejection. Tissues that display significant antigenic differences are *histoincompatible* and induce an immune response that leads to tissue rejection. The various antigens that determine histocompatibility are encoded by more than 40 different loci, but the loci responsible for the most vigorous allograft-rejection reactions are located within the **major histocompatibility complex (MHC)**. The organization of the MHC—called the H-2 complex in mice and the HLA complex in humans—was described in Chapter 7 (see Figure 7-1). Because the MHC loci are closely linked, they are usually inherited as a complete set, called a **haplotype**, from each parent.



VISUALIZING CONCEPTS

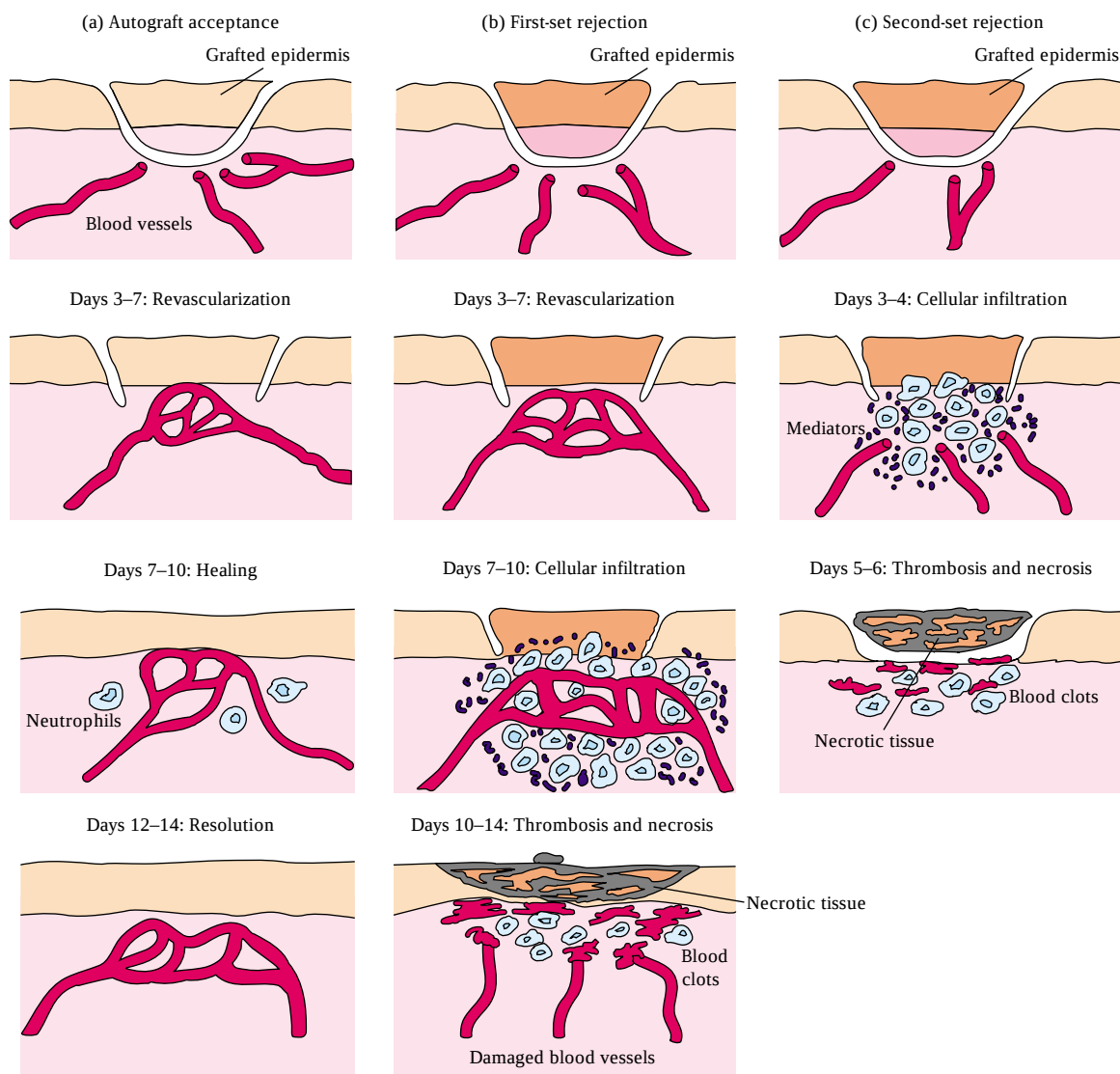


FIGURE 21-1 Schematic diagrams of the process of graft acceptance and rejection. (a) Acceptance of an autograft is completed within 12–14 days. (b) First-set rejection of an allograft begins 7–10 days after grafting, with full rejection occurring by

10–14 days. (c) Second-set rejection of an allograft begins within 3–4 days, with full rejection by 5–6 days. The cellular infiltrate that invades an allograft (b, c) contains lymphocytes, phagocytes, and other inflammatory cells.

Within an inbred strain of mice, all animals are homozygous at each MHC locus. When mice from two different inbred strains, with haplotypes *b* and *k*, for example, are mated, all the F_1 progeny inherit one haplotype from each parent (see Figure 7-2a). These F_1 offspring have the MHC type *b/k* and

can accept grafts from either parent. Neither of the parental strains, however, can accept grafts from the F_1 offspring because each parent lacks one of the F_1 haplotypes. MHC inheritance in outbred populations is more complex, because the high degree of polymorphism exhibited at each MHC locus

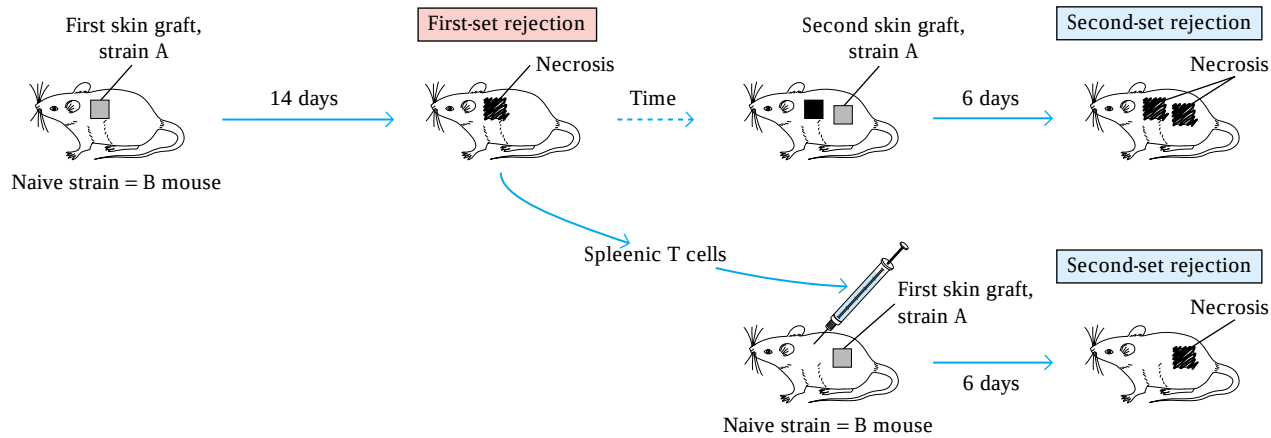


FIGURE 21-2 Experimental demonstration that T cells can transfer allograft rejection. When T cells derived from an allograft-primed mouse are transferred to an unprimed syngeneic mouse, the recipi-

ent mounts a second-set rejection to an initial allograft from the original allogeneic strain.

gives a high probability of heterozygosity at most loci. In matings between members of an outbred species, there is only a 25% chance that any two offspring will inherit identical MHC haplotypes (see Figure 7-2c), unless the parents share one or more haplotypes. Therefore, for purposes of organ or bone-marrow grafts, it can be assumed that there is a 25% chance of identity within the MHC between siblings. With parent-to-child grafts, the donor and recipient will always have one haplotype in common but are nearly always mismatched for the haplotype inherited from the other parent.

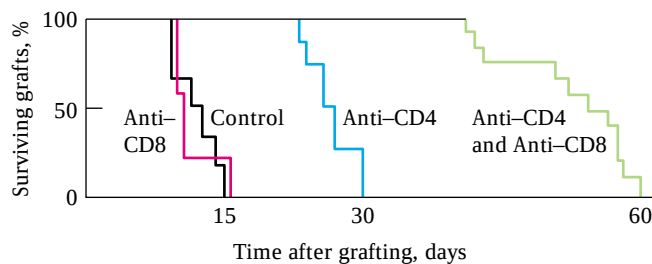


FIGURE 21-3 The role of CD4⁺ and CD8⁺ T cells in allograft rejection is demonstrated by the curves showing survival times of skin grafts between mice mismatched at the MHC. Animals in which the CD8⁺ T cells were removed by treatment with an anti-CD8 monoclonal antibody (red) showed little difference from untreated control mice (black). Treatment with monoclonal anti-CD4 (blue) improved graft survival significantly, and treatment with both anti-CD4 and anti-CD8 antibody prolonged graft survival most dramatically (green). [Adapted from S. P. Cobbold et al., 1986, *Nature* **323**:165.]

Graft Donors and Recipients Are Typed for RBC and MHC Antigens

Since differences in blood group and major histocompatibility antigens are responsible for the most intense graft-rejection reactions, various tissue-typing procedures to identify these antigens have been developed to screen potential donor and recipient cells. Initially, donor and recipient are screened for ABO blood-group compatibility. The blood-group antigens are expressed on RBCs, epithelial cells, and endothelial cells. Antibodies produced in the recipient to any of these antigens that are present on transplanted tissue will induce antibody-mediated complement lysis of the incompatible donor cells.

HLA typing of potential donors and a recipient can be accomplished with a microcytotoxicity test (Figure 21-4a, b). In this test, white blood cells from the potential donors and recipient are distributed into a series of wells on a microtiter plate, and then antibodies specific for various class I and class II MHC alleles are added to different wells. After incubation, complement is added to the wells, and cytotoxicity is assessed by the uptake or exclusion of various dyes (e.g., trypan blue or eosin Y) by the cells. If the white blood cells express the MHC allele for which a particular monoclonal antibody is specific, then the cells will be lysed upon addition of complement, and these dead cells will take up a dye such as trypan blue. HLA typing based on antibody-mediated microcytotoxicity can thus indicate the presence or absence of various MHC alleles.

Even when a fully HLA-compatible donor is not available, transplantation may be successful. In this situation, a one-way mixed-lymphocyte reaction (MLR) can be used to quantify the degree of class II MHC compatibility between potential

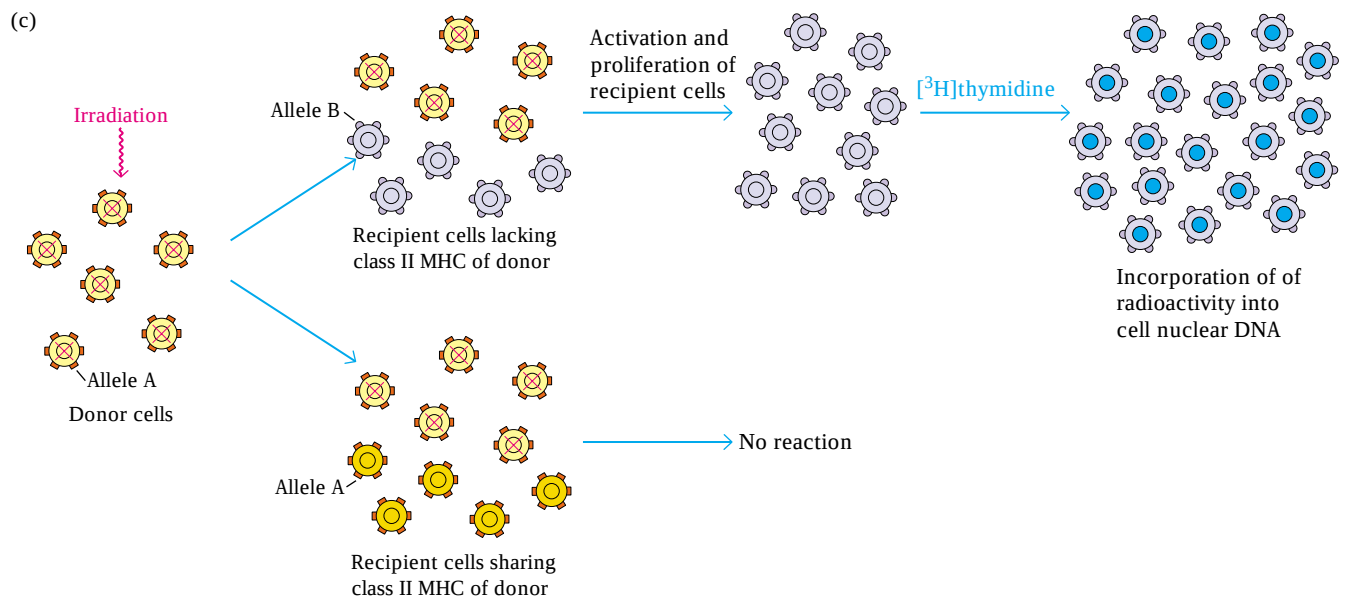
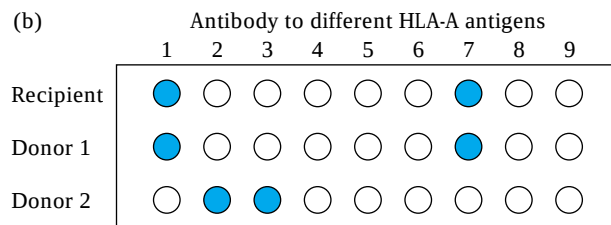
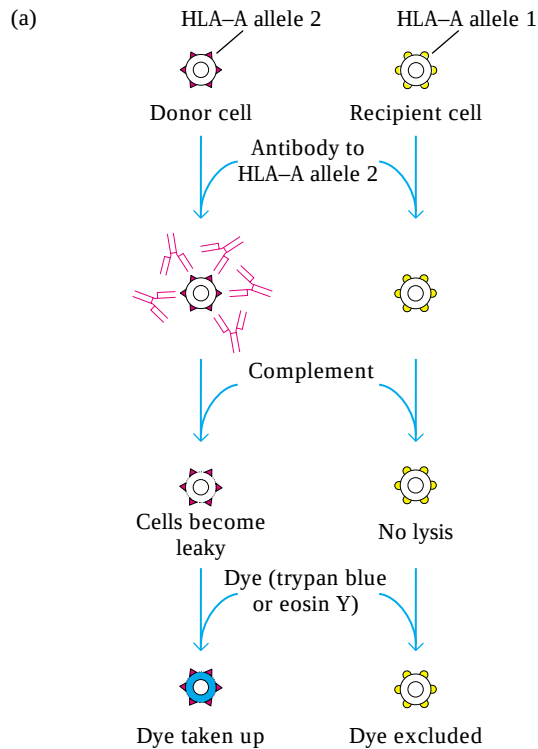


FIGURE 21-4 Typing procedures for HLA antigens. (a, b) HLA typing by microcytotoxicity. (a) White blood cells from potential donors and the recipient are added to separate wells of a microtiter plate. The example depicts the reaction of donor and recipient cells with a single antibody directed against an HLA-A antigen. The reaction sequence shows that if the antigen is present on the lymphocytes, addition of complement will cause them to become porous and unable to exclude the added dye. (b) Because cells express numerous HLA antigens, they are tested separately with a battery of antibodies specific for various HLA-A antigens. Here, donor 1 shares HLA-A antigens recognized by antisera in wells 1 and 7 with the recipient, whereas donor 2 has none of HLA-A antigens in common with the recipient. (c) Mixed lymphocyte reaction to determine identity of class II HLA antigens between a potential donor and recipient. Lymphocytes from the donor are irradiated or treated with mitomycin C to prevent cell division and then added to cells from the recipient. If the class II antigens on the two cell populations are different, the recipient cells will divide rapidly and take up large quantities of radioactive nucleotides into the newly synthesized nuclear DNA. The amount of radioactive nucleotide uptake is roughly proportionate to the MHC class II differences between the donor and recipient lymphocytes.

donors and a recipient (Figure 21-4c). Lymphocytes from a potential donor that have been x-irradiated or treated with mitomycin C serve as the stimulator cells, and lymphocytes from the recipient serve as responder cells. Proliferation of the recipient T cells, which indicates T-cell activation, is measured by the uptake of [³H]thymidine into cell DNA. The greater the class II MHC differences between the donor and recipient cells, the more [³H]thymidine uptake will be observed in an MLR assay. Intense proliferation of the recipient lymphocytes indicates a poor prognosis for graft survival. The advantage of the MLR over microcytotoxicity typing is that it gives a better indication of the degree of T_H-cell activation generated in response to the class II MHC antigens of the potential graft. The disadvantage of the MLR is that it takes several days to run the assay. If the potential donor is a cadaver, for example, it is not possible to wait for the results of the MLR, because the organ must be used soon after removal from the cadaver. In that case, the microcytotoxicity test, which can be performed within a few hours, must be relied on.

The importance of MHC matching for acceptance of allografts is confirmed by data gathered from recipients of kidney transplants. The data in Figure 21-5 reveal that survival of kidney grafts depends primarily on donor-recipient matching of the HLA class II antigens. Matching or mismatching of the class I antigens has a lesser effect on graft survival unless there also is mismatching of the class II antigens. A two-year survival rate of 90% is seen for kidney transplants in which one or two class I HLA loci are mismatched, while transplanted kidneys with differences in the class II MHC have only a 70% chance of lasting for this period. Those with greater numbers of mismatches have a very low survival rate at one year after transplant. As described below, HLA matching is most important for kidney and bone-marrow transplants; liver and heart transplants may survive with greater mismatching.

Current understanding of the killer-inhibitory receptors (KIR) on the NK cell (see Chapter 14) suggests that absence of a class I antigen recognized by the KIR molecules could lead to killing of the foreign cell. Rejection was observed in experimental bone-marrow transplants where the class I molecule recognized by the recipient NK-inhibitory receptor is absent on donor cells. The effects of such class I mismatching on solid organ grafts may be less marked.

MHC identity of donor and host is not the sole factor determining tissue acceptance. When tissue is transplanted between genetically different individuals, even if their MHC antigens are identical, the transplanted tissue can be rejected because of differences at various **minor histocompatibility loci**. As described in Chapter 10, the major histocompatibility antigens are recognized directly by T_H and T_C cells, a phenomenon termed *alloreactivity*. In contrast, minor histocompatibility antigens are recognized only when they are presented in the context of self-MHC molecules. The tissue rejection induced by minor histocompatibility differences

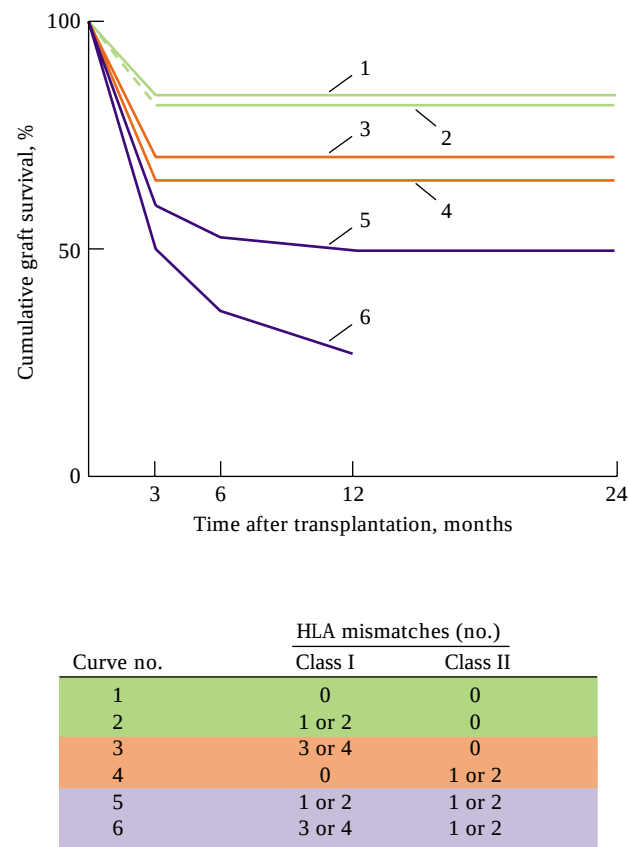


FIGURE 21-5 The effect of HLA class I and class II antigen matching on survival of kidney grafts. Mismatching of one or two class I (HLA-A or HLA-B) antigens has little effect on graft survival. A single class II difference (line 4) has the same effect as 3 or 4 differences in class I antigens (line 3). When both class I and class II antigens are mismatched, rejection is accelerated. [Adapted from T. Moen et al., 1980, N. Engl. J. Med. 303:850.]

is usually less vigorous than that induced by major histocompatibility differences. Still, reaction to these minor tissue differences often results in graft rejection. For this reason, successful transplantation even between HLA-identical individuals requires some degree of immune suppression.

Cell-Mediated Graft Rejection Occurs in Two Stages

Graft rejection is caused principally by a cell-mediated immune response to alloantigens (primarily, MHC molecules) expressed on cells of the graft. Both delayed-type hypersensitive and cell-mediated cytotoxicity reactions have been implicated. The process of graft rejection can be divided into two stages: (1) a sensitization phase, in which antigen-reactive lymphocytes of the recipient proliferate in response to allo-

antigens on the graft, and (2) an effector stage, in which immune destruction of the graft takes place.

SENSITIZATION STAGE

During the sensitization phase, $CD4^+$ and $CD8^+$ T cells recognize alloantigens expressed on cells of the foreign graft and proliferate in response. Both major and minor histocompatibility alloantigens can be recognized. In general, the response to minor histocompatibility antigens is weak, although the combined response to several minor differences can sometimes be quite vigorous. The response to major histocompatibility antigens involves recognition of both the donor MHC molecule and an associated peptide ligand in the cleft of the MHC molecule. The peptides present in the groove of allogeneic class I MHC molecules are derived from proteins synthesized within the allogeneic cell. The peptides present in the groove of allogeneic class II MHC molecules are generally proteins taken up and processed through the endocytic pathway of the allogeneic antigen-presenting cell.

A host T_H cell becomes activated when it interacts with an antigen-presenting cell (APC) that both expresses an appropriate antigenic ligand–MHC molecule complex and provides the requisite co-stimulatory signal. Depending on the tissue, different populations of cells within a graft may function as APCs. Because dendritic cells are found in most tissues and because they constitutively express high levels of class II MHC molecules, dendritic cells generally serve as the major APC in grafts. APCs of host origin can also migrate into a graft and endocytose the foreign alloantigens (both major and minor histocompatibility molecules) and present them as processed peptides together with self-MHC molecules.

In some organ and tissue grafts (e.g., grafts of kidney, thymus, and pancreatic islets), a population of donor APCs called *passenger leukocytes* has been shown to migrate from the graft to the regional lymph nodes. These passenger leukocytes are dendritic cells, which express high levels of class II MHC molecules (together with normal levels of class I MHC molecules) and are widespread in mammalian tissues, with the chief exception of the brain. Because passenger leukocytes express the allogeneic MHC antigens of the donor graft, they are recognized as foreign and therefore can stimulate immune activation of T lymphocytes in the lymph node. In some experimental situations, the passenger cells have been shown to induce tolerance to their surface antigens by deletion of thymic T-cell populations with receptors specific for them. Consistent with the notion that exposure to donor cells can induce tolerance are data showing that blood transfusions from the donor prior to transplantation can aid acceptance of the graft.

Passenger leukocytes are not the only cells involved in immune stimulation. For example, they do not seem to play any role in skin grafts. Other cell types that have been implicated in alloantigen presentation to the immune system include

Langerhans cells and endothelial cells lining the blood vessels. Both of these cell types express class I and class II MHC antigens.

Recognition of the alloantigens expressed on the cells of a graft induces vigorous T-cell proliferation in the host. This proliferation can be demonstrated *in vitro* in a mixed-lymphocyte reaction (see Figure 21-4c). Both dendritic cells and vascular endothelial cells from an allogeneic graft induce host T-cell proliferation. The major proliferating cell is the $CD4^+$ T cell, which recognizes class II alloantigens directly or alloantigen peptides presented by host antigen-presenting cells. This amplified population of activated T_H cells is thought to play a central role in inducing the various effector mechanisms of allograft rejection.

EFFECTOR STAGE

A variety of effector mechanisms participate in allograft rejection (Figure 21-6). The most common are cell-mediated reactions involving delayed-type hypersensitivity and CTL-mediated cytotoxicity; less common mechanisms are antibody-plus-complement lysis and destruction by antibody-dependent cell-mediated cytotoxicity (ADCC). The hallmark of graft rejection involving cell-mediated reactions is an influx of T cells and macrophages into the graft. Histologically, the infiltration in many cases resembles that seen during a delayed-type hypersensitive response, in which cytokines produced by T_{DTH} cells promote macrophage infiltration (see Figure 14-15). Recognition of foreign class I alloantigens on the graft by host $CD8^+$ cells can lead to CTL-mediated killing (see Figure 14-4). In some cases, $CD4^+$ T cells that function as class II MHC-restricted cytotoxic cells mediate graft rejection.

In each of these effector mechanisms, cytokines secreted by T_H cells play a central role (see Figure 21-6). For example, IL-2, IFN- γ , and TNF- β have each been shown to be important mediators of graft rejection. IL-2 promotes T-cell proliferation and generally is necessary for the generation of effector CTLs (see Figure 14-1). IFN- γ is central to the development of a DTH response, promoting the influx of macrophages into the graft and their subsequent activation into more destructive cells. TNF- β has been shown to have a direct cytotoxic effect on the cells of a graft. A number of cytokines promote graft rejection by inducing expression of class I or class II MHC molecules on graft cells. The interferons (α , β , and γ), TNF- α , and TNF- β all increase class I MHC expression, and IFN- γ increases class II MHC expression as well. During a rejection episode, the levels of these cytokines increase, inducing a variety of cell types within the graft to express class I or class II MHC molecules. In rat cardiac allografts, for example, dendritic cells are initially the only cells that express class II MHC molecules. However, as an allograft reaction begins, localized production of IFN- γ in the graft induces vascular endothelial cells and myocytes to express class II MHC molecules as well, making these cells targets for CTL attack.

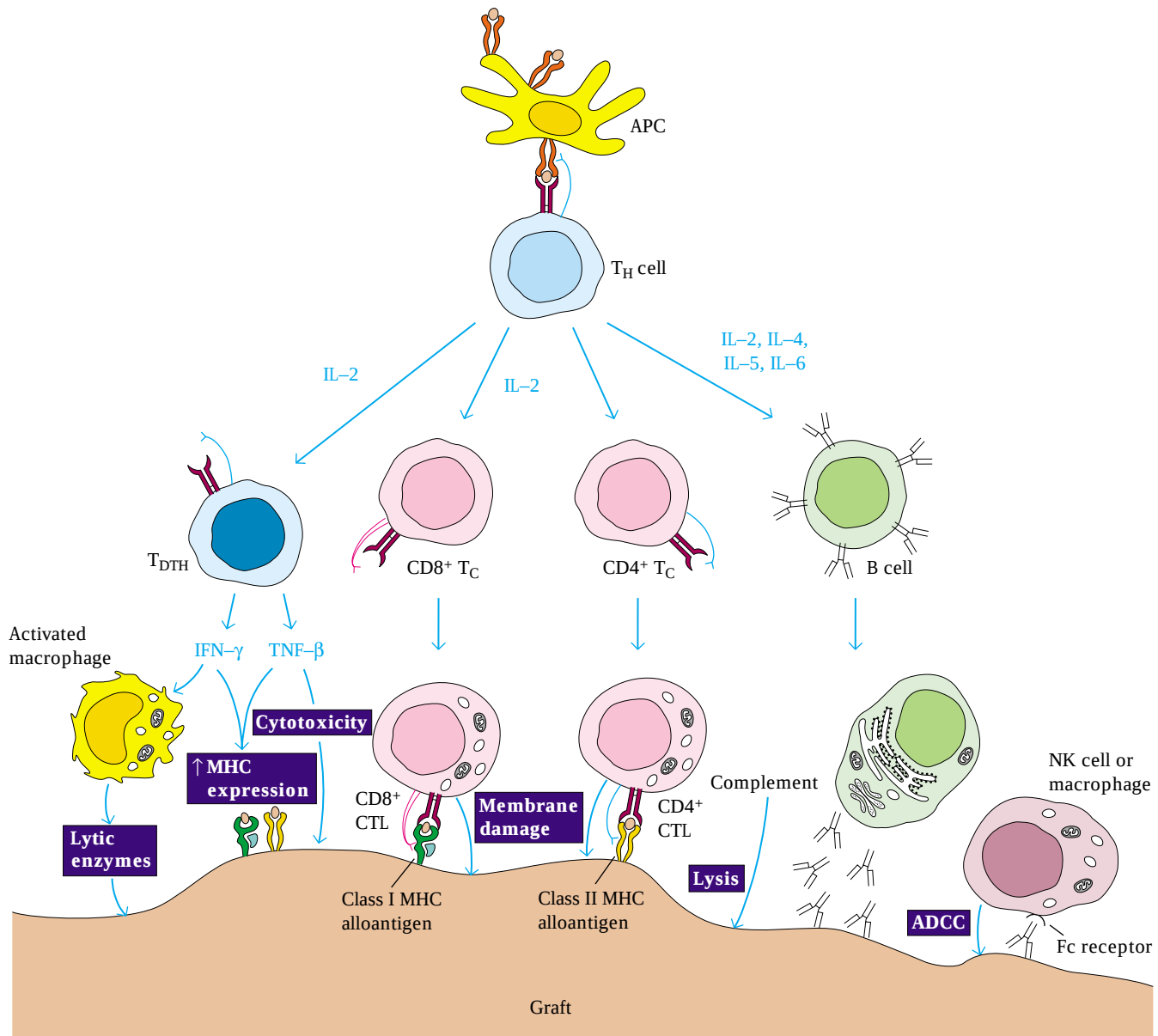


FIGURE 21-6 Effector mechanisms (purple blocks) involved in allograft rejection. The generation or activity of various effector cells

depends directly or indirectly on cytokines (blue) secreted by activated T_H cells. ADCC = antibody-dependent cell-mediated cytotoxicity.

Clinical Manifestations of Graft Rejection

Graft-rejection reactions have various time courses depending upon the type of tissue or organ grafted and the immune response involved. Hyperacute rejection reactions occur within the first 24 hours after transplantation; acute rejection reactions usually begin in the first few weeks after transplantation; and chronic rejection reactions can occur from months to years after transplantation.

Pre-Existing Recipient Antibodies Mediate Hyperacute Rejection

In rare instances, a transplant is rejected so quickly that the grafted tissue never becomes vascularized. These hyperacute reactions are caused by preexisting host serum antibodies specific for antigens of the graft. The antigen-antibody complexes that form activate the complement system, resulting in an intense infiltration of neutrophils into the grafted tissue. The ensuing inflammatory reaction causes massive blood clots within the capillaries, preventing vascularization of the graft (Figure 21-7).

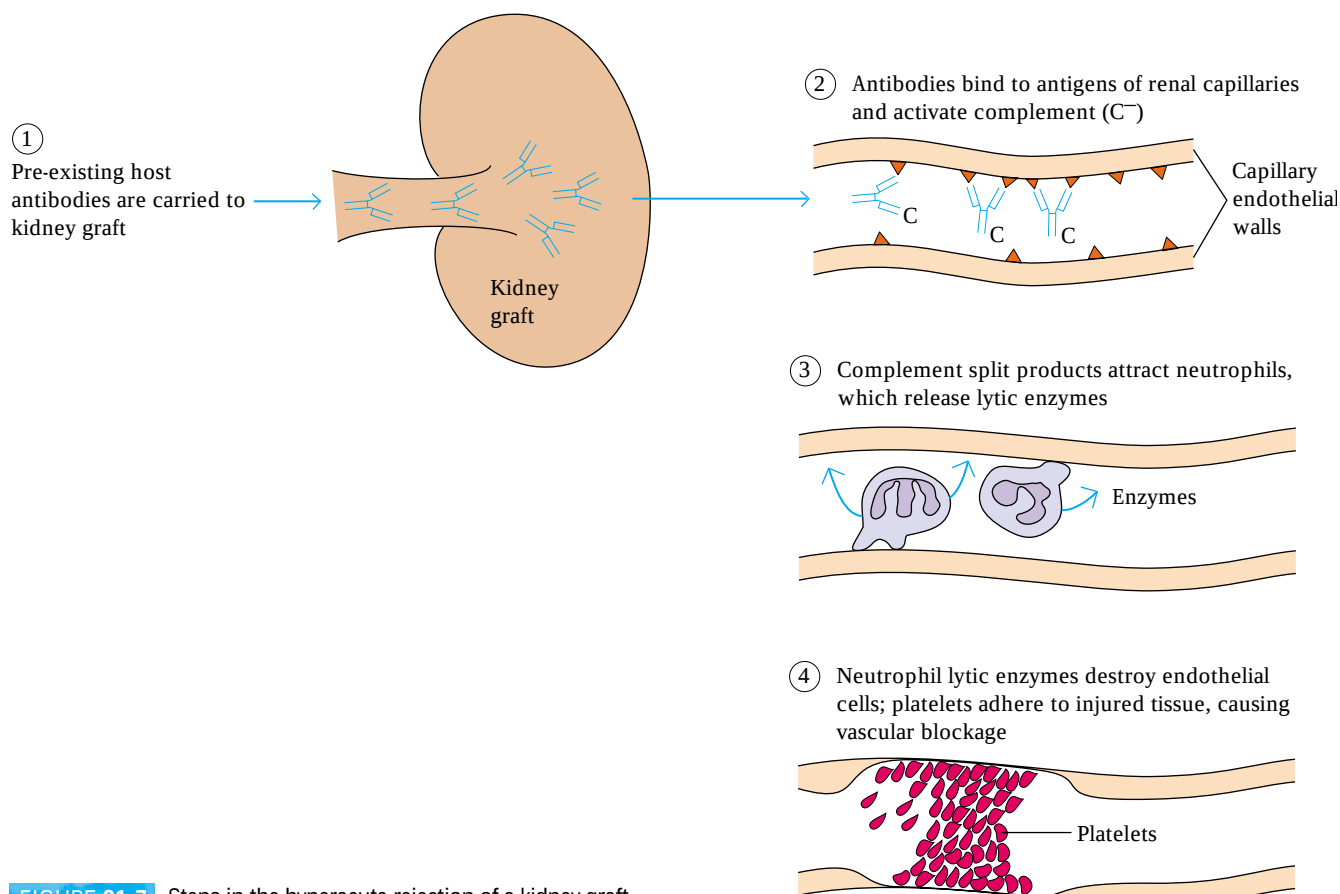


FIGURE 21-7 Steps in the hyperacute rejection of a kidney graft.

Several mechanisms can account for the presence of pre-existing antibodies specific for allogeneic MHC antigens. Recipients of repeated blood transfusions sometimes develop significant levels of antibodies to MHC antigens expressed on white blood cells present in the transfused blood. If some of these MHC antigens are the same as those on a subsequent graft, then the antibodies can react with the graft, inducing a hyperacute rejection reaction. With repeated pregnancies, women are exposed to the paternal alloantigens of the fetus and may develop antibodies to these antigens. Finally, individuals who have had a previous graft sometimes have high levels of antibodies to the allogeneic MHC antigens of that graft.

In some cases, the preexisting antibodies participating in hyperacute graft rejection may be specific for blood-group antigens in the graft. If tissue typing and ABO blood-group typing are performed prior to transplantation, these preexisting antibodies can be detected and grafts that would result in hyperacute rejection can be avoided. Xenotransplants are often rejected in a hyperacute manner because of antibodies to cellular antigens of the donor species that are not present in the recipient species. Such an antigen is discussed in the Clinical Focus section of this chapter.

In addition to the hyperacute rejection mediated by pre-existing antibodies, there is a less frequent form of rejection

termed *accelerated rejection* caused by antibodies that are produced immediately after transplantation.

Acute Rejection Is Mediated by T-Cell Responses

Cell-mediated allograft rejection manifests as an acute rejection of the graft beginning about 10 days after transplantation (see Figure 21-1b). Histopathologic examination reveals a massive infiltration of macrophages and lymphocytes at the site of tissue destruction, suggestive of T_H -cell activation and proliferation. Acute graft rejection is effected by the mechanisms described previously (see Figure 21-6).

Chronic Rejection Occurs Months or Years Post-Transplant

Chronic rejection reactions develop months or years after acute rejection reactions have subsided. The mechanisms of chronic rejection include both humoral and cell-mediated responses by the recipient. While the use of immunosuppressive drugs and the application of tissue-typing methods to obtain optimum match of donor and recipient have dramatically increased survival of allografts during the first years after engraftment, little



CLINICAL FOCUS

Is There a Clinical Future for Xenotransplantation?

Unless organ

donations increase drastically, most of the 72,000 U.S. patients on the waiting list for a transplant will not receive one. The majority (47,000) need a kidney, but last year only 12,500 kidneys were transplanted. A solution to this shortfall is to utilize animal organs. Some argue that xenografts bring the risk of introducing pathogenic retroviruses into the human population; others object based on ethical grounds relating to animal rights. Nevertheless, the use of pigs to supply organs for humans is under serious consideration. Pigs breed rapidly, have large litters, can be housed in pathogen-free environments, and share considerable anatomic and physiologic similarity with humans. In fact, pigs have served as donors of cardiac valves for humans for years. Primates are more closely related to humans than pigs are, but the availability of large primates as transplant donors is, and will continue to be, extremely limited.

Balancing the advantages of pig donors are serious difficulties. For example, if a pig kidney were implanted into a human by techniques standard for human transplants, it would likely fail in a rapid and dramatic fashion due to hyperacute rejection. This antibody-mediated rejection is due to the presence on the pig cells

(and those of most mammals other than humans and the highest nonhuman primates) of a disaccharide antigen (galactosyl-1,3- α -galactose) that is not present on human cells. The presence of this antigen on many microorganisms means that nearly everyone has been exposed to it and has formed antibodies against it. The preexisting antibodies react with pig cells, which are then lysed rapidly by complement. The absence of human regulators of complement activity on the pig cells, including human decay-accelerating factor (DAF) and human membrane-cofactor protein (MCP), intensifies the complement lysis cycle. (See Chapter 13 for descriptions of DAF and MCP.)

How can this major obstacle be circumvented? Being tested are strategies for absorbing the antibodies from the circulation on solid supports, and using soluble gal-gal disaccharides to block antibody reactions. A more elegant solution involves genetically engineering pigs to knock out the gene for the enzyme that synthesizes galactosyl-1,3- α -galactose. Solving the immediate rejection reaction by interfering with the specific reaction against this antigen may not prevent all antibody-mediated rejection. Certainly other antigenic differences to which human recipients have antibodies will be present in some if not all donor/recipient pairs. However, any antibody attack on the pig

cells may be blunted if human DAF is present on the targeted cell to dampen the complement reaction. The lack of human DAF is remedied by producing transgenic pigs that express this protein. Addition of human complement regulators to the pig represents a universal solution, in that any cell that might become a target in the transplant will resist complement lysis.

An additional concern is that pig endogenous retroviruses will be introduced into humans as a result of xenotransplantation and cause disease. Opponents of xenotransplantation raise the specter of another HIV-type epidemic resulting from human infection by a new animal retrovirus. Recently, a Boston-based company announced development of pigs free of endogenous pig retroviruses, reducing the possibility of this bleak outcome.

Will we see the use of pig kidneys in humans in the near future? The increasing demand for organs is driving the commercial development of colonies of pigs suitable to become organ donors. While kidneys are the most sought-after organ at present, other organs and cells from the specially bred and engineered animals will find use if they are proven to be safe and effective. A statement from the American Society of Transplantation and the American Society of Transplant Surgeons endorses the use of xenotransplants if certain conditions are met (*Xenotransplantation* 7:235). These include the demonstration of feasibility in a nonhuman primate model, proven benefit to the patient, and lack of infectious-disease risk. Barriers remain to the clinical use of xenotransplants, but serious efforts are in motion to overcome them.

progress has been made in long-term survival. The use of immunosuppressive drugs, which are described below, greatly increases the short-term survival of the transplant, but chronic rejection is not prevented in most cases. Data for rejection of kidney transplants since 1975 indicates an increase from 40% to over 80% in one-year survival of grafts. However, in the same period long-term survival has risen only slightly; as in 1975, about 50% of transplanted kidneys are still functioning at 10 years after transplant. Chronic rejection reactions are difficult to manage with immunosuppressive drugs and may necessitate another transplantation.

General Immunosuppressive Therapy

Allogeneic transplantation requires some degree of immunosuppression if the transplant is to survive. Most of the immunosuppressive treatments that have been developed have the disadvantage of being nonspecific; that is, they result in generalized immunosuppression of responses to all antigens, not just those of the allograft, which places the recipient at increased risk of infection. In addition, many

immunosuppressive measures are aimed at slowing the proliferation of activated lymphocytes. However, because any rapidly dividing nonimmune cells (e.g., epithelial cells of the gut or bone-marrow hematopoietic stem cells) are also affected, serious or even life-threatening complications can occur. Patients on long-term immunosuppressive therapy are at increased risk of cancer, hypertension, and metabolic bone disease.

Mitotic Inhibitors Thwart T-Cell Proliferation

Azathioprine (Imuran), a potent mitotic inhibitor, is often given just before and after transplantation to diminish T-cell proliferation in response to the alloantigens of the graft. Azathioprine acts on cells in the S phase of the cell cycle to block synthesis of inosinic acid, which is a precursor of the purines adenylic and guanylic acid. Both B-cell and T-cell proliferation is diminished in the presence of azathioprine. Functional immune assays such as the MLR, CML, and skin test show a significant decline after azathioprine treatment, indicating an overall decrease in T-cell numbers.

Two other mitotic inhibitors that are sometimes used in conjunction with other immunosuppressive agents are cyclophosphamide and methotrexate. Cyclophosphamide is an alkylating agent that inserts into the DNA helix and becomes cross-linked, leading to disruption of the DNA chain. It is especially effective against rapidly dividing cells and therefore is sometimes given at the time of grafting to block T-cell proliferation. Methotrexate acts as a folic-acid antagonist to block purine biosynthesis. The fact that the mitotic inhibitors act on all rapidly dividing cells and not specifically on those involved in immune response against the allograft can lead to deleterious side reactions by thwarting division of other functional cells in the body.

Corticosteroids Suppress Inflammation

As described at the end of Chapter 15, corticosteroids, such as prednisone and dexamethasone, are potent anti-inflammatory agents that exert their effects at many levels of the immune response. These drugs are often given to transplant recipients together with a mitotic inhibitor such as azathioprine to prevent acute episodes of graft rejection.

Certain Fungal Metabolites Are Immunosuppressants

Cyclosporin A (CsA), FK506 (tacrolimus), and rapamycin (sirolimus) are fungal metabolites with immunosuppressive properties. Although chemically unrelated, CsA and FK506 have similar actions. Both drugs block activation of resting T cells by inhibiting the transcription of genes encoding IL-2 and the high-affinity IL-2 receptor (IL-2R), which are essential for activation. CsA and FK506 exert this effect by binding to cytoplasmic proteins called immunophilins, forming a complex that blocks the phosphatase activity of calcineurin. This prevents the formation and nuclear translocation of the

cytoplasmic subunit NFATc and its subsequent assembly into NFAT, a DNA-binding protein necessary for transcription of the genes encoding a number of molecules important to T-cell activation (see Figure 10-11). Rapamycin is structurally similar to FK506 and also binds to an immunophilin. However, the rapamycin-immunophilin complex does not inhibit calcineurin activity; instead, it blocks the proliferation and differentiation of activated T_H cells in the G_1 phase of the cell cycle. All three drugs, by inhibiting T_H -cell proliferation and thus T_H -cell cytokine expression, reduce the subsequent activation of various effector populations involved in graft rejection, including T_H cells, T_C cells, NK cells, macrophages, and B cells.

The profound immunosuppressive properties of these three agents have made them a mainstay of heart, liver, kidney, and bone-marrow transplantation. Cyclosporin A has been shown to prolong graft survival in kidney, liver, heart, and heart-lung transplants. In one study of 209 kidney transplants from cadaver donors, the 1-year survival rate was 64% among recipients receiving other immunosuppressive treatments and 80% among those receiving cyclosporin A. Similar results have been obtained with liver transplants (Figure 21-8). Despite these impressive results, CsA does have some negative side effects, the most notable of which is toxicity to the kidneys. Acute nephrotoxicity is quite common, in some cases progressing to chronic nephrotoxicity and drug-induced kidney failure. FK506 and rapamycin are 10–100 times more potent as immune suppressants than CsA, and therefore can be administered at lower doses and with fewer side effects than CsA.

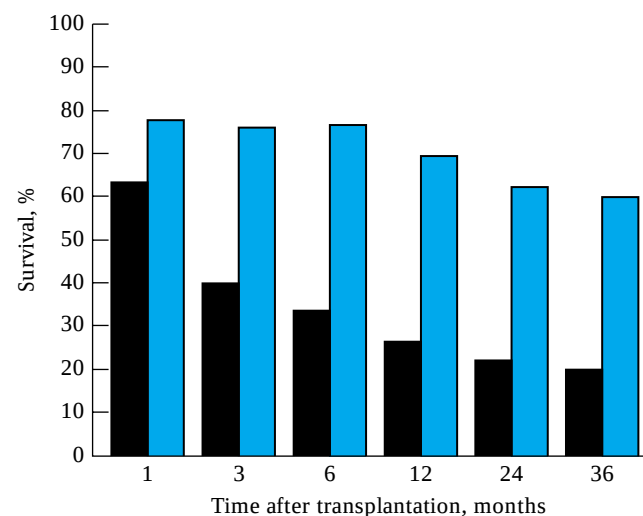


FIGURE 21-8 Comparison of the survival rate of liver transplants in 84 patients who were immunosuppressed with azathioprine and corticosteroids (black) with the survival rate in 55 patients who were immunosuppressed with cyclosporin A and corticosteroids (blue). [Adapted from S. M. Sabesin and J. W. Williams, 1987, *Hosp. Pract.* 15(July):75.]

Total Lymphoid Irradiation Eliminates Lymphocytes

Because lymphocytes are extremely sensitive to x-rays, x-irradiation can be used to eliminate them in the transplant recipient just before grafting. In total lymphoid x-irradiation, the recipient receives multiple x-ray exposures to the thymus, spleen, and lymph nodes before the transplant surgery. The typical protocol is daily x-irradiation treatments of about 200 rads per day for several weeks until a total of 3400 rads has been administered. The recipient is grafted in this immunosuppressed state. Because the bone marrow is not x-irradiated, lymphoid stem cells proliferate and renew the population of recirculating lymphocytes. These newly formed lymphocytes appear to be more tolerant to the antigens of the graft.

Specific Immunosuppressive Therapy

In addition to harmful side effects peculiar to the various immunosuppressive treatments described above, a major limitation common to all is that they lack specificity, thus producing a more-or-less generalized immunosuppression and increasing the recipient's risk for infection. What is needed ideally is an antigen-specific immunosuppressant that reduces the immune response to the alloantigens of the graft while preserving the recipient's ability to respond to other foreign antigens. Although this goal has not yet been achieved in human transplants, recent successes in animal experiments indicate that it may be possible. Specific immunosuppression to allografts has been achieved in animal experiments using antibodies or soluble ligands reactive with cell-surface molecules.

Monoclonal Antibodies Can Suppress Graft-Rejection Responses

Monoclonal antibodies directed against various surface molecules on cells of the immune system have been used successfully to suppress T-cell activity in general or to suppress the activity of subpopulations of T cells. Results from studies with animal models suggest further that certain monoclonals may be used to suppress only T cells that are activated. Successes with animal models and trials with humans give reason to believe that two types of strategies involving antibodies to suppress rejection will find broad clinical use. Monoclonal antibodies may be used to deplete the recipient of a certain broad or specific cell population; alternatively, they may be used to block co-stimulatory signals. In the latter case, a state of anergy is induced in those T cells that react to antigens present on the allograft.

A strategy to deplete immune cells involves use of a monoclonal antibody to the CD3 molecule of the TCR complex. Injection of such monoclonal antibodies results in a rapid

depletion of mature T cells from the circulation. This depletion appears to be caused by binding of antibody-coated T cells to Fc receptors on phagocytic cells, which then phagocytose and clear the T cells from the circulation. In a further refinement of this strategy, a cytotoxic agent such as diphtheria toxin is coupled with the antibody. The cell with which the antibody reacts internalizes the toxin, causing its death. Another depletion strategy used to increase graft survival uses monoclonal antibodies specific for the high-affinity IL-2 receptor (anti-TAC). Since the high-affinity IL-2 receptor is expressed only on activated T cells, exposure to anti-TAC after the graft specifically blocks proliferation of T cells activated in response to the alloantigens of the graft.

Monoclonal-antibody therapy, which was initially employed to deplete T cells in graft recipients, also has been used to treat donors' bone marrow before it is transplanted. Such treatment is designed to deplete the immunocompetent T cells in the bone-marrow transplant; these are the cells that react with the recipient tissues, causing graft-versus-host disease (described below). Monoclonal antibodies with isotypes that activate the complement system are most effective in all cell-depletion strategies.

The CD3 receptor and the high-affinity IL-2 receptor are targets present on all activated T cells; molecules present on particular T-cell subpopulations may also be targeted for immunosuppressive therapy. For example, a monoclonal antibody to CD4 has been shown to prolong graft survival. In one study, monkeys were given a single large dose of anti-CD4 just before they received a kidney transplant. Graft survival in the treated animals was markedly increased over that in untreated control animals. Interestingly, the anti-CD4 did not reduce the CD4⁺ T-cell count, but instead appeared to induce the T cells to enter an immunosuppressed state. This is an example of a nondepleting antibody.

Other targets for monoclonal-antibody therapy are the cell-surface adhesion molecules. Simultaneous treatment with monoclonal antibodies to the adhesion molecules ICAM-1 and LFA-1 for 6 days after transplantation has permitted indefinite survival of cardiac grafts between allogeneic mice. However, when either monoclonal antibody was administered alone, the cardiac transplant was rejected. The requirement that both monoclonal antibodies be given at the same time probably reflects redundancy of the adhesion molecules: LFA-1 is known to bind to ICAM-2 in addition to ICAM-1; and ICAM-1 is known to bind to Mac-1 and CD43 in addition to LFA-1. Only when all possible pairings among these adhesins are blocked at the same time is adhesion and signal transduction through this ligand pair blocked.

A practical difficulty with using monoclonal antibodies to prolong graft survival in humans is that they are generally of mouse origin. Many recipients develop an antibody response to the mouse monoclonal antibody, rapidly clearing it from the body. This limitation has been overcome by the construction of human monoclonal antibodies and mouse-human chimeric antibodies (see Figure 5-25 and Clinical Focus in Chapter 5).

Because cytokines appear to play an important role in allograft rejection, another strategy for prolonging graft survival is to inject animals with monoclonal antibodies specific for the implicated cytokines, particularly TNF- α , IFN- γ , and IL-2. Monoclonal antibodies to TNF- α have been shown to prolong bone-marrow transplants in mice and to reduce the incidence of graft-versus-host disease. Monoclonal antibodies to IFN- γ and to IL-2 have each been reported in some cases to prolong cardiac transplants in rats.

Blocking Co-Stimulatory Signals Can Induce Anergy

As described in Chapter 10, T_H-cell activation requires a co-stimulatory signal in addition to the signal mediated by the T-cell receptor. The interaction between the B7 molecule on the membrane of antigen-presenting cells and the CD28 or CTLA-4 molecule on T cells provides one such signal (see Figure 10-13). Lacking a co-stimulatory signal, antigen-activated T cells become anergic (see Figure 10-15). CD28 is expressed on both resting and activated T cells and binds B7 with a moderate affinity; CTLA-4 is expressed at much lower levels and only on activated T cells but binds B7 with a 20-fold higher affinity. A second pair of co-stimulatory molecules required for T-cell activation are CD40, which is present on the APC, and CD40 ligand (CD40L or CD154), which is present on the T cell.

D. J. Lenschow, J. A. Bluestone, and colleagues demonstrated that blocking the B7-mediated co-stimulatory signal with CTLA-4 after transplantation would cause the host's

T cells directed against the grafted tissue to become anergic, thus enabling it to survive. In their experiment, human pancreatic islets were transplanted into mice injected with CTLA-4Ig, a soluble fusion protein consisting of the extracellular domains of CTLA4 and the constant region of the IgG1 heavy chain (see Figure 10-14). Including the IgG1 heavy-chain constant region increases the half-life of the soluble fusion protein. The xenogeneic graft exhibited long-term survival in treated mice but was quickly rejected in untreated controls. The fact that the soluble form of the CTLA-4 receptor was able to block the rejection of the human tissue transplant in the recipient mice is evidence that blocking co-stimulatory signals *in vivo* is a viable strategy (Figure 21-9).

These exciting results were extended to transplantation of kidneys mismatched for class I and class II antigens in monkeys by Allan Kirk, David Harlan, and their colleagues. The recipients were treated for about 4 weeks after transplantation with either CTLA4-Ig or a monoclonal antibody directed against CD40L, or both in combination. Untreated control animals rejected the mismatched kidneys within 5–8 days; those treated with a single agent retained their grafts for 20–98 days. However, the animals given both reagents showed no evidence of rejection at 150 days after transplantation. This suppression of allograft rejection did not lead to a state of general immunosuppression; peripheral T-cell counts remained normal and other immune functions were present, including mixed lymphocyte reactivity between donor and recipients. Human clinical trials of the procedures developed for monkeys are planned; if successful, they could revolutionize clinical transplantation procedures. The ability to block

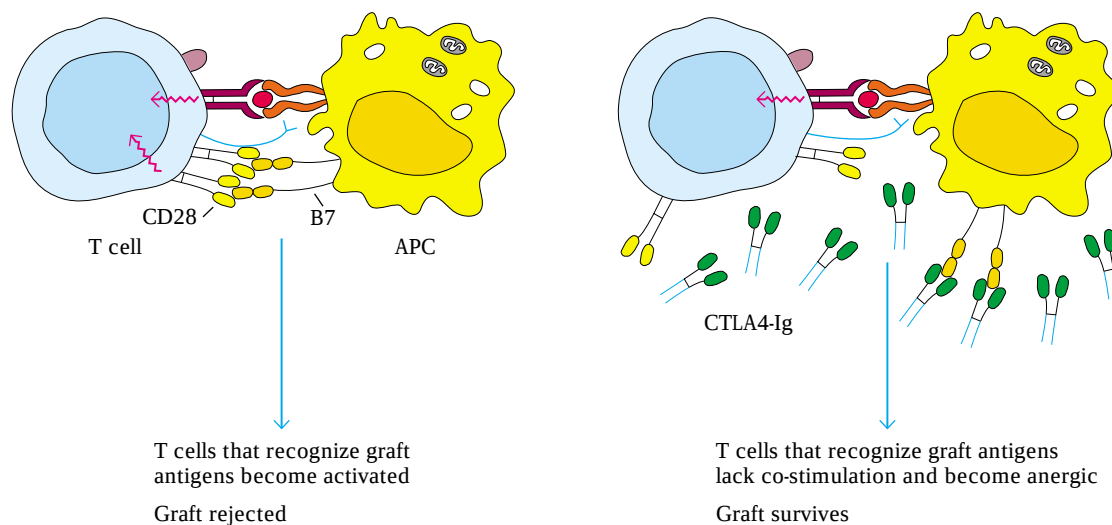


FIGURE 21-9 Blocking co-stimulatory signals at the time of transplantation can cause anergy instead of activation of the T cells reactive against the graft. T-cell activation requires both the interaction of the TCR with its ligand and the reaction of co-stimulatory receptors with their ligands (a). In (b), contact between one of the co-stimulatory re-

ceptors, CD28 on the T cell, and its ligand, B7 on the APC, is blocked by reaction of B7 with the soluble ligand CTLA4-Ig. The CTLA4 is coupled to an Ig H chain, which slows its clearance from the circulation. This process specifically suppresses graft rejection without inhibiting the immune response to other antigens.

allograft rejection without general immunosuppression and without the deleterious side effects of suppressive drugs would enable recipients to lead normal lives.

Immune Tolerance to Allografts

There are instances in which an allograft may be accepted without the use of immunosuppressive measures. Obviously, in the case of tissues that lack alloantigens, such as cartilage or heart valves, there is no immunologic barrier to transplantation. However, there are also instances in which the strong predicted response to an allograft does not occur. There are two general cases in which an allograft may be accepted. One is when cells or tissue are grafted to a so-called privileged site that is sequestered from immune-system surveillance. The second is when a state of tolerance has been induced biologically, usually by previous exposure to the antigens of the donor in a manner that causes immune tolerance rather than sensitization in the recipient. Each of these exceptions is considered below.

Privileged Sites Accept Antigenic Mismatches

In immunologically privileged sites, an allograft can be placed without engendering a rejection reaction. These sites include the anterior chamber of the eye, the cornea, the uterus, the testes, and the brain. The cheek pouch of the Syrian hamster is a privileged site used in experimental situations. Each of these sites is characterized by an absence of lymphatic vessels and in some cases by an absence of blood vessels as well. Consequently, the alloantigens of the graft are not able to sensitize the recipient's lymphocytes, and the graft has an increased likelihood of acceptance even when HLA antigens are not matched.

The privileged location of the cornea has allowed corneal transplants to be highly successful. The brain is an immunologically privileged site because the blood-brain barrier prevents the entry or exit of many molecules, including antibodies. The successful transplantation of allogeneic pancreatic islet cells into the thymus in a rat model of diabetes suggests that the thymus may also be an immunologically privileged site.

Immunologically privileged sites fail to induce an immune response because they are effectively sequestered from the cells of the immune system. This suggests the possibility of physically sequestering grafted cells. In one study, pancreatic islet cells were encapsulated in semipermeable membranes (fabricated from an acrylic copolymer) and then transplanted into diabetic mice. The islet cells survived and produced insulin. The transplanted cells were not rejected, because the recipient's immune cells could not penetrate the membrane. This novel transplant method enabled the diabetic mice to produce normal levels of insulin and may have application for treatment of human diabetics.

Early Exposure to Alloantigens Can Induce Specific Tolerance

In 1945, Ray Owen reported that nonidentical twins in cattle retained the ability to accept cells or tissue from the genetically distinct sibling throughout their lives, unlike nonidentical twins of other mammalian species. A shared placenta in cattle allows free circulation of cells from one twin to the other throughout the embryonic period. Although the twins may have inherited distinct paternal and maternal antigens, they do not recognize those of their placental partner as foreign and can accept grafts from them.

Experimental support for the notion that tolerance comes from exposure of the developing organism to alloantigens came from mouse experiments. If neonates of mouse strain A are injected with cells from strain C they will accept grafts from C strain as adults. Immunocompetence of the injected A-strain mice and specificity of the tolerance is shown by the fact that they reject grafts from other strains as rapidly as their untreated littermates. While no human experimental data demonstrate such specific tolerance, anecdotal data suggests that it may operate in humans as well. There are examples in which allografts, mismatched at a single HLA locus are accepted with little or no immune suppression. In cases where the mismatched antigen is expressed by the mother, but not inherited by the offspring, there is the possibility that perinatal exposure induced subsequent tolerance to this antigen. Because human maternal cells do not normally cross the placental barrier, such specific tolerance to noninherited maternal antigens would be an exception rather than a commonplace event.

Clinical Transplantation

For a number of illnesses, a transplant is the only means of therapy. Figure 21-10 summarizes the major organ and cell transplants being performed at the present time. In addition, certain combinations of organs, such as heart and lung or kidney and pancreas, are being transplanted simultaneously with increasing frequency. Since the first kidney transplant was performed in the 1950s, approximately 400,000 kidneys have been transplanted worldwide. The next most frequently transplanted solid organ is the liver (52,000), followed by the heart (42,000) and, more distantly, by the lung (6,000) and pancreas (2,000). Bone-marrow transplants number around 80,000. Although the clinical results of transplantation of various cells, tissues, and organs in humans have improved considerably in the past few years, major obstacles to the use of this treatment exist. As explained above, the use of immunosuppressive drugs greatly increases the short-term survival of the transplant, but medical problems arise from use of these drugs, and chronic rejection is not prevented in most cases. The need for additional transplants after rejection exacerbates the shortage of organs which is a major obstacle to the



VISUALIZING CONCEPTS

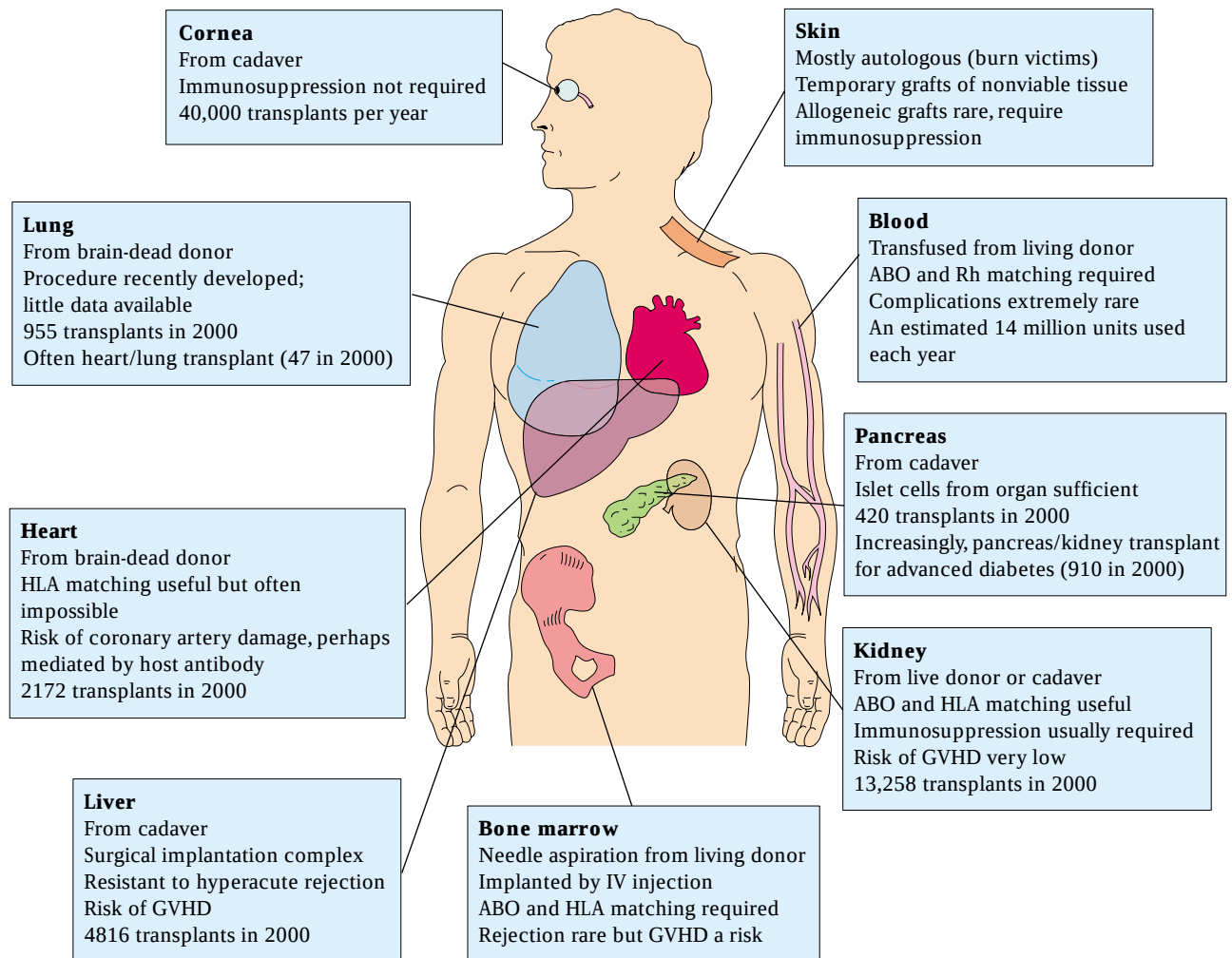


FIGURE 21-10 Transplantations routinely used in clinical practice. For the solid organs, the number of transplants performed in

the United States in 2000 is indicated. Estimates are included for other transplants if available.

widespread use of transplantation. Several of the organ systems for which transplantation is a common treatment are considered below. The frequency with which a given organ or tissue is transplanted depends on a number of factors:

- Clinical situations in which transplantation is indicated
- Availability of tissue or organs
- Difficulty in performing transplantation and caring for post-transplantation patients

- Specific factors that aid or hinder acceptance of the particular transplant

The urgency of the transplantation may depend on the affected organ. In the case of the heart, lung, and liver, few alternative procedures can keep the patient alive when these organs cease to function. Although dialysis may be used to maintain a patient awaiting a kidney transplant, there are no comparable measures for the heart or lungs if the allograft fails. Research on artificial organs is ongoing but there are no reports of long-term successes.

The Most Commonly Transplanted Organ Is the Kidney

As mentioned above, the most commonly transplanted organ is the kidney; in 2000, there were 13,258 kidney transplants performed in the United States. Major factors contributing to this number are the numerous clinical indications for kidney transplantation. Many common diseases, such as diabetes and various types of nephritis, result in kidney failure that can be alleviated by transplantation. With respect to availability, kidneys can be obtained not only from cadavers but also from living relatives or volunteers, because it is possible to donate a kidney and live a normal life with the remaining kidney. In 1999, 4457 of the 12,483 kidneys transplanted in the U.S. came from living donors. Surgical procedures for transplantation are straightforward; technically, the kidney is simpler to reimplant than the liver or heart. Because many kidney transplants have been done, patient-care procedures have been worked out in detail. Matching of blood and histocompatibility groups is advantageous in kidney transplantation because the organ is heavily vascularized, but the kidney presents no special problems that promote rejection or graft-versus-host disease (GVHD), as the bone marrow or liver do.

Two major problems are faced by patients waiting for a kidney. One is the short supply of available organs, and the second is the increasing number of sensitized recipients. The latter problem stems from rejection of a first transplant, which then sensitizes the individual and leads to the formation of antibodies and activation of cellular mechanisms directed against kidney antigens. Any subsequent graft containing antigens in common with the first would be quickly rejected. Therefore, detailed tissue typing procedures must be used to ascertain that the patient has no antibodies or active cellular mechanisms directed against the potential donor's kidney. In many cases, patients can never again find a match after one or two rejection episodes. It is almost always necessary to maintain kidney-transplant patients on some form of immunosuppression, usually for their entire lives. Unfortunately, this gives rise to complications, including risks of cancer and infection as well as other side effects such as hypertension and metabolic bone disease.

Bone-Marrow Transplants Are Used for Leukemia, Anemia, and Immunodeficiency

After the kidney, bone marrow is the most frequent transplant. Since the early 1980s, bone-marrow transplantation has been increasingly adopted as a therapy for a number of malignant and nonmalignant hematologic diseases, including leukemia, lymphoma, aplastic anemia, thalassemia major, and immunodeficiency diseases, especially severe combined immunodeficiency, or SCID (see Chapter 19). The bone marrow, which is obtained from a living donor by multiple needle aspirations, consists of erythroid, myeloid, monocytoid, megakaryocytic, and lymphocytic lineages. The graft, usually about 10^9 cells per kilogram of host body weight, is

injected intravenously into the recipient. The first successful bone-marrow transplantations were performed between identical twins. However, development of the tissue-typing procedures described earlier now makes it possible to identify allogeneic donors who have HLA antigens identical or near-identical to those of the recipients. While the supply of bone marrow for transplantation is not a problem, finding a matched donor may be one.

In the usual procedure, the recipient of a bone-marrow transplant is immunologically suppressed before grafting. Leukemia patients, for example, are often treated with cyclophosphamide and total-body irradiation to kill all cancerous cells. The immune-suppressed state of the recipient makes graft rejection rare; however, because the donor bone marrow contains immunocompetent cells, the graft may reject the host, causing **graft-versus-host disease (GVHD)**. GVHD affects 50%–70% of bone-marrow-transplant patients; it develops as donor T cells recognize alloantigens on the host cells. The activation and proliferation of these T cells and the subsequent production of cytokines generate inflammatory reactions in the skin, gastrointestinal tract, and liver. In severe cases, GVHD can result in generalized erythroderma of the skin, gastrointestinal hemorrhage, and liver failure.

Various treatments are used to prevent GVHD in bone-marrow transplantation. The transplant recipient is usually placed on a regimen of immunosuppressive drugs, often including cyclosporin A and methotrexate, in order to inhibit the immune responses of the donor cells. In another approach, the donor bone marrow is treated with anti-T-cell antisera or monoclonal antibodies specific for T cells before transplantation, thereby depleting the offending T cells. Complete T-cell depletion from donor bone marrow, however, increases the likelihood that the marrow will be rejected, and so the usual procedure now is a partial T-cell depletion. Apparently, a low level of donor T-cell activity, which results in a low-level GVHD, is actually beneficial because the donor cells kill any host T cells that survive the immunosuppression treatment. This prevents residual recipient cells from becoming sensitized and causing rejection of the graft. In leukemia patients, low-level GVHD also seems to result in destruction of host leukemic cells, thus making it less likely for the leukemia to recur.

Heart Transplantation Is a Challenging Operation

Perhaps the most dramatic form of transplantation is that of the heart; once the damaged heart has been removed, the patient must be kept alive by wholly artificial means until the transplanted heart is in place and beating. Heart-lung machines are available to circulate and aerate the patient's blood after the heart is removed. The donor's heart must be maintained in such a manner that it will begin beating when it is placed in the recipient. It has been found that a human heart can be kept viable for a limited period in ice-cold buffer solutions that effectively short circuit the electric impulses

that control the rhythmic beating, which could damage the isolated organ. The surgical methods of implanting a heart have been available for a number of years. The first heart transplant was carried out in South Africa by Dr. Christian Barnard, in 1964. Since then, the one-year survival rate for transplantation of the heart has become greater than 80%. In 2000, 2172 heart transplants were performed in the United States and about 3500 worldwide. An issue peculiar to heart transplantation has been a new type of atherosclerotic disease in the coronary arteries of the implanted organ. There is some possibility that host antibodies mediate injury to the vessels in the donated heart.

Although a heart transplant may greatly benefit patients with various types of heart disease or damage, there is obviously a strict limit on the number of available hearts. Accident victims who are declared brain dead but have an intact circulatory system and a functioning heart are the normal source of these organs. HLA matching is desirable but not often possible, because of the limited supply of hearts and the urgency of the procedure.

Lung Transplants Are on the Increase

In recent years, lung transplantation, either by itself or in conjunction with heart transplantation, has been used to treat diseases such as cystic fibrosis and emphysema or acute damage to the lungs such as that caused by smoke inhalation. In 2000, 945 lung and 47 heart/lung transplants were performed. First-year survival rate for lung transplants is reported at about 60%.

Liver Transplants Treat Congenital Defects and Damage from Viral or Chemical Agents

The liver is a large organ that performs a number of functions related to clearance and detoxification of chemical and biological substances. Liver malfunction can be caused by damage to the organ from viral diseases such as hepatitis or by exposure to harmful chemicals, as in chronic alcoholism. Damage to the liver may correct itself and the damaged tissue can regenerate after the causative injurious agent is cleared. If the liver tissue does not regenerate, damage may be fatal. The majority of liver transplants are used as a therapy for congenital abnormalities of the liver. Because the liver is large and has a complicated circulation, re-implantation of the liver initially posed a technical problem. Techniques have been developed to overcome this major surgical challenge, and the recent one-year survival rate has risen to approximately 65%. In 2000, 4816 livers were transplanted in the United States. Increasingly, a liver from a single donor may be split and given to two recipients; normally, a child will receive the smaller portion and an adult the larger.

The immunology of liver transplantation is interesting because the organ appears to resist rejection by hyperacute antibody-mediated mechanisms. It has been shown that even transplantation across blood-group barriers, which would

be expected to trigger hyperacute rejection, can be successful in the short term. However, leukocytes within the donor organ together with anti-blood-group antibodies can mediate antibody-dependent hemolysis of recipient red blood cells if there is a mismatch of the blood groups. In addition, manifestations of GVHD have occurred in liver transplants even when donor and recipient are blood-group compatible. These reactions are obviously caused by donor lymphocytes carried by the transplanted liver.

Pancreas Transplantation Offers a Cure for Diabetes Mellitus

One of the more common diseases in the United States is diabetes mellitus. This disease is caused by malfunction of insulin-producing islet cells in the pancreas. Transplantation of a pancreas could provide the appropriately regulated levels of insulin necessary to make the diabetic individual normal. Recently, one-year success rates for pancreas transplantation of about 55% have been reported. Transplantation of the complete pancreas is not necessary to restore the function needed to produce insulin in a controlled fashion; transplantation of the islet cells alone could restore function. Kidney failure is a frequent complication of advanced diabetes occurring in about 30% of diabetics, therefore kidney and pancreas transplants are indicated. In 2000, there were 420 pancreas transplants and 904 simultaneous kidney/pancreas transplants. A group at the University of Wisconsin reports that they have overcome surgical and medical barriers to the dual transplant and have achieved survival rates of 87% at one year and 78% at five years for the 381 cases in their study. Whether it is better to carry out simultaneous kidney-pancreas transplants or to transplant separately remains an issue to be resolved on a case-to-case basis.

Skin Grafts Are Used to Treat Burn Victims

Most skin transplantation in humans is done with autologous tissue. However, in cases of severe burn, grafts of foreign skin thawed from frozen deposits in tissue banks may be used. These grafts generally act as biologic dressings, because the cellular elements are no longer viable and the graft does not grow in the new host; the grafts are left in place for several days but are regularly replaced. True allogeneic skin grafting using fresh viable donor skin has been undertaken in some cases, but rejection must be prevented by the use of immunosuppressive therapy. This is not desirable because a major problem with burn victims is the high risk of infection, and immunosuppressive therapy accentuates this risk.

The above list of common transplants is by no means all-inclusive and is expected to grow in future years. For example, intracerebral neural-cell grafts have restored functionality in victims of Parkinson's disease. In studies conducted thus far, the source of neural donor cells was human embryos; the possibility of using those from other animal species is being tested.

Xenotransplantation May Be the Answer to the Shortage of Donor Organs

While the immune system represents a formidable barrier to the use of transplantation, there has been significant progress in overcoming this obstacle. However, there has not been comparable progress in solving the complex problem of finding organs for those who need them. The insufficient supply of available organs means that a large percentage of patients die while waiting for a transplant. The need for an alternative source of donor organs has focused attention on xenotransplantation. The larger nonhuman primates (chimpanzees and baboons) have served as the main transplant donors, and, as discussed in the Clinical Focus section, the use of the pig as a source of organs is under serious consideration.

The earliest transplants of chimpanzee kidneys into humans date back to 1964. Since that time, sporadic attempts at kidney, heart, liver, and bone-marrow transplantation from primates into humans have been made. No attempt has met with great success but several have received some attention. In 1993, T. E. Starzl performed two liver transplants from baboons into patients suffering from liver failure. Both patients died, one after 26 days and the other after 70 days. In 1994, a pig liver was transplanted into a 26-year-old suffering from acute hepatic failure. The liver functioned only 30 hours before it was rejected by a hyperacute rejection reaction. In 1995, baboon bone marrow was infused into an HIV-infected man with the aim of boosting his weakened immune system with the baboon immune cells, which do not become infected with the virus. Although there were no complications from the transplant, the baboon bone marrow did not appear to establish itself in the recipient.

A major problem with xenotransplants is that immune rejection is often quite vigorous, even when recipients are treated with potent immunosuppressive drugs such as FK506 or rapamycin. The major response involves the action of humoral antibody and complement, leading to the development of a hyperacute rejection reaction. In addition to the problem of rejection, there is general concern that xenotransplantation has the potential of spreading pathogens from the donor to the recipient. These pathogens could potentially cause diseases, called zoonoses, that are fatal for humans. For example, certain viruses, including close relatives of HIV-1 found in chimpanzees and HIV-2 and herpesvirus B, which occur in several primate species, cause limited pathogenesis in their primate hosts but can lead to deadly infections in humans. In addition, there is the fear that primate retroviruses (see Chapter 19), such as SIV, may recombine with human variants to produce new agents of disease. The possibility of introducing new viruses into humans may be greater for transplants from closely related species, such as primates, and less in the case of more distantly related species, such as pigs, because viruses are less likely to replicate in cells from unrelated species.

SUMMARY

- Graft rejection is an immunologic response displaying the attributes of specificity, memory, and self/nonself recognition. There are three major types of rejection reactions:
 - Hyperacute rejection mediated by preexisting host antibodies to graft antigens.
 - Acute graft rejection in which T_H cells and/or CTLs mediate tissue damage.
 - Chronic rejection, which involves both cellular and humoral immune components.
- The immune response to tissue antigens encoded within the major histocompatibility complex is the strongest force in rejection.
- The match between a recipient and potential graft donors is assessed by typing MHC class I and class II tissue antigens.
- The process of graft rejection can be divided into a sensitization stage, in which T cells are stimulated, and an effector stage, in which they attack the graft.
- In most clinical situations, graft rejection is suppressed by nonspecific immunosuppressive agents or by total lymphoid x-irradiation.
- Experimental approaches using monoclonal antibodies offer the possibility of specific immunosuppression. These antibodies may act by:
 - Deleting populations of reactive cells.
 - Inhibiting co-stimulatory signals leading to anergy in specifically reactive cells.
- Certain sites in the body, including the cornea of the eye, brain, testes, and uterus, do not reject transplants despite genetic mismatch between donor and recipient.
- Specific tolerance to alloantigens is induced by exposure to them in utero or by injection of neonates.
- A major complication in bone-marrow transplantation is graft-versus-host reaction mediated by the lymphocytes contained within the donor marrow.
- The critical shortage of organs available for transplantation may be solved in the future by using organs from nonhuman species (xenotransplants).

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USEFUL WEB SITES

<http://www.transweb.org>

Links to hundreds of sites giving information on all aspects of organ transplantation.

<http://www.unos.org>

United Network for Organ Sharing site has information concerning solid-organ transplantation for patients, families, doctors, and teachers.

<http://www.marrow.org>

The National Marrow Donor Program Web site contains information about all aspects of bone-marrow transplantation.

Study Questions

CLINICAL FOCUS QUESTION What features would you include in an ideal animal donor for xenotransplantation? How would you test your model prior to doing clinical trials in humans?

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.
 - a. Acute rejection is mediated by preexisting host antibodies specific for antigens on the grafted tissue.

- b. Second-set rejection is a manifestation of immunologic memory.
 - c. Passenger leukocytes are host dendritic cells that migrate into grafted tissue and act as antigen-presenting cells.
 - d. All allografts between individuals with identical HLA haplotypes will be accepted.
 - e. Cytokines produced by host T_H cells activated in response to alloantigens play a major role in graft rejection.
2. You are a pediatrician treating a child who needs a kidney transplant. The child does not have an identical twin, but both parents and several siblings are willing to donate a kidney if the MHC match with the patient is good.
 - a. What is the best possible MHC match that could be achieved in this situation?
 - b. In which relative(s) might you find it? Why?
 - c. What test(s) would you perform in order to find the best-matched kidney?
 3. Indicate in the Response column in the table on page 500 whether a skin graft from each donor to each recipient listed would result in a rejection (R) or an acceptance (A) response. If you believe a rejection reaction would occur, then indicate in the right-hand column whether it would be a first-set rejection (FSR), occurring in 12–14 days, or a second-set rejection (SSR), occurring in 5–6 days. All the mouse strains listed in the table have different H-2 haplotypes.
 4. Graft-versus-host disease (GVHD) frequently develops after certain types of transplantations.
 - a. Briefly outline the mechanisms involved in GVHD.
 - b. Under what conditions is GVHD likely to occur?
 - c. Some researchers have found that GVHD can be diminished by prior treatment of the graft with monoclonal antibody plus complement or with monoclonal antibody conjugated with toxins. List at least two cell-surface antigens to which monoclonal antibodies could be prepared and used for this purpose, and give the rationale for your choices.
 5. A child who requires a kidney transplant has been offered a kidney from both parents and from five siblings.
 - a. Cells from the potential donors are screened with monoclonal antibodies to the HLA-A, -B, and -C antigens in a microcytotoxicity assay. In addition, ABO blood-group typing is performed. Based on the results in the table on page 500, a kidney graft from which donor(s) is most likely to survive?
 - b. Now a one-way MLR is performed using various combinations of mitomycin-treated lymphocytes. The results, expressed as counts per minute of [³H]thymidine incorporated, are shown in the table on page 500; the stimulation index (ratio of the experimental value to the control in which identical leukocytes are mixed) is listed below in parentheses. Based on these data, a graft from which donor(s) is most likely to be accepted?
 6. What is the biologic basis for attempting to use soluble CTLA4 or anti-CD40L to block allograft rejection? Why might this be better than treating a graft recipient with CsA or FK506?



For use with Question 3:

Donor	Recipient	Response	Type of rejection
BALB/c	C3H		
BALB/c	Rat		
BALB/c	Nude mouse		
BALB/c	C3H, had previous BALB/c graft		
BALB/c	C3H, had previous C57BL/6 graft		
BALB/c	BALB/c		
BALB/c	(BALB/c × C3H) F_1		
BALB/c	(C3H × C57BL/6) F_1		
(BALB/c × C3H) F_1	BALB/c		
(BALB/c × C3H) F_1	BALB/c, had previous F_1 graft		

For use with Question 5a:

	ABO type	HLA-A type	HLA-B type	HLA-C type
Recipient	O	A1/A2	B8/B12	Cw3
Potential donors				
Mother	A	A1/A2	B8/B12	Cw1/Cw3
Father	O	A2	B12/B15	Cw3
Sibling A	O	A1/A2	B8/B15	Cw3
Sibling B	O	A2	B12	Cw1/Cw3
Sibling C	O	A1/A2	B8/B12	Cw3
Sibling D	A	A1/A2	B8/B12	Cw3
Sibling E	O	A1/A2	B8/B15	Cw3

For use with Question 5b:

Respondent cells	Mytomycin C-treated stimulator cells					
	Patient	Sibling A	Sibling B	Sibling C	Sibling D	Sibling E
Patient	1,672 (1.0)	1,800 (1.1)	13,479 (8.1)	5,210 (3.1)	13,927 (8.3)	13,808 (8.3)
Sibling A	1,495 (1.6)	933 (1.0)	11,606 (12.4)	8,443 (9.1)	11,708 (12.6)	13,430 (14.4)
Sibling B	25,418 (9.9)	26,209 (10.2)	2,570 (1.0)	13,170 (5.1)	19,722 (7.7)	4,510 (1.8)
Sibling C	10,722 (6.2)	10,714 (5.9)	13,032 (7.5)	1,731 (1.0)	1,740 (1.0)	14,365 (8.3)
Sibling D	15,988 (5.1)	13,492 (4.2)	18,519 (5.9)	3,300 (1.1)	3,151 (1.0)	18,334 (5.9)
Sibling E	5,777 (6.5)	8,053 (9.1)	2,024 (2.3)	6,895 (7.8)	10,720 (12.1)	888 (1.0)

chapter 22

Cancer and the Immune System

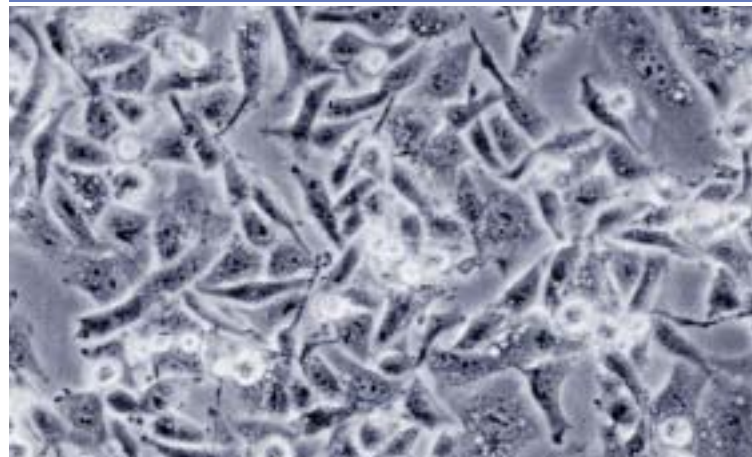
AS THE DEATH TOLL FROM INFECTIOUS DISEASE has declined in the Western world, cancer has become the second-ranking cause of death there, led only by heart disease. Current estimates project that one person in three in the United States will develop cancer, and that one in five will die from it. From an immunologic perspective, cancer cells can be viewed as altered self-cells that have escaped normal growth-regulating mechanisms. This chapter examines the unique properties of cancer cells, paying particular attention to those properties that can be recognized by the immune system. The immune responses that develop to cancer cells, as well as the methods by which cancers manage to evade those responses, are then described. The final section describes current clinical and experimental immunotherapies for cancer.

Cancer: Origin and Terminology

In most organs and tissues of a mature animal, a balance is usually maintained between cell renewal and cell death. The various types of mature cells in the body have a given life span; as these cells die, new cells are generated by the proliferation and differentiation of various types of stem cells. Under normal circumstances, the production of new cells is regulated so that the number of any particular type of cell remains constant. Occasionally, though, cells arise that no longer respond to normal growth-control mechanisms. These cells give rise to clones of cells that can expand to a considerable size, producing a tumor, or **neoplasm**.

A tumor that is not capable of indefinite growth and does not invade the healthy surrounding tissue extensively is **benign**. A tumor that continues to grow and becomes progressively invasive is **malignant**; the term *cancer* refers specifically to a malignant tumor. In addition to uncontrolled growth, malignant tumors exhibit **metastasis**; in this process, small clusters of cancerous cells dislodge from a tumor, invade the blood or lymphatic vessels, and are carried to other tissues, where they continue to proliferate. In this way a primary tumor at one site can give rise to a secondary tumor at another site (Figure 22-1).

Malignant tumors or cancers are classified according to the embryonic origin of the tissue from which the tumor is derived. Most (>80%) are **carcinomas**, tumors that arise from endodermal or ectodermal tissues such as skin or the epithelial lining of internal organs and glands. The majority



Cancerous melanoma cells.

- Cancer: Origin and Terminology
- Malignant Transformation of Cells
- Oncogenes and Cancer Induction
- Tumors of the Immune System
- Tumor Antigens
- Immune Response to Tumors
- Tumor Evasion of the Immune System
- Cancer Immunotherapy

of cancers of the colon, breast, prostate, and lung are carcinomas. The **leukemias** and **lymphomas** are malignant tumors of hematopoietic cells of the bone marrow and account for about 9% of cancer incidence in the United States. Leukemias proliferate as single cells, whereas lymphomas tend to grow as tumor masses. **Sarcomas**, which arise less frequently (around 1% of the incidence in the United States), are derived from mesodermal connective tissues such as bone, fat, and cartilage.

Malignant Transformation of Cells

Treatment of normal cultured cells with chemical carcinogens, irradiation, and certain viruses can alter their morphology and growth properties. In some cases this process, referred to as **transformation**, makes the cells able to produce tumors when they are injected into animals. Such cells



VISUALIZING CONCEPTS

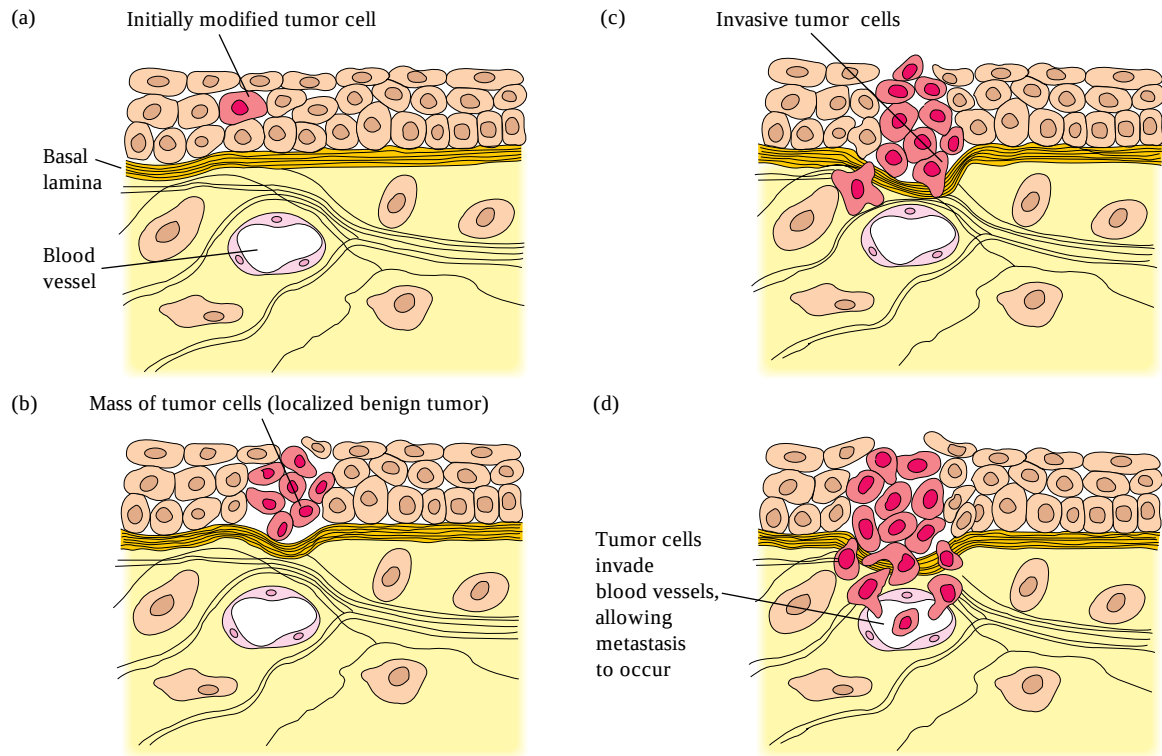


FIGURE 22-1 Tumor growth and metastasis. (a) A single cell develops altered growth properties at a tissue site. (b) The altered cell proliferates, forming a mass of localized tumor cells, or benign tumor. (c) The tumor cells become progressively more invasive, invading the underlying basal lamina. The tumor is now

classified as malignant. (d) The malignant tumor metastasizes by generating small clusters of cancer cells that dislodge from the tumor and are carried by the blood or lymph to other sites in the body. [Adapted from J. Darnell et al., 1990, *Molecular Cell Biology*, 2d ed., Scientific American Books.]

are said to have undergone malignant transformation, and they often exhibit properties *in vitro* similar to those of cancer cells. For example, they have decreased requirements for growth factors and serum, are no longer anchorage-dependent, and grow in a density-independent fashion. Moreover, both cancer cells and transformed cells can be subcultured indefinitely; that is, for all practical purposes, they are immortal. Because of the similar properties of cancer and transformed cells, the process of malignant transformation has been studied extensively as a model of cancer induction.

Various chemical agents (e.g., DNA-alkylating reagents) and physical agents (e.g., ultraviolet light and ionizing radiation) that cause mutations have been shown to induce transformation. Induction of malignant transformation with chemical or

physical carcinogens appears to involve multiple steps and at least two distinct phases: initiation and promotion. Initiation involves changes in the genome but does not, in itself, lead to malignant transformation. After initiation, promoters stimulate cell division and lead to malignant transformation.

The importance of mutagenesis in the induction of cancer is illustrated by diseases such as xeroderma pigmentosum. This rare disorder is caused by a defect in the gene that encodes a DNA-repair enzyme called UV-specific endonuclease. Individuals with this disease are unable to repair UV-induced mutations and consequently develop skin cancers.

A number of DNA and RNA viruses have been shown to induce malignant transformation. Two of the best-studied are SV40 and polyoma. In both cases the viral genomes,

which integrate randomly into the host chromosomal DNA, include several genes that are expressed early in the course of viral replication. SV40 encodes two early proteins called large T and little T, and polyoma encodes three early proteins called large T, middle T, and little T. Each of these proteins plays a role in the malignant transformation of virus-infected cells.

Most RNA viruses replicate in the cytosol and do not induce malignant transformation. The exceptions are retroviruses, which transcribe their RNA into DNA by means of a reverse-transcriptase enzyme and then integrate the transcript into the host's DNA. This process is similar in the cytopathic retroviruses such as HIV-1 and HIV-2 and in the transforming retroviruses, which induce changes in the host cell that lead to malignant transformation. In some cases, retrovirus-induced transformation is related to the presence of **oncogenes**, or “cancer genes,” carried by the retrovirus.

One of the best-studied transforming retroviruses is the **Rous sarcoma virus**. This virus carries an oncogene called *v-src*, which encodes a 60-kDa protein kinase (v-Src) that catalyzes the addition of phosphate to tyrosine residues on proteins. The first evidence that oncogenes alone could induce malignant transformation came from studies of the *v-src* oncogene from Rous sarcoma virus. When this oncogene was cloned and transfected into normal cells in culture, the cells underwent malignant transformation.

Oncogenes and Cancer Induction

In 1971, Howard Temin suggested that oncogenes might not be unique to transforming viruses but might also be found in normal cells; indeed, he proposed that a virus might acquire oncogenes from the genome of an infected cell. He called these cellular genes **proto-oncogenes**, or **cellular oncogenes** (*c-onc*), to distinguish them from their viral counterparts (*v-onc*). In the mid-1970s, J. M. Bishop and H. E. Varmus identified a DNA sequence in normal chicken cells that is homologous to *v-src* from Rous sarcoma virus. This cellular oncogene was designated *c-src*. Since these early discoveries, numerous cellular oncogenes have been identified.

Sequence comparisons of viral and cellular oncogenes reveal that they are highly conserved in evolution. Although most cellular oncogenes consist of a series of exons and introns, their viral counterparts consist of uninterrupted coding sequences, suggesting that the virus might have acquired the oncogene through an intermediate RNA transcript from which the intron sequences had been removed during RNA processing. The actual coding sequences of viral oncogenes and the corresponding proto-oncogenes exhibit a high degree of homology; in some cases, a single point mutation is all that distinguishes a viral oncogene from the corresponding proto-oncogene. It has now become apparent that most, if not all, oncogenes (both viral and cellular) are derived from cellular genes that encode various growth-controlling proteins. In addition, the proteins encoded by a particular onco-

gene and its corresponding proto-oncogene appear to have very similar functions. As described below, the conversion of a proto-oncogene into an oncogene appears in many cases to accompany a change in the level of expression of a normal growth-controlling protein.

Cancer-Associated Genes Have Many Functions

Homeostasis in normal tissue is maintained by a highly regulated process of cellular proliferation balanced by cell death. If there is an imbalance, either at the stage of cellular proliferation or at the stage of cell death, then a cancerous state will develop. Oncogenes and tumor suppressor genes have been shown to play an important role in this process, by regulating either cellular proliferation or cell death. Cancer-associated genes can be divided into three categories that reflect these different activities, summarized in Table 22-1.

INDUCTION OF CELLULAR PROLIFERATION

One category of proto-oncogenes and their oncogenic counterparts encodes proteins that induce cellular proliferation. Some of these proteins function as growth factors or growth-factor receptors. Included among these are *sis*, which encodes a form of platelet-derived growth factor, and *fms*, *erbB*, and *neu*, which encode growth-factor receptors. In normal cells, the expression of growth factors and their receptors is carefully regulated. Usually, one population of cells secretes a growth factor that acts on another population of cells that carries the receptor for the factor, thus stimulating proliferation of the second population. Inappropriate expression of either a growth factor or its receptor can result in uncontrolled proliferation.

Other oncogenes in this category encode products that function in signal-transduction pathways or as transcription factors. The *src* and *abl* oncogenes encode tyrosine kinases, and the *ras* oncogene encodes a GTP-binding protein. The products of these genes act as signal transducers. The *myc*, *jun*, and *fos* oncogenes encode transcription factors. Overactivity of any of these oncogenes may result in unregulated proliferation.

INHIBITION OF CELLULAR PROLIFERATION

A second category of cancer-associated genes—called **tumor-suppressor genes**, or anti-oncogenes—encodes proteins that inhibit excessive cell proliferation. Inactivation of these results in unregulated proliferation. The prototype of this category of oncogenes is *Rb*, the retinoblastoma gene. Hereditary retinoblastoma is a rare childhood cancer, in which tumors develop from neural precursor cells in the immature retina. The affected child has inherited a mutated *Rb* allele; somatic inactivation of the remaining *Rb* allele leads to tumor growth. Probably the single most frequent genetic abnormality in human cancer is mutation in *p53*, which encodes a nuclear phosphoprotein. Over 90% of small-cell lung cancers and

TABLE 22-1 Functional classification of cancer-associated genes

Type/name	Nature of gene product
CATEGORY I: GENES THAT INDUCE CELLULAR PROLIFERATION	
Growth factors	
<i>sis</i>	A form of platelet-derived growth factor (PDGF)
Growth-factor receptors	
<i>fms</i>	Receptor for colony-stimulating factor 1 (CSF-1)
<i>erbB</i>	Receptor for epidermal growth factor (EGF)
<i>neu</i>	Protein (HER2) related to EGF receptor
<i>erbA</i>	Receptor for thyroid hormone
Signal transducers	
<i>src</i>	Tyrosine kinase
<i>abl</i>	Tyrosine kinase
Ha-ras	GTP-binding protein with GTPase activity
N-ras	GTP-binding protein with GTPase activity
K-ras	GTP-binding protein with GTPase activity
Transcription factors	
<i>jun</i>	Component of transcription factor AP1
<i>fos</i>	Component of transcription factor AP1
<i>myc</i>	DNA-binding protein
CATEGORY II: TUMOR-SUPPRESSOR GENES, INHIBITORS OF CELLULAR PROLIFERATION*	
<i>Rb</i>	Suppressor of retinoblastoma
p53	Nuclear phosphoprotein that inhibits formation of small-cell lung cancer and colon cancers
<i>DCC</i>	Suppressor of colon carcinoma
<i>APC</i>	Suppressor of adenomatous polyposis
<i>NF1</i>	Suppressor of neurofibromatosis
<i>WT1</i>	Suppressor of Wilm's tumor
CATEGORY III: GENES THAT REGULATE PROGRAMMED CELL DEATH	
<i>bcl-2</i>	Suppressor of apoptosis

* The activity of the normal products of the category II genes inhibits progression of the cell cycle. Loss of a gene or its inactivation by mutation in an indicated tumor-suppressor gene is associated with development of the indicated cancers.

over 50% of breast and colon cancers have been shown to be associated with mutations in *p53*.

REGULATION OF PROGRAMMED CELL DEATH

A third category of cancer-associated genes regulates programmed cell death. These genes encode proteins that either block or induce apoptosis. Included in this category of oncogenes is *bcl-2*, an anti-apoptosis gene. This oncogene was originally discovered because of its association with B-cell follicular lymphoma. Since its discovery, *bcl-2* has been shown to play an important role in regulating cell survival during hematopoiesis and in the survival of selected B cells and T cells during maturation. Interestingly, the Epstein-Barr virus contains a gene that has sequence homology to *bcl-2* and may act in a similar manner to suppress apoptosis.

Proto-Oncogenes Can Be Converted to Oncogenes

In 1972, R. J. Huebner and G. J. Todaro suggested that mutations or genetic rearrangements of proto-oncogenes by carcinogens or viruses might alter the normally regulated function of these genes, converting them into potent cancer-causing oncogenes (Figure 22-2). Considerable evidence supporting this hypothesis accumulated in subsequent years. For example, some malignantly transformed cells contain multiple copies of cellular oncogenes, resulting in increased production of oncogene products. Such amplification of cellular oncogenes has been observed in cells from various types of human cancers. Several groups have identified *c-myc* oncogenes in homogeneously staining regions (HSRs) of chromosomes from can-

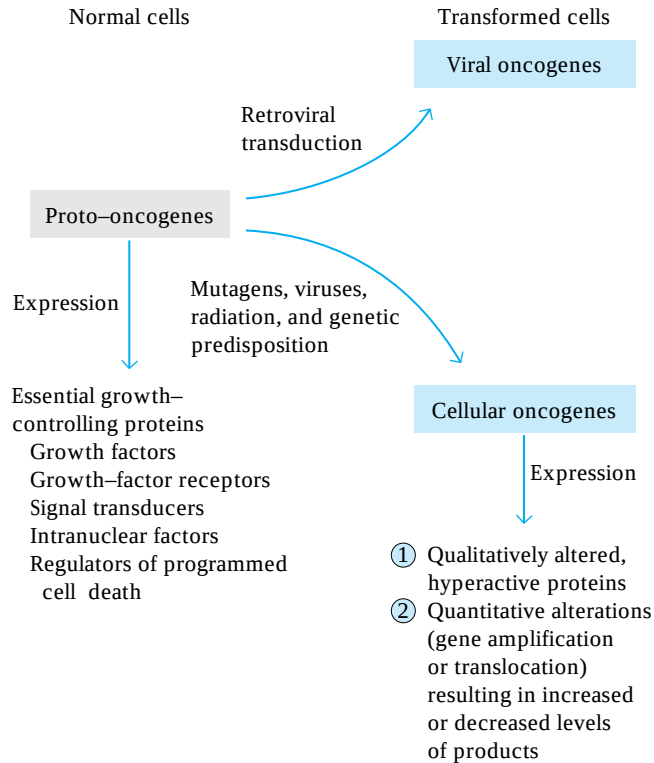


FIGURE 22-2 Conversion of proto-oncogenes into oncogenes can involve mutation, resulting in production of qualitatively different gene products, or DNA amplification or translocation, resulting in increased or decreased expression of gene products.

cer cells; these HSRs represent long tandem arrays of amplified genes.

In addition, some cancer cells exhibit chromosomal translocations, usually the movement of a proto-oncogene from one chromosomal site to another (Figure 22-3). In many cases of Burkitt's lymphoma, for example, *c-myc* is moved from its normal position on chromosome 8 to a position near the immunoglobulin heavy-chain enhancer on chromosome 14. As a result of this translocation, synthesis of the c-Myc protein, which functions as a transcription factor, increases.

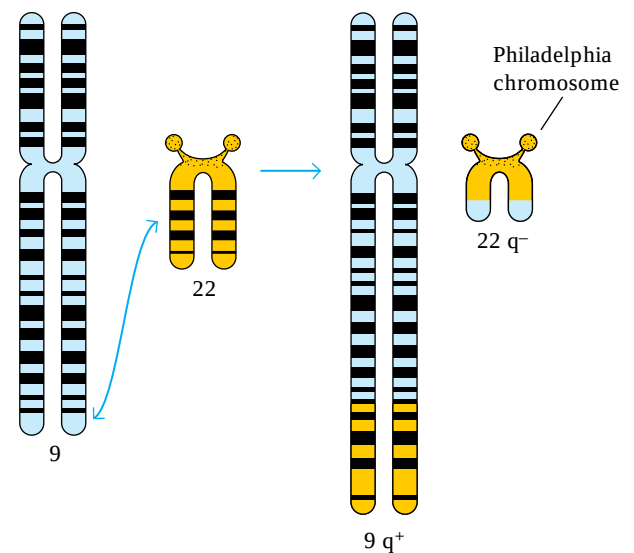
Mutation in proto-oncogenes also has been associated with cellular transformation, and it may be a major mechanism by which chemical carcinogens or x-irradiation convert a proto-oncogene into a cancer-inducing oncogene. For instance, single-point mutations in *c-ras* have been detected in a significant fraction of several human cancers, including carcinomas of the bladder, colon, and lung. Some of these mutations appear to reduce the ability of Ras to associate with GTPase-stimulating proteins, thus prolonging the growth-activated state of Ras.

Viral integration into the host-cell genome may in itself serve to convert a proto-oncogene into a transforming onco-

gene. For example, avian leukosis virus (ALV) is a retrovirus that does not carry any viral oncogenes and yet is able to transform B cells into lymphomas. This particular retrovirus has been shown to integrate within the *c-myc* proto-oncogene, which contains three exons. Exon 1 of *c-myc* has an unknown function; exons 2 and 3 encode the Myc protein. Insertion of AVL between exon 1 and exon 2 has been shown in some cases to allow the provirus promoter to increase transcription of exons 2 and 3, resulting in increased synthesis of c-Myc.

A variety of tumors have been shown to express significantly increased levels of growth factors or growth-factor

(a) Chronic myelogenous leukemia



(b) Burkitt's lymphoma

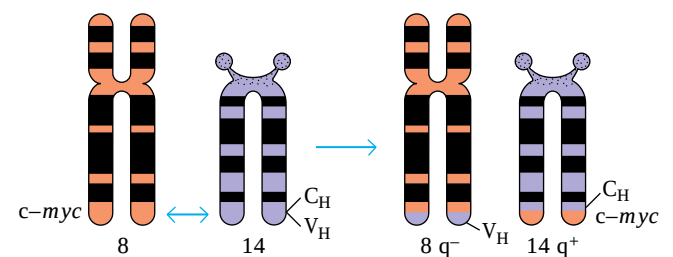


FIGURE 22-3 Chromosomal translocations in (a) chronic myelogenous leukemia (CML) and (b) Burkitt's lymphoma. Leukemic cells from all patients with CML contain the so-called Philadelphia chromosome, which results from a translocation between chromosomes 9 and 22. Cancer cells from some patients with Burkitt's lymphoma exhibit a translocation that moves part of chromosome 8 to chromosome 14. It is now known that this translocation involves *c-myc*, a cellular oncogene. Abnormalities such as these are detected by banding analysis of metaphase chromosomes. Normal chromosomes are shown on the left, and translocated chromosomes on the right.

receptors. Expression of the receptor for epidermal growth factor, which is encoded by *c-erbB*, has been shown to be amplified in many cancer cells. And in breast cancer, increased synthesis of the growth-factor receptor encoded by *c-neu* has been linked with a poor prognosis.

The Induction of Cancer Is a Multistep Process

The development from a normal cell to a cancerous cell is usually a multistep process of clonal evolution driven by a series of somatic mutations that progressively convert the cell from normal growth to a precancerous state and finally a cancerous state.

The presence of myriad chromosomal abnormalities in precancerous and cancerous cells lends support to the role of multiple mutations in the development of cancer. This has been demonstrated in human colon cancer, which progresses in a series of well-defined morphologic stages (Figure 22-4). Colon cancer begins as small, benign tumors called adenomas in the colorectal epithelium. These precancerous tumors grow, gradually becoming increasingly disorganized in their intracellular organization until they acquire the malignant phenotype. These well-defined morphologic stages of colon cancer have been correlated with a sequence of gene changes involving inactivation or loss of three tumor-suppressor genes (*APC*, *DCC*, and *p53*) and activation of one cellular proliferation oncogene (*K-ras*).

Studies with transgenic mice also support the role of multiple steps in the induction of cancer. Transgenic mice expressing high levels of Bcl-2 develop a population of small resting B cells, derived from secondary lymphoid follicles, that have greatly extended life spans. Gradually these transgenic mice develop lymphomas. Analysis of lymphomas from these mice has shown that approximately half have a *c-myc* translocation to the immunoglobulin H-chain locus. The synergism of Myc and Bcl-2 is highlighted in double-transgenic mice produced

by mating the *bcl-2*⁺ transgenic mice with *myc*⁺ transgenic mice. These mice develop leukemia very rapidly.

Tumors of the Immune System

Tumors of the immune system are classified as lymphomas or leukemias. Lymphomas proliferate as solid tumors within a lymphoid tissue such as the bone marrow, lymph nodes, or thymus; they include Hodgkin's and non-Hodgkin's lymphomas. Leukemias tend to proliferate as single cells and are detected by increased cell numbers in the blood or lymph. Leukemia can develop in lymphoid or myeloid lineages.

Historically, the leukemias were classified as acute or chronic according to the clinical progression of the disease. The acute leukemias appeared suddenly and progressed rapidly, whereas the chronic leukemias were much less aggressive and developed slowly as mild, barely symptomatic diseases. These clinical distinctions apply to untreated leukemias; with current treatments, the acute leukemias often have a good prognosis, and permanent remission can often be achieved. Now the major distinction between acute and chronic leukemias is the maturity of the cell involved. Acute leukemias tend to arise in less mature cells, whereas chronic leukemias arise in mature cells. The acute leukemias include **acute lymphocytic leukemia (ALL)** and **acute myelogenous leukemia (AML)**; these diseases can develop at any age and have a rapid onset. The chronic leukemias include **chronic lymphocytic leukemia (CLL)** and **chronic myelogenous leukemia (CML)**; these diseases develop slowly and are seen in adults.

A number of B- and T-cell leukemias and lymphomas involve a proto-oncogene that has been translocated into the immunoglobulin genes or T-cell receptor genes. One of the best characterized is the translocation of *c-myc* in Burkitt's lymphoma and in mouse plasmacytomas. In 75% of Burkitt's lymphoma patients, *c-myc* is translocated from chromosome 8 to the Ig heavy-chain gene cluster on chromosome 14 (see

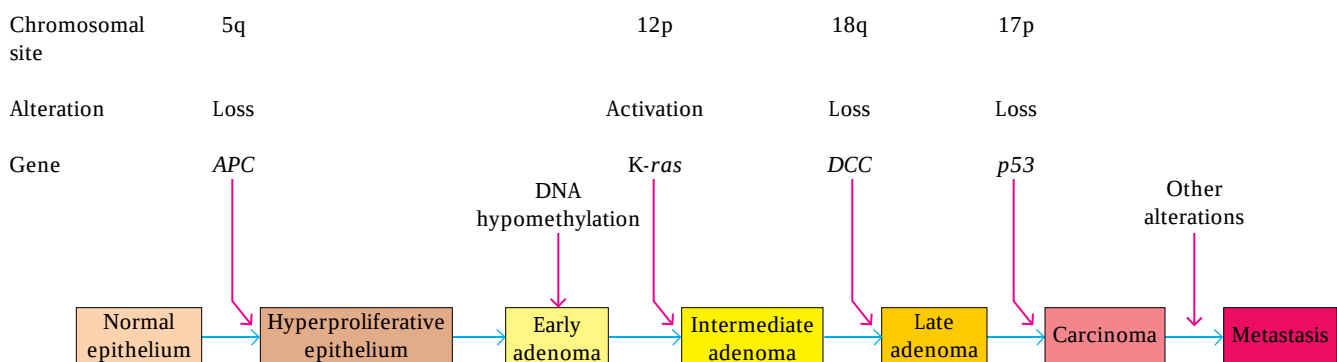


FIGURE 22-4 Model of sequential genetic alterations leading to metastatic colon cancer. Each of the stages indicated at the bottom is morphologically distinct, allowing researchers to determine the se-

quence of genetic alterations. [Adapted from B. Vogelstein and K. W. Kinzler, 1993, Trends Genet. 9:138.]

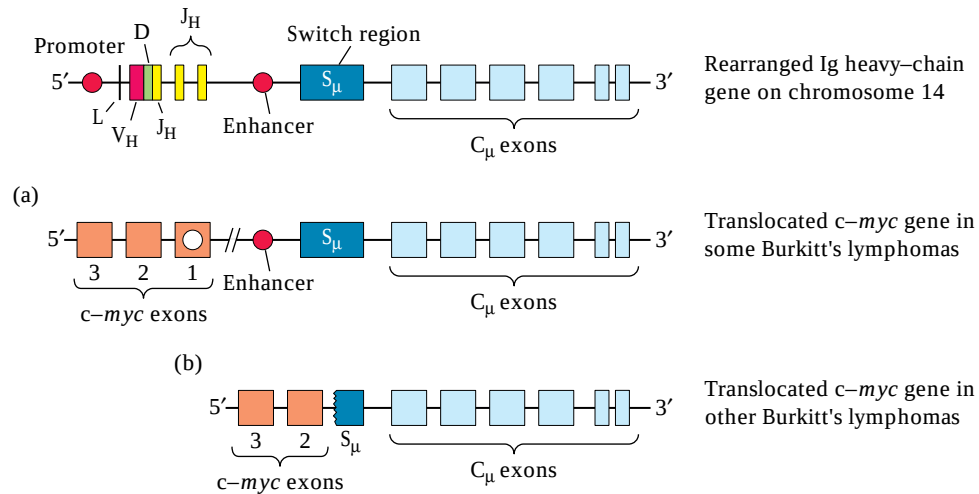


FIGURE 22-5 In many patients with Burkitt's lymphoma, the *c-myc* gene is translocated to the immunoglobulin heavy-chain gene cluster on chromosome 14. In some cases, the entire *c-myc* gene is inserted near the heavy-chain enhancer (a), but in other cases, only the coding

exons (2 and 3) of *c-myc* are inserted at the S_μ switch site (b). Only exons 2 and 3 of *c-myc* are coding exons. Translocation may lead to overexpression of c-Myc.

Figure 22-3b). In the remaining patients, *c-myc* remains on chromosome 8 and the κ or γ light-chain genes are translocated to a region 3' of *c-myc*. Kappa-gene translocations from chromosome 2 to chromosome 8 occur 9% of the time, and γ -gene translocations from chromosome 22 to chromosome 8 occur 16% of the time.

Translocations of *c-myc* to the Ig heavy-chain gene cluster on chromosome 14 have been analyzed, and, in some cases, the entire *c-myc* gene is translocated head-to-head to a region near the heavy-chain enhancer. In other cases, exons 1, 2, and 3 or exons 2 and 3 of *c-myc* are translocated head-to-head to the S_μ or S_α switch site (Figure 22-5). In each case, the translocation removes the *myc* coding exons from the regulatory mechanisms operating in chromosome 8 and places them in the immunoglobulin-gene region, a very active region that is expressed constitutively in these cells. The consequences of enhancer-mediated high levels of constitutive *myc* expression in lymphoid cells have been investigated in transgenic mice. In one study, mice containing a transgene consisting of all three *c-myc* exons and the immunoglobulin heavy-chain enhancer were produced. Of 15 transgenic pups born, 13 developed lymphomas of the B-cell lineage within a few months of birth.

Tumor Antigens

The subdiscipline of tumor immunology involves the study of antigens on tumor cells and the immune response to these antigens. Two types of tumor antigens have been identified on tumor cells: **tumor-specific transplantation antigens (TSTAs)** and **tumor-associated transplantation antigens**

(TATAs). Tumor-specific antigens are unique to tumor cells and do not occur on normal cells in the body. They may result from mutations in tumor cells that generate altered cellular proteins; cytosolic processing of these proteins would give rise to novel peptides that are presented with class I MHC molecules, inducing a cell-mediated response by tumor-specific CTLs (Figure 22-6). Tumor-associated antigens, which are not unique to tumor cells, may be proteins that are expressed on normal cells during fetal development when the immune system is immature and unable to respond but that normally are not expressed in the adult. Reactivation of the embryonic genes that encode these proteins in tumor cells results in their expression on the fully differentiated tumor cells. Tumor-associated antigens may also be proteins that are normally expressed at extremely low levels on normal cells but are expressed at much higher levels on tumor cells. It is now clear that the tumor antigens recognized by human T cells fall into one of four major categories:

- Antigens encoded by genes exclusively expressed by tumors
- Antigens encoded by variant forms of normal genes that have been altered by mutation
- Antigens normally expressed only at certain stages of differentiation or only by certain differentiation lineages
- Antigens that are overexpressed in particular tumors

Many tumor antigens are cellular proteins that give rise to peptides presented with MHC molecules; typically, these antigens have been identified by their ability to induce the proliferation of antigen-specific CTLs or helper T cells.

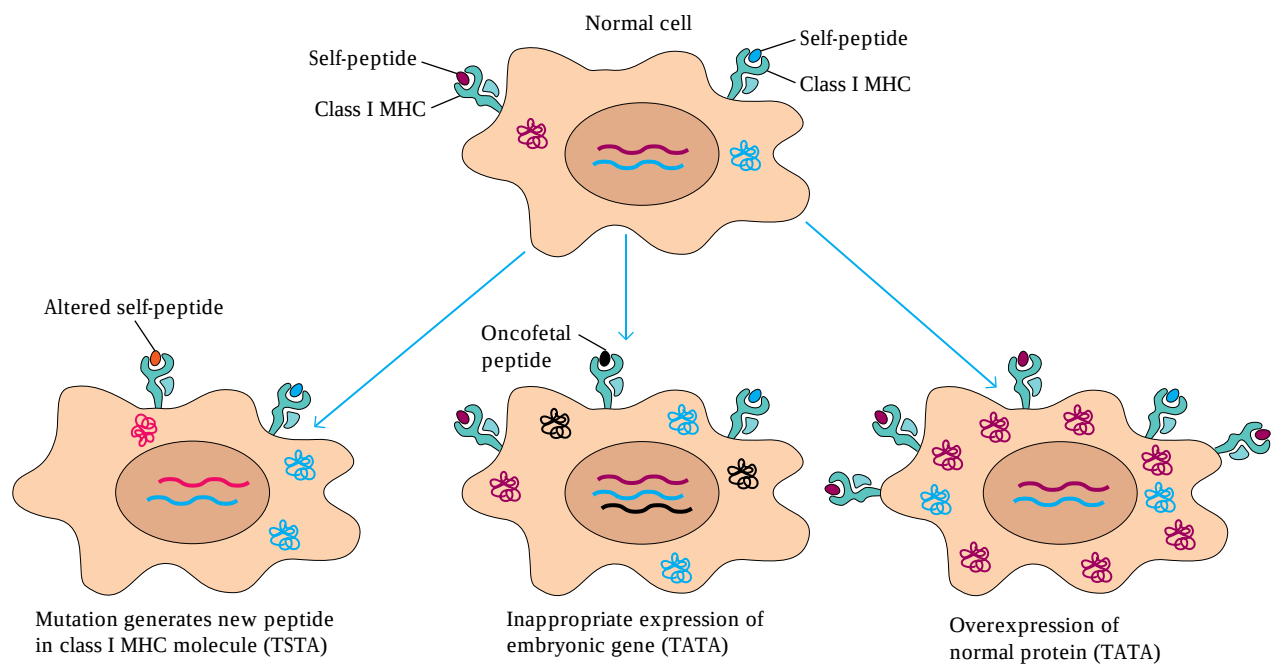


FIGURE 22-6 Different mechanisms generate tumor-specific transplantation antigens (TSTAs) and tumor-associated transplantation antigens (TATAs). The latter are more common.

Some Antigens Are Tumor-Specific

Tumor-specific antigens have been identified on tumors induced with chemical or physical carcinogens and on some virally induced tumors. Demonstrating the presence of tumor-specific antigens on spontaneously occurring tumors is particularly difficult because the immune response to such tumors eliminates all of the tumor cells bearing sufficient numbers of the antigens and in this way selects for cells bearing low levels of the antigens.

CHEMICALLY OR PHYSICALLY INDUCED TUMOR ANTIGENS

Methylcholanthrene and ultraviolet light are two carcinogens that have been used extensively to generate lines of tumor cells. When syngeneic animals are injected with killed cells from a carcinogen-induced tumor-cell line, the animals develop a specific immunologic response that can protect against later challenge by live cells of the same line but not other tumor-cell lines (Table 22-2). Even when the same chemical carcinogen induces two separate tumors at different sites in the same animal, the tumor antigens are distinct and the immune response to one tumor does not protect against the other tumor.

The tumor-specific transplantation antigens of chemically induced tumors have been difficult to characterize because they cannot be identified by induced antibodies but only by their T-cell-mediated rejection. One experimental approach that has allowed identification of genes encoding some TSTAs is outlined in Figure 22-7. When a mouse tumorigenic cell line

(tum⁺), which gives rise to progressively growing tumors, is treated in vitro with a chemical mutagen, some cells are mutated so that they no longer are capable of growing into a

TABLE 22-2 Immune response to methyl-cholanthrene (MCA) or polyoma virus (PV)*

Transplanted killed tumor cells	Live tumor cells for challenge	Tumor growth
CHEMICALLY INDUCED		
MCA-induced sarcoma A	MCA-induced sarcoma A	–
MCA-induced sarcoma A	MCA-induced sarcoma B	+
VIRALLY INDUCED		
PV-induced sarcoma A	PV-induced sarcoma A	–
PV-induced sarcoma A	PV-induced sarcoma B	–
PV-induced sarcoma A	SV40-induced sarcoma C	+

*Tumors were induced either with MCA or PV, and killed cells from the induced tumors were injected into syngeneic animals, which were then challenged with live cells from the indicated tumor-cell lines. The absence of tumor growth after live challenge indicates that the immune response induced by tumor antigens on the killed cells provided protection against the live cells.

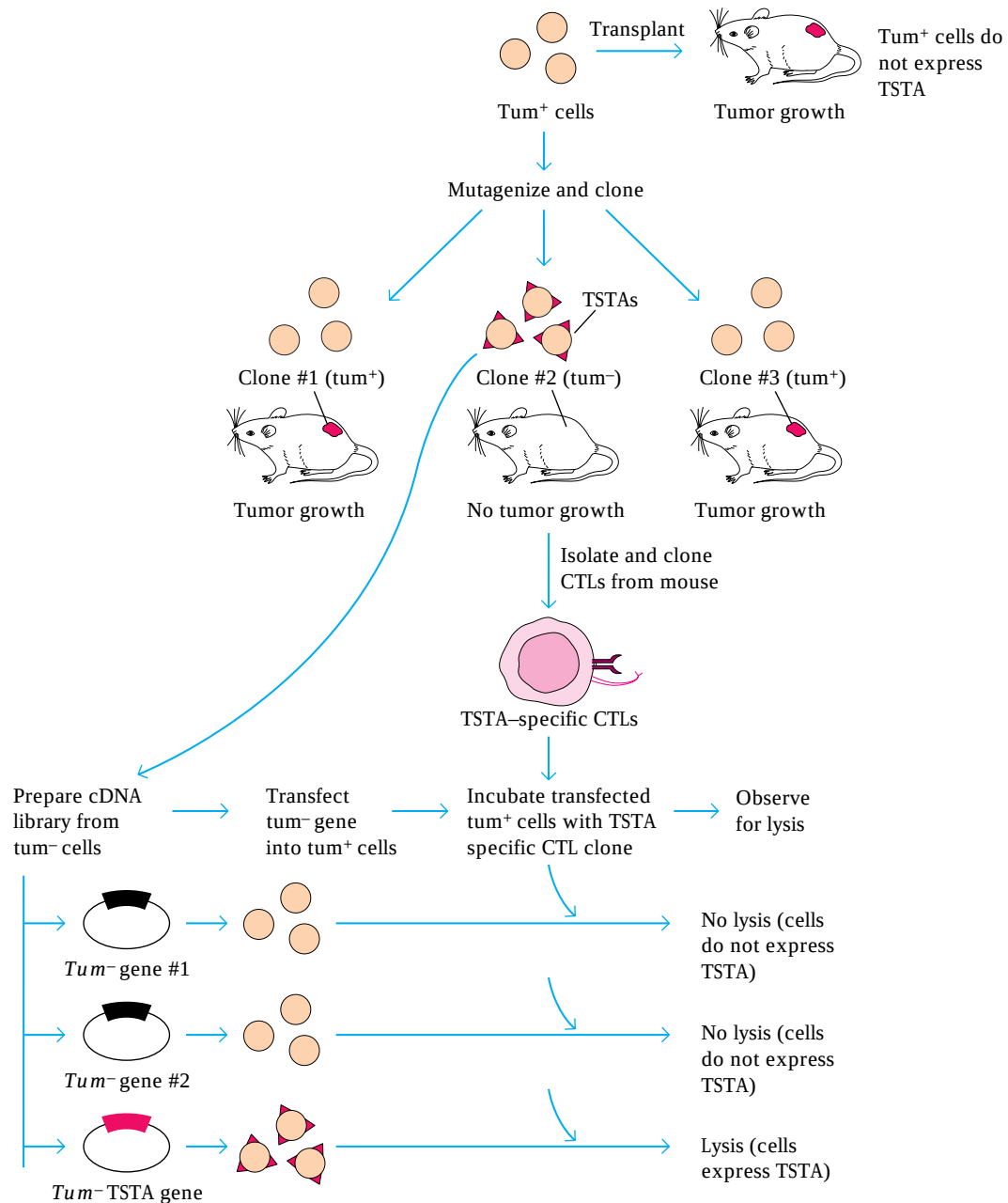


FIGURE 22-7 One procedure for identifying genes encoding tumor-specific transplantation antigens (TSTAs). Most TSTAs can be detected only by the cell-mediated rejection they elicit. In the first part of this procedure, a nontumorigenic (tum⁻) cell line is generated; this cell line expresses a TSTA that is recognized by syngeneic mice, which mount a

cell-mediated response against it. To isolate the gene encoding the TSTA, a cosmid gene library is prepared from the tum⁻ cell line, the genes are transfected into tumorigenic tum⁺ cells, and the transfected cells are incubated with TSTA-specific CTLs.

tumor in syngeneic mice. These mutant tumor cells are designated as tum⁻ variants. Most tum⁻ variants have been shown to express TSTAs that are not expressed by the original tum⁺ tumor-cell line. When tum⁻ cells are injected into syngeneic mice, the unique TSTAs that the tum⁻ cells express are recognized by specific CTLs. The TSTA-specific CTLs destroy the tum⁻ tumor cells, thus preventing tumor growth. To identify

the genes encoding the TSTAs that are expressed on a tum⁻ cell line, a cosmid DNA library is prepared from the tum⁻ cells. Genes from the tum⁻ cells are transfected into the original tum⁺ cells. The transfected tum⁺ cells are tested for the expression of the tum⁻ TSTAs by their ability to activate cloned CTLs specific for the tum⁻ TSTA. A number of diverse TSTAs have been identified by this method.

In the past few years, two methods have facilitated the characterization of TSTAs (Figure 22-8). In one method, peptides bound to class I MHC molecules on the membranes of the tumor cells are eluted with acid and purified by high-pressure liquid chromatography (HPLC). In some cases, sufficient peptide is eluted to allow its sequence to be deduced by Edman degradation. In a second approach, cDNA libraries are prepared from tumor cells. These cDNA libraries are transfected transiently into COS cells, which are monkey kidney cells transfected with the gene that codes for the SV40 large-T antigen. When these cells are later transfected with plasmids containing both the tumor-cell cDNA and an SV40 origin of replication, the large-T antigen stimulates plasmid replication, so that up to 10^4 – 10^5 plasmid copies are produced per cell. This results in high-level expression of the tumor-cell DNA.

The genes that encode some TSTAs have been shown to differ from normal cellular genes by a single point mutation. Further characterization of TSTAs has demonstrated that many of them are not cell-membrane proteins; rather, as indicated already, they are short peptides derived from cytosolic proteins that have been processed and presented together with class I MHC molecules.

Tumor Antigens May Be Induced by Viruses

In contrast to chemically induced tumors, virally induced tumors express tumor antigens shared by all tumors induced

by the same virus. For example, when syngeneic mice are injected with killed cells from a particular polyoma-induced tumor, the recipients are protected against subsequent challenge with live cells from any polyoma-induced tumors (see Table 22-2). Likewise, when lymphocytes are transferred from mice with a virus-induced tumor into normal syngeneic recipients, the recipients reject subsequent transplants of all syngeneic tumors induced by the same virus. In the case of both SV40- and polyoma-induced tumors, the presence of tumor antigens is related to the neoplastic state of the cell. In humans, Burkitt's-lymphoma cells have been shown to express a nuclear antigen of the Epstein-Barr virus that may indeed be a tumor-specific antigen for this type of tumor. Human papilloma virus (HPV) E6 and E7 proteins are found in more than 80% of invasive cervical cancers—the clearest example of a virally encoded tumor antigen. Consequently, there is great interest in testing as vaccine candidates the HPVs that are strongly linked to cervical cancer, such as HPV-16.

The potential value of these virally induced tumor antigens can be seen in animal models. In one experiment, mice immunized with a preparation of genetically engineered polyoma virus tumor antigen were shown to be immune to subsequent injections of live polyoma-induced tumor cells. In another experiment, mice were immunized with a vaccinia-virus vaccine engineered with the gene encoding the polyoma-tumor antigen. These mice also developed immunity, rejecting later injections of live polyoma-induced tumor cells (Figure 22-9).

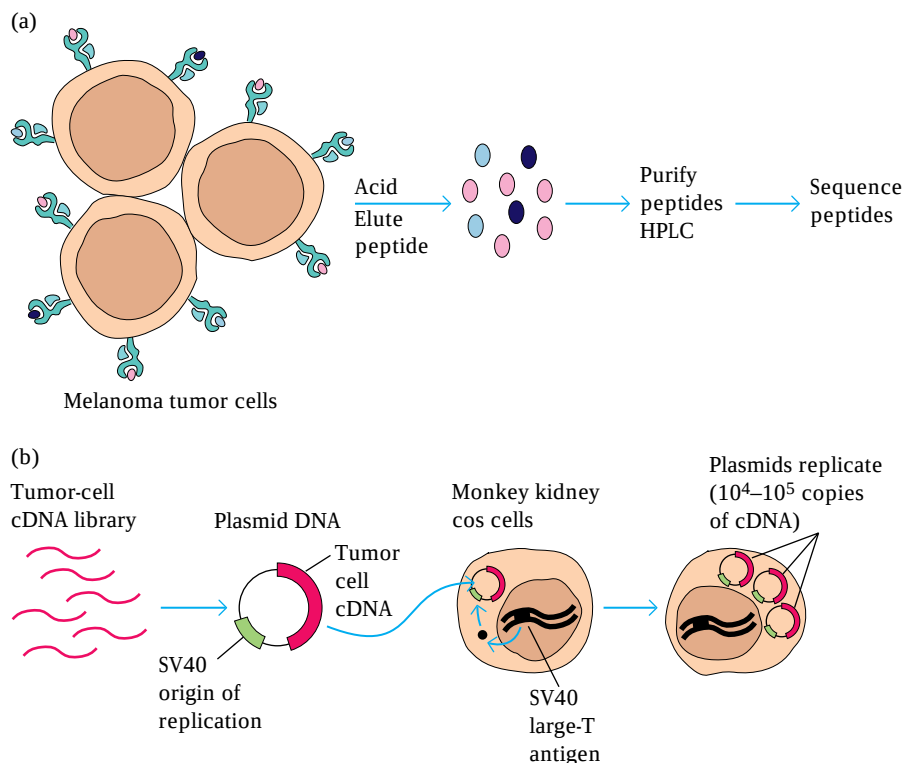


FIGURE 22-8 Two methods used to isolate tumor antigens that induce tumor-specific CTLs. See text for details.

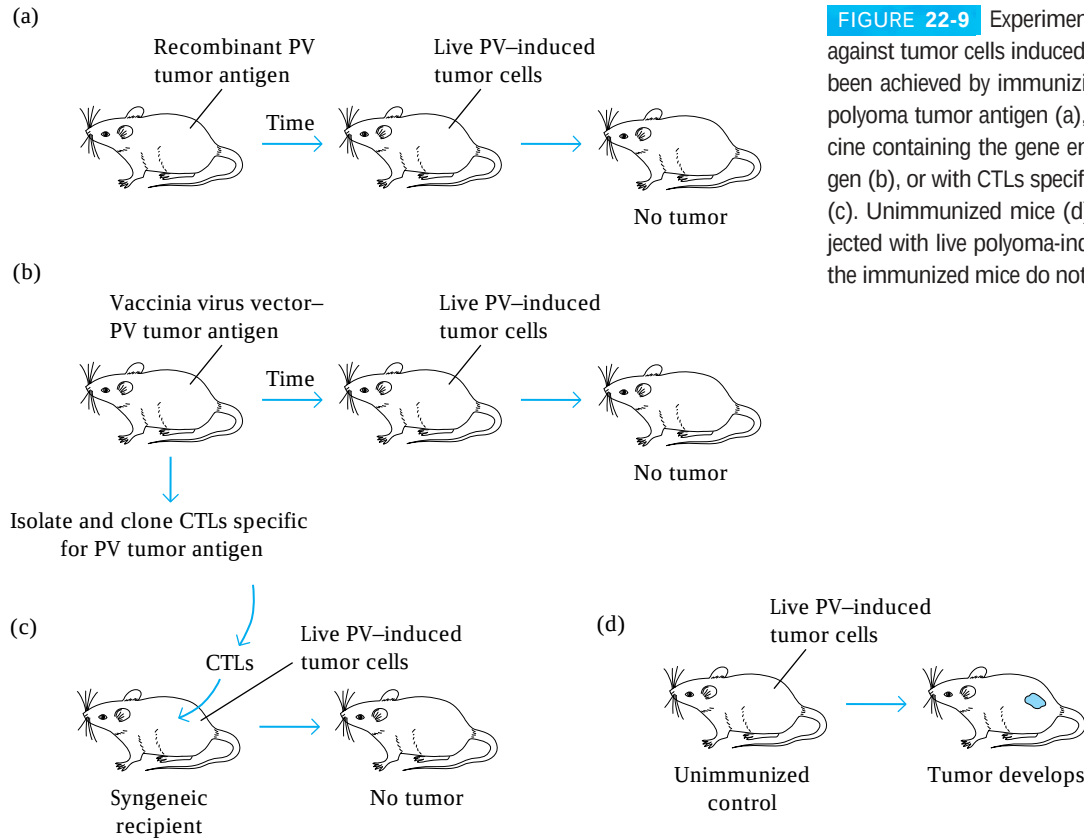


FIGURE 22-9 Experimental induction of immunity against tumor cells induced by polyoma virus (PV) has been achieved by immunizing mice with recombinant polyoma tumor antigen (a), with a vaccinia vector vaccine containing the gene encoding the PV tumor antigen (b), or with CTLs specific for the PV tumor antigen (c). Unimmunized mice (d) develop tumors when injected with live polyoma-induced tumor cells, whereas the immunized mice do not.

Most Tumor Antigens Are Not Unique to Tumor Cells

The majority of tumor antigens are not unique to tumor cells but also are present on normal cells. These tumor-associated transplantation antigens may be proteins usually expressed only on fetal cells but not on normal adult cells, or they may be proteins expressed at low levels by normal cells but at much higher levels by tumor cells. The latter category includes growth factors and growth-factor receptors, as well as oncogene-encoded proteins.

Several growth-factor receptors are expressed at significantly increased levels on tumor cells and can serve as tumor-associated antigens. For instance, a variety of tumor cells express the epidermal growth factor (EGF) receptor at levels 100 times greater than that in normal cells. An example of an over-expressed growth factor serving as a tumor-associated antigen is a transferrin growth factor, designated p97, which aids in the transport of iron into cells. Whereas normal cells express less than 8,000 molecules of p97 per cell, melanoma cells express 50,000–500,000 molecules of p97 per cell. The gene that encodes p97 has been cloned, and a recombinant vaccinia virus vaccine has been prepared that carries the cloned gene. When this vaccine was injected into mice, it induced both humoral and cell-mediated immune responses, which protected the mice against live melanoma cells ex-

pressing the p97 antigen. Results such as this highlight the importance of identifying tumor antigens as potential targets of tumor immunotherapy.

ONCOFETAL TUMOR ANTIGENS

Oncofetal tumor antigens, as the name implies, are found not only on cancerous cells but also on normal fetal cells. These antigens appear early in embryonic development, before the immune system acquires immunocompetence; if these antigens appear later on cancer cells, they are recognized as nonself and induce an immunologic response. Two well-studied oncofetal antigens are **alpha-fetoprotein (AFP)** and **carcinoembryonic antigen (CEA)**.

Although the serum concentration of AFP drops from milligram levels in fetal serum to nanogram levels in normal adult serum, elevated AFP levels are found in a majority of patients with liver cancer (Table 22-3). CEA is a membrane glycoprotein found on gastrointestinal and liver cells of 2- to 6-month-old fetuses. Approximately 90% of patients with advanced colorectal cancer and 50% of patients with early colorectal cancer have increased levels of CEA in their serum; some patients with other types of cancer also exhibit increased CEA levels. However, because AFP and CEA can be found in trace amounts in some normal adults and in some noncancerous disease states, the presence of these oncofetal antigens is not diagnostic of

TABLE 22-3

Elevation of Alpha-fetoprotein (AFP) and carcinoembryonic antigen (CEA) in serum of patients with various diseases

Disease	No. of patients tested	% of patients with high AFP or CEA levels*
AFP > 400 µg/mL		
Alcoholic cirrhosis	NA	0
Hepatitis	NA	1
Hepatocellular carcinoma	NA	69
Other carcinoma	NA	0
CEA > 10 ng/ml		
CEA > 10 µg/mL		
Cancerous		
Breast carcinoma	125	14
Colorectal carcinoma	544	35
Gastric carcinoma	79	19
Noncarcinoma carcinoma	228	2
Pancreatic carcinoma	55	35
Pulmonary carcinoma	181	26
Noncancerous		
Alcoholic cirrhosis	120	2
Cholecystitis	39	1
Nonmalignant disease	115	0
Pulmonary emphysema	49	4
Rectal polyps	90	1
Ulcerative colitis	146	5

*Although trace amounts of both AFP and CEA can be found in some healthy adults, none would have levels greater than those indicated in the table.

tumors but rather serves to monitor tumor growth. If, for example, a patient has had surgery to remove a colorectal carcinoma, CEA levels are monitored after surgery. An increase in the CEA level is an indication of resumed tumor growth.

ONCOGENE PROTEINS AS TUMOR ANTIGENS

A number of tumors have been shown to express tumor-associated antigens encoded by cellular oncogenes. These antigens are also present in normal cells encoded by the corresponding proto-oncogene. In many cases, there is no qualitative difference between the oncogene and proto-oncogene products; instead, the increased levels of the oncogene product can be recognized by the immune system. For example, as noted earlier, human breast-cancer cells exhibit elevated expression of the oncogene-encoded Neu protein, a growth-factor receptor, whereas normal adult cells express only trace

amounts of Neu protein. Because of this difference in the Neu level, anti-Neu monoclonal antibodies can recognize and selectively eliminate breast-cancer cells without damaging normal cells.

TATAs ON HUMAN MELANOMAS

Several tumor-associated transplantation antigens have been identified on human melanomas. Five of these—MAGE-1, MAGE-3, BAGE, GAGE-1, GAGE-2—are oncofetal-type antigens. Each of these antigens is expressed on a significant proportion of human melanoma tumors, as well as on a number of other human tumors, but not on normal differentiated tissues except for the testis, where it is expressed on germ-line cells. In addition, a number of differentiation antigens expressed on normal melanocytes—including tyrosinase, gp100, Melan-A or MART-1, and gp75—are overexpressed by melanoma cells, enabling them to function as tumor-associated transplantation antigens.

Several of the human melanoma tumor antigens are shared by a number of other tumors. About 40% of human melanomas are positive for MAGE-1, and about 75% are positive for MAGE-2 or 3. In addition to melanomas, a significant percentage of glioma cell lines, breast tumors, non-small-cell lung tumors, and head or neck carcinomas express MAGE-1, 2, or 3. These shared tumor antigens could be exploited for clinical treatment. It might be possible to produce a tumor vaccine expressing the shared antigen for treatment of a number of these tumors, as described at the end of this chapter.

Tumors Can Induce Potent Immune Responses

In experimental animals, tumor antigens can be shown to induce both humoral and cell-mediated immune responses that result in the destruction of the tumor cells. In general, the cell-mediated response appears to play the major role. A number of tumors have been shown to induce tumor-specific CTLs that recognize tumor antigens presented by class I MHC on the tumor cells. However, as discussed below, expression of class I MHC molecules is decreased in a number of tumors, thereby limiting the role of specific CTLs in their destruction.

NK Cells and Macrophages Are Important in Tumor Recognition

The recognition of tumor cells by NK cells is not MHC restricted. Thus, the activity of these cells is not compromised by the decreased MHC expression exhibited by some tumor cells. In some cases, Fc receptors on NK cells can bind to antibody-coated tumor cells, leading to ADCC. The importance of NK cells in tumor immunity is suggested by the mutant mouse strain called beige and by **Chediak-Higashi syndrome** in humans, as described in the Clinical Focus in Chapter 14. In each case, a genetic defect causes marked impairment of NK cells and an associated increase in certain types of cancer.



CLINICAL FOCUS

Cancer Vaccines Promise Hope for the Future

THE realization that the vertebrate immune system evolved to distinguish self from non-self led to the notion that our immune system could recognize a tumor as foreign. In fact, a major research effort amongst cancer immunologists during the latter half of the 20th century was the identification and characterization of tumor specific molecules, the so-called tumor antigens. This area of research was met with skepticism. First of all, the existence of tumor-specific antigens was questionable; many antigens were identified as tumor specific only to find that other cells also expressed these antigens. Secondly, early investigations in the field of tumor immunology necessarily employed animal models that may or may not be relevant to human cancers. However, with advances in biotechnology, genomics, and proteomics, coupled with our increased understanding of the cellular interactions in the immune system, tumor immunology now offers us the promise of new drugs that will aid in the treatment of cancer. We now understand that tumor-associated antigens do exist and that focusing the cellular arm of the immune system toward the recognition these proteins is a rational approach to the development of a cancer vaccine.

One of the best-studied tumor-immunity models is melanoma. Melanoma has evolved as a model system for several reasons. First of all and paradoxically, most human cancers are difficult to establish in tissue culture, making it difficult to develop *in vitro* systems for experimental manipulation. Melanoma is relatively easy to adapt to tissue culture, which has led to the identification of several tumor-associated antigens,

some of which are unique to melanoma (see Table 22-5). These observations are enhanced by the ability to create cDNA libraries (see Chapter 23) from tumor cells. The cDNAs can be transfected into target cells expressing the appropriate MHC molecules and then used as targets for CTL-mediated killing. Once CTL reactivity is recognized, the transfected cDNA can be isolated and identified as a potential tumor antigen. The ability to isolate genes encoding tumor-associated antigens provides us with the opportunity to use these proteins as immunogens for the induction of tumor-specific responses. Additionally, the identification of tumor-associated proteins allows us to identify peptides that elicit anti-tumor responses.

Over the past few years, several biotech companies have devised strategies for the development of vaccines against melanoma as well as other cancers. These strategies have one thing in the common; the induction of a cell-mediated response to tumor-associated antigens. Antigens are derived from individual patient tumors or established tumor cell lines. The use of patient-derived tumors is appealing for obvious reasons. The response to that tumor should, in theory, be uniquely directed only at tumor antigens and not other, potentially allotypic, determinants. However, such individualized therapy could be very expensive and time-consuming. In this scenario, the tumor would be biopsied or surgically removed, placed into culture, and then used as an immunogen. Establishing a primary tumor in culture is not easy, even for melanoma, and the procedure can take several weeks. The time factor, coupled with the realization that many tumors are not easy to grow *in vitro*,

places this into the category of “designer therapies” that may or may not be feasible under the reality of managed health care today.

The use of established tumors as the source of the immunogen is much more accessible in cost and practicality. Samples from several tumors can be grown in culture and protein extracts prepared and frozen, providing a source of immunogen for many patients. In addition to reduced costs, this strategy also allows careful assessment of the immunogenicity of the tumor antigens found in the cultured cells. It is possible that some tumors may express higher levels of tumor-associated antigens and be more immunogenic than others. Indeed one biotech company in California, CancerVax (www.CancerVax.com), has derived three cell lines that express high levels of over 20 tumor-associated antigens. Additionally, these cells express MHC class I alleles which are represented in the majority of individuals in the population, meaning that intracellular antigens will be presented properly. Cells are irradiated to render them incapable of cell division and used as irradiated whole cells for immunization. The advantages of this approach lies in the ability to standardize the immunogen as well as reducing the cost.

Antigen presentation is a critical feature of any immunization strategy and one way to enhance to immunization against tumor antigens is to manipulate the fashion in which the antigen is presented. Professional antigen-presenting cells such as dendritic cells are excellent candidates to employ in vaccination protocols. Several companies have developed novel uses for dendritic cells in cancer therapy. Dendreon (www.Dendreon.com), a Seattle-based company, first isolates dendritic-cell precursors from patient blood, then introduces the immunogen into the dendritic cells and returns the antigen-pulsed dendritic cells to the bloodstream of the cancer patient. This company, through genomics-based drug discovery, has identified tumor-associated antigens prevalent on a wide variety of cancers. Thus the

(continued)



CLINICAL FOCUS (continued)

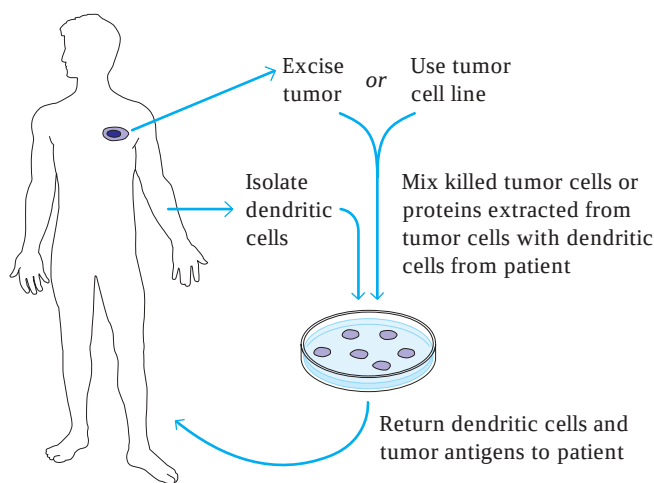
dendritic-cell therapy can be tailored to a variety of different tumors. A variation on this theme currently is being tested by Genzyme Molecular Oncology (www.genzymemolecularoncology.com). Their approach also uses dendritic cells, but rather than employing already-defined antigens, clinical trials are underway where dendritic cells from the patient are fused, using polyethylene glycol, with inactivated tumor cells taken from the same patient. The advantage of this technique is that the hybrid cell has the antigen-presenting capability of a dendritic cell but also contains the antigens from the patient's tumor cells. The dendritic cell processes these tumor antigens and efficiently presents the processed antigen to the immune system of the patient.

A different but equally promising approach to the design of cancer vaccines comes from observations made many years ago that tumor cells are immunogenic—animals injected with killed tumor cells do not grow tumors when challenged with live tissue. When the basis of this protective immunogenicity was explored, it was found that heat-shock proteins (HSPs) are critical in providing protection. Furthermore HSPs were found to carry immunogenic peptides, thus acting as molecular chaperones. But how do HSP/peptide complexes in tumor tissue prime the host immune system? Recent data demonstrate that HSPs bind CD91, a receptor found primarily on APCs such as dendritic cells as well as on macrophages. In this scenario, HSP/peptide complexes from tumor cells bind the CD91 on

APCs, are internalized, and unexpectedly, the antigen is processed and is thought to re-emerge as peptide/MHC class I complexes on the APC, resulting in the priming of a CD8⁺ T-cell response. This would not be the predicted response, since the exogenous antigens are almost uniformly presented by class II MHC molecules. However, an impressive amount of experimental data demonstrates that HSPs isolated and purified from tumor tissue are a potent inducer of tumor-specific CTLs. These observations have led to phase III clinical trials conducted by Antigenics (www.antigenics.com) of HSP/antigen com-

plexes as immunogens for kidney cancer as well as melanoma. The mechanism by which HSP/antigen complexes bound to CD91 are delivered to the class I presentation machinery is not well understood, but it is clear that HSP/antigen complexes, when presented to APCs, result in the vigorous activation of CD8⁺ T cells.

The promise for cancer vaccines appears very bright. Genomics and proteomic methodologies provide novel tools for identifying tumor antigens. Additionally, there is a variety of approaches available to engage the immune system to respond to tumor antigens. The past decade has seen a rapid increase in the number of biotech companies directed at identifying cancer vaccines, and the number of companies in phase II or phase III clinical trials invites an air of optimism about this area of clinical research.



Cancer vaccine design. Tumor cells are removed from the patient and placed in culture. Alternatively, established tumor-cell lines are chosen and placed into culture. Tumor cells are inactivated and mixed with dendritic cells from the patient and injected back into the patient as immunogens. An alternate approach is to prepare extracts or antigens from the tumor cells and inject these, in addition to dendritic cells, into the patient.

Numerous observations indicate that activated macrophages also play a significant role in the immune response to tumors. For example, macrophages are often observed to cluster around tumors, and their presence is often correlated with tumor regression. Like NK cells, macrophages are not

MHC restricted and express Fc receptors, enabling them to bind to antibody on tumor cells and mediate ADCC. The antitumor activity of activated macrophages is probably mediated by lytic enzymes and reactive oxygen and nitrogen intermediates. In addition, activated macrophages secrete a

cytokine called tumor necrosis factor (TNF- α) that has potent antitumor activity. When TNF- α is injected into tumor-bearing animals, it has been found to induce hemorrhage and necrosis of the tumor.

IMMUNE SURVEILLANCE THEORY

The immune surveillance theory was first conceptualized in the early 1900s by Paul Ehrlich. He suggested that cancer cells frequently arise in the body but are recognized as foreign and eliminated by the immune system. Some 50 years later, Lewis Thomas suggested that the cell-mediated branch of the immune system had evolved to patrol the body and eliminate cancer cells. According to these concepts, tumors arise only if cancer cells are able to escape immune surveillance, either by reducing their expression of tumor antigens or by an impairment in the immune response to these cells.

Among the early observations that seemed to support the immune surveillance theory was the increased incidence of cancer in transplantation patients on immunosuppressive drugs. Other findings, however, were difficult to reconcile with this theory. Nude mice, for example, lack a thymus and consequently lack functional T cells. According to the immune surveillance theory, these mice should show an increase in cancer, instead, nude mice are no more susceptible to cancer than other mice. Furthermore, although individuals on immunosuppressive drugs do show an increased incidence of cancers of the immune system, other common cancers (e.g., lung, breast, and colon cancer) are not increased in these individuals, contrary to what the theory predicts. One possible explanation for the selective increase in immune-system cancers is that the immunosuppressive agents themselves may exert a direct carcinogenic effect on immune cells.

Experimental data concerning the effect of tumor-cell dosage on the ability of the immune system to respond also are incompatible with the immune surveillance theory. For example, animals injected with very low or very high doses of tumor cells develop tumors, whereas those injected with intermediate doses do not. The mechanism by which a low dose of tumor cells “sneaks through” is difficult to reconcile with the immune surveillance theory. Finally, this theory assumes that cancer cells and normal cells exhibit qualitative antigen differences. In fact, as stated earlier, many types of tumors do not express tumor-specific antigens, and any immune response that develops must be induced by quantitative differences in antigen expression by normal cells and tumor cells. However, tumors induced by viruses would be expected to express some antigens encoded by the viral genome. These antigens are qualitatively different from those expressed by normal tissues and would be expected to attract the attention of the immune system. In fact, there are many examples of specific immune responses to virally induced tumors.

Nevertheless, apart from tumors caused by viruses, the basic concept of the immune surveillance theory—that malignant tumors arise only if the immune system is somehow impaired or if the tumor cells lose their immunogenicity, enabling them to escape immune surveillance—at this time

remains unproved. In spite of this, it is clear that an immune response can be generated to tumor cells, and therapeutic approaches aimed at increasing that response may serve as a defense against malignant cells.

Tumor Evasion of the Immune System

Although the immune system clearly can respond to tumor cells, the fact that so many individuals die each year from cancer suggests that the immune response to tumor cells is often ineffective. This section describes several mechanisms by which tumor cells appear to evade the immune system.

Anti-Tumor Antibodies Can Enhance Tumor Growth

Following the discovery that antibodies could be produced to tumor-specific antigens, attempts were made to protect animals against tumor growth by active immunization with tumor antigens or by passive immunization with antitumor antibodies. Much to the surprise of the researchers, these immunizations did not protect against tumor growth; in many cases, they actually enhanced growth of the tumor.

The tumor-enhancing ability of immune sera subsequently was studied in cell-mediated lympholysis (CML) reactions *in vitro*. Serum taken from animals with progressive tumor growth was found to block the CML reaction, whereas serum taken from animals with regressing tumors had little or no blocking activity. K. E. and I. Hellstrom extended these findings by showing that children with progressive neuroblastoma had high levels of some kind of blocking factor in their sera and that children with regressive neuroblastoma did not have such factors. Since these first reports, blocking factors have been found to be associated with a number of human tumors.

In some cases, antitumor antibody itself acts as a blocking factor. Presumably the antibody binds to tumor-specific antigens and masks the antigens from cytotoxic T cells. In many cases, the blocking factors are not antibodies alone but rather antibodies complexed with tumor antigens. Although these immune complexes have been shown to block the CTL response, the mechanism of this inhibition is not known. The complexes also may inhibit ADCC by binding to Fc receptors on NK cells or macrophages and blocking their activity.

Antibodies Can Modulate Tumor Antigens

Certain tumor-specific antigens have been observed to disappear from the surface of tumor cells in the presence of serum antibody and then to reappear after the antibody is no longer present. This phenomenon, called antigenic modulation, is readily observed when leukemic T cells are injected into mice previously immunized with a leukemic T-cell antigen (TL antigen). These mice develop high titers of anti-TL

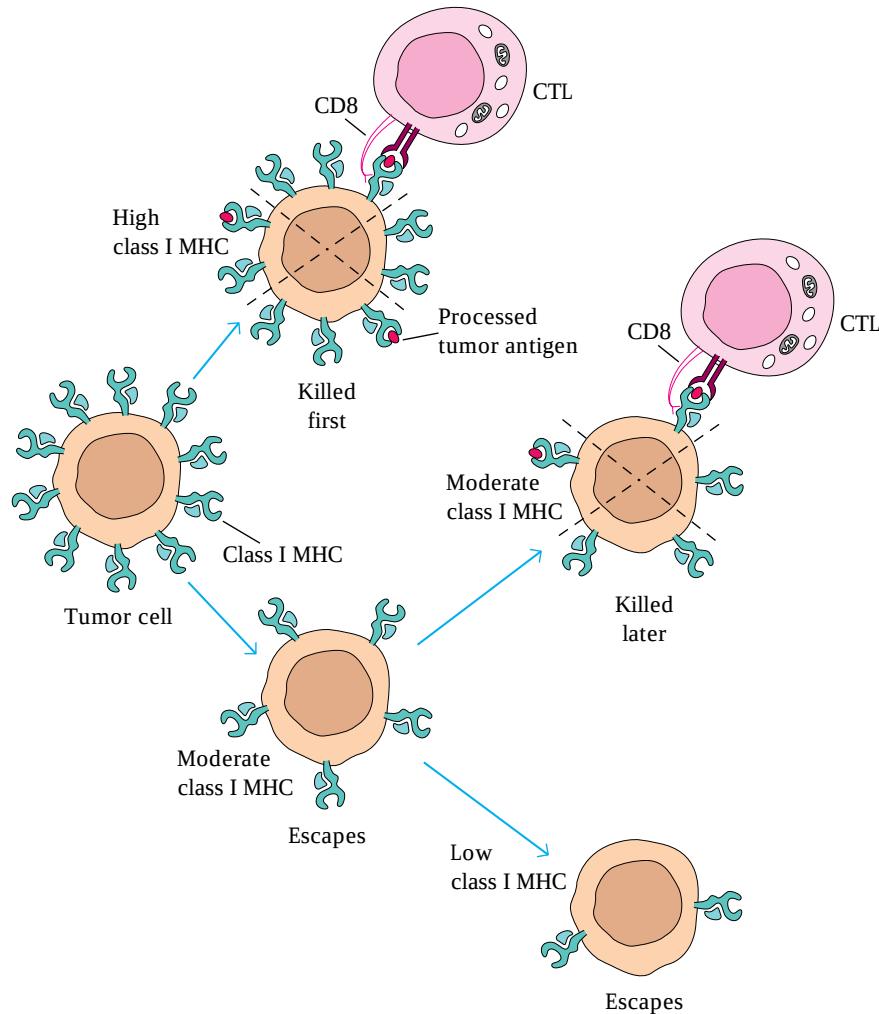


FIGURE 22-10 Down-regulation of class I MHC expression on tumor cells may allow a tumor to escape CTL-mediated recognition. The immune response may play a role in selecting for tumor cells expressing lower levels of class I MHC molecules by preferentially elim-

inating those cells expressing high levels of class I molecules. With time, malignant tumor cells may express progressively fewer MHC molecules and thus escape CTL-mediated destruction.

antibody, which binds to the TL antigen on the leukemic cells and induces capping, endocytosis, and/or shedding of the antigen-antibody complex. As long as antibody is present, these leukemic T cells fail to display the TL antigen and thus cannot be eliminated.

Tumor Cells Frequently Express Low Levels of Class I MHC Molecules

Since CD8⁺ CTLs recognize only antigen associated with class I MHC molecules, any alteration in the expression of class I MHC molecules on tumor cells may exert a profound effect on the CTL-mediated immune response. Malignant transformation of cells is often associated with a reduction (or even a complete loss) of class I MHC molecules, and a number of tumors have been shown to express decreased lev-

els of class I MHC molecules. The decrease in class I MHC expression can be accompanied by progressive tumor growth, and so the absence of MHC molecules on a tumor is generally an indication of a poor prognosis. As illustrated in Figure 22-10, the immune response itself may play a role in selecting tumor cells with decreased class I MHC expression.

Tumor Cells May Provide Poor Co-Stimulatory Signals

T-cell activation requires an activating signal, triggered by recognition of a peptide–MHC molecule complex by the T-cell receptor, and a co-stimulatory signal, triggered by the interaction of B7 on antigen-presenting cells with CD28 on the T cells. Both signals are needed to induce IL-2 production and proliferation of T cells. The poor immunogenicity of

many tumor cells may be due in large part to lack of the co-stimulatory molecules. Without sufficient numbers of antigen-presenting cells in the immediate vicinity of a tumor, the T cells will receive only a partial activating signal, which may lead to clonal anergy.

Cancer Immunotherapy

Although various immune responses can be generated to tumor cells, the response frequently is not sufficient to prevent tumor growth. One approach to cancer treatment is to augment or supplement these natural defense mechanisms. Several types of cancer immunotherapy in current use or under development are described in this concluding section.

Manipulation of Co-Stimulatory Signals Can Enhance Immunity

Several research groups have demonstrated that tumor immunity can be enhanced by providing the co-stimulatory signal necessary for activation of CTL precursors (CTL-Ps). When mouse CTL-Ps are incubated with melanoma cells in vitro, antigen recognition occurs, but in the absence of a co-stimulatory signal, the CTL-Ps do not proliferate and differentiate into effector CTLs. However, when the melanoma cells are transfected with the gene that encodes the B7 ligand, then the CTL-Ps differentiate into effector CTLs.

These findings offer the possibility that B7-transfected tumor cells might be used to induce a CTL response in vivo. For instance, when P. Linsley, L. Chen, and their colleagues injected melanoma-bearing mice with B7⁺ melanoma cells, the melanomas completely regressed in more than 40% of the mice. S. Townsend and J. Allison used a similar approach to vaccinate mice against malignant melanoma. Normal mice were first immunized with irradiated, B7-transfected melanoma cells and then challenged with unaltered malignant melanoma cells. The “vaccine” was found to protect a high percentage of the mice (Figure 22-11a). It is hoped that a similar vaccine might prevent metastasis after surgical removal of a primary melanoma in human patients.

Because human melanoma antigens are shared by a number of different human tumors, it might be possible to generate a panel of B7-transfected melanoma cell lines that are typed for tumor-antigen expression and for HLA expression. In this approach, the tumor antigen(s) expressed by a patient's tumor would be determined, and then the patient would be vaccinated with an irradiated B7-transfected cell line that expresses similar tumor antigen(s).

Enhancement of APC Activity Can Modulate Tumor Immunity

Mouse dendritic cells cultured in GM-CSF and incubated with tumor fragments, then reinfused into the mice, have been shown to activate both TH cells and CTLs specific for

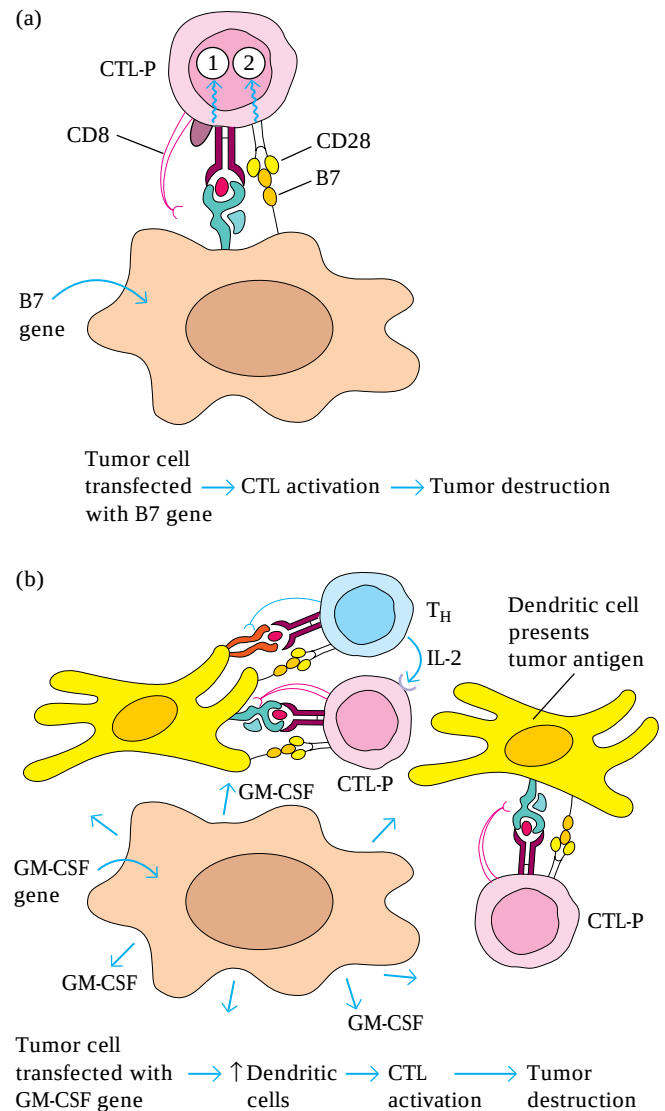


FIGURE 22-11 Use of transfected tumor cells for cancer immunotherapy. (a) Tumor cells transfected with the B7 gene express the co-stimulatory B7 molecule, enabling them to provide both activating signal (1) and co-stimulatory signal (2) to CTL-Ps. As a result of the combined signals, the CTL-Ps differentiate into effector CTLs, which can mediate tumor destruction. In effect, the transfected tumor cell acts as an antigen-presenting cell. (b) Transfection of tumor cells with the gene encoding GM-CSF allows the tumor cells to secrete high levels of GM-CSF. This cytokine will activate dendritic cells in the vicinity of the tumor, enabling the dendritic cells to present tumor antigens to both TH cells and CTL-Ps.

the tumor antigens. When the mice were subsequently challenged with live tumor cells, they displayed tumor immunity. These experiments have led to a number of approaches aimed at expanding the population of antigen-presenting cells, so that these cells can activate TH cells or CTLs specific for tumor antigens.

One approach that has been tried is to transfect tumor cells with the gene encoding GM-CSF. These engineered tumor cells, when reinfused into the patient, will secrete GM-CSF, enhancing the differentiation and activation of host antigen-presenting cells, especially dendritic cells. As these dendritic cells accumulate around the tumor cells, the GM-CSF secreted by the tumor cells will enhance the presentation of tumor antigens to TH cells and CTLs by the dendritic cells (Figure 22-11b).

Another way to expand the dendritic-cell population is to culture dendritic cells from peripheral-blood progenitor cells in the presence of GM-CSF, TNF- α , and IL-4. These three cytokines induce the generation of large numbers of dendritic cells. There is some hope that, if these dendritic cells are pulsed with tumor fragments and then reintroduced into the patient, they can activate TH and TC cells specific for the tumor antigens. Whether these hopes are justified will be determined by further investigation.

A number of adjuvants, including the attenuated strains of *Mycobacterium bovis* called bacillus Calmette-Guerin (BCG) and *Corynebacterium parvum*, have been used to boost tumor immunity. These adjuvants activate macrophages, increasing their expression of various cytokines, class II MHC molecules, and the B7 co-stimulatory molecule. These activated macrophages are better activators of TH cells, resulting in generalized increases in both humoral and cell-mediated responses. Thus far, adjuvants have shown only modest therapeutic results.

Cytokine Therapy Can Augment Immune Responses to Tumors

The isolation and cloning of the various cytokine genes has facilitated their large-scale production. A variety of experimental and clinical approaches have been developed to use recombinant cytokines, either singly or in combination, to augment the immune response against cancer. Among the cytokines that have been evaluated in cancer immunotherapy are IFN- α , β , and γ ; IL-1, IL-2, IL-4, IL-5, and IL-12; GM-CSF; and TNF. Although these trials have produced occasional encouraging results, many obstacles remain to the successful use of this type of cancer immunotherapy.

The most notable obstacle is the complexity of the cytokine network itself. This complexity makes it very difficult to know precisely how intervention with a given recombinant cytokine will affect the production of other cytokines. And since some cytokines act antagonistically, it is possible that intervention with a recombinant cytokine designed to enhance a particular branch of the immune response may actually lead to suppression. In addition, cytokine immunotherapy is plagued by the difficulty of administering the cytokines locally. In some cases, systemic administration of high levels of a given cytokine has been shown to lead to serious and even life-threatening consequences. Although the results of several experimental and clinical trials of cytokine therapy for cancer are discussed

here, it is important to keep in mind that this therapeutic approach is still in its infancy.

INTERFERONS

Large quantities of purified recombinant preparations of the interferons, IFN- α , IFN- β , and IFN- γ , are now available, each of which has shown some promise in the treatment of human cancer. To date, most of the clinical trials have involved IFN- α . Daily injections of recombinant IFN- α have been shown to induce partial or complete tumor regression in some patients with hematologic malignancies such as leukemias, lymphomas, and myelomas and with solid tumors such as melanoma, Kaposi's sarcoma, renal cancer, and breast cancer.

Interferon-mediated antitumor activity may involve several mechanisms. All three types of interferon have been shown to increase class I MHC expression on tumor cells; IFN- γ has also been shown to increase class II MHC expression on macrophages. Given the evidence for decreased levels of class I MHC molecules on malignant tumors, the interferons may act by restoring MHC expression, thereby increasing CTL activity against tumors. In addition, the interferons have been shown to inhibit cell division of both normal and malignant transformed cells in vitro. It is possible that some of the antitumor effects of the interferons are related to this ability to directly inhibit tumor-cell proliferation. Finally, IFN- γ directly or indirectly increases the activity of TC cells, macrophages, and NK cells, all of which play a role in the immune response to tumor cells.

TUMOR NECROSIS FACTORS

In some instances, the tumor necrosis factors TNF- α and TNF- β have been shown to exhibit direct antitumor activity, killing some tumor cells and reducing the rate of proliferation of others while sparing normal cells (Figure 22-12). In the presence of TNF- α or TNF- β , a tumor undergoes visible hemorrhagic necrosis and regression. TNF- α has also been shown to inhibit tumor-induced vascularization (angiogenesis) by damaging the vascular endothelial cells in the vicinity of a tumor, thereby decreasing the flow of blood and oxygen that is necessary for progressive tumor growth.

IN VITRO-ACTIVATED LAK AND TIL CELLS

Animal studies have shown that lymphocytes can be activated against tumor antigens in vitro by culturing them with x-irradiated tumor cells in the presence of IL-2 and added tumor antigens. These activated lymphocytes mediate more effective tumor destruction than untreated lymphocytes when they are reinjected into the original tumor-bearing animal. It is difficult, however, to activate in vitro enough lymphocytes with antitumor specificity to be useful in cancer therapy.

While sensitizing lymphocytes to tumor antigens by this method, S. Rosenberg discovered that, in the presence of high

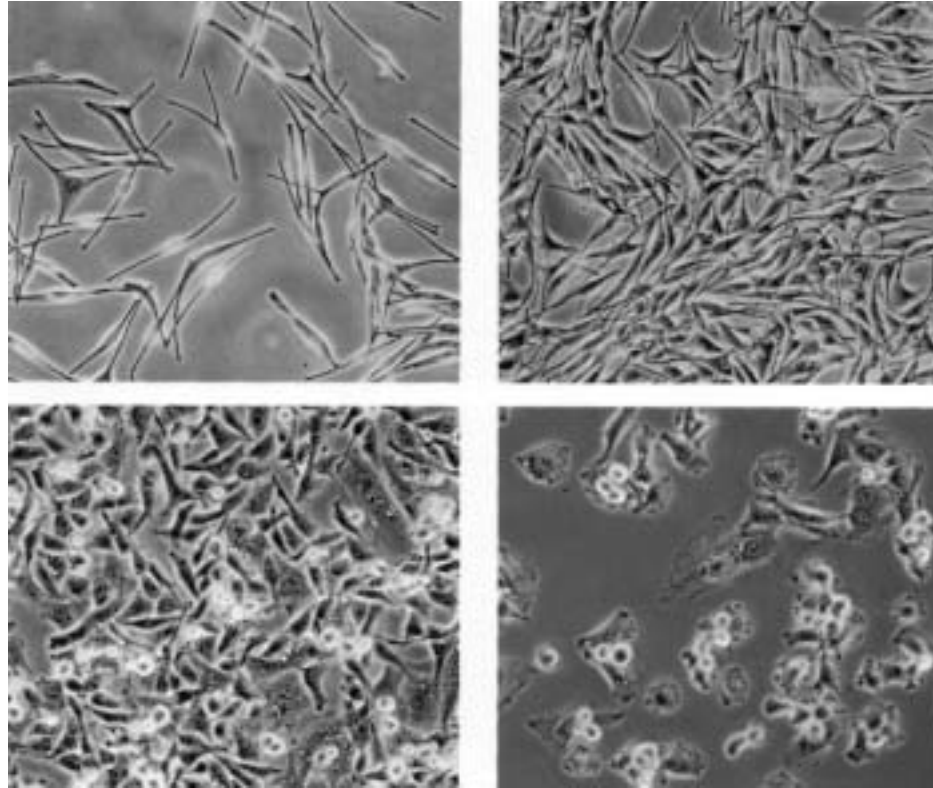


FIGURE 22-12 Photomicrographs of cultured normal melanocytes (*top*) and cultured cancerous melanoma cells (*bottom*) in the presence (*left*) and absence (*right*) of tumor necrosis factor (TNF- α). Note

that, in the presence of TNF- α , the cancer cells stop proliferating, whereas TNF- α has no inhibitory effect on proliferation of the normal cells. [From L. J. Old, 1988, *Sci. Am.* **258**(5):59.]

concentrations of cloned IL-2 but without the addition of tumor antigens, large numbers of activated lymphoid cells were generated that could kill fresh tumor cells but not normal cells. He called these cells **lymphokine-activated killer (LAK) cells**. In one study, for example, Rosenberg found that infusion of LAK cells plus recombinant IL-2 into tumor-bearing animals mediated effective tumor-cell destruction (Figure 22-13). LAK-cell populations are typically >90% activated NK cells. However, small numbers of TCR-bearing cells are present in LAK populations and it is possible that these may also contribute to their tumoricidal activity.

Because large numbers of LAK cells can be generated in vitro and because these cells are active against a wide variety of tumors, their effectiveness in human tumor immunotherapy has been evaluated in several clinical trials. In these trials, peripheral-blood lymphocytes were removed from patients with various advanced metastatic cancers and were activated in vitro to generate LAK cells. In an early study, patients were then infused with their autologous LAK cells together with IL-2. In this trial, which involved 25 patients, cancer regression was seen in some patients. Subsequently, a more extensive trial with 222 patients resulted in complete regression in

16 patients. However, a number of undesirable side effects are associated with the high levels of IL-2 required for LAK-cell activity. The most noteworthy is vascular leak syndrome, in which lymphoid cells and plasma emigrate from the peripheral blood into the tissues, leading to shock.

Tumors contain lymphocytes that have infiltrated the tumor and presumably are taking part in an antitumor response. By taking small biopsy samples of tumors, one can obtain a population of these lymphocytes and expand it in vitro with IL-2. These activated **tumor-infiltrating lymphocytes** are called **TILs**. Many TILs have a wide range of antitumor activity and appear to be indistinguishable from LAK cells. However, some TILs cells have specific cytolytic activity against their autologous tumor. These tumor-specific TILs are of interest because they have increased antitumor activity and require 100-fold lower levels of IL-2 for their activity than LAK cells do. In one study, TIL populations were expanded in vitro from biopsy samples taken from patients with malignant melanoma, renal-cell carcinoma, and small-cell lung cancer. The expanded populations of TILs were reinjected into autologous patients together with continuous infusions of recombinant IL-2. Renal-cell carcinomas and malignant

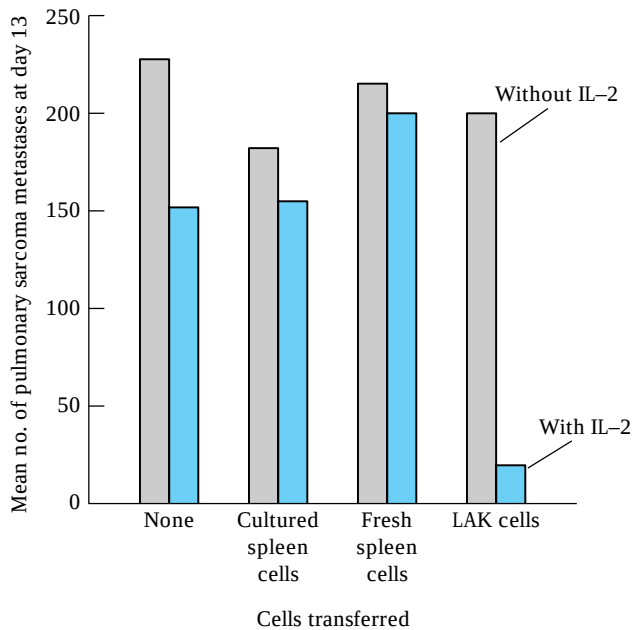


FIGURE 22-13 Experimental demonstration of tumor-destroying activity of LAK cells plus IL-2. Spleen cells or LAK cells, in the presence or absence of recombinant IL-2, were infused into mice with pulmonary sarcoma. The animals were evaluated 13 days later for the number of pulmonary sarcoma metastases. The LAK cells were prepared by isolating lymphocytes from tumor-bearing animals and incubating them *in vitro* with high concentrations of IL-2. Note that LAK cells caused tumor regression only when IL-2 was also infused. [Data from S. Rosenberg *et al.*, 1988, *Ann. Int. Med.*, **108**:853.]

melanomas showed partial regression in 29% and 23% of the patients, respectively.

Monoclonal Antibodies Are Effective in Treating Some Tumors

Monoclonal antibodies have been used in various ways as experimental immunotherapeutic agents for cancer. For example, anti-idiotypic monoclonal antibodies have been used with some success in treating human B-cell lymphomas and T-cell leukemias. In one remarkable study, R. Levy and his colleagues successfully treated a 64-year-old man with terminal B-cell lymphoma. At the time of treatment, the lymphoma had metastasized to the liver, spleen, bone marrow, and peripheral blood. Because this was a B-cell cancer, the membrane-bound antibody on all the cancerous cells had the same idiotype. By the procedure outlined in Figure 22-14, these researchers produced mouse monoclonal antibody specific for the B-lymphoma idiotype. When this mouse monoclonal anti-idiotypic antibody was injected into the patient, it bound specifically to the B-lymphoma cells, because these cells expressed that particular idiotype. Since B-lymphoma

cells are susceptible to complement-mediated lysis, the monoclonal antibody activated the complement system and lysed the lymphoma cells without harming other cells. After four injections with this anti-idiotypic monoclonal antibody, the tumors began to shrink, and this patient entered an unusually long period of complete remission.

However, this approach requires that a custom monoclonal antibody be raised for each lymphoma patient. This is prohibitively expensive and cannot be used as a general therapeutic approach for the thousands of patients diagnosed each year with B lymphoma. Recently, Levy and his colleagues have used direct immunization to recruit the immune systems of patients to an attack against their B lymphoma. In a clinical trial with 41 B-cell lymphoma patients, the genes encoding the rearranged immunoglobulin genes of the lymphomas of each patient were isolated and used to encode the synthesis of recombinant immunoglobulin that bore the idiotype typical of the patient's tumor. Each of these Igs was coupled to keyhole limpet hemocyanin (KLH), a mollusk protein that is often used as a carrier protein because of its efficient recruitment of T-cell help. The patients were immunized with their own tumor-specific antigens, the idiotypically unique immunoglobulins produced by their own lymphomas. About 50% of the patients developed anti-idiotypic antibodies against their tumors. Significantly, improved clinical outcomes were seen in the 20 patients with anti-idiotypic responses, but not in the others. In fact, 2 of these 20 experienced complete remission.

Despite its promise, the anti-idiotypic approach is by its very nature patient-specific. A more general monoclonal-antibody therapy for B-cell lymphoma is based on the fact that most B cells, whether normal or cancerous, bear lineage-distinctive antigens. One such determinant, CD20, has been the target of intensive efforts; a monoclonal antibody to it, raised in mice and engineered to contain mostly human sequences, has been useful in the treatment of B-cell lymphoma (see Clinical Focus, Chapter 5). Aside from CD20, a number of tumor-associated antigens (Table 22-4) are being tested in clinical trials for their suitability as targets for antibody-mediated anti-tumor therapy.

A variety of tumors express significantly increased levels of growth-factor receptors, which are promising targets for anti-tumor monoclonal antibodies. For example, in 25 to 30 percent of women with metastatic breast cancer, a genetic alteration of the tumor cells results in the increased expression of HER2, an epidermal-growth-factor-like receptor. An anti-HER2 monoclonal antibody was raised in mice and the genes encoding it were isolated. Except for the sequences encoding the antibody's CDRs, the mouse Ig sequences were replaced with human Ig counterparts. This prevents the generation of human anti-mouse antibodies (HAMAs) and allows the patient to receive repeated doses of the "humanized" anti-HER2 in large amounts (100 milligrams or more). Preparations of this antibody, called Herceptin, are now commercially available for the treatment of HER2-receptor-bearing breast cancers (see Clinical Focus, Chapter 5). Monoclonal antibodies also have been used to prepare tumor-

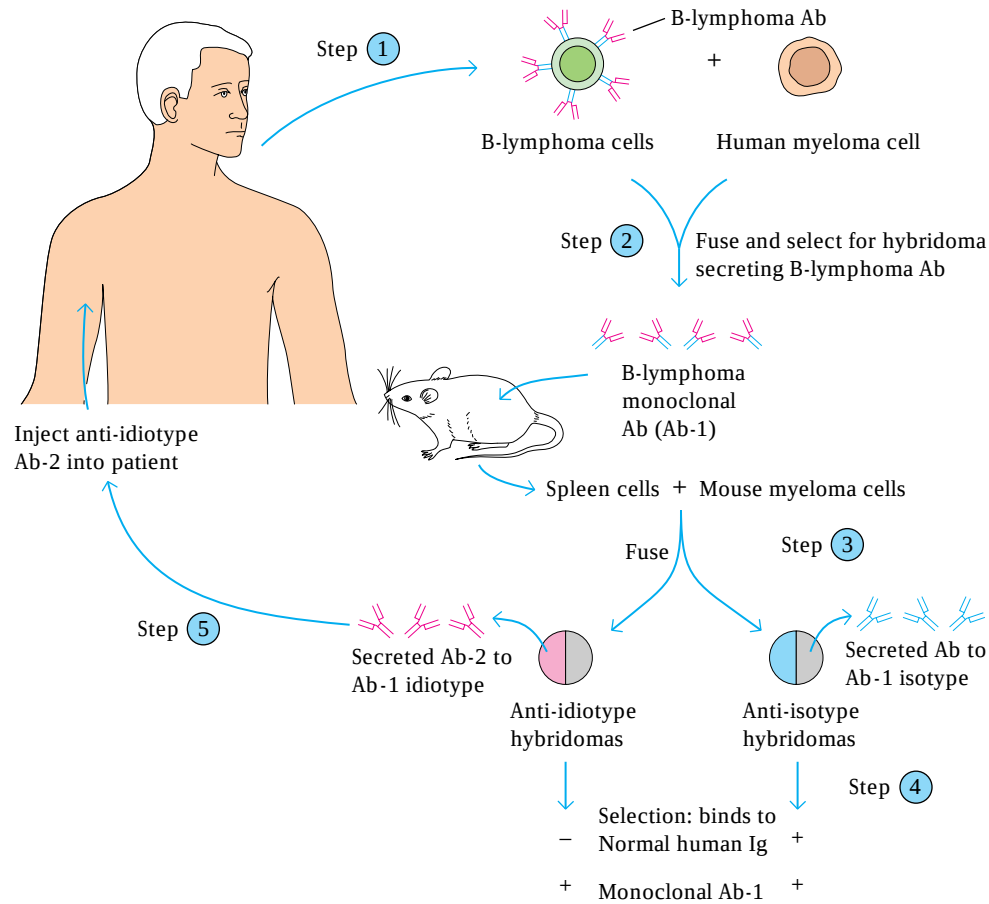


FIGURE 22-14 Treatment of B-cell lymphoma with monoclonal antibody specific for idiotypic determinants on the cancer cells. Because all the lymphoma cells are derived from a single transformed B cell, they all express membrane-bound antibody (Ab-1) with the same idiotype (i.e., the same antigenic specificity). In the procedure illustrated, monoclonal

anti-idiotypic antibody (Ab-2) against the B-lymphoma membrane-bound antibody was produced (steps 1–4). When this anti-idiotypic antibody was injected into the patient (step 5), it bound selectively to B-lymphoma cells, which then were susceptible to complement-plus-antibody lysis.

specific anti-tumor agents. In this approach, antibodies to tumor-specific or tumor-associated antigens are coupled with radioactive isotopes, chemotherapy drugs, or potent toxins of biological origin. In such “guided missile” therapies, the toxic agents are delivered specifically to tumor cells. This focuses the toxic effects on the tumor and spares normal tissues. Reagents known as **immunotoxins** have been constructed by coupling the inhibitor chain of a toxin (e.g., diphtheria toxin) to an antibody against a tumor-specific or tumor-associated antigen (see Figure 4-23). In vitro studies have demonstrated that these “magic bullets” can kill tumor cells without harming normal cells. Immunotoxins specific for tumor antigens in a variety of cancers (e.g., melanoma, colorectal carcinoma, metastatic breast carcinoma, and various lymphomas and leukemias) have been evaluated in phase I or phase II clinical trials. In a number of trials, significant numbers of leukemia and lymphoma patients exhibited partial or complete remission. However in a number of cases, the clinical responses in patients with larger tumor masses were disap-

pointing. In some of these patients, the sheer size of the tumor may render most of its cells inaccessible to the immunotoxin.

SUMMARY

- Tumor cells differ from normal cells in numerous ways. In particular, changes in the regulation of growth of tumor cells allow them to proliferate indefinitely, then invade the underlying tissue, and eventually metastasize to other tissues (see Figure 22-1). Normal cells can be transformed in vitro by chemical and physical carcinogens and by transforming viruses. Transformed cells exhibit altered growth properties and are sometimes capable of inducing cancer when they are injected into animals.
- Proto-oncogenes encode proteins involved in control of normal cellular growth. The conversion of proto-oncogenes to oncogenes is one of the key steps in the induction of most human cancer. This conversion may result from mutation in

TABLE 22-4

Some tumor-associated antigens under examination as potential targets for monoclonal-antibody therapy

Tumor antigen	Tumor type	Target antigen
LYMPHOID CELL-SURFACE MARKERS		
T-cell marker	T-cell leukemia/lymphoma	CD5
B-cell marker	B-cell lymphoma	CD20
Hematopoietic-cell marker	Acute myeloblastic leukemia	CD45
Anti-idiotypic	B-cell lymphoma	Immunoglobulin
NONLYMPHOID TISSUE MARKERS		
Cell-surface antigens		
Carcinoembryonic antigen (CEA)	Colon cancer (some others)	Glycoprotein
MUC1	Breast cancer	Glycoprotein
Gangliosides such as GD2 and GD3	Neuroectodermal tumors	Glycolipids associated with neural tissue
Growth-factor receptors		
Epidermal growth-factor receptor (EGFR)	Some lung, head, neck, and breast tumors	EGF-binding cell surface protein
HER2 (and EFG-like receptor)	Breast and ovarian tumors	Cell-surface EGF-binding protein with homology to EGFR

SOURCE: Adapted from Scott and Welt, 1997, *Curr. Opin. Immunol.* 9:717.

an oncogene, its translocation, or its amplification (see Figure 22-2).

- A number of B- and T-cell leukemias and lymphomas are associated with translocated proto-oncogenes. In its new site, the translocated gene may come under the influence of enhancers or promoters that cause its transcription at higher levels than usual (see Figure 22-5).
- Tumor cells display tumor-specific antigens and the more common tumor-associated antigens. Among the latter are oncofetal antigens, (see Table 22-3) and increased levels of normal oncogene products (see Figure 22-6). In contrast to tumor antigens induced by chemicals or radiation, virally encoded tumor antigens are shared by all tumors induced by the same virus.
- The tumor antigens recognized by T cells fall into one of four major categories: antigens encoded by genes with tumor-specific expression; antigens encoded by variant forms of normal genes that have been altered by mutation; certain antigens normally expressed only at certain stages of differentiation or differentiation lineages; antigens that are overexpressed in particular tumors.
- The use of a variety of genetic, biochemical, and immunological approaches has allowed the identification of several tumor-associated antigens (see Table 22-4). In many cases the antigen is expressed on more than one type of tumor.

Common tumor antigens offer hope for the design of better therapies, detection, and monitoring, and have important implications for the possibility of anti-tumor immunization.

- The immune response to tumors includes CTL-mediated lysis, NK-cell activity, macrophage-mediated tumor destruction, and destruction mediated by ADCC. Several cytotoxic factors, including TNF- α and TNF- β , help to mediate tumor-cell killing. Tumors may evade the immune response by modulating their tumor antigens, by reducing their expression of class I MHC molecules, and by antibody-mediated or immune complex-mediated inhibition of CTL activity.
- Experimental cancer immunotherapy is exploring a variety of approaches. Some of these are the enhancement of the co-stimulatory signal required for T-cell activation (see Figure 22-11a), genetically engineering tumor cells to secrete cytokines that may increase the intensity of the immune response against them (see Figure 22-11b), the therapeutic use of cytokines (see Figure 22-12), and ways of increasing the activity of antigen-presenting cells.
- A number of encouraging clinical results have been obtained with therapy using monoclonal antibodies against tumor-associated and (in a few cases) tumor-specific antigens (see Figure 22-14). Coupling of antibodies against

TABLE 22-5 Tumor-associated and tumor-specific antigen peptides recognized by human T cells

Human tumor	Protein	Peptide
Many melanomas, esophageal carcinomas, non small-cell lung carcinomas and hepatocellular carcinomas	MAGE-1	EADPTGHSY and SAYGEPRKL
Melanoma	Tyrosinase	MLLAVLYCL, YMNGTMSQV, YMDGTMSQV, and others
Colon cancer	Carcinoembryonic antigen (CEA)	YLSGANLNL
Breast and ovarian cancer	HER2/NEU	KIFGSLAFL
Head and neck squamous-cell carcinoma	Caspase 8	FPSDSWCYF
Chronic myeloid leukemia	<i>bcr-abl</i> fusion protein (product of a fusion of an Ig gene with the <i>abl</i> gene)	ATGFKQSSKALQRPVAS
Prostatic cancer	PSA	FLTPKKKLQCV and VISNDVCAQV

SOURCE: Adapted from B. Van Den Eynde and P. van der Bruggen, 1996, *Curr. Opin. Immunol.* **9**:684.

tumor antigens with toxins, chemotherapeutic agents, or radioactive elements is being examined. The expectation is that such strategies will focus the toxic effects of these agents on the tumor and spare normal cells their deleterious effects.

- Key elements in the design of strategies for vaccination against cancer are the identification of significant tumor antigens by genetic or biochemical approaches; the development of strategies for the effective presentation of tumor antigens; and the generation of activated populations of helper or cytotoxic T cells.

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USEFUL WEB SITES

<http://www.oncolink.upenn.edu/>

Oncolink is a site that offers comprehensive information about many types of cancer. It is a good source of information about cancer research and advances in cancer therapy. The site is regularly updated and it includes many useful links to other resources.

http://www.cancer.org/index_4up.html

This is the Web site of the American Cancer Society. It contains a great deal of information on the incidence, treatment, prevention of cancer. The site also highlights significant achievements in cancer research.

<http://www.cytopathnet.org/>

A good resource for information on the cytological examination of tumors and on matters related to staining patterns that are typical of the cell populations found in a number of cancers.

Study Questions

CLINICAL FOCUS QUESTION You are an oncologist and wish to treat a patient with one of the newly available cancer vaccines, but the only tumor from this patient is preserved in formaldehyde. Can you still use a vaccine? If so, what type of vaccine is available for your use? If you have a tumor sample containing living cells, are there other types of vaccines available?

1. Indicate whether each of the following statements is true or false. If you think a statement is false, explain why.

- Hereditary retinoblastoma results from overexpression of a cellular oncogene.
- Translocation of *c-myc* gene is found in many patients with Burkitt's lymphoma.
- Multiple copies of cellular oncogenes are sometimes observed in cancer cells.
- Viral integration into the cellular genome may convert a proto-oncogene into a transforming oncogene.
- All oncogenic retroviruses carry viral oncogenes.
- The immune response against a virus-induced tumor protects against another tumor induced by the same virus.
- LAK cells are tumor specific.

2. You are a clinical immunologist studying acute lymphoblastic leukemia (ALL). Leukemic cells from most patients with ALL have the morphology of lymphocytes but do not express cell-surface markers characteristic of mature B or T cells. You have isolated cells from ALL patients that do not express membrane Ig but do react with monoclonal antibody against a normal pre-B-cell marker (B-200). You therefore suspect that these leukemic cells are pre-B cells. How would you use genetic analysis to confirm that the leukemic cells are committed to the B-cell lineage?

3. In a recent experiment, melanoma cells were isolated from patients with early or advanced stages of malignant melanoma. At the same time, T cells specific for tetanus-toxoid antigen were isolated and cloned from each patient.

- When early-stage melanoma cells were cultured together with tetanus-toxoid antigen and the tetanus toxoid-specific T-cell clones, the T-cell clones were observed to proliferate. This proliferation was blocked by addition of chloroquine or by addition of monoclonal antibody to HLA-DR. Proliferation was not blocked by addition of monoclonal antibody to HLA-A, -B, -DQ, or -DP. What might these findings indicate about the early-stage melanoma cells in this experimental system?
- When the same experiment was repeated with advanced-stage melanoma cells, the tetanus-toxoid T-cell clones failed to proliferate in response to the tetanus-toxoid antigen. What might this indicate about advanced-stage melanoma cells?
- When early and advanced malignant melanoma cells were fixed with paraformaldehyde and incubated with processed tetanus toxoid, only the early-stage melanoma cells could induce proliferation of the tetanus-toxoid-T-cell clones. What might this indicate about early-stage melanoma cells?
- How might you confirm your hypothesis experimentally?

4. What are three likely sources of tumor antigens?

5. Various cytokines have been evaluated for use in tumor immunotherapy. Describe four mechanisms by which cytokines mediate antitumor effects and the cytokines that induce each type of effect.

6. Infusion of transfected melanoma cells into cancer patients is a promising immunotherapy.

- Which two genes have been transfected into melanoma cells for this purpose? What is the rationale behind use of each of these genes?
- Why might use of such transfected melanoma cells also be effective in treating other types of cancers?

Experimental Systems

EXPERIMENTAL SYSTEMS OF VARIOUS TYPES ARE used to unravel the complex cellular interactions of the immune response. In vivo systems, which involve the whole animal, provide the most natural experimental conditions. However, in vivo systems have a myriad of unknown and uncontrollable cellular interactions that add ambiguity to the interpretation of data. At the other extreme are in vitro systems, in which defined populations of lymphocytes are studied under controlled and consequently repeatable conditions; in vitro systems can be simplified to the extent that individual cellular interactions can be studied effectively. Yet they have their own limitations, the most notable of which is their artificiality. For example, providing antigen to purified B cells in vitro does not stimulate maximal antibody production unless T cells are present. Therefore a study of antibody production in an artificial in vitro system that lacks T cells could lead to the incorrect conclusion that B cells do not synthesize high levels of antibodies. One must ask whether a cellular response observed in vitro reflects reality or is a product of the unique conditions of the in vitro system itself.

This chapter describes some of the experimental systems routinely used to study the immune system. It also covers some recombinant DNA techniques that have revolutionized the study of the immune system in the past decade or so. Other chapters also cover experimental systems and techniques in detail. Table 23-1 lists them and directs the reader to the appropriate location for a description.

Experimental Animal Models

The study of the immune system in vertebrates requires suitable animal models. The choice of an animal depends on its suitability for attaining a particular research goal. If large amounts of antiserum are sought, a rabbit, goat, sheep, or horse might be an appropriate experimental animal. If the goal is development of a protective vaccine, the animal chosen must be susceptible to the infectious agent so that the efficacy of the vaccine can be assessed. Mice or rabbits can be used for vaccine development if they are susceptible to the pathogen. But if growth of the infectious agent is limited to humans and primates, vaccine development may require the use of monkeys, chimpanzees, or baboons.

For most basic research in immunology, mice have been the experimental animal of choice. They are easy to handle,

ART TK

Addition of Expression Profile of Diffuse Large B-cell Lymphoma.

- [Experimental Animal Models](#)
- [Cell-Culture Systems](#)
- [Protein Biochemistry](#)
- [Recombinant DNA Technology](#)
- [Analysis of DNA Regulatory Sequences](#)
- [Gene Transfer into Mammalian Cells](#)
- [Microarrays—An Approach for Analyzing Patterns of Gene Expression](#)

are genetically well characterized, and have a rapid breeding cycle. The immune system of the mouse has been characterized more extensively than that of any other species. The value of basic research in the mouse system is highlighted by the enormous impact this research has had on clinical intervention in human disease.

Inbred Strains Can Reduce Experimental Variation

To control experimental variation caused by differences in the genetic backgrounds of experimental animals, immunologists often work with inbred strains—that is, genetically identical animals produced by inbreeding. The rapid breeding cycle of mice makes them particularly well suited for the production of inbred strains, which are developed by repeated inbreeding between brother and sister littermates. In this way the heterozygosity of alleles that is normally found in randomly outbred mice is replaced by homozygosity at all

TABLE 23-1

Immunological methods described in other chapters

Method	Location
Bone-marrow transplantation	Ch. 2 Clinical Focus
Preparation of immunotoxins	Fig. 4-22
Genetic engineering of chimeric mouse-human monoclonal antibodies	Fig. 5-20 and Ch 5 Clinical Focus
Determination of antibody affinity by equilibrium dialysis	Fig. 6.2
Precipitation reactions	Fig. 6.4
Immunodiffusion and immunoelectrophoresis	Figs. 6.5 and 6.6
Hemagglutination	Fig. 6.7
Radioimmunoassay (RIA)	Fig. 6.9
ELISA assays	Fig. 6.10
ELISPOT assay	Fig. 6.11
Western blotting	Fig. 6.12
Immunoprecipitation	Fig. 6.13
Immunofluorescence	Fig. 6.14
Flow cytometry	Fig. 6.15
Production of congenic mice	Fig. 7-3
Mixed lymphocyte reaction (MLR)	Fig. 14-16
Cell-mediated lympholysis (CML)	Fig. 14-17
Production of vaccinia vector vaccine	Fig. 18-5
Production of multivalent subunit vaccines	Fig. 18-7
HLA typing	Fig. 21-4

loci. Repeated inbreeding for 20 generations usually yields an inbred strain whose progeny are homozygous at more than 98% of all loci. More than 150 different inbred strains of mice are available, each designated by a series of letters and/or numbers (Table 23-2). Most strains can be purchased by immunologists from such suppliers as the Jackson Laboratory in Bar Harbor, Maine. Inbred strains have also been produced in rats, guinea pigs, hamsters, rabbits, and domestic fowl. Because inbred strains of animals are genetically identical (**syngeneic**) within that strain, their immune responses can be studied in the absence of variables introduced by individual genetic differences—an invaluable property. With inbred strains, lymphocyte subpopulations isolated from one animal can be injected into another animal of the same strain without eliciting a rejection reaction. This type of experimental system permitted immunologists to first demonstrate that lymphocytes from an antigen-primed animal could transfer immunity to an unprimed syngeneic recipient.

Adoptive-Transfer Systems Permit the *in Vivo* Examination of Isolated Cell Populations

In some experiments, it is important to eliminate the immune responsiveness of the syngeneic host so that the response of only the transferred lymphocytes can be studied in isolation. This can be accomplished by a technique called **adoptive transfer**: first, the syngeneic host is exposed to x-rays that kill its lymphocytes; then the donor immune cells are introduced. Subjecting a mouse to high doses of x-rays (650–750 rads) can kill 99.99% of its lymphocytes, after which the activities of lymphocytes transplanted from the spleen of a syngeneic donor can be studied without interference from host lymphocytes. If the host's hematopoietic cells might influence an adoptive-transfer experiment, then higher x-ray levels (900–1000 rads) are used to eliminate the entire hematopoietic system. Mice irradiated with such doses will die unless reconstituted with bone marrow from a syngeneic donor.

The adoptive-transfer system has enabled immunologists to study the development of injected lymphoid stem cells in various organs of the recipient, and have facilitated the study of various populations of lymphocytes and of the cellular interactions required to generate an immune response. Such experiments, for instance, first enabled immunologists to show that a T helper cell is necessary for B-cell activation in the humoral response. In these experiments, adoptive transfer of purified B cells or purified T cells did not produce antibody in the irradiated host. Only when both cell populations were transferred was antibody produced in response to antigen.

SCID Mice and SCID-Human Mice Are a Valuable Animal Model for Immunodeficiency

An autosomal recessive mutation resulting in **severe combined immunodeficiency disease (SCID)** developed spontaneously in a strain of mice called CB-17. These CB-17 SCID mice fail to develop mature T and B cells and consequently are severely compromised immunologically. This defect is due to a failure in V(D)J recombination. SCID mice must be housed in a sterile (germ-free) environment, because they cannot fight off microorganisms of even low pathogenicity. The absence of functional T and B cells enables these mice to accept foreign cells and grafts from other strains of mice or even from other species.

Apart from their lack of functional T and B cells, SCID mice appear to be normal in all respects. When normal bone-marrow cells are injected into SCID mice, normal T and B cells develop, and the mice are cured of their immunodeficiency. This finding has made SCID mice a valuable model system for the study of immunodeficiency and the process of differentiation of bone-marrow stem cells into mature T or B cells.

Interest in SCID mice mushroomed when it was found that they could be used to study the human immune system. In this system, portions of human fetal liver, adult thymus,

TABLE 23-2 Some inbred mouse strains commonly used in immunology

Strain	Common substrains	Characteristics
A	A/He A/J A/WySn	High incidence of mammary tumors in some substrains
AKR	AKR/J AKR/N AKR/Cum	High incidence of leukemia <i>Thy 1.2</i> allele in AKR/Cum, and <i>Thy 1.1</i> allele in other substrains (<i>Thy</i> gene encodes a T-cell surface protein)
BALB/c	BALB/cj BALB/c AnN BALB/cBy	Sensitivity to radiation Used in hybridoma technology Many myeloma cell lines were generated in these mice
CBA	CBA/J CBA/H CBA/N	Gene (<i>rd</i>) causing retinal degeneration in CBA/J Gene (<i>xid</i>) causing X-linked immunodeficiency in CBA/N
C3H	C3H/He C3H/HeJ C3H/HeN	Gene (<i>rd</i>) causing retinal degeneration High incidence of mammary tumors in many substrains (these carry a mammary-tumor virus that is passed via maternal milk to offspring)
C57BL/6	C57BL/6J C57BL/6By C57BL/6N	High incidence of hepatomas after irradiation High complement activity
C57BL/10	C57BL/10J C57BL/10ScSn C57BL/10N	Very close relationship to C57BL/6 but differences in at least two loci Frequent partner in preparation of congenic mice
C57BR	C57BR/cdj	High frequency of pituitary and liver tumors Very resistant to x-irradiation
C57L	C57L/J C57L/N	Susceptibility to experimental autoimmune encephalomyelitis (EAE) High frequency of pituitary and reticular cell tumors
C58	C58/J C58/LwN	High incidence of leukemia
DBA/1	DBA/1J DBA/1N	High incidence of mammary tumors
DBA/2	DBA/2J DBA/2N	Low immune response to some antigens Low response to pneumococcal polysaccharide type III
HRS	HRS/J	Hairless (<i>hr</i>) gene, usually in heterozygous state
NZB	NZB/BINJ NZB/N	High incidence of autoimmune hemolytic anemia and lupus-like nephritis Autoimmune disease similar to systemic lupus erythematosus (SLE) in F ₁ progeny from crosses with NZW
NZW	NZW/N	SLE-type autoimmune disease in F ₁ progeny from crosses with NZB
P	P/J	High incidence of leukemia
SJL	SJL/J	High level of aggression and severe fighting to the point of death, especially in males Tendency to develop certain autoimmune diseases, most susceptible to EAE
SWR	SWR/J	Tendency to develop several autoimmune diseases, especially EAE
129	129/J 129/SvJ	High incidence of spontaneous teratocarcinoma

SOURCE: Adapted from Federation of American Societies for Experimental Biology, 1979, *Biological Handbooks*, Vol. III: Inbred and Genetically Defined Strains of Laboratory Animals.

and adult lymph nodes are implanted into SCID mice (Figure 23-1). Because the mice lack mature T and B cells of their own, they do not reject the transplanted human tissue. The implanted human fetal liver contains immature lymphocytes (stem cells), which migrate to the implanted human tissues,

where they mature into T and B cells, producing a **SCID-human mouse**. Because the human lymphocytes are exposed to mouse antigens while they are still immature, they later recognize mouse cells as self and do not mount an immunologic response against the mouse host.

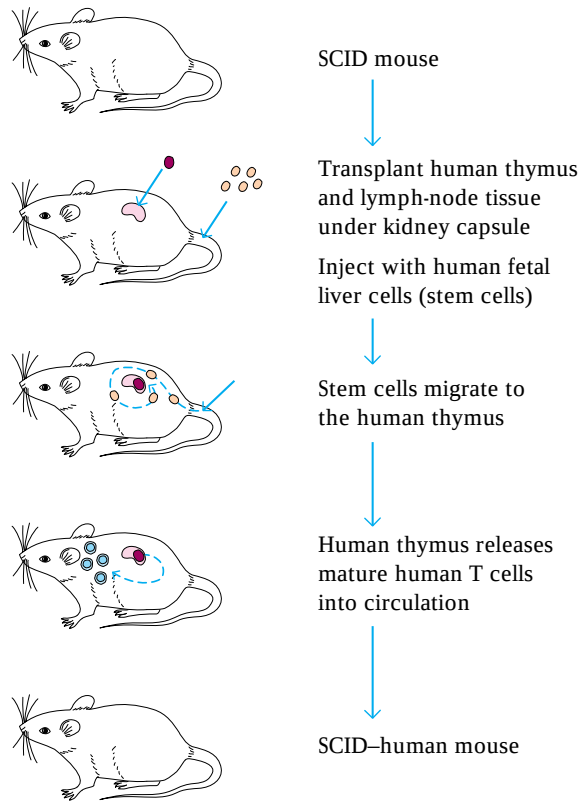


FIGURE 23-1 Production of SCID-human mouse. This system permits study of human lymphocytes within an animal model. In this example, human T-cells are transferred to SCID mouse, but B-cells also can be transferred by the use of bone-marrow precursors.

The beauty of the SCID-human mouse is that it enables one to study human lymphocytes within an animal model. This valuable system has proved useful in research on the development of various lymphoid cells and also as an important animal model in AIDS research, since mouse lymphocytes cannot be infected with HIV, whereas the lymphocytes of a SCID-human mouse are readily infected.

Cell-Culture Systems

The complexity of the cellular interactions that generate an immune response has led immunologists to rely heavily on various types of in vitro cell-culture systems. A variety of cells can be cultured, including primary lymphoid cells, cloned lymphoid cell lines, and hybrid cells.

Primary Lymphoid Cell Cultures

Primary lymphoid cell cultures can be obtained by isolating lymphocytes directly from blood or lymph or from various lymphoid organs by tissue dispersion. The lymphocytes can

then be grown in a chemically defined basal medium (containing saline, sugars, amino acids, vitamins, trace elements, and other nutrients) to which various serum supplements are added. For some experiments, serum-free culture conditions are employed. Because in vitro culture techniques require from 10- to 100-fold fewer lymphocytes than do typical in vivo techniques, they have enabled immunologists to assess the functional properties of minor subpopulations of lymphocytes. It was by means of cell-culture techniques, for example, that immunologists were first able to define the functional differences between $CD4^+$ T helper cells and $CD8^+$ T cytotoxic cells.

Cell-culture techniques have also been used to identify numerous cytokines involved in the activation, growth, and differentiation of various cells involved in the immune response. Early experiments showed that media conditioned, or modified, by the growth of various lymphocytes or antigen-presenting cells would support the growth of other lymphoid cells. Conditioned media contain the secreted products from actively growing cells. Many of the individual cytokines that characterized various conditioned media have subsequently been identified and purified, and in many cases the genes that encode them have been cloned. These cytokines, which play a central role in the activation and regulation of the immune response, are described in Chapter 12 and elsewhere.

Cloned Lymphoid Cell Lines

A primary lymphoid cell culture comprises a heterogeneous group of cells that can be propagated only for a limited time. This heterogeneity can complicate the interpretation of experimental results. To avoid these problems, immunologists use cloned lymphoid cell lines and hybrid cells.

Normal mammalian cells generally have a finite life span in culture; that is, after a number of population doublings characteristic of the species and cell type, the cells stop dividing. In contrast, tumor cells or normal cells that have undergone **transformation** induced by chemical carcinogens or viruses can be propagated indefinitely in tissue culture; thus, they are said to be immortal. Such cells are referred to as **cell lines**.

The first cell line—the mouse fibroblast L cell—was derived in the 1940s from cultured mouse subcutaneous connective tissue by exposing the cultured cells to a chemical carcinogen, methylcholanthrene, over a 4-month period. In the 1950s, another important cell line, the HeLa cell, was derived by culturing human cervical cancer cells. Since these early studies, hundreds of cell lines have been established, each consisting of a population of genetically identical (syngeneic) cells that can be grown indefinitely in culture.

Table 23-3 lists some of the cell lines used in immunologic research and briefly describes their properties. Some were derived from spontaneously occurring tumors of lymphocytes, macrophages, or other accessory cells involved in the immune response. In other cases, the cell line was induced by transformation of normal lymphoid cells with viruses such as Abelson's murine leukemia virus (A-MLV), simian virus 40

TABLE 23-3

Cell lines commonly used in immunologic research

Cell line	Description
L-929	Mouse fibroblast cell line; often used in DNA transfection studies and to assay tumor necrosis factor (TNF)
SP2/0	Nonsecreting mouse myeloma; often used as a fusion partner for hybridoma secretion
P3X63-Ag8.653	Nonsecreting mouse myeloma; often used as a fusion partner for hybridoma secretion
MPC 11	Mouse IgG2b-secreting myeloma
P3X63-Ag8	Mouse IgG1-secreting myeloma
MOPC 315	Mouse IgA-secreting myeloma
J558	Mouse IgA-secreting myeloma
7OZ/3	Mouse pre-B-cell lymphoma; used to study early events in B-cell differentiation
BCL 1	Mouse B-cell leukemia lymphoma that expresses membrane IgM and IgD and can be activated with mitogen to secrete IgM
CTLL-2	Mouse T-cell line whose growth is dependent on IL-2; often used to assay IL-2 production
Jurkat	Human T-cell leukemia that secretes IL-2
DO11.10	Mouse T-cell hybridoma with specificity for ovalbumin
PU 5-1.8	Mouse monocyte-macrophage line
P338 D1	Mouse monocyte-macrophage line that secretes high levels of IL-1
WEHI 265.1	Mouse monocyte line
P815	Mouse mastocytoma cells; often used as target to assess killing by cytotoxic T lymphocytes (CTLs)
YAC-1	Mouse lymphoma cells; often used as target for NK cells
HL-60	Human myeloid-leukemia cell line
COS-1	African green monkey kidney cells transformed by SV40; often used in DNA transfection studies

(SV40), Epstein-Barr virus (EBV), or human T-cell leukemia virus type 1 (HTLV-1).

Lymphoid cell lines differ from primary lymphoid cell cultures in several important ways: They survive indefinitely in tissue culture, show various abnormal growth properties, and often have an abnormal number of chromosomes. Cells with more or less than the normal diploid number of chromosomes for a species are said to be aneuploid. The big advantage of cloned lymphoid cell lines is that they can be

grown for extended periods in tissue culture, enabling immunologists to obtain large numbers of homogeneous cells in culture.

Until the late 1970s, immunologists did not succeed in maintaining normal T cells in tissue culture for extended periods. In 1978, a serendipitous finding led to the observation that conditioned medium containing a T-cell growth factor was required. The essential component of the conditioned medium turned out to be interleukin 2 (IL-2). By culturing normal T lymphocytes with antigen in the presence of IL-2, clones of antigen-specific T lymphocytes could be isolated. These individual clones could be propagated and studied in culture and even frozen for storage. After thawing, the clones continued to grow and express their original antigen-specific functions.

Development of cloned lymphoid cell lines has enabled immunologists to study a number of events that previously could not be examined. For example, research on the molecular events involved in activation of naive lymphocytes by antigen was hampered by the low frequency of naive B and T cells specific for a particular antigen; in a heterogeneous population of lymphocytes, the molecular changes occurring in one responding cell could not be detected against a background of 10^3 – 10^6 nonresponding cells. Cloned T- and B-cell lines with known antigenic specificity have provided immunologists with large homogeneous cell populations in which to study the events involved in antigen recognition. Similarly, the genetic changes corresponding to different maturational stages can be studied in cell lines that appear to be “frozen” at different stages of differentiation. Cell lines have also been useful in studying the soluble factors produced by lymphoid cells. Some cell lines secrete large quantities of various cytokines; other lines express membrane receptors for particular cytokines. These cell lines have been used by immunologists to purify various cytokines and their receptors and eventually to clone their genes.

With the advantages of lymphoid cell lines come a number of limitations. Variants arise spontaneously in the course of prolonged culture, necessitating frequent subcloning to limit the cellular heterogeneity that can develop. If variants are selected in subcloning, it is possible that two subclones derived from the same parent clone may represent different subpopulations. Moreover, any cell line derived from tumor cells or transformed cells may have unknown genetic contributions characteristic of the tumor or of the transformed state; thus, researchers must be cautious when extrapolating results obtained with cell lines to the normal situation in vivo. Nevertheless, transformed cell lines have made a major contribution to the study of the immune response, and many molecular events discovered in experiments with transformed cell lines have been shown to take place in normal lymphocytes.

Hybrid Lymphoid Cell Lines

In somatic-cell hybridization, immunologists fuse normal B or T lymphocytes with tumor cells, obtaining hybrid cells, or

heterokaryons, containing nuclei from both parent cells. Random loss of some chromosomes and subsequent cell proliferation yield a clone of cells that contain a single nucleus with chromosomes from each of the fused cells; such a clone is called a **hybridoma**.

Historically, cell fusion was promoted with Sendai virus, but now it is generally done with polyethylene glycol. Normal antigen-primed B cells can be fused with cancerous plasma cells, called **myeloma cells** (Figure 23-2). The hybridoma thus formed continues to express the antibody genes of the normal B lymphocyte but is capable of unlimited growth, a characteristic of the myeloma cell. B-cell hybridomas that secrete antibody with a single antigenic specificity, called monoclonal antibody, in reference to its derivation from a single clone, have revolutionized not only immunology but biomedical research as well as the clinical laboratory. Chapter 4 describes the production and uses of monoclonal antibodies in detail (see Figures 4-21).

T-cell hybridomas can also be obtained by fusing T lymphocytes with cancerous T-cell lymphomas. Again, the resulting hybridoma continues to express the genes of the normal T cell but acquires the immortal-growth properties of the cancerous T lymphoma cell. Immunologists have generated a number of stable hybridoma cell lines representing T-helper and T-cytotoxic lineages.

Protein Biochemistry

The structures and functions of many important molecules of the immune system have been determined with the techniques of protein biochemistry, and many of these techniques are in constant service in experimental immunology. For example, fluorescent and radioactive labels allow immunologists to localize and visualize molecular activities, and the ability to determine such biochemical characteristics of a protein as its size, shape, and three-dimensional structure has provided essential information for understanding the functions of immunologically important molecules.

Radiolabeling Techniques Allow Sensitive Detection of Antigens or Antibodies

Radioactive labels on antigen or antibody are extremely sensitive markers for detection and quantification. There are a number of ways to introduce radioactive isotopes into proteins or peptides. For example, tyrosine residues may be labeled with radioiodine by chemical or enzymatic procedures. These reactions attach an iodine atom to the phenol ring of the tyrosine molecule. One of the enzymatic iodination techniques, which uses lactoperoxidase, can label proteins on the plasma membrane of a live cell without labeling proteins in the cytoplasm, allowing the study of cell-surface proteins without isolating them from other cell constituents.

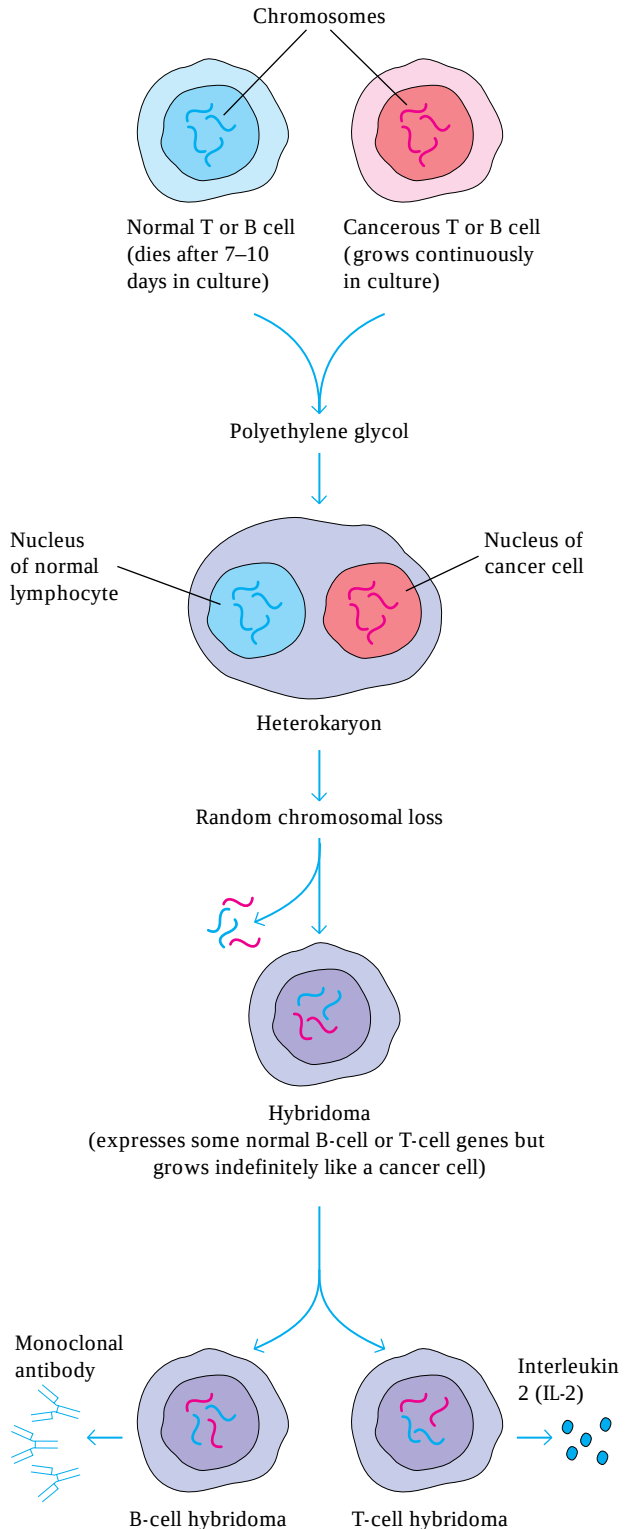


FIGURE 23-2 Production of B-cell and T-cell hybridomas by somatic-cell hybridization. The resulting hybridomas express some of the genes of the original normal B or T cell but also exhibit the immortal-growth properties of the tumor cell. This procedure is used to produce B-cell hybridomas that secrete monoclonal antibody and T-cell hybridomas that secrete various growth factors.

TABLE 23-4

Radioisotopes commonly used in immunology laboratories

Isotope	Half-life	Radiation type*	Autoradiography†
^{125}I	60.0 da	γ	+
^{131}I	6.8 da	γ	+
^{51}Cr	27.8 da	γ	–
^{32}P	14.3 da	β	+
^{35}S	87.4 da	β	+
^{14}C	57.30 yrs	β	+
^3H	12.35 yrs	β	–

* γ (gamma) radiation may be detected in a solid scintillation counter.

β (beta) radiation is detected in a liquid scintillation counter by its ability to convert energy to photons of light in a solution containing phosphorescent compounds.

† Radiation may also be detected by exposure to x-ray film. ^{35}S and ^{14}C must be placed in direct contact with film for detection. ^3H cannot be detected by normal autoradiographic techniques.

A general radiolabeling of cell proteins may be carried out by growing the cells in a medium that contains one or more radiolabeled amino acids. The amino acids selected for this application are those most resistant to metabolic modification during cell growth so that the radioactive label will appear in the cell protein rather than in all cell constituents. Leucine marked with ^{14}C or ^3H , and cysteine or methionine labeled with ^{35}S , are the most commonly used amino acids for metabolic labeling of proteins. Table 23-4 lists some properties of the radioisotopes used in immunologic research.

Biotin Labels Facilitate Detection of Small Amounts of Proteins

In some instances direct labeling of proteins, especially with enzymes or other large molecules, as described in Chapter 6,

may cause denaturation and loss of activity. A convenient labeling system has been developed which may be used in conjunction with the ELISA and ELISPOT assays described in Chapter 6. This labeling technique exploits the high affinity of the reaction between the vitamin biotin and avidin, a large molecule that may be labeled with radioactive isotopes, with fluorescent molecules, or with enzymes. Biotin is a small molecule (mol. wt. 244) that can be coupled to an antibody (or to any protein molecule) by a gentle chemical reaction that causes no loss of antibody activity. After the biotin-coupled antibody has reacted in the assay system, the labeled avidin is introduced and binding is measured by detecting the label on the avidin molecule (Figure 23-3). The reaction between biotin and avidin is highly specific and of such high affinity that the bond between the two molecules under most assay conditions is virtually irreversible.

Gel Electrophoresis Separates Proteins by Size and Charge

When subjected to an electric field in an electrophoresis chamber, a charged molecule will move toward the oppositely charged electrode. The rate at which a charged molecule moves in a stable field (its electrophoretic mobility) depends upon two factors specific to the molecule: one is the sign and magnitude of its net electrical charge, and the other is its size and shape. All other factors being equal, if molecules are of equal size the one with higher net charge will move faster in an applied electrical field due to the molecular sieving properties of the solid medium. It also follows that small molecules will move faster than large ones of the same net charge. Although there are exceptions in which the shape of a molecule may increase or decrease its frictional drag and cause atypical migration behavior, these general principles underlie all electrophoretic separations.

Most electrophoretic separations are not conducted in free solution but rather in a stable supporting medium, such as a gel. The most popular in research laboratories is a polymerized and crosslinked form of acrylamide. Separation on polyacrylamide gels, commonly referred to as polyacrylamide

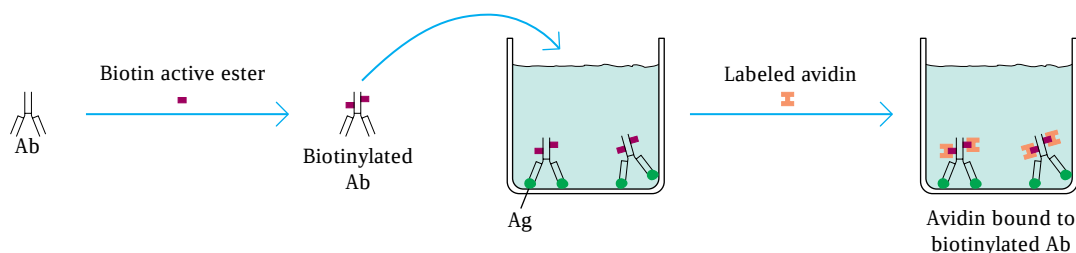


FIGURE 23-3 Labeling of antibody with biotin. An antibody preparation is mixed with a biotin ester, which reacts with the antibody. The biotin-labeled antibody can be used to detect antigens on a solid substrate such as the well of a microtiter plate. After washing away unbound

antibody, the bound antibody can be detected with labeled avidin. The avidin can be radioactively labeled or linked to an enzyme that catalyzes a color reaction, as in ELISA procedures (see Figure 6-10).

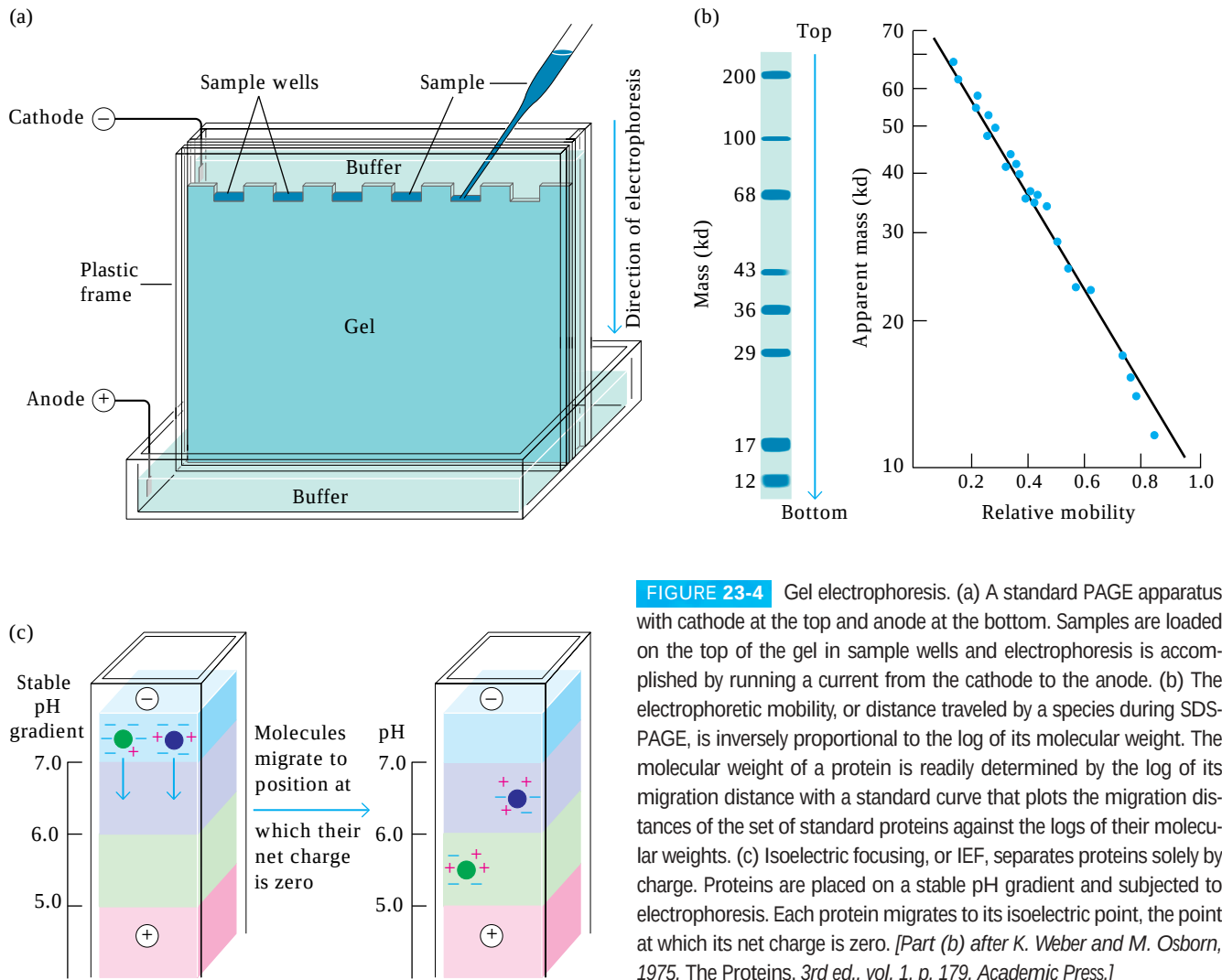


FIGURE 23-4 Gel electrophoresis. (a) A standard PAGE apparatus with cathode at the top and anode at the bottom. Samples are loaded on the top of the gel in sample wells and electrophoresis is accomplished by running a current from the cathode to the anode. (b) The electrophoretic mobility, or distance traveled by a species during SDS-PAGE, is inversely proportional to the log of its molecular weight. The molecular weight of a protein is readily determined by the log of its migration distance with a standard curve that plots the migration distances of the set of standard proteins against the logs of their molecular weights. (c) Isoelectric focusing, or IEF, separates proteins solely by charge. Proteins are placed on a stable pH gradient and subjected to electrophoresis. Each protein migrates to its isoelectric point, the point at which its net charge is zero. [Part (b) after K. Weber and M. Osborn, 1975, *The Proteins*, 3rd ed., vol. 1, p. 179. Academic Press.]

gel electrophoresis (PAGE), may be used for analysis of proteins or nucleic acids (Figure 23-4a).

In one common application, the electrophoresis of proteins through a polyacrylamide gel is carried out in the presence of the detergent sodium dodecyl sulfate (SDS). This method, known as SDS-PAGE, provides a relatively simple and highly effective means of separating mixtures of proteins on the basis of size. SDS is a negatively charged detergent that binds to protein in amounts proportional to the length of the protein. This binding destroys the characteristic tertiary and secondary structure of the protein, transforming it into a negatively charged rod. A protein binds so many negatively charged SDS molecules that its own intrinsic charge becomes insignificant by comparison with the net charge of the SDS molecules. Therefore, treatment of a mixture of proteins with SDS transforms them into a collection of rods whose electric charges are proportional to their molecular weights. This has two extremely useful consequences. First, it is possible to sep-

arate the components of a mixture of proteins according to molecular weight. Second, because the electrophoretic mobility, or distance traveled by a species during SDS-PAGE, is inversely proportional to the logarithm of its molecular weight, that distance is a measure of its molecular weight. The gel is stained with a dye that reacts with protein to visualize the locations of the proteins. The migration distance of a protein in question is then compared with a plot of the distances migrated by a set of standard proteins (Figure 23-4b).

Another electrophoretic technique, isoelectric focusing (IEF), separates proteins solely on the basis of their charge. This method is based on the fact that a molecule will move in an electric field as long as it has a net positive or negative charge; molecules that bear equal numbers of positive and negative charges and therefore have a net charge of zero will not move. At most pH values, proteins (which characteristically bear a number of both positive and negative charges) have either a net negative or a net positive charge. However,

for each protein there is a particular pH, called its isoelectric point (pI), at which that protein has equal numbers of positive and negative charges. Isoelectric focusing makes use of a gel containing substances, called carrier ampholytes, that arrange themselves into a continuous pH gradient when subjected to an electric field. When a mixture of proteins is applied to such a gel and subjected to electrophoresis, each protein moves until it reaches that point in the gradient where the pH of the gel is equal to its isoelectric point. It then stops moving because it has a net charge of zero. Isoelectric focusing is an extremely gentle and effective way of separating different proteins (Figure 23-4c).

A method known as two-dimensional gel electrophoresis (2D gel electrophoresis) combines the advantages of SDS-PAGE and isoelectric focusing in one of the most sensitive and discriminating ways of analyzing a mixture of proteins. In this method, one first subjects the mixture to isoelectric focusing on an IEF tube gel, which separates the molecules on the basis of their isoelectric points without regard to molecular weight. This is the first dimension. In the next step, one places the IEF gel lengthwise across the top of an SDS-polyacrylamide slab (that is, in place of the sample wells in Figure 23-4a) and runs SDS-PAGE. Preparatory to this step, all proteins have been reacted with SDS and therefore migrate out of the IEF gel and through the SDS-PAGE slab according to their molecular weights. This is the second dimension. The position of the proteins in the resulting 2D gel can be visualized in a number of ways. In the least sensitive the gel is stained with a protein-binding dye (such as Coomassie blue). If the proteins have been radiolabeled, the more sensitive method of autoradiography can be used. Alternatively, silver staining is a method of great sensitivity that takes advantage of the capacity of proteins to reduce silver ions to an easily visualized deposit of metallic silver. Finally, immunoblotting—blotting of proteins onto a membrane and detection with antibody (see Figure 6-13)—can be used as a way of locating the position of specific proteins on 2D gels if an appropriate antibody is available. Figure 23-5 shows an autoradiograph of a two-dimensional gel of labeled proteins from murine thymocytes.

X-Ray Crystallography Provides Structural Information

A great deal of information about the structure of cells, parts of cells, and even molecules has been obtained by light microscopy. The microscope uses a lens to focus radiation to form an image after it has passed through a specimen. However, a practical limitation of light microscopy is the limit of resolution. Radiation of a given wavelength cannot resolve structural features less than about 1/2 its wavelength. Since the shortest wavelength of visible light is around 400 nm, even the very best light microscopes have a theoretical limit of resolution of no less than 200 nm.

Because of the much shorter wavelength (0.004 nm) of the electron at the voltages normally used in the electron

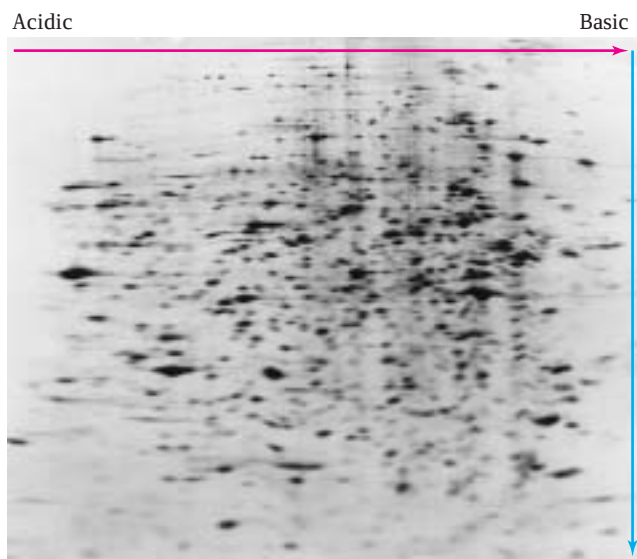


FIGURE 23-5 Two-dimensional gel electrophoresis of ^{35}S -methionine labeled total cell proteins from murine thymocytes. These proteins were first subjected to isoelectric focusing (direction of migration indicated by red arrow) and then the focused proteins were separated by SDS-PAGE (direction of migration indicated by blue arrow). The gel was exposed to x-ray film to detect the labeled proteins. [Courtesy of B. A. Osborne.]

microscope, the theoretical limit of resolution of the electron microscope is about 0.002 nm. If it were possible to build an instrument that could actually approach this limit, the electron microscope could readily be used to determine the detailed atomic arrangement of biological molecules, since the constituent atoms are separated by distances of 0.1 nm to 0.2 nm. In practice, aberrations inherent in the operation of the magnetic lenses that are used to image the electron beam limit the resolution to about 0.1 nm (1 Å). This practical limit can be reached in the examination of certain specimens, particularly metals. Other considerations, however, such as specimen preparation and contrast, limit the resolution for biological materials to about 2 nm (20 Å). To determine the arrangement of a molecule's atoms, then, we must turn to x-rays, a form of electromagnetic radiation that is readily generated in wavelengths on the order of size of interatomic distances. Even though there are no microscopes with lenses that can focus x-rays into images, x-ray crystallography can reveal molecular structure at an extraordinary level of detail.

X-ray crystallography is based on the analysis of the diffraction pattern produced by the scattering of an x-ray beam as it passes through a crystal. The degree to which a particular atom scatters x-rays depends upon its size. Atoms such as carbon, oxygen, or nitrogen, scatter x-rays more than do hydrogen atoms, and larger atoms, such as iron, iodine, or mercury give intense scattering. X-rays are a form of electromagnetic waves;

as the scattered waves overlap, they alternately interfere with and reinforce each other. An appropriately placed detector records a pattern of spots (the diffraction pattern) whose distribution and intensities are determined by the structure of the diffracting crystal. This relationship between crystal structure and diffraction pattern is the basis of x-ray crystallographic analysis. Here is an overview of the procedures used:

OBTAIN CRYSTALS OF THE PROTEIN OF INTEREST. To those who have not experienced the frustrations of crystallizing proteins, this may seem a trivial and incidental step of an otherwise highly sophisticated process. It is not. There is great variation from protein to protein in the conditions required to produce crystals that are of a size and geometrical formation appropriate for x-ray diffraction analysis. For example, myoglobin formed crystals over the course of several days at pH 7 in a 3 M solution of ammonium sulfate, but 1.5 M ammonium sulfate at pH 4 worked well for a human IgG1. There is no set formula that can be applied, and those who are consistently successful are persistent, determined, and, like great chefs, have a knack for making just the right “sauce.”

SELECTION AND MOUNTING. Crystal specimens must be at least 0.1 mm in the smallest dimension and rarely exceed a few millimeters in any dimension. Once chosen, a crystal is harvested into a capillary tube along with the solution from which the crystal was grown (the “mother liquor”). This keeps the crystal from drying and maintains its solvent content, an important consideration for maintaining the internal order of the specimen. The capillary is then mounted in the diffraction apparatus.

GENERATING AND RECORDING A DIFFRACTION PATTERN.

The precisely positioned crystal is then irradiated with x-rays of a known wavelength produced by accelerating electrons against the copper target of an x-ray tube. When the x-ray beam strikes the crystal, some of it goes straight through and some is scattered; sensitive detectors record the position and intensity of the scattered beam as a pattern of spots (Figure 23-6a,b).

INTERPRETING THE DIFFRACTION PATTERN. The core of diffraction analysis is the mathematical deduction of the detailed structure that would produce the diffraction pattern observed. One must calculate to what extent the waves scattered by each atom have combined to reinforce or cancel each other to produce the net intensity observed for each spot in the array. A difficulty arises in the interpretation of complex diffraction patterns because the waves differ with respect to phase, the timing of the period between maxima and minima. Since the pattern observed is the net result of the interaction of many waves, information about phase is critical to calculating the distribution of electron densities that is responsible. The solution of this “phase problem” looms as a major obstacle to the derivation of a high-resolution structure of any complex molecule.

The problem is solved by derivatizing the protein—modifying it by adding heavy atoms, such as mercury, and then obtaining crystals that have the same geometry as (are iso-

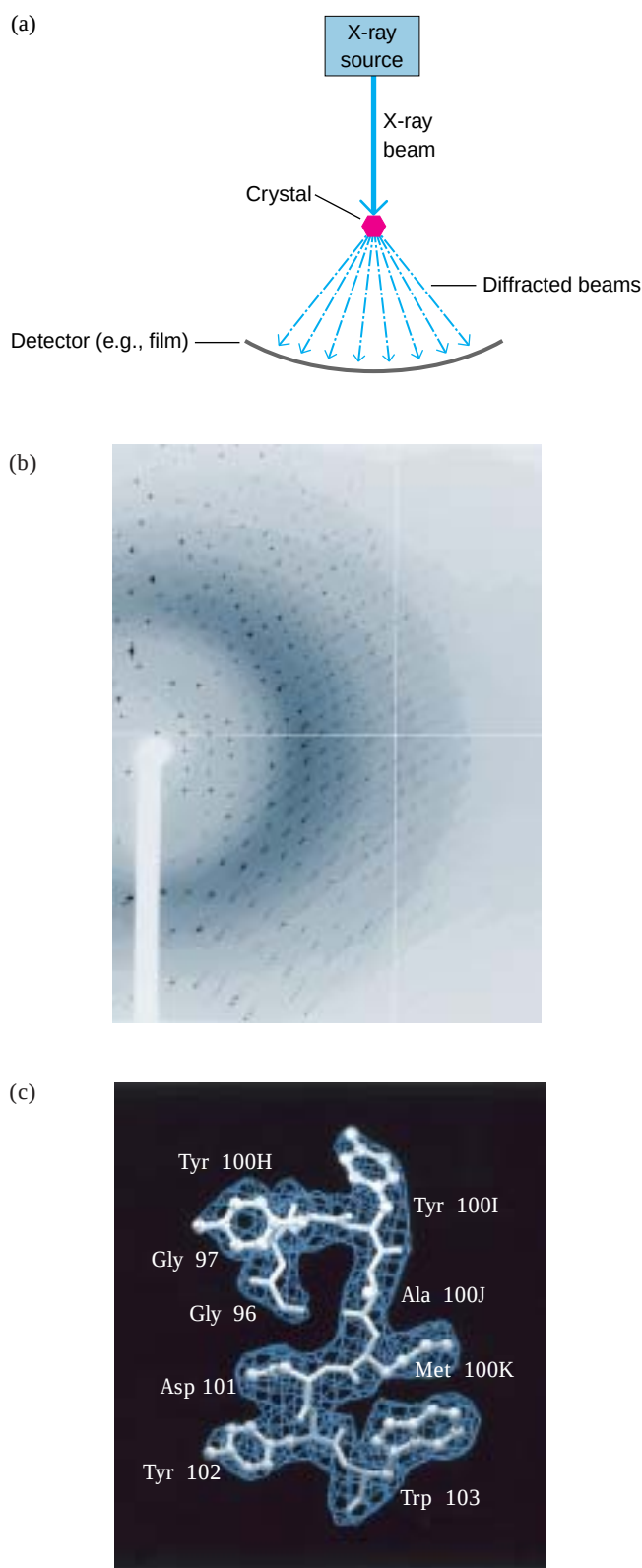


FIGURE 23-6 X-ray crystallography. (a) Schematic diagram of an x-ray crystallographic experiment in which an x-ray beam bombards the crystal and diffracted rays are detected. (b) Section of x-ray diffraction pattern of a crystal of murine IgG2a. (c) Section from the electron-density map of murine IgG2a. [Part (a) from L. Stryer, 1995, *Biochemistry*, 4th ed.; parts (b) and (c) courtesy of A. McPherson.]

morphous with) those of the underivatized protein. The diffraction pattern of the isomorphous crystal is obtained and compared with that of the native protein. Usually, armed with a knowledge of the diffraction patterns of two or more isomorphous heavy-atom derivatives, the phases for the native protein can be calculated by reference to the characteristic diffraction patterns generated by heavy-atom landmarks. The phases established, it is possible to move on to a calculation of the distribution of electron density. This is accomplished by Fourier synthesis, a mathematical treatment particularly suited to the analysis of periodic phenomena such as those involving waves. In this case, it is used to compute the distribution of electron density along the *x*, *y*, and *z* axes within a unit cell of the crystal. The deduced electron density can then be visualized on a computer (Figure 23-6c).

DERIVATION OF THE STRUCTURE. The resolution of a model depends upon a number of factors. First of all, the ultimate resolution possible is set by the quality of the crystal and the internal order of the crystal. Even the highest-quality crystals have a degree of internal disorder that establishes a limit of resolution of about 2 Å. Second, a factor of paramount importance is the number of intensities fed into the Fourier synthesis. A relatively small number of spots may produce a low-resolution (6 Å) image that traces the course of the polypeptide chain but provides little additional structural information. On the other hand, the processing of data provided by tens of thousands of spots allows the tracing of very detailed electron-density maps. Provided one knows the amino-acid sequence of the protein, such maps can guide the construction of high-resolution, three-dimensional models. Amino-acid sequence data is necessary because it can be difficult, and in some cases impossible, to unambiguously distinguish among some amino-acid side chains on even the most detailed electron-density maps.

Since 1960, when the first detailed structures of proteins were deduced, the structures of many thousands of proteins have been solved. These range from small and (relatively) simple proteins such as lysozyme, consisting of a single polypeptide chain, to poliovirus, an 8,500,000 dalton, stunningly complex nucleoprotein made up of RNA encased by multiple copies of four different polypeptide subunits. Of particular importance to immunologists are the large number of immunologically relevant molecules for which detailed crystal structures are now available. These include many immunoglobulins, most of the major and minor proteins involved in the MHC and T-cell–receptor complexes, and many other important immunological macromolecules, with new structures and structural variants appearing every month.

Recombinant DNA Technology

The various techniques called recombinant DNA technology have had an impact on every area of immunologic research. Genes can be cloned, DNA can be sequenced, and recombinant proteins can be produced, supplying immunologists with

defined components for study of the structure and function of the immune system at the molecular level. This section briefly describes some of the recombinant DNA techniques commonly employed in immunologic research; examples of their use have been presented throughout the book.

Restriction Enzymes Cleave DNA at Precise Sequences

A variety of bacteria produce enzymes, called *restriction endonucleases*, that degrade foreign DNA (e.g., bacteriophage DNA) but spare the bacterial-cell DNA, which contains methylated residues. The discovery of these bacterial enzymes in the 1970s opened the way to a major technological advance in the field of molecular biology. Before the discovery of restriction endonucleases, double-stranded DNA (dsDNA) could be cut only with DNases. These enzymes do not recognize defined sites and therefore randomly cleave DNA into a variable series of small fragments, which are impossible to sort by size or sequence. In contrast, restriction endonucleases recognize and cleave DNA at specific sites, called restriction sites, which are short double-stranded segments of specific sequence containing four to eight nucleotides (Table 23-5).

TABLE 23-5 Some restriction enzymes and their recognition sequences		
Microorganism source	Abbreviation	Sequence* 5'→3' 3'→5'
<i>Bacillus amyloliquefaciens</i> H	BamHI	G <u>G A T C</u> C C C T A <u>G</u> G
<i>Escherichia coli</i> RY 13	EcoRI	G <u>A A T T</u> C C T T A A <u>G</u>
<i>Haemophilus aegyptius</i>	HaeIII	G G C C C C G G
<i>Haemophilus influenzae</i> Rd	HindIII	A <u>A G C T</u> T T T C G A <u>A</u>
<i>Haemophilus parainfluenzae</i>	HpaI	G T T A A C C A A T T G
<i>Nocardia otitidis-caviarum</i>	NotI	G C <u>G G C C</u> G C C G C C <u>G</u> G C G
<i>Providencia stuartii</i> 164	PstI	G T G C A <u>G</u> G <u>A C G T</u> C
<i>Staphylococcus aureus</i> 3A	Sau3A	<u>G A T C</u> C T A <u>G</u>

*Blue lines indicate locations of single-strand cuts within the restriction site. Enzymes that make off-center cuts produce fragments with short single-stranded extensions at their ends.
SOURCE: New England Biolabs, <http://www.neb.com>.

A restriction endonuclease cuts both DNA strands at a specific point within its restriction site. Some enzymes, such as *HpaI*, cut on the central axis and thus generate blunt-ended fragments. Other enzymes, such as *EcoRI*, cut the DNA at staggered points in the recognition site. In this case, the end of each cleaved fragment is a short segment of single-stranded DNA, called a *sticky end*. When two different DNA molecules are cut with the same restriction enzyme that makes staggered cuts, the sticky ends of the fragments are complementary; under appropriate conditions, fragments from the two molecules can be joined by base pairing to generate a recombinant DNA molecule. Several hundred different restriction endonucleases have been isolated and many are available commercially, allowing researchers to purchase enzymes that cut DNA at defined restriction sites.

Cloning of DNA Sequences

The development of DNA-cloning technology in the 1970s provided a means of amplifying a given DNA fragment to such an extent that unlimited amounts of identical DNA fragments (cloned DNA) could be produced.

Cloning Vectors Are Useful to Replicate Defined Sequences of DNA

In DNA cloning, a given DNA fragment is inserted into an autonomously replicating DNA molecule, called a cloning vector, so that the inserted DNA is replicated with the vector. A number of different viruses have been used as vectors, including bacterial viruses, insect viruses, and mammalian retroviruses. A common bacterial virus used as a vector is bacteriophage λ . If a gene is inserted into bacteriophage λ and the resulting recombinant λ phage is used to infect *E. coli*, the inserted gene will be expressed by the bacteria.

Retroviruses, which can infect virtually any type of mammalian cell, are a common vector used to clone DNA in mammalian cells. Retroviruses are RNA viruses that contain reverse transcriptase, an enzyme that catalyzes conversion of the viral RNA genome into DNA. The viral DNA then integrates into the host chromosomal DNA, where it is retained as a provirus, replicating along with the host chromosomal DNA at each cell division. When a retrovirus is used as a vector in research, most of the retroviral genes are removed so that the vector cannot produce viral particles; the retroviral genes that are left include a strong promoter region, located at the 5' end of the viral genome, in a sequence called the *long terminal repeat* (LTR). If a gene is inserted into such a retroviral vector and the vector is then used to infect mammalian cells, expression of the gene will be under the control of the retroviral promoter region.

Plasmids are another common type of cloning vector. A plasmid is a small, circular, extrachromosomal DNA molecule that can replicate independently in a host cell; the most common host used in DNA cloning is *E. coli*. In general, the DNA to be cloned is inserted into a plasmid that contains an

antibiotic-resistance gene. After the recombinant plasmid is incubated with bacterial cells, the cells containing the recombinant plasmid can be selected by their ability to grow in the presence of the antibiotic.

Another type of vector that is often used for cloning is called a cosmid vector. This type of vector is a plasmid that has been genetically engineered to contain the COS sites of λ -phage DNA, a drug-resistance gene, and a replication origin. COS sites are DNA sequences that allow any DNA up to 50 kb in length to be packaged into the λ -phage head.

Cloning of cDNA and Genomic DNA Allows the Isolation of Defined Sequences

Messenger RNA (mRNA) isolated from cells can be transcribed into complementary DNA (cDNA) with the enzyme reverse transcriptase. The cDNA can be cloned by inserting it into a plasmid vector carrying a gene that confers resistance to an antibiotic, such as ampicillin. The resulting recombinant plasmid DNA is subsequently transferred into specially treated *E. coli* cells by one of several techniques; the transfer process is called **transfection**. If the foreign DNA is incorporated into the host cell and expressed, the cell is said to be **transformed**. When the cells are cultured on agar plates containing ampicillin, only transformed cells containing the ampicillin-resistance gene will survive and grow (Figure 23-7). A collection of DNA sequences within plasmid vectors representing all the mRNA sequences derived from a cell or tissue is called a *cDNA library*. A cDNA library differs from a genomic library (see Figure 23-8) by virtue of the fact that it contains only the sequences derived from mRNA, the sequences that represent expressed genes.

Genomic cloning, cloning of the entire genome of an animal, requires specialized vectors. *E. coli* plasmid vectors are impractical for cloning of all the genomic DNA fragments that constitute a large genome because of the low efficiency of *E. coli* transformation and the small number of transformed colonies that can be detected on a typical petri dish. Instead, cloning vectors derived from bacteriophage λ are used to clone genomic DNA fragments obtained by cleaving chromosomal DNA with restriction enzymes (Figure 23-8). Bacteriophage λ DNA is 48.5 kb long and contains a central section of about 15 kb that is not necessary for λ replication in *E. coli* and can therefore be replaced with foreign genomic DNA. As long as the recombinant DNA does not exceed the length of the original λ -phage DNA by more than 5%, it can be packaged into the λ -phage head and propagated in *E. coli*. This means that somewhat more than 1.5×10^4 base pairs can be cloned in one particle of λ phage. A collection of λ clones that includes all the DNA sequences of a given species is called a *genomic library*. It has been calculated that about 1 million different recombinant λ -phage particles would be needed to form a genomic DNA library representing an entire mammalian genome, which contains about 3×10^9 base pairs.

Often the 20–25 kb stretch of DNA that can be cloned in bacteriophage λ is not long enough to include the regulatory

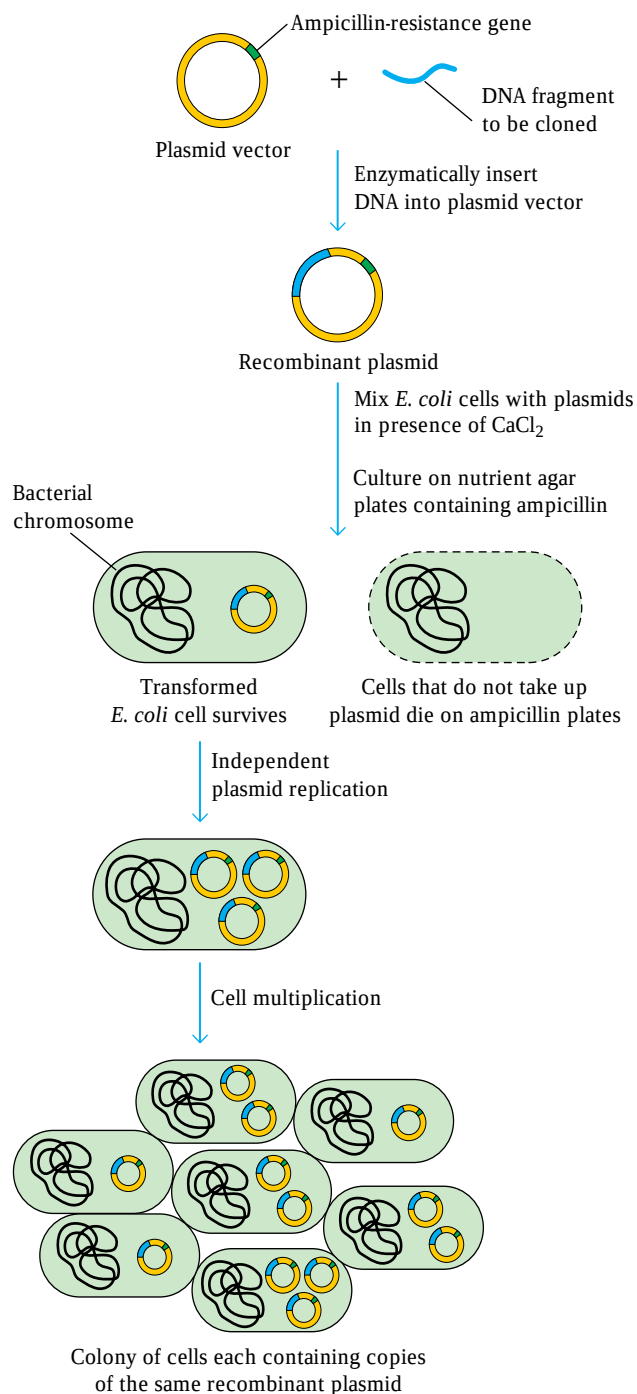


FIGURE 23-7 cDNA cloning using a plasmid vector. A plasmid containing a replication origin and an ampicillin-resistance gene is cut with a restriction endonuclease that produces blunt ends. After addition of a poly-C tail to the 3' ends of the cDNA and of a complementary poly-G tail to the 3' ends of the cut plasmid, the two DNAs are mixed, annealed, and joined by DNA ligase, forming the recombinant plasmid. Uptake of the recombinant plasmid into *E. coli* cells is stimulated by high concentrations of CaCl_2 . Transformation occurs with a low frequency, but the transformed cells can be selected in the presence of ampicillin. [Adapted from H. Lodish et al., 1995, Molecular Cell Biology, 3rd ed. Scientific American Books.]

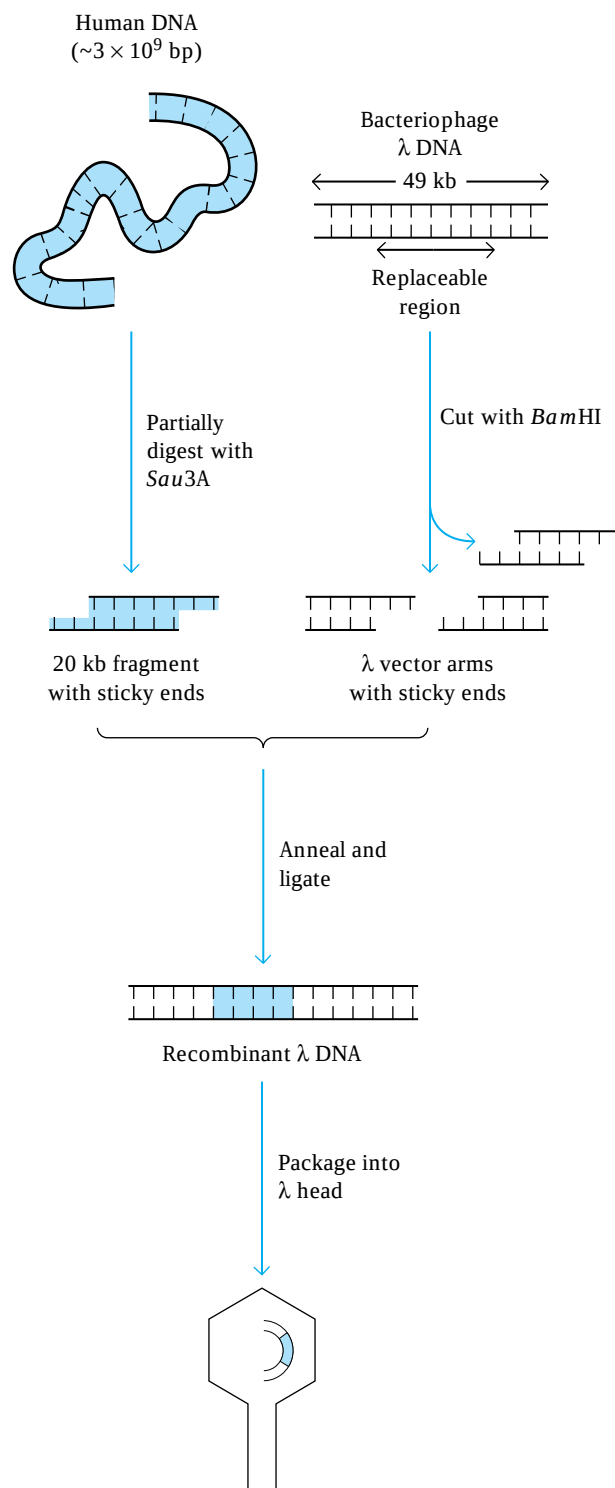


FIGURE 23-8 Genomic DNA cloning using bacteriophage λ as the vector. Genomic DNA is partly digested with *Sau*3A, producing fragments with sticky ends. The central 15-kb region of the λ -phage DNA is cut out with *Bam*HI and discarded. These two restriction enzymes produce complementary sticky ends, so the genomic and DNA fragments can be annealed and ligated. After the resulting recombinant DNA is packaged into a λ -phage head, it can be propagated in *E. coli*.

TABLE 23-6

Vectors and maximum length of DNA that they can carry

Vector type	Maximum length of cloned DNA (kb)
Plasmid	20
Bacteriophage λ	25
Cosmid	45
Bacteriophage P1	100
Bacterial artificial chromosome (BAC)	100–300
Yeast artificial chromosome (YAC)	>1000

sequences that lie outside the 5' and 3' ends of the direct coding sequences of a gene. As noted already, larger genomic DNA fragments—between 30 and 50 kb in length—can be cloned in a cosmid vector. A recombinant cosmid vector, although not a fully functional bacteriophage, can infect *E. coli* and replicate as a plasmid, generating a cosmid library. Recently, a larger *E. coli* virus, called bacteriophage P1, has been used to package DNA fragments up to 100 kb long. Even larger DNA fragments, greater than a megabase (1000 kb) in length, can be cloned in yeast artificial chromosomes (YACs), which are linear DNA segments that can replicate in yeast cells (Table 23-6). The BAC, or bacterial artificial chromosome, is another useful vector. BACs can accept pieces of DNA up to 100–300 kb in length. Although YACs accept larger inserts of foreign DNA,

BACs are much easier to propagate and are the vector of choice for many large-scale cloning efforts.

Selection of DNA Clones

Once a cDNA or genomic DNA library has been prepared, it can be screened to identify a particular DNA fragment by a technique called *in situ hybridization*. The cloned bacterial colonies, yeast colonies, or phage plaques containing the recombinant DNA are transferred onto nitrocellulose or nylon filters by replica plating (Figure 23-9). The filter is then treated with NaOH, which both lyses the bacteria and denatures the DNA, allowing single-stranded DNA (ssDNA) to bind to the filter. The filter with bound DNA is then incubated with a radioactive probe specific for the gene of interest. The probe will hybridize with DNA in the colonies or plaques on the filter that contain the sought-after gene, and they can be identified by autoradiography. The position of the positive colonies or plaques on the filter shows where the corresponding clones can be found on the original agar plate.

Various radioactive probes can be used to screen a library. In some cases, radiolabeled mRNA or cDNA serves as the probe. If the protein encoded by the gene of interest has been purified and partly sequenced, it is possible to work backward from the amino-acid sequence to determine the probable nucleotide sequence of the corresponding gene. A known sequence of five or six amino-acid residues is all that is needed to synthesize radiolabeled oligonucleotide probes with which to screen a cDNA or genomic library for a particular gene. To cope with the degeneracy of the genetic code, peptide segments containing amino acids encoded by a limited number of codons are usually chosen. Oligonucleotides representing

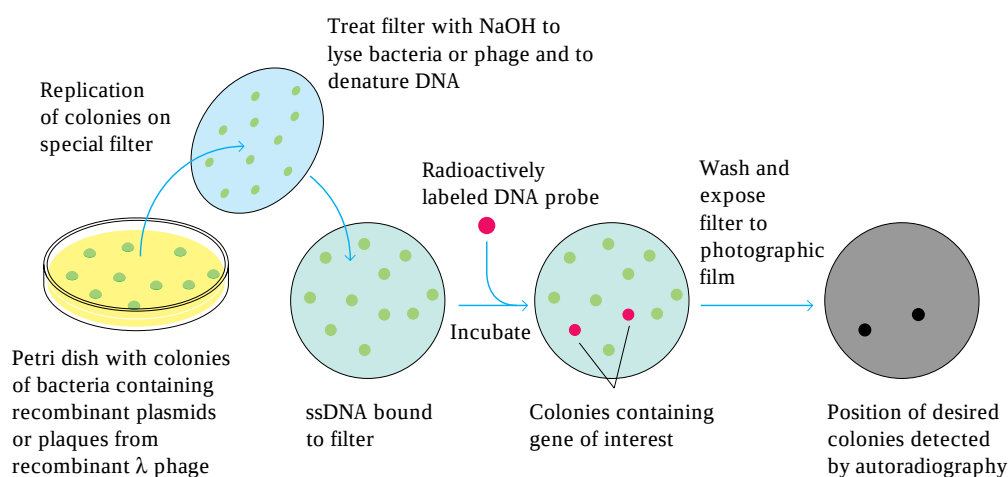


FIGURE 23-9 Selection of specific clones from a cDNA or genomic DNA library by *in situ* hybridization. A nitrocellulose or nylon filter is placed against the plate to pick up the bacterial colonies or phage plaques containing the cloned genes. After the filter is placed

in a NaOH solution and heated, the denatured ssDNA becomes fixed to the filter. A radioactive probe specific for the gene of interest is incubated with the filter. The position of the colonies or plaques containing the desired gene is revealed by autoradiography.

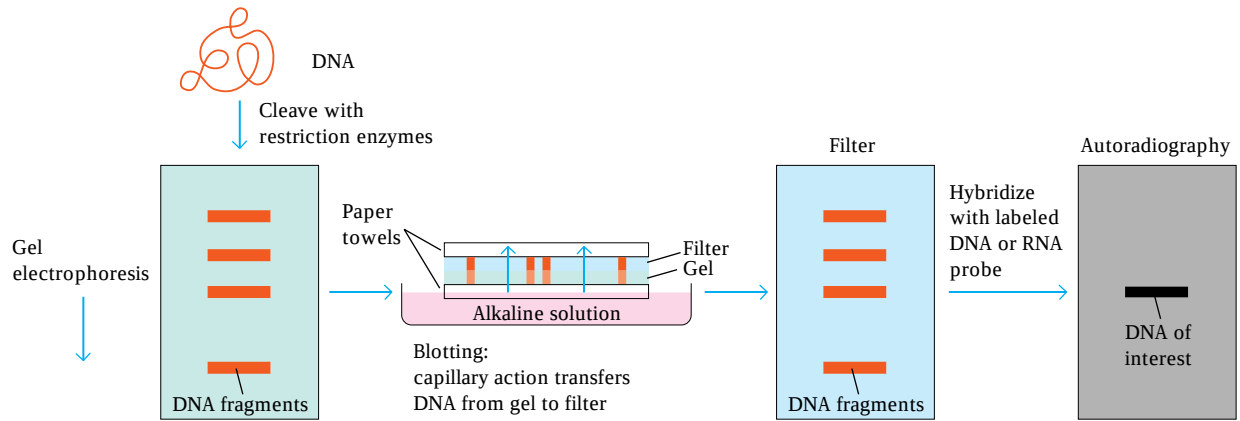


FIGURE 23-10 The Southern-blot technique for detecting specific sequences in DNA fragments. The DNA fragments produced by restriction-enzyme cleavage are separated by size by agarose gel electrophoresis. The agarose gel is overlaid with a nitrocellulose or nylon filter and a thick stack of paper towels. The gel is then placed in an alkaline salt solution, which denatures the DNA. As the paper towels

soak up the moisture, the solution is drawn through the gel into the filter, transferring each ssDNA band to the filter. This process is called blotting. After heating, the filter is incubated with a radiolabeled probe specific for the sequence of interest; DNA fragments that hybridize with the probe are detected by autoradiography. [Adapted from J. Darnell *et al.*, 1990, *Molecular Cell Biology*, 2nd ed., *Scientific American Books*.]

all possible codons for the peptide are then synthesized and used as probes to screen the DNA library.

Southern Blotting Detects DNA of a Given Sequence

DNA fragments generated by restriction-endonuclease cleavage can be separated on the basis of length by agarose gel electrophoresis. The shorter a fragment is, the faster it moves in the gel. An elegant technique developed by E. M. Southern can be used to identify any band containing fragments with a given gene sequence (Figure 23-10). In this technique, called **Southern blotting**, DNA is cut with restriction enzymes and the fragments are separated according to size by electrophoresis on an agarose gel. Then the gel is soaked in NaOH to denature the dsDNA, and the resulting ssDNA fragments are transferred onto a nitrocellulose or nylon filter by capillary action. After transfer, the filter is incubated with an appropriate radiolabeled probe specific for the gene of interest. The probe hybridizes with the ssDNA fragment containing the gene of interest, and the position of the band containing these hybridized fragments is determined by autoradiography. Southern-blot analysis played a critical role in unraveling the mechanism by which diversity of antibodies and T-cell receptors is generated (see Figures 5-2 and 9-2).

Northern Blotting Detects mRNA

Northern blotting (named for its similarity to Southern blotting) is used to detect the presence of specific mRNA molecules. In this procedure the mRNA is first denatured to

ensure that it is in an unfolded, linear form. The mRNA molecules are then separated according to size by electrophoresis and transferred to a nitrocellulose filter, to which the mRNAs will adhere. The filter is then incubated with a labeled DNA probe and subjected to autoradiography. Northern-blot analysis is often used to determine how much of a specific mRNA is expressed in cells under different conditions. Increased levels of mRNA will bind proportionally more of the labeled DNA probe.

Polymerase Chain Reaction Amplifies Small Quantities of DNA

The polymerase chain reaction (PCR) is a powerful technique for amplifying specific DNA sequences even when they are present at extremely low levels in a complex mixture (Figure 23-11). The procedure requires that the DNA sequences that flank the desired DNA sequence be known, so that short oligonucleotide primers can be synthesized. The DNA mixture is denatured into single strands by a brief heat treatment. The DNA is then cooled in the presence of an excess of the oligonucleotide primers, which hybridize with the complementary ssDNA. A temperature-resistant DNA polymerase is then added, together with the four deoxyribonucleotide triphosphates, and each strand is copied. The newly synthesized DNA duplex is separated by heating and the cycle is repeated. In each cycle there is a doubling of the desired DNA sequence; in only 25 cycles the desired DNA sequence can be amplified about a million-fold.

The DNA amplified by the PCR can be further characterized by Southern blotting, restriction-enzyme mapping, and

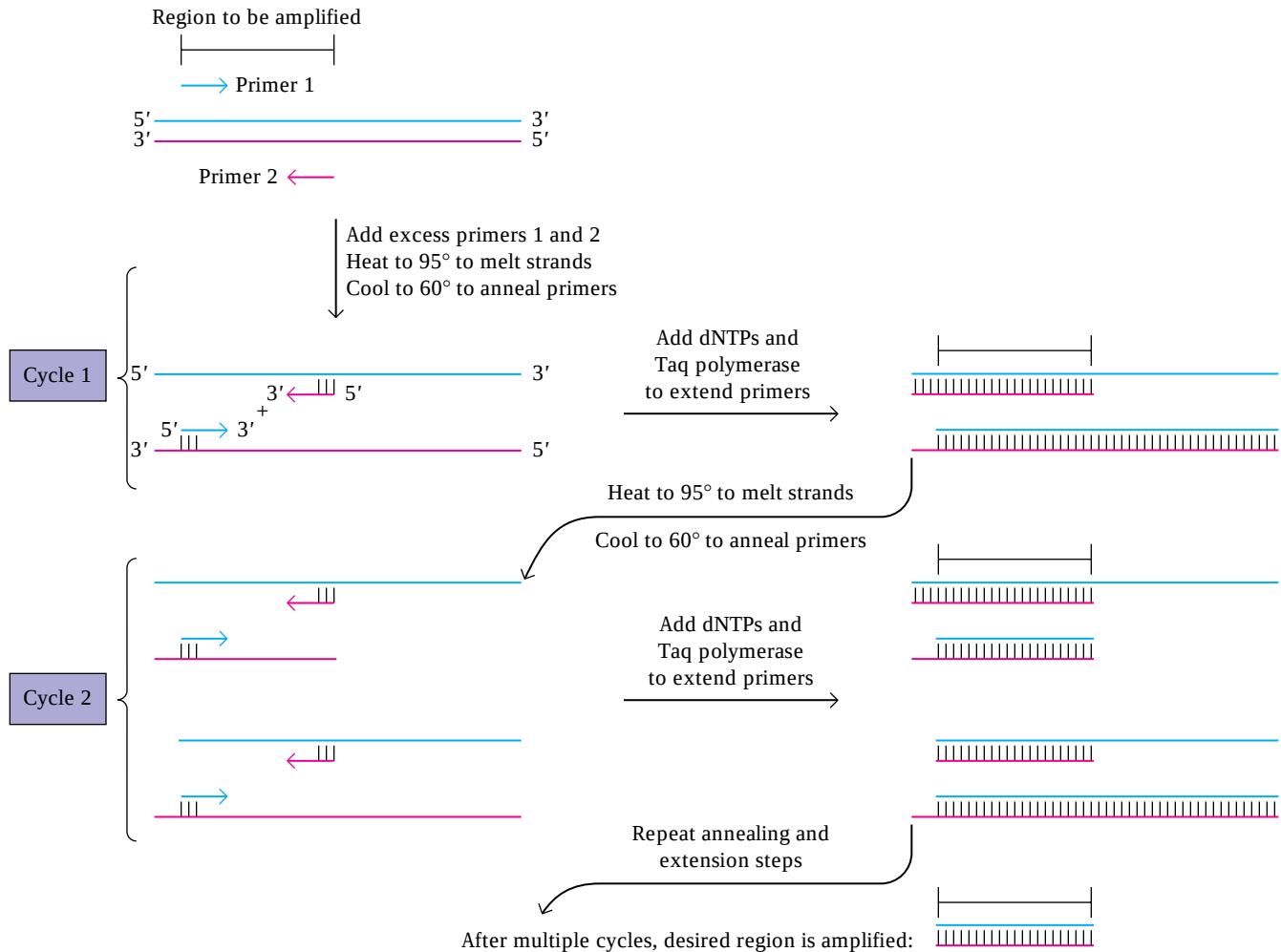


FIGURE 23-11 The polymerase chain reaction (PCR). DNA is denatured into single strands by a brief heat treatment and is then cooled in the presence of an excess of oligonucleotide primers complementary to the DNA sequences flanking the desired DNA segment. A heat-resistant DNA polymerase is used to copy the DNA

from the 3' ends of the primers. Because all of the reaction components are heat stable, the heating and cooling cycle can be repeated many times, resulting in alternate DNA melting and synthesis, and rapid amplification of a given sequence. [Adapted from H. Lodish et al., 1995, *Molecular Cell Biology*, 3rd ed., Scientific American Books.]

direct DNA sequencing. The PCR technique has enabled immunologists to amplify genes encoding proteins that are important in the immune response, such as MHC molecules, the T-cell receptor, and immunoglobulins.

Analysis of DNA Regulatory Sequences

The transcriptional activity of genes is regulated by promoter and enhancer sequences. These sequences are *cis-acting*, meaning that they regulate only genes on the same DNA molecule. The promoter sequence lies upstream from the gene it regulates and includes a TATA box, where the general transcrip-

tion machinery, including RNA polymerase II, binds and begins transcription. The enhancer sequence confers a high rate of transcription on the promoter. Unlike the promoter, which always lies upstream from the gene it controls, the enhancer element can be located anywhere with respect to the gene (5' of the promoter, 3' of the gene, or even in an intron of the gene).

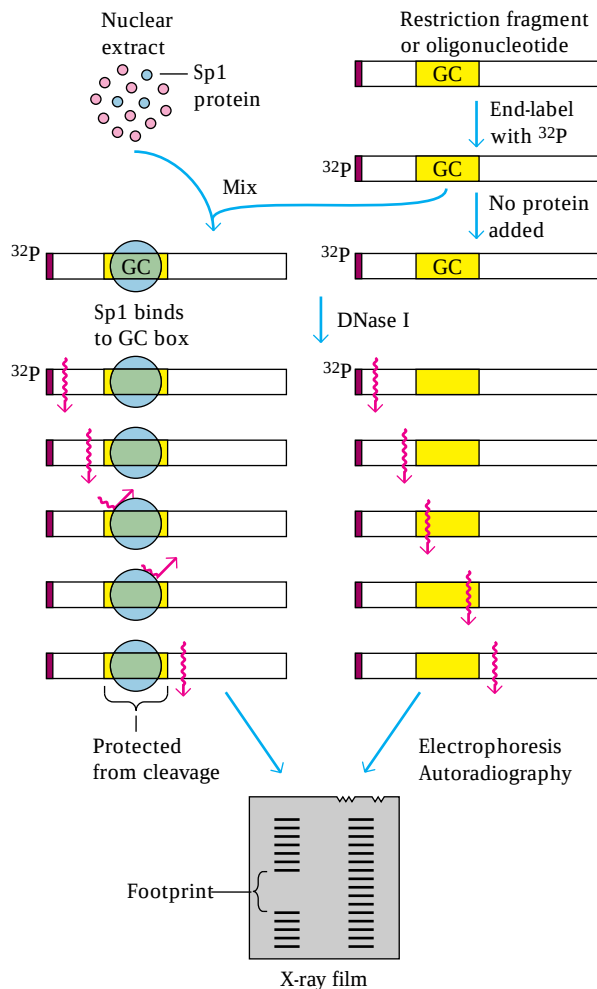
The activity of enhancer and promoter sequences is controlled by transcription factors, which are DNA-binding proteins. These proteins bind to specific nucleotide sequences within promoters and enhancers and act either to enhance or suppress their activity. Enhancer and promoter sequences and their respective DNA-binding proteins have been identified by a variety of techniques, including DNA footprinting, gel-shift analysis, and the CAT assay.

DNA Footprinting Identifies the Sites Where Proteins Bind DNA

The binding sites for DNA-binding proteins on enhancers and promoters can be identified by a technique called DNA footprinting (Figure 23-12a). In this technique, a cloned DNA fragment containing a putative enhancer or promoter sequence is first radiolabeled at the 5' end with ^{32}P . The labeled DNA is then divided into two fractions: one fraction is incubated with a nuclear extract containing a DNA-binding protein; the other DNA fraction is not incubated with the extract. Both DNA samples are then digested with a nuclease

or a chemical that makes random cuts in the phosphodiester bonds of the DNA, and the strands are separated. The resulting DNA fragments are run on a gel to separate fragments of different sizes. In the absence of DNA-binding proteins, a complete ladder of bands is obtained on the electrophoretic gel. When a protein that binds to a site on the DNA fragment is present, it covers some of the nucleotides, protecting that stretch of the DNA from digestion. The electrophoretic pattern of such protected DNA will contain blank regions (or footprints). Each footprint represents the site within an enhancer or promoter that binds a particular DNA-binding protein.

(a) DNA footprinting



(b) Gel-shift analysis

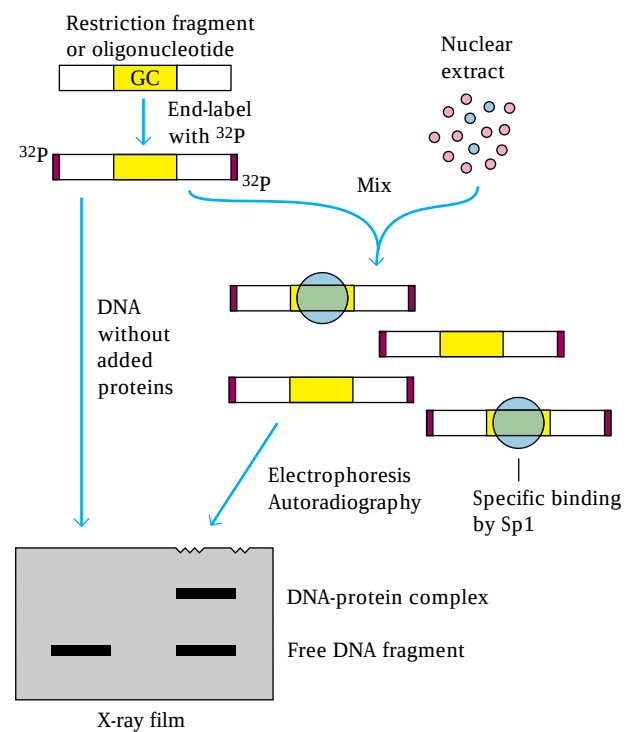


FIGURE 23-12 Identification of DNA sequences that bind protein by DNA-footprinting and gel-shift analysis. (a) In the footprinting technique, labeled DNA fragments containing a putative promoter or enhancer sequence are incubated in the presence and absence of a DNA-binding protein (e.g., Sp1 protein, which binds to a “GC box,” a GC-rich region of DNA). After the samples are treated with DNase and the strands separated, the resulting fragments are electrophoresed; the

gel then is subjected to autoradiography. A blank region (footprint) in the gel pattern indicates that protein has bound to the DNA. (b) In gel-shift analysis, a labeled DNA fragment is incubated with a cellular extract containing transcription factors. The electrophoretic mobility of the DNA-protein complex is slower than that of free DNA fragments. [Adapted from J. D. Watson *et al.*, 1992, *Recombinant DNA*, 2nd ed., W. H. Freeman and Company.]

Gel-Shift Analysis Identifies DNA-Protein Complexes

When a protein binds to a DNA fragment, forming a DNA-protein complex, the electrophoretic mobility of the DNA fragment in a gel is reduced, producing a shift in the position of the band containing that fragment. This phenomenon is the basis of gel-shift analysis. In this technique, radioactively labeled cloned DNA containing an enhancer or a promoter sequence is incubated with a nuclear extract containing a DNA-binding protein (Figure 23-12b). The DNA-protein complex is then electrophoresed and its electrophoretic mobility is compared with that of the cloned DNA alone. A shift in the mobility indicates that a protein is bound to the DNA, retarding its migration on the electrophoretic gel.

CAT Assays Measure Transcriptional Activity

One way to assess promoter activity is to engineer and clone a DNA sequence containing a *reporter gene* attached to the promoter that is being assessed. When this sequence, or construct, is introduced into eukaryotic cells, transcription will be initiated from the promoter if it is active, and the reporter gene will be transcribed and its protein product synthesized. Measuring the amount of this protein produced is thus a way to determine the activity of the promoter.

Most reporter genes are chosen because they encode proteins that can be easily measured, such as the enzyme chloramphenicol acetyltransferase (CAT), which transfers the acetyl group from acetyl-CoA to the antibiotic chloramphenicol (Figure 23-13). The more active the promoter, the more CAT will be produced within the transfected cell. By introducing mutations into promoter sequences and then assaying for promoter activity with the corresponding reporter gene, conserved sequence motifs have been identified within promoters. Another reporter gene, the firefly luciferase gene, is also convenient and easy to use. Luciferase activity is analyzed by the emission of light, which is detected by a luminometer.

Gene Transfer into Mammalian Cells

A variety of genes involved in the immune response have been isolated and cloned by use of recombinant DNA techniques. The expression and regulation of these genes has been studied by introducing them into cultured mammalian cells and, more recently, into the **germ line** of animals.

Cloned Genes Transferred into Cultured Cells Allow in Vitro Analysis of Gene Function

Diverse techniques have been developed for transfecting genes into cells. A common technique involves the use of a retrovirus in which a viral structural gene has been replaced with the cloned gene to be transfected. The altered retrovirus is then

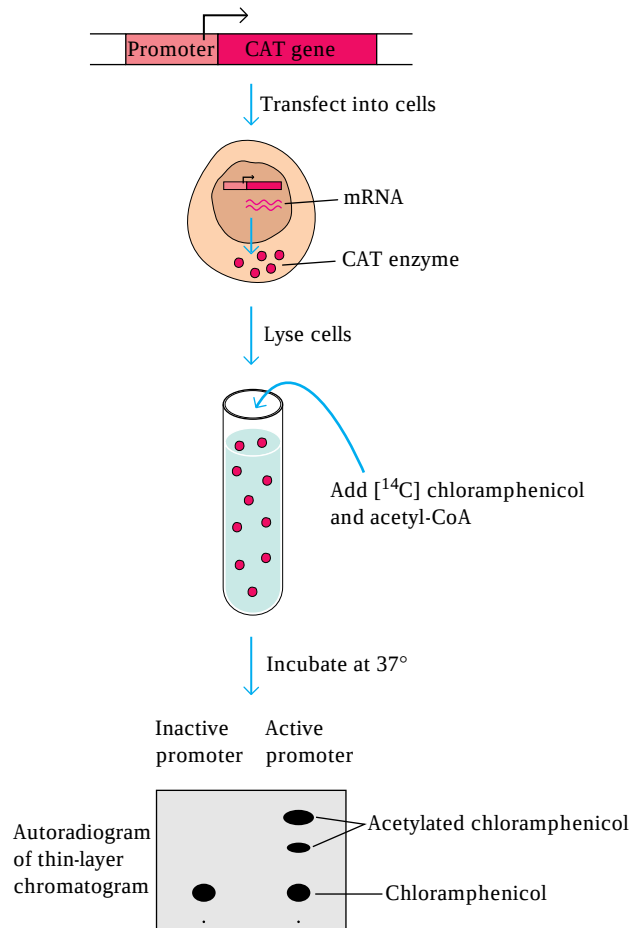


FIGURE 23-13 CAT assay for assessing functional activity of a promoter sequence. In this assay, a DNA construct consisting of the promoter of interest and the reporter gene encoding chloramphenicol acetyltransferase (CAT) is introduced (transfected) into eukaryotic cells. If the promoter is active, the CAT gene will be transcribed and the CAT enzyme will be produced within the transfected cell. The presence of the enzyme can easily be detected by lysing the cell and incubating the cell lysate with [¹⁴C] chloramphenicol and acetyl-CoA. If present, the CAT enzyme will transfer the acetyl group from acetyl-CoA to the chloramphenicol, forming acetylated chloramphenicol, which can be easily detected by thin-layer chromatography. [Adapted from J. D. Watson et al., 1992, *Recombinant DNA*, 2nd ed., W. H. Freeman and Company.]

used as a vector for introducing the cloned gene into cultured cells. Because of the properties of retroviruses, the recombinant DNA integrates into the cellular genome with a high frequency. In an alternative method, the cloned gene of interest is complexed with calcium phosphate. The calcium-phosphate-DNA complex is slowly precipitated onto the cells and the DNA is taken up by a small percentage of them. In another transfection method, called electroporation, an electric current creates pores in cell membranes through which the cloned DNA is taken up. In both of these latter methods, the trans-

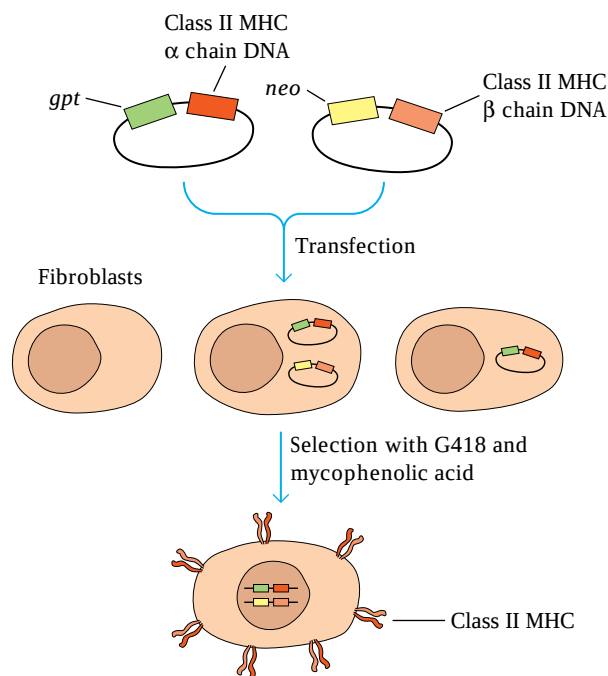


FIGURE 23-14 Transfection of the genes encoding the class II MHC α chain and β chain into mouse fibroblast L cells, which do not normally produce these proteins. Two constructs containing one of the MHC genes and a selectable gene were engineered: the α-chain gene with the guanine phosphoribosyl transferase gene (*gpt*), which confers resistance to the drug G418, and the β-chain gene with a neomycin gene (*neo*), which confers resistance to mycophenolic acid. After transfection, the cells are placed in medium containing both G418 and mycophenolic acid. Only those fibroblasts containing both the *neo* and *gpt* genes (and consequently the genes encoding the class II MHC α and β chains) will survive this selection. These fibroblasts will express both class II MHC chains on their membranes.

fected DNA integrates, apparently at random sites, into the DNA of a small percentage of treated cells.

Generally, the cloned DNA being transfected is engineered to contain a selectable marker gene, such as one that confers resistance to neomycin. After transfection, the cells are cultured in the presence of neomycin. Because only the transfected cells are able to grow, the small number of transfected cells in the total cell population can be identified and selected.

Transfection of cloned genes into cells has proved to be highly effective in immunologic research. By transfecting genes involved with the immune response into cells that lack those genes, the product of a specific gene can be studied apart from interacting proteins encoded by other genes. For example, transfection of MHC genes, under the control of appropriate promoters, into a mouse fibroblast cell line (L929, or simply L cells) has enabled immunologists to study the role of MHC molecules in antigen presentation to T cells (Figure 23-14). Transfection of the gene that encodes the T-cell receptor has

provided information about the antigen-MHC specificity of the T-cell receptor.

Cloned Genes Tranferred into Mouse Embryos Allow in Vivo Analysis of Gene Function

Development of techniques to introduce cloned foreign genes (called **transgenes**) into mouse embryos has permitted immunologists to study the effects of immune-system genes in vivo. If the introduced gene integrates stably into the germ-line cells, it will be transmitted to the offspring. Two techniques for producing transgenic mice are described in this section; one of these has been used to produce **knockout mice**, which cannot express a particular gene product (Table 23-7).

Transgenic Mice Aid in the Analysis of Gene Function

The first step in producing transgenic mice is injection of foreign cloned DNA into a fertilized egg. In this technically demanding process, fertilized mouse eggs are held under suction at the end of a pipet and the transgene is microinjected into one of the pronuclei with a fine needle. The transgene integrates into the chromosomal DNA of the pronucleus and is passed on to the daughter cells of eggs that survive the process. The eggs then are implanted in the oviduct of “pseudo-pregnant” females, and transgenic pups are born after 19 or

TABLE 23-7 Comparison of transgenic and knockout mice		
Characteristic	Transgenic mice	Knockout mice
Cells receiving DNA	Zygote	Embryonic stem (ES) cells
DNA constructs used	Natural gene or cDNA	Mutated gene
Means of delivery	Microinjection into zygote and implantation into foster mother	Transfer of ES cells to blastocyst and implantation into foster mother
Outcome	Gain of a gene	Loss of gene

20 days of gestation (Figure 23-15). In general the efficiency of this procedure is low, with only one or two transgenic mice produced for every 100 fertilized eggs collected.

With transgenic mice, immunologists have been able to study the expression of a given gene in a living animal. Although all the cells in a transgenic animal contain the transgene, differences in the expression of the transgene in different tissues has shed light on mechanisms of tissue-specific gene expression. By constructing a transgene with a particular promoter, researchers can control the expression of a given transgene. For example, the metallothionein promoter is activated by zinc. Transgenic mice carrying a transgene linked to a metallothionein promoter express the transgene only if zinc is added to their water supply. Other promoters are functional only in certain tissues; the insulin promoter, for instance, promotes transcription only in pancreatic cells. Transgenic mice carrying a transgene linked to the insulin promoter, therefore, will express the transgene in the pancreas but not in other tissues.

Because a transgene is integrated into the chromosomal DNA within the one-celled mouse embryo, it will be integrated into both somatic cells and germ-line cells. The resulting transgenic mice thus can transmit the transgene to their offspring as a Mendelian trait. In this way, it has been possible to produce lines of transgenic mice in which every member of a line contains the same transgene. A variety of such transgenic lines are currently available and are widely used in immunologic research. Included among these are lines carrying transgenes that encode immunoglobulin, T-cell receptor, class I and class II MHC molecules, various foreign antigens, and a number of cytokines. Several lines carrying oncogenes as transgenes also have been produced.

Gene-Targeted Knockout Mice Assess the Contribution of a Particular Gene

One of the limitations with transgenic mice is that the transgene is integrated randomly within the genome. This means that some transgenes insert in regions of DNA that are not transcriptionally active, and hence the gene is not expressed. To circumvent this limitation, researchers have developed a technique in which a desired gene is targeted to specific sites within the germ line of a mouse. The primary use of this technique has been to replace a normal gene with a mutant allele or a disrupted form of the gene, thus knocking out the gene's function. Transgenic mice that carry such a disrupted gene, called *knockout mice*, have been extremely helpful to immunologists trying to understand how the removal of a particular gene product affects the immune system. Various knockout mice are being used in immunologic research, including mice that lack particular cytokines or MHC molecules.

Production of gene-targeted knockout mice involves the following steps:

- Isolation and culturing of embryonic stem (ES) cells from the inner cell mass of a mouse blastocyst

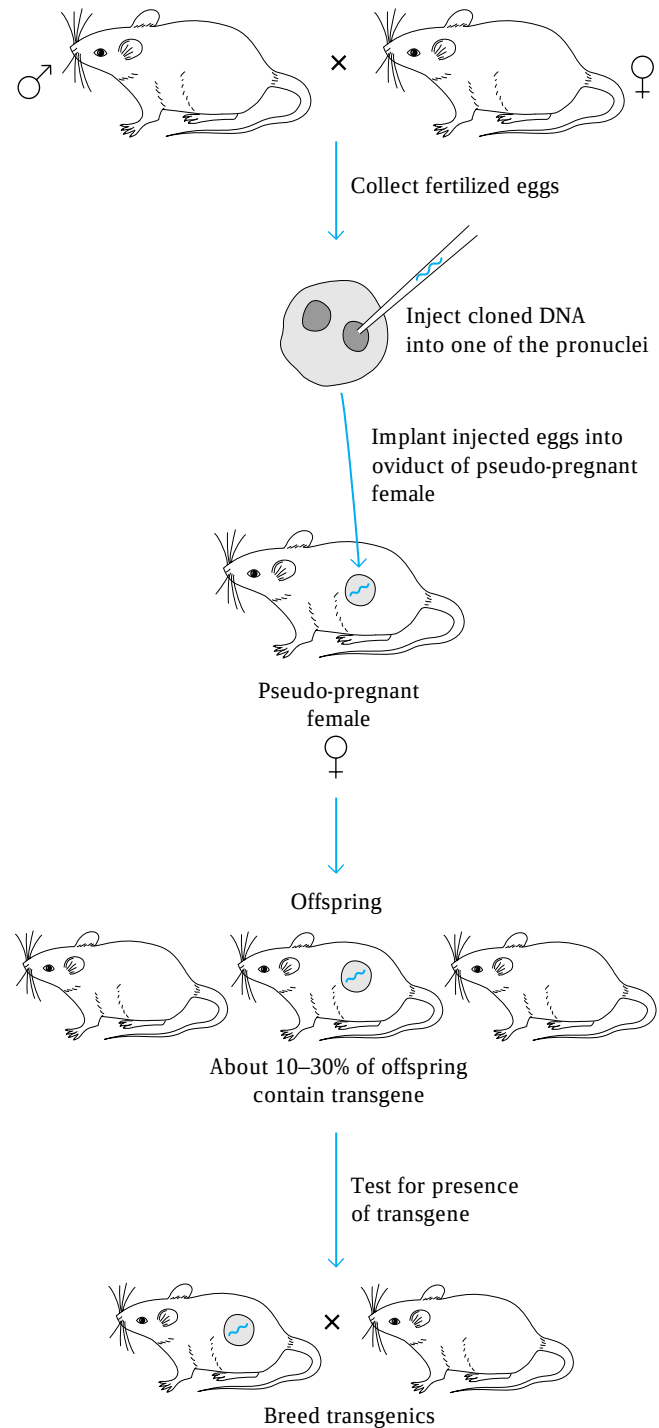
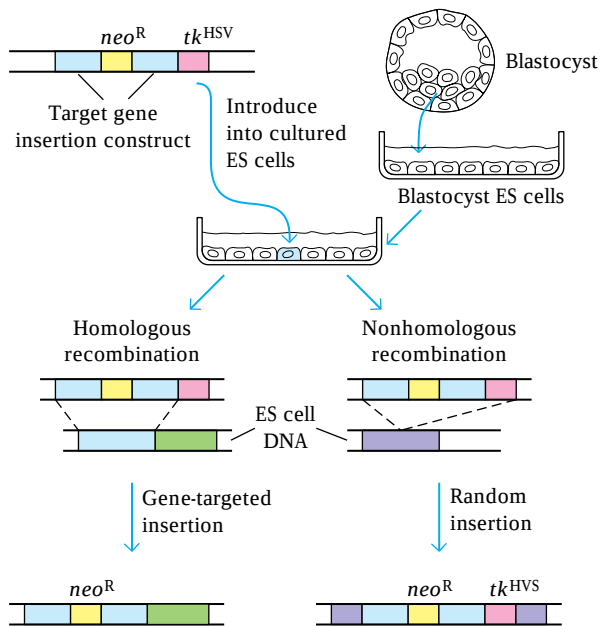


FIGURE 23-15 General procedure for producing transgenic mice. Fertilized eggs are collected from a pregnant female mouse. Cloned DNA (referred to as the transgene) is microinjected into one of the pronuclei of a fertilized egg. The eggs are then implanted into the oviduct of pseudopregnant foster mothers (obtained by mating normal females with a sterile male). The transgene will be incorporated into the chromosomal DNA of about 10%–30% of the offspring and will be expressed in all of their somatic cells. If a tissue-specific promoter is linked to a transgene, then tissue-specific expression of the transgene will result.

(a) Formation of recombinant ES cells



(b) Selection of ES cell carrying knockout gene

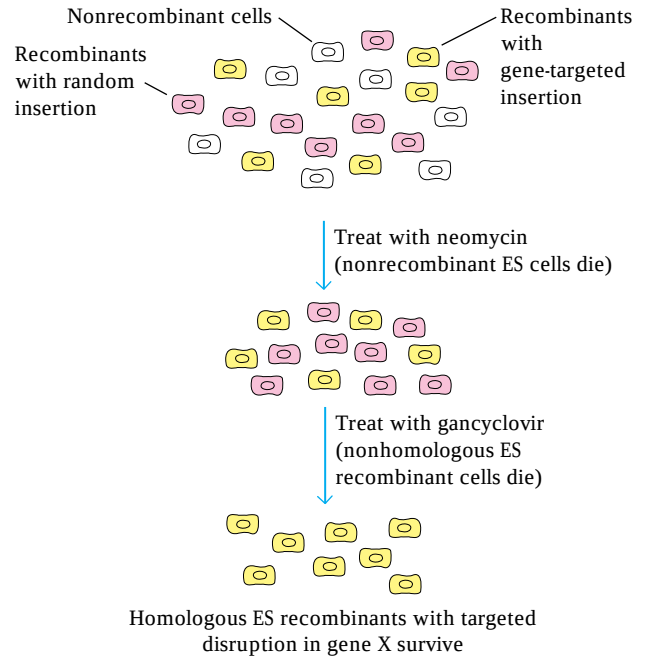


FIGURE 23-16 Formation and selection of mouse recombinant ES cells in which a particular target gene is disrupted. (a) In the engineered insertion construct, the target gene is disrupted with the *neo^R* gene, and the thymidine kinase *tk^{HSV}* gene is located outside the target gene. The construct is transfected into cultured ES cells. If homologous recombination occurs, only the target gene and the *neo^R* gene will be inserted into the chromosomal DNA of the ES cells. If nonhomologous recombination occurs, all three genes will be inserted. Recombination occurs

in only about 1% of the cells, with nonhomologous recombination much more frequent than homologous recombination. (b) Selection with the neomycin-like drug G418 will kill any nonrecombinant ES cells because they lack the *neo^R* gene. Selection with gancyclovir will kill the nonhomologous recombinants carrying the *tk^{HSV}* gene, which confers sensitivity to gancyclovir. Only the homologous ES recombinants will survive this selection scheme. [Adapted from H. Lodish et al., 1995, *Molecular Cell Biology*, 3rd ed., Scientific American Books.]

- Introduction of a mutant or disrupted gene into the cultured ES cells and selection of homologous recombinant cells in which the gene of interest has been knocked out (i.e., replaced by a nonfunctional form of the gene)
- Injection of homologous recombinant ES cells into a recipient mouse blastocyst and surgical implantation of the blastocyst into a pseudo-pregnant mouse
- Mating of chimeric offspring heterozygous for the disrupted gene to produce homozygous knockout mice

The ES cells used in this procedure are obtained by culturing the inner cell mass of a mouse blastocyst on a feeder layer of fibroblasts or in the presence of leukemia-inhibitory factor. Under these conditions, the stem cells grow but remain pluripotent and capable of later differentiating in a variety of directions, generating distinct cellular lineages (e.g., germ cells, myocardium, blood vessels, myoblasts, nerve cells). One of the advantages of ES cells is the ease with which they can be genetically manipulated. Cloned DNA containing a desired

gene can be introduced into ES cells in culture by various transfection techniques. The introduced DNA will be inserted by recombination into the chromosomal DNA of a small number of ES cells.

The insertion constructs introduced into ES cells contain three genes: the target gene of interest and two selection genes, such as *neo^R*, which confers neomycin resistance, and the thymidine kinase gene from herpes simplex virus (*tk^{HSV}*), which confers sensitivity to gancyclovir, a cytotoxic nucleotide analog (Figure 23-16a). The construct often is engineered with the target-gene sequence disrupted by the *neo^R* gene and with the *tk^{HSV}* gene at one end, beyond the sequence of the target gene. Most constructs will insert at random by nonhomologous recombination rather than by gene-targeted insertion through homologous recombination. As illustrated in Figure 23-16b, a two-step selection scheme is used to obtain those ES cells that have undergone homologous recombination, whereby the disrupted gene replaces the target gene.

The ES cells obtained by this procedure are heterozygous for the knockout mutation in the target gene. These cells are clonally expanded in cell culture and then injected into a

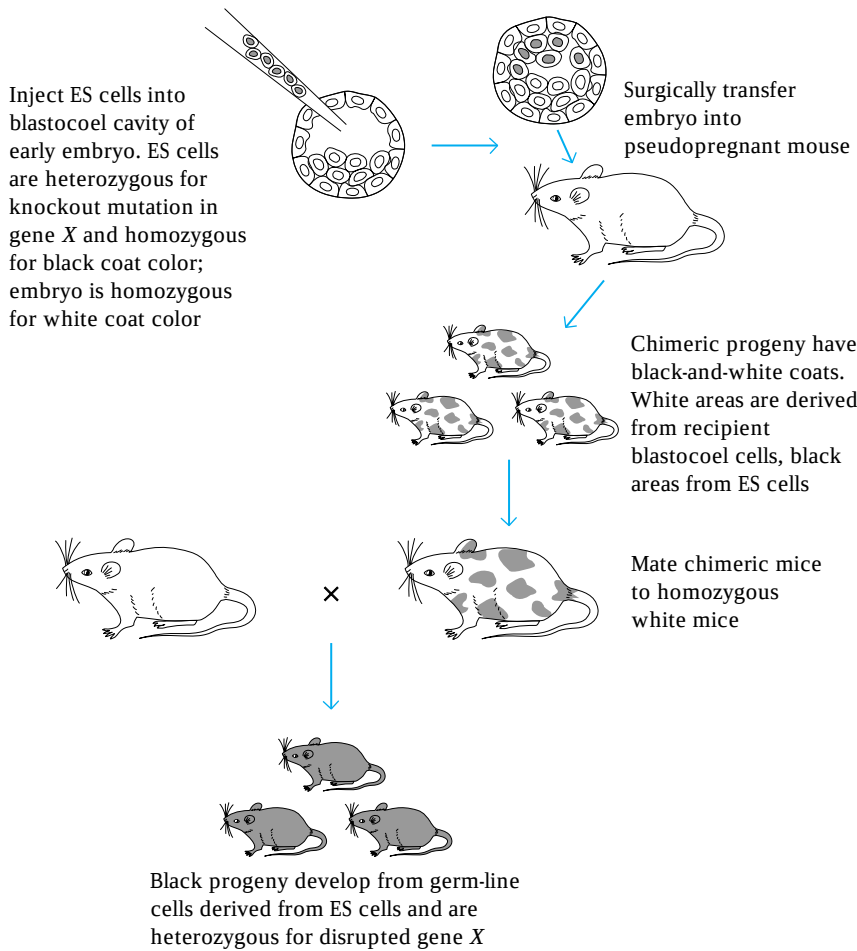


FIGURE 23-17 General procedure for producing homozygous knockout mice. ES cells homozygous for a marker gene (e.g., black coat color) and heterozygous for a disrupted target gene (see Figure 23-18) are injected into an early embryo homozygous for an alternate marker (e.g., white coat color). The chimeric transgenic offspring, which have black-and-white coats, then are mated with homozygous white mice. The all-black progeny from this mating have ES-derived cells in their germ line, which are heterozygous for the disrupted target gene. Mating of these mice with each other produces animals homozygous for the disrupted target gene, that is, knockout mice. [Adapted from M. R. Capecchi, 1989, *Trends Genet.* 5:70.]

mouse blastocyst, which subsequently is implanted into a pseudo-pregnant female. The transgenic offspring that develop are chimeric, composed of cells derived from the genetically altered ES cells and cells derived from normal cells of the host blastocyst. When the germ-line cells are derived from the genetically altered ES cells, the genetic alteration can be passed on to the offspring. If the recombinant ES cells are homozygous for black coat color (or other visible marker) and they are injected into a blastocyst homozygous for white coat color, then the chimeric progeny that carry the heterozygous knockout mutation in their germ line can be easily identified (Figure 23-17). When these are mated with each other, some of the offspring will be homozygous for the knockout mutation.

“Knock-In” Technology Allows the Replacement of an Endogenous Gene

In addition to deleting a gene of choice, it also is possible to replace the endogenous gene with a mutated form of that gene. As in the strategy for knocking out a gene, DNA constructs that carry mutations in a particular gene can be exchanged for the endogenous gene. It also is possible to replace all of an endogenous gene with a DNA sequence of

choice. In a recent report, for example, the CD4 gene was replaced with the one for β -galactosidase. In these experiments, the CD4 promoter was left intact to drive the expression of β -galactosidase, which catalyzes the color change of certain reporter chemicals to blue. Because the CD4 promoter drove the expression of β -galactosidase, only those thymic cells destined to express CD4 turned blue in the presence of the reporter chemicals. Data from these experiments were useful in tracing CD4/CD8 lineage commitment in developing T cells.

Inducible Gene Targeting, the Cre/Lox System, Targets Gene Deletion

In addition to the deletion of genes by gene targeting, recent experimental strategies have been developed that allow the specific deletion of a gene of interest in precisely the tissue of choice. These technologies rely on the use of site-specific recombinases from bacteria or yeast. The most commonly used recombinase is Cre, isolated from bacteriophage P1. Cre recognizes a specific 34-bp site in DNA known as *loxP* and, upon recognition, catalyzes a recombination. Therefore, DNA sequences that are flanked by *loxP* are recognized by Cre and

the recombinational event results in the deletion of the intervening DNA sequences. In other words, animals that ubiquitously express Cre recombinase will delete all *loxP*-flanked sequences. The real innovation of this technique is that expression of the Cre recombinase gene can be controlled by the use of a tissue specific promoter. This allows tissue-specific expression of the recombinase protein and thus tissue-specific deletion of DNA flanked by *loxP*. For example, one could ex-

press Cre in B cells using the immunoglobulin promoter, and this would result in the targeted deletion of *loxP*-flanked DNA sequences only in B cells.

This technology is particularly useful when the targeted deletion of a particular gene is lethal. For example the DNA polymerase β gene is required for embryonic development. In experiments designed to test the Cre/*lox* system, scientists flanked the mouse DNA polymerase β gene with *loxP* and mated these mice with mice carrying a Cre transgene under the control of a T-cell promoter (Figure 23-18a). The results of this mating are offspring that express the Cre recombinase specifically in T cells. Using such mice, the scientists were able to examine the effects of deleting the enzyme DNA polymerase β specifically in T cells. The effects of the deletion of this gene could not be examined in a conventional gene-targeting experiment, because deletion of DNA polymerase β throughout the animal would be lethal. However, with the Cre/*lox* system, it now is possible to examine the effects of the deletion of this gene in a specific tissue of the immune system.

The Cre/*lox* system also can be used to turn on gene expression in a particular tissue. Just as the lack of a particular gene may be lethal during embryonic development, the expression of a gene can be toxic. To examine tissue-specific expression of such a gene, it is possible to insert a translational stop sequence flanked by *loxP* into an intron at the beginning of the gene (Figure 23-18b). Using a tissue-specific promoter driving Cre expression, the stop sequence may be deleted in the tissue of choice and the expression of the potentially toxic gene examined in this tissue. These modifications of gene-targeting technology have been very useful in determining the effects of particular genes in cells and tissues of the immune system.

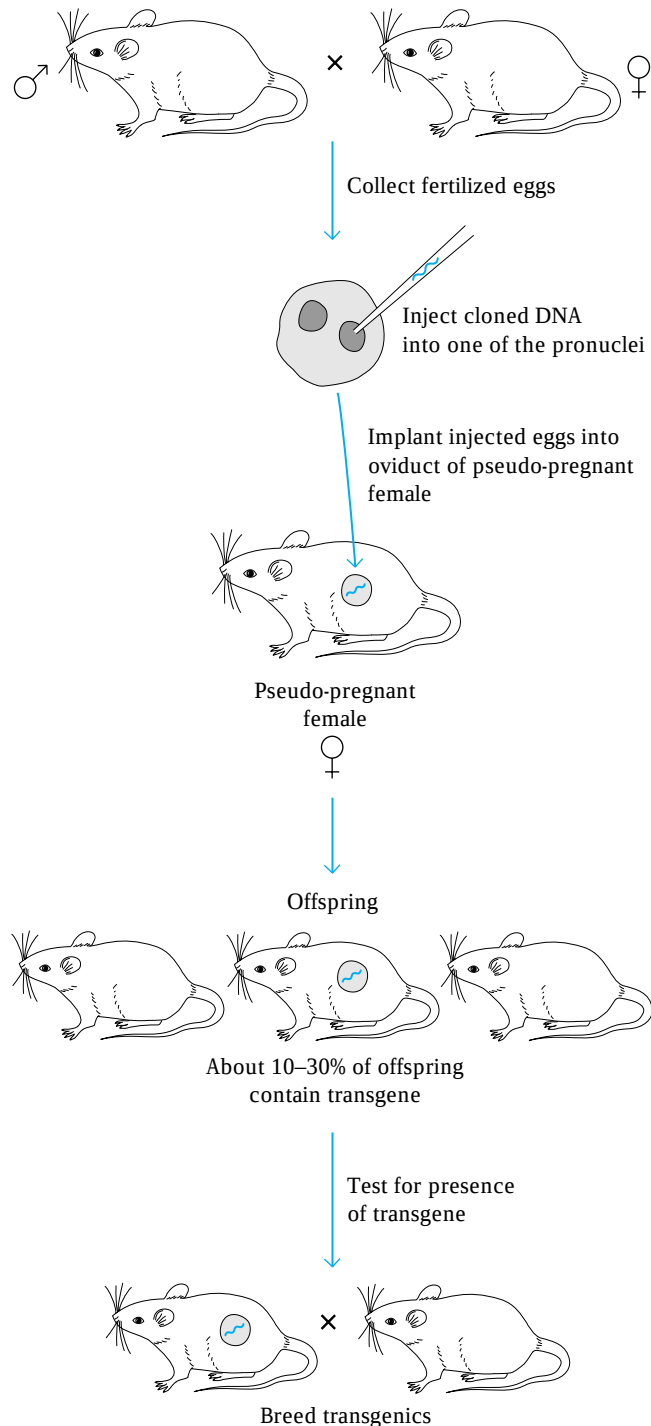


FIGURE 23-18 Gene targeting with Cre/*loxP* (a) Conditional deletion by Cre recombinase. The targeted DNA polymerase β gene is modified by flanking the gene with *loxP* sites (for simplicity, only one allele is shown). Mice are generated from ES cells by standard procedures. Mating of the *loxP*-modified-mice with a Cre transgenic will generate double transgenic mice in which the *loxP*-flanked DNA polymerase β gene will be deleted in the tissue where Cre is expressed. In this example, Cre is expressed in thymus tissue (striped) so that deletion of the *loxP*-flanked gene occurs only in the thymus (white) of the double transgenic. Other tissues and organs still express the *loxP*-flanked gene (orange). (b) Activation of gene expression using Cre/*lox*. A *loxP*-flanked translational STOP cassette is inserted between the promoter and the potentially toxic gene, and mice are generated from ES cells using standard procedures. These mice are mated to a transgenic line carrying the Cre gene driven by a tissue-specific promoter. In this example, Cre is expressed in the thymus, so that mating results in expression of the toxic gene (blue) solely in the thymus. Using this strategy, it is possible to determine the effects of expression of the potentially toxic gene in a tissue-specific fashion. [Adapted from B. Sauer, 1998, *Methods* **14**:381.]

Microarrays—An Approach for Analyzing Patterns of Gene Expression

In the past few years, a new approach has emerged designed to assess differences in gene expression between various cell types or the same cells treated in different fashions. This technology, referred to as microarray technology or *gene profiling*, has the ability to rapidly and reliably scan large numbers of different mRNAs. The principle is simple and is derived from what we already know about RNA and DNA hybridization. mRNA is isolated from a given sample. Then, when cDNA synthesis is initiated the first strand of the cDNA is labeled with the tag. This forms the pool of target sequences.

The next step is to hybridize the labeled cDNA to a microarray. There are many microarrays commercially available, which fall mainly into two classes; those composed of cDNA, and those composed of oligonucleotides. Microarrays of cDNAs are, as the name suggests, a collection of cDNA that have been arranged, or arrayed, on a solid substrate in defined locations. The substrate varies but usually is a nylon membrane or a glass slide. If a very small amount of cDNA is used, the spots of cDNA arrayed on the substrate can be as small as 100–300 μm in size; it is relatively simple to array as many as 30,000 cDNAs on a single microscope slide (Figure 23-19a). The actual process of arraying the cDNA is usually accomplished using robotics. The cDNAs are most frequently obtained from available cDNA libraries and, in some cases, are PCR products amplified from the cDNA library using primers specific for certain known genes.

The oligonucleotide arrays are usually a collection of oligos 20–25 nucleotides long (Figure 23-19b). The advantage of this type of array is that one only needs sequences of genes of interest. No cDNA library is needed. However, the cost of assembling such an array is high, since the oligos have to be made and then spotted onto the filter or glass slide. Another problem with this approach is, depending upon the length of the oligo, there can be a degree of non-specific hybridization that hinders the final analysis of the data. This problem can be avoided by making longer oligos—which further increases the cost. For these reasons, oligo arrays are used most often by large pharmaceutical or biotechnology companies.

Although the source of the targets used for both cDNA and oligo arrays are cDNA, the preparation of the target differs depending upon the microarray. The target preparation for cDNA arrays involves labeling the cDNA with different fluorescent dyes such as Cy3 and Cy5 (Figure 28-19a). Cy3 and Cy5 are cyanine-based dyes that are easily conjugated to nucleic acids and are highly stable and emit less background fluorescence than conventional fluorescent dyes. Suppose you wish to compare two different cell types, or one cell type in two different states of activation. cDNA from one population is prepared using mRNA as a template. First strand synthesis of the mRNA is performed using one nucleotide con-

jugated to Cy3. Then, using mRNA from the second cell population, cDNA is prepared using a nucleotide conjugated to Cy5. These two populations of cDNA, one marked with Cy3 and the other with Cy5, are hybridized to the microarray. If one of the targets hybridizes to a cDNA on the array, a green (Cy3) or red (Cy5) fluorescence emission is detected. If both hybridize to the cDNA, yellow fluorescence is detected (the combination of the red and green emissions from both dyes). The arrays are analyzed by scanning the array at two different wavelengths to distinguish between the Cy3 and Cy5 signals. Once scanned at two wavelengths, the signals are compared and the signal intensity of each dye is determined and compared. The results are presented as a ratio between the two samples.

In the case of oligo-based microarrays, the usual approach is to label the target cDNA with a biotin-labeled nucleotide during first-strand synthesis of the mRNA. The biotin-labeled cDNA is hybridized to the oligo array and detected by the use of the fluorescent streptavidin (Figure 28-19b). The procedure is then repeated with cDNA from the other cell type and another microarray is used. The resultant microarrays are analyzed by either phosphoimaging or fluorescent-based scanning. This is most commonly accomplished using specialized scanners developed for scanning microarrays.

The difference between this procedure and the cDNA-based array described above is that two microarrays are used. This is possible since the method for producing the oligo-based microarrays is more precise and it is possible to ensure that the same oligo will be present in precisely the same position on two separate microarrays. This is not possible with the technology used to prepare cDNA microarrays. Therefore, both targets must be hybridized to the same array to derive an accurate comparison. There is an advantage to using two microarrays. Quantitation of expression levels is easier when using one labeled target per microarray. When two targets are hybridized to the same array, it is always necessary to “subtract” the fluorescence of one target from the other before it is possible to obtain quantitative data. Since only one target is hybridized to a single oligo microarray, subtraction is not necessary.

The application of microarray technology to immunology is apparent. One could easily ask what is the difference between T cells and B cells. Or what is the difference between an activated T cell and a resting T cell? The list of possible comparisons is immense. To begin to answer some of the interesting immunology questions, Louis Staudt and co-workers at the NIH have developed an array they term “Lymphochip.” The Lymphochip is an array that consists of more than 10,000 human genes and is enriched in genes expressed in lymphoid cells. It also includes genes from normal as well as transformed lymphocytes. This particular microarray has provided a great deal of useful information, including a profile of T cells compared to B cells, plasma cells compared to germinal center B cells, and gene expression patterns induced by various signaling pathways. The Lymphochip and other clinical applications of microarrays are described in the Clinical Focus box.

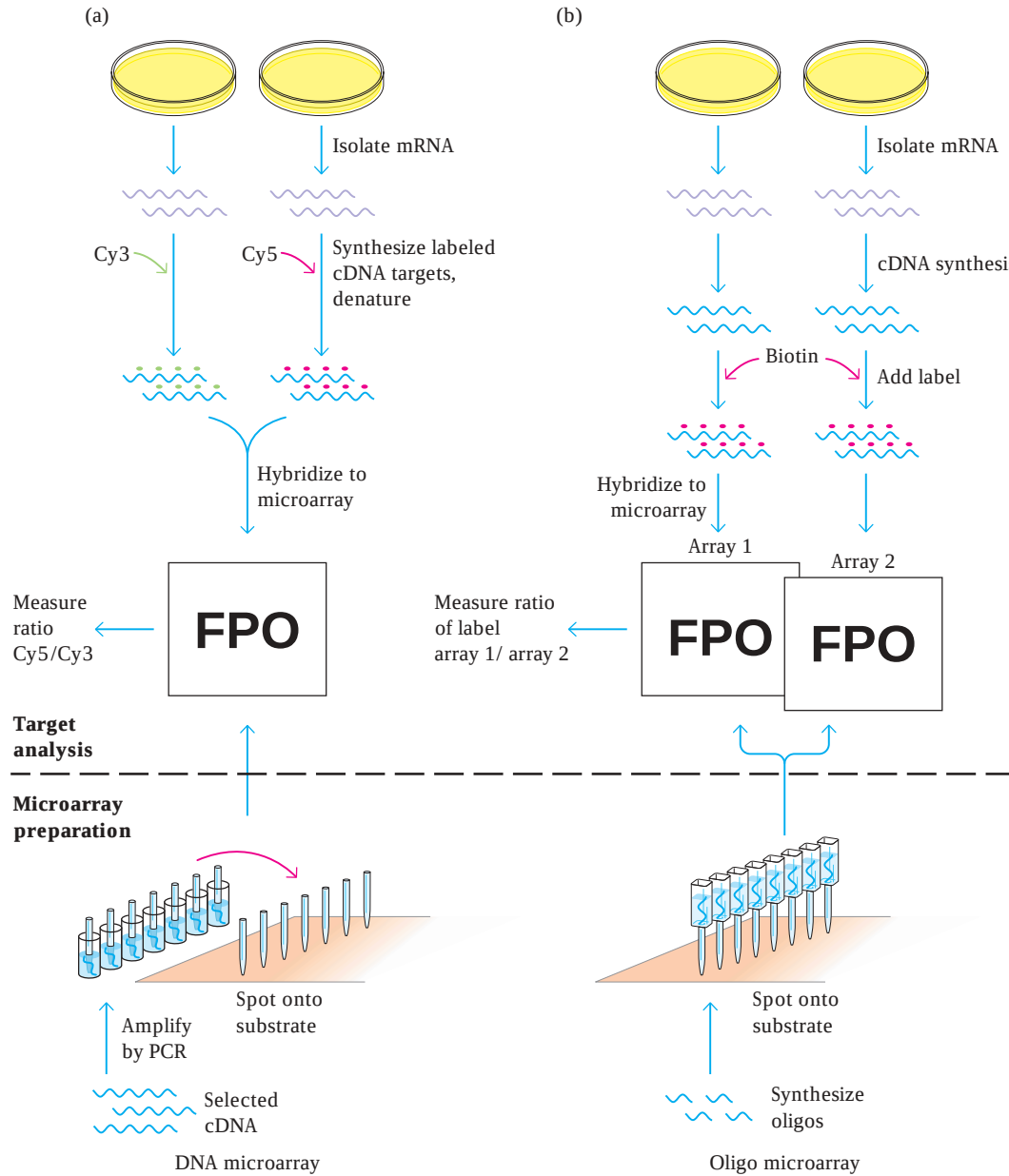


FIGURE 23-19 DNA microarray analysis using cDNA microarrays (a) or high-density oligonucleotide microarrays (b). As described in the text, microarray analysis relies on the isolation of RNA from the tissues or cells to be analyzed, the conversion of RNA into cDNA,

and the subsequent labeling of DNA during target preparation. The labeled target sequences are hybridized to either a cDNA microarray (a) or an oligo microarray (b).

SUMMARY

- Inbred mouse strains allow immunologists to work routinely with syngeneic, or genetically identical, animals. With these strains, aspects of the immune response can be studied uncomplicated by unknown variables that could be introduced by genetic differences between animals.
- In adoptive-transfer experiments, lymphocytes are transferred from one mouse to a syngeneic recipient mouse that

has been exposed to a sublethal (or potentially lethal) dose of x-rays. The irradiation inactivates the immune cells of the recipient, so that one can study the response of only the transferred cells.

- With in vitro cell-culture systems, populations of lymphocytes can be studied under precisely defined conditions. Such systems include primary cultures of lymphoid cells, cloned lymphoid cell lines, and hybrid lymphoid cell lines. Unlike primary cultures, cell lines are immortal and homogeneous.



CLINICAL FOCUS

Microarray Analysis as a Diagnostic Tool for Human Diseases

It is almost impossible to distinguish visually between B and T cells without molecular analysis. Similarly, it can be quite difficult to distinguish one tumor from another. Two of the best-known acute leukemias are AML, which arises from a myeloid precursor (hence the name, acute *myeloid leukemia*) and ALL (acute *lymphoid leukemia*), which arises from lymphoid precursors. Both leukemias are derived from hematopoietic stem cells, but the prognosis and treatment for the two diseases are quite different. Until recently, the two diseases could be diagnosed with some degree of confidence using a combination of surface phenotyping, karyotypic analysis, and histochemical analysis, but no single test was conclusive; reliable diagnosis depended upon the expertise of the clinician.

The difference between an ALL diagnosis and an AML diagnosis can mean the difference between life and death. ALL responds best to corticosteroids and chemotherapeutics such as vincristine and methotrexate. AML is usually treated with daunorubicin and cytarabine. The cure rates are dramatically diminished if the less appropriate treatment is delivered due to misdiagnosis.

In 1999, a breakthrough in diagnosis of these two leukemias was achieved using microarray technology. Todd Golub, Eric Lander, and their colleagues isolated

RNA from 38 samples of acute leukemia, labeled the RNA with biotin, and hybridized the biotinylated RNA to commercial high-density microarrays that contained oligonucleotides corresponding to some 6817 human genes. Whenever the biotin-labeled RNA recognized a homologous oligonucleotide, hybridization occurred. Analysis revealed a group of 50 genes that were highly associated with either AML or ALL when compared with control samples. These 50 genes were then used to sample nucleic acid from 34 independent leukemias as well as samples from 24 presumed-normal human bone-marrow or blood samples. The result? A set of markers that clearly classified a tumor as ALL or AML.

The results of the microarray analysis further suggested that the treatments for AML and ALL can be targeted more precisely. For example, an AML expressing genes *x*, *y*, and *z* might respond to one treatment modality better than an AML that expresses *a*, *b*, and *c*. Several pharmaceutical companies have established research groups to evaluate different treatments for tumors based on the tumor's microarray profile. This designer-approach to oncology is expected to produce much more effective treatments of individual tumors, and ultimately, enhanced survival rates.

Microarray analysis is likely to be very useful in the diagnosis of tumors of the immune system. Most notably, a labora-

tory at the National Institutes of Health (NIH) has developed a specialized DNA microarray containing more than 10,000 human cDNAs that are enriched for genes expressed in lymphocytes. Some of these cDNAs are from genes of known function, others are unknown cDNAs derived from normal or malignantly transformed lymphocyte cDNA libraries. This specialized array is called the "Lymphochip" because the lymphocyte cDNAs are arrayed on a silicon wafer. The group at NIH asked whether they could use the Lymphochip to divide the B-cell leukemia known as diffuse large B-cell lymphoma (DLBCL) into subgroups, an important question because this type of lymphoma has a highly variable clinical course, with some patients responding well to treatment while others respond poorly. Earlier attempts to define subgroups within this group had been unsuccessful. A definition of subgroups within DLBCL could be useful in designing more effective treatments. Using the Lymphochip, the group at NCI identified two genotypically distinct subgroups of DLBCL. One group was comprised of tumors expressing genes characteristic of germinal-center B cells and was called "germinal-center-B-like DLBCL (see Figure). The other group more resembled activated B cells and was termed "activated B-like DLBCL." Significantly, patients with germinal-center-B-like DLBCL had a higher survival rate than those with activated B-like DLBCL. Normally all patients with DLBCL receive multi-agent chemotherapy. Patients who do not respond well to chemotherapy are then considered for bone-marrow transplantation. The data obtained from this study suggests that patients with activated B-like DLBCL will not respond as well to chemotherapy and may be better served

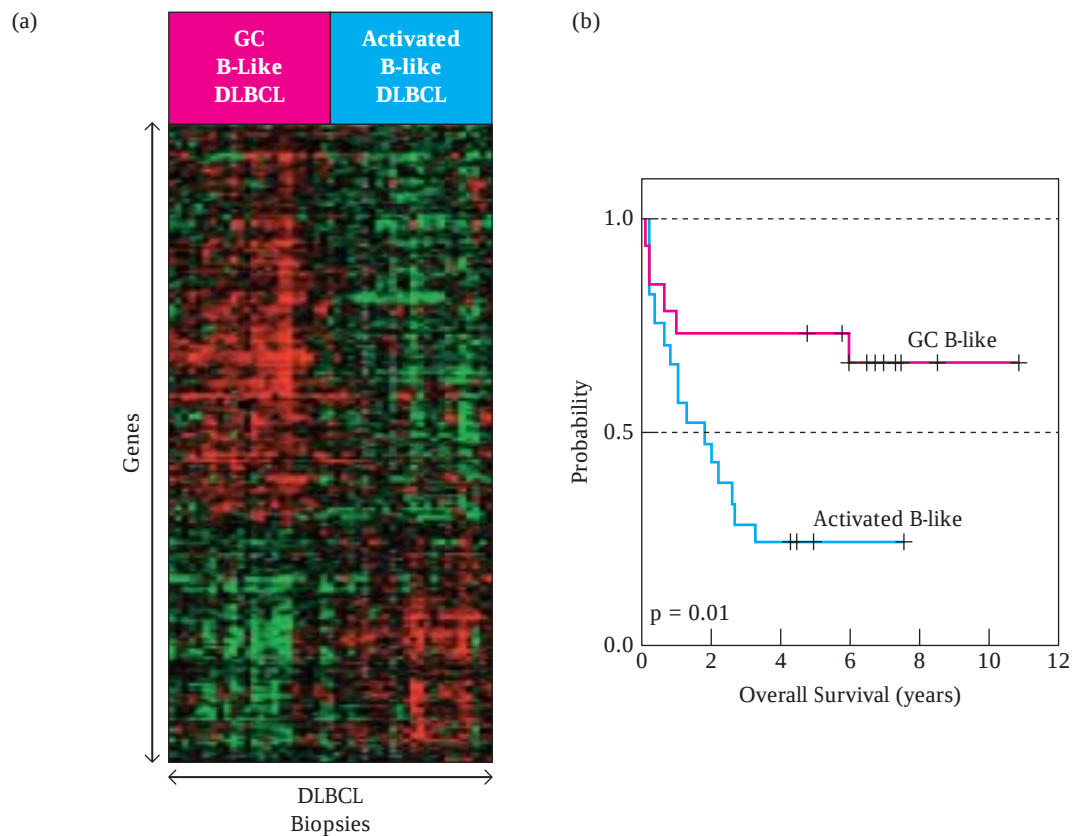
- Biochemical techniques provide tools for labeling important proteins of the immune system. Labeling antibodies with molecules such as biotin and avidin allows accurate determination of the level of antibody response. Gel electrophoresis is a convenient tool for separating and determining the molecular weight of a protein.
- The ability to identify, clone, and sequence immune-system genes using recombinant DNA techniques has revolutionized the study of all aspects of the immune response. Both cDNA, which is prepared by transcribing mRNA with reverse transcriptase, and genomic DNA can be cloned. Generally, cDNA is cloned using a plasmid vector; the re-

by bone-marrow transplantation shortly after diagnosis. As a direct result of this work, ongoing clinical trials are evaluating how best to treat patients with activated B-like DLBCL.

Gene profiling is not restricted to diagnosis of cancer. This technology pro-

vides us with a unique opportunity to examine differences between any distinct populations of cells. One can compare which genes are expressed in common or differentially in a naïve T cell and a memory T cell. What is the difference between a normal T cell and a T cell dy-

ing by apoptosis? Comparisons like these will be a rich source of insight into differences in cell populations. The key to using this valuable information will be the development of tools to analyze the vast quantities of data that can be obtained from this new approach.



Diffuse large B-cell lymphoma (DLBCL) is at least two distinct diseases. (a) Shown are differences in gene expression between samples taken from patients with either germinal center B-like DLBCL (left, orange) or activated B-like DLBCL (right, blue). Relative expression of the 100 genes (y axis) that discriminate most significantly between the two DLBCL types is depicted over a 16-fold range using the graded color scale at bottom. Note the strikingly different gene

expression profiles of the two diseases. (b) Plot of overall DLBCL patient survival following chemotherapy. Gene expression profiles of tumor-biopsy samples allow the assignment of patients to the correct prognostic categories and may aid in the treatment of this complex disease. [Adapted from L. M. Staudt, 2002. *Gene expression profiling of lymphoid malignancies*. *Annu. Rev. Med.* **53**:303-318.]

combinant DNA containing the gene to be cloned is propagated in *E. coli* cells. Genomic DNA can be cloned within a bacteriophage vector or a cosmid vector, both of which are propagated in *E. coli*. Even larger genomic DNA fragments can be cloned within bacteriophage P1 vectors, which can replicate in *E. coli*, or yeast artificial chromo-

somes, which can replicate in yeast cells. Polymerase chain reaction (PCR) is a convenient tool for amplifying small quantities of DNA.

- Transcription of genes is regulated by promoter and enhancer sequences; the activity of these sequences is

controlled by DNA-binding proteins. Footprinting and gel-shift analysis can be used to identify DNA-binding proteins and their binding sites within the promoter or enhancer sequence. Promoter activity can be assessed by the CAT assay.

- Cloned genes can be transfected (transferred) into cultured cells by several methods. Commonly, immune-system genes are transfected into cells that do not normally express the gene of interest. Cloned genes also can be incorporated into the germ-line cells of mouse embryos, yielding transgenic mice, which can transmit the incorporated transgene to their offspring. Expression of a chosen gene can then be studied in a living animal. Knockout mice are transgenics in which a particular target gene has been replaced by a nonfunctional form of the gene, so the gene product is not expressed. The Cre/lox system provides a mechanism that allows tissue-specific expression or deletion of a particular gene.
- Microarrays are a powerful approach for the examination of tissue-specific gene expression and comparison of gene expression in different cells. It has already begun to revolutionize the study of gene regulation and gene expression.

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USEFUL WEB SITES

<http://www.biomednet.com/db/mkmd>

Access to all known knockouts in mice, updated regularly.

<http://www.jax.org/>

Home page for The Jackson Laboratory, the major repository of inbred mice in the world.

<http://www.neb.com/>

Home page for New England Biolabs, a molecular biology company. Useful information concerning restriction enzymes is found at this site, under Technical Resources.

http://www.public.iastate.edu/~pedro/research_tools.html

A very useful site for molecular biology, containing links to many informative sites. Updated regularly.

Study Questions

CLINICAL FOCUS QUESTION How has microarray technology changed disease diagnosis and how is it likely to influence treatment of diseases in the future?

1. Explain why the following statements are false.
 - a. The amino-acid sequence of a protein can be determined from the nucleotide sequence of a genomic clone encoding the protein.
 - b. Transgenic mice can be prepared by microinjection of DNA into a somatic-cell nucleus.
 - c. Primary lymphoid cultures can be propagated indefinitely and are useful in studies of specific subpopulations of lymphocytes.
2. Fill in the blanks in the following statements with the most appropriate terms:
 - a. In inbred mouse strains, all or nearly all genetic loci are _____; such strains are said to be _____.

- b. SCID mice have a genetic defect that prevents development of functional _____ and _____ cells.
- c. B-cell hybridomas are formed by fusion of _____ with _____. They are capable of _____ growth and are used to produce _____.
- d. A normal lymphoid cell that undergoes _____ can give rise to a cell line, which has an _____ life span.
- 3.** The gene diagrammed below contains one leader (L), three exons (E), and three introns (I). Illustrate the primary transcript, mRNA, and the protein product that could be generated from such a gene.
- 4.** The term *transfection* refers to which of the following?
- Synthesis of mRNA from a DNA template
 - Synthesis of protein based on an mRNA sequence
 - Introduction of foreign DNA into a cell
 - The process by which a normal cell becomes malignant
 - Transfer of a signal from outside a cell to inside a cell
- 5.** Which of the following are required to carry out the PCR?
- Short oligonucleotide primers
 - Thermostable DNA polymerase
 - Antibodies directed against the encoded protein
 - A method for heating and cooling the reaction mixture periodically
 - All of the above
- 6.** Why is it necessary to include a selectable marker gene in transfection experiments?
- 7.** What would be the result if a transgene were injected into one cell of a four-cell mouse zygote rather than into a fertilized mouse egg before it divides?
- 8.** A circular plasmid was cleaved with *EcoRI*, producing a 5.4-kb band on a gel. A 5.4-kb band was also observed when the plasmid was cleaved with *HindIII*. Cleaving the plasmid with both enzymes simultaneously resulted in a single band 2.7 kb in size. Draw a diagram of this plasmid showing the relative location of its restriction sites. Explain your reasoning.
- 9.** DNA footprinting is a suitable technique for identifying which of the following?
- Particular mRNAs in a mixture
 - Particular tRNAs in a mixture
 - Introns within a gene
 - Protein-binding sites within DNA
 - Specific DNA sites at which restriction endonucleases cleave the nucleotide chain
- 10.** Explain briefly how you might go about cloning a gene for interleukin 2 (IL-2). Assume that you have available a monoclonal antibody specific for IL-2.
- 11.** You have a sample of a mouse DNA-binding protein and of the mRNA that encodes it. Assuming you have a mouse genomic library available, briefly describe how you could select a clone carrying a DNA fragment that contains the gene that encodes the binding protein.
- 12.** What are the major differences between transgenic mice and knockout mice and in the procedures for producing them?
- 13.** How does a knock-in mouse differ from a knockout mouse?
- 14.** How does the Cre/*lox* technology enhance knockout and knock-in strategies?
- 15.** For each term related to recombinant DNA technology (a–i), select the most appropriate description (1–10) listed below. Each description may be used once, more than once, or not at all.

Terms

- _____ Yeast artificial chromosome
- _____ Restriction endonuclease
- _____ cDNA
- _____ COS sites
- _____ Retrovirus
- _____ Plasmid
- _____ cDNA library
- _____ Sticky ends
- _____ Genomic library

Descriptions

- (1) Cleaves mRNA at specific sites.
- (2) Cleaves double-stranded DNA at specific sites.
- (3) Circular genetic element that can replicate in *E. coli* cells.
- (4) Used to clone DNA in mammalian cells.
- (5) Formed from action of reverse transcriptase.
- (6) Collection of DNA sequences within plasmid vectors representing all of the mRNA sequences derived from a cell.
- (7) Produced by action of certain DNA-cleaving enzymes.
- (8) Used to clone very large DNA sequences.
- (9) Used to introduce larger-than-normal DNA fragments in λ -phage vectors.
- (10) Collection of λ clones that includes all the DNA sequences of a given species.